# External Reviewer Packet

Adam Okulicz-Kozaryn

Tuesday 28th April, 2026    16:13

## Contents

ADAM OKULICZ-KOZARYN [updated: 2026:4]

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https://theaok.github.io
http://scholar.google.com/citations?hl=en&user=pz9RYIoAAAAJ

## EMPLOYMENT

Associate Professor of Public Policy, Rutgers University, the State University of New Jersey, Camden, 2018-??

Assistant Professor of Public Policy, Rutgers University, the State University of New Jersey, Camden, 2012-18

Research Scientist at Texas Schools Project (2010-11) and Clinical Assistant Professor, UT-Dallas, 2010-12

## EDUCATION

Post-doctoral Fellow, Institute for Quantitative Social Science, Harvard University, 2008-10

PhD Public Policy and Political Economy, The University of Texas at Dallas, 2008

MS Corporate and Economic Policy, Poznan University of Economics (Poland), 2003

BS Banking and Finance, Poznan School of Banking (Poland), 2001

## MAJOR PUBLICATIONS (all on Google Scholar; publications with students listed under service; AOK=Adam Okulicz-Kozaryn; click text for PDF)

2026 Golden L, Valente RR, AOK, Mikhaeil E, Do Unpredictable and Unstable Work Hours Reduce Subjective Wellbeing? Social Indicators Research

Urbanization and Happiness. A Research Agenda for Happiness. Why we want to know, what we know, and what we want to know additionally. Editors: Jan Ott and Ruut Veenhoven. Edward Elgar Publishing

2025 AOK, Martinez L Colombia: Unlivable but Happy. Fool's Paradise? (No, a Real Paradise, Better than the US) Applied Research in Quality of Life

2024 D'Italia M, AOK, Constructing General Human Agency Indicators (GHAIs) and a General Personal Agency Scale (GPAS) Social Indicators Research

Urban Regrets (Unhappy Metros: Satisfaction With Life Scale (SWLS)) Heliyon

2023 Magdy Zaki E, AOK, Valente RR, Subjective Well-being and Urbanization in Egypt Cities

AOK, Everett B, Magdy Zaki E, Happiness is In The Air if It Grows (Growing Places are Happier than Shrinking ones) Urban Affairs Review

2022 Unhappy Metros: Panel Evidence Applied Research in Quality of Life

AOK, Valente R, Misanthropolis: Do Cities Promote Misanthropy? Cities

2021 AOK, Valente R, Urban unhappiness is common Cities

AOK, Morawski L, A similar effect of volunteering and pensions on subjective wellbeing of elderly Economics and Sociology

2020 Valente R, AOK, Religiosity and Trust: Evidence from the United States Review of Religious Research

AOK, Valente R, The Perennial Dissatisfaction of Urban Upbringing Cities

AOK, Morawski L, Effect of volunteering and pensions on subjective wellbeing of elderly–are there cross-country differences? Applied Research in Quality of Life

2019 Are we happier among our own race? Economics and Sociology

AOK, Altman M, The Energy Paradox: Energy Use And Happiness Applied Research in Quality of Life

2018 AOK, Valente R, No Urban Malaise for Millennials Regional Studies

2017 AOK, Valente R, Livability and Subjective Wellbeing Across European Cities Applied Research in Quality of Life

Strzelecka M, AOK, Is tourism conducive to residents' social trust? Evidence from large-scale surveys. Tourism Review

AOK, Valente R, Life Satisfaction of Career Women and Housewives Applied Research in Quality of Life

AOK, Valente R, City Life: Glorification, Desire, and the Unconscious Size Fetish, ed. by I Kapoor, in Psychoanalysis and the Global, University of Nebraska Press

AOK, Golden L, Happiness is flextime Applied Research in Quality of Life

2016 AOK, Strzelecka M, Happy Tourists, Unhappy Locals Social Indicators Research

Happiness Research for Public Policy and Administration Transforming Government: People, Process and Policy

Unhappy Metropolis (When American City Is Too Big) Cities

AOK, Mazelis J M, More Unequal in Income, More Unequal in WellBeing Social Indicators Research

AOK, Mazelis J M, Urbanism and Happiness: A Test of Wirth's Theory of Urban Life Urban Studies

2015 Freedom and Life Satisfaction in Transition Society and Economy

AOK, Nash T, Tursi N O, Luxury Car Owners Are Not Happier Than Frugal Car Owners International Review of Economics

The More Religiosity, the Less Creativity Across US Counties Business Creativity & the Creative Economy

Income Inequality and Wellbeing Applied Research in Quality of Life

!!: `this book is excluded from the promotion packet (this is a popular science book)` Happiness and Place: Cities v Nature. Why Life is Better Outside of the City? Palgrave Macmillan

2014 AOK, Holmes IV O, Avery D R, The Subjective Well-Being Political Paradox: Happy Welfare States and Unhappy Liberals Journal of Applied Psychology

Winners and Losers in Transition: Preferences For Redistribution and Nostalgia For Communism In Eastern Europe Kyklos

'Freedom From' and 'Freedom To' across countries Social Indicators Research

Natural Sprawl Administration & Society (Disputatio Sine Fine section) (section of the journal reviewed by editor only)

2013 Cluttered Writing: Adjectives and Adverbs in Academia Scientometrics

City Life: Rankings (Livability) vs Perceptions (Satisfaction) Social Indicators Research

2012 Does Religious Diversity Make Us Unhappy? Mental Health, Religion & Culture

Income and Well-being across European Provinces Social Indicators Research

2011 Berry B J L, AOK, An Urban-Rural Happiness Gradient Urban Geography

Geography of European Life Satisfaction Social Indicators Research

Europeans Work To Live and Americans Live To Work (Who is Happy to Work More: Americans or Europeans?) Journal of Happiness Studies

2010 Religiosity and Life Satisfaction Across Nations Mental Health, Religion & Culture

2009 Berry B J L, AOK, Dissatisfaction with City Life: A New Look at Some Old Questions Cities

EXTERNAL GRANTS

190k PLN (∼$50k) PI, "The effect of social transfers and social capital on happiness of elderly." Vistula University, Warsaw Poland, Polish National Science Foundation (Narodowe Centrum Nauki NCN) 2016/21/B/HS4/03058 .. 2017-2019

$1.5k Consultant (summer salary), "Health information inequity in South Jersey-Vineland" South Jersey Institute for Population Health (SJIPH) Catalyst Grant `https://sjiph.org/projects/health-information-inequity-in-south-jersey-v` 2026

INTERNAL GRANTS (AWARDS and FELLOWSHIPS)

$12k faculty advisor, Rutgers Camden chancellor's dissertation completion award for Marlo Rossi .................... 2026

$30k faculty advisor, Rutgers Camden urban innovation fund grant to Ojobo Agbo Eje ........................ 2024-2026

$15k PI, Rutgers Camden chancellor's interdisciplinary research collaboration grant .......................... 2023-2024

$1k Rutgers-Camden Research Council Grant Award .................................................. 2016-2017

$1k Rutgers-Camden Digital Teaching Fellow ........................................................ 2016-2017

$4k Rutgers-Camden Rand Institute Faculty Fellow .................................................. 2013-2014

$1k Rutgers-Camden Civic Engagement Faculty Fellow ................................................ 2013-2014

TEACHING (note curriculum development: [developed/designed] or [re-designed])

2025 recipient of department's Jay A Sigler award for "outstanding service and dedication to students" (awarded by students); the plaque says "this award is given to the faculty member who exhibits a commitment to student's academic development and personal growth."

GIS for data science `https://theaok.github.io/gisPy/` [developed/designed] .................... 2023-present

Data Processing (data management for data science) `https://theaok.github.io/datManPy` [developed/designed] 2023-present

Data Visualization `https://theaok.github.io/vis` [developed/designed] ......................... 2023-present

Coping with Extremes: Ecological resilience & human well-being [innovative class co-taught with biology dept] [co-developed/designed] ... 2019

Community & WellBeing (Happiness and Place) [developed/designed] ... 2018-present

Data Management [developed/designed] ... 2016-present

GIS for Public Sector/Urban mapping [developed/designed] ... 2012-present

Quantitative Methods 2 [re-designed] ... 2013-2017, 2024

Research Methods [re-designed] ... 2012-present

Introduction to Regional and Economic Development [re-designed] ... 2012-2014

Quantitative Methods 1 [re-designed] ... 2012-2013

## SERVICE

### SERVICE TO THE FIELD/PROFESSION:

Review papers (approx 3-4/mo or 36-48/year as of Jan 2026; "top journals" underlined; starting in 2022: last 2 digits of year of review shown):

- <informal/for a colleague> [24]
- Acta Sociologica
- Applied Economics [23,23]
- Applied Research in Quality of Life [23,23,23,24,24,24,24,26]
- Basic and Applied Social Psychology [22]
- BMC Psychology [26]
- BMJ Open
- British Journal of Political Science [25]
- Cities [22,23,23,23,23,23,24,24,24,24,24,25,25,25,25,25,25,25,25,25]
- City, Culture and Society
- City, Territory and Architecture [24]
- Cogent Arts & Humanities [23]
- Cognition and Emotion
- Communications Psychology [23]
- Comparative Economic Studies [26]
- Discover Public Health [25,26]
- Ecological Economics [24]
- Environment and Planning A
- Economics and Politics
- European Journal of Social Psychology
- European Journal of Political Economy [23,23,24]
- European Societies
- European Urban and Regional Studies [22,22]
- Frontiers in Psychology [23]
- Global Transitions [24]

- Humanities and Social Sciences Communications [23,23,23,24,24,25,25]
- Heliyon [23,24]
- International Journal of Comparative Sociology [23,24,24,25]
- International Journal of Environmental Research and Public Health
- International Journal of Psychology
- International Journal of Social Economics [22]
- International Journal of Wellbeing [25]
- International Perspectives in Psychology: Research, Practice, Consultation
- International Sociology
- Journal of Comparative Economics
- Journal of Contemporary Religion
- Journal of Economic Growth
- Journal of Economic Psychology
- Journal of Economic and Social Geography
- Journal of Personality and Social Psychology: Personality Processes and Individual Differences[16]
- Journal of Personality and Social Psychology: Attitudes and Social Cognition[26]
- Journal of Religion and Health
- Journal of Regional Science
- Journal of Happiness Studies [22,23,24,26,26,26]
- Journal of Hospitality and Tourism Management
- Journal of Public and Nonprofit Affairs
- Journal of Public Policy [23,23]
- Journal of the American Planning Association
- Journal of Urbanism
- Land Use Policy [23,24,24]
- Local Development and Society [25]
- Mental Health, Religion and Culture
- Millah: Journal of Religious Studies [25]
- Nature Communications [24]
- Open Psychology Journal
- Quality of Life Research
- Papers in Regional Science [22,22]
- PLOS ONE [22]
- Political Behavior [22]
- PNAS Nexus [24]
- Problems of Post-Communism [22]
- Psychology of Religion and Spirituality
- Public Administration Review [24,25]
- Regional Studies [22,22]
- Review of Political Economy
- Routledge (book publisher) [22]
- Rural Sociology [22]
- Urban Geography
- Urban Studies [25]

- Urban Affairs Review
- Utilities Policy [25,25]
- Science of the Total Environment
- Scientific Reports [24]
- Scienze Regionali Italian Journal of Regional Science
- Scientometrics [22,22]
- Semestre Economico (Universidad de Medellín) [22]
- Social Forces
- Social Psychological and Personality Science
- Social Indicators Research [as reviewer] [22,22,23,23,23,23,24,24,24,24,25,25,25,25,25]
- Social Indicators Research [as co-editor][1] [ 25, 25, 25, 25, 25 ]
- Social Science And Medicine [24,24,25,25,25,25,25]
- Social Science Research [23]
- Social Sciences & Humanities Open [23]
- Sociological Spectrum
- Sociology of Religion
- Sustainability
- Sustainable Cities and Society
- Sustainable Futures [25]
- The Lancet Planetary Health (IF $>$10)
- Third World Quarterly [22,23]
- Wellbeing, Space & Society [22,25]

Reviewed books chapters:
- 3 chapters in "A Research Agenda for Happiness. Why we want to know, what we know, and what we want to know additionally." Editors: Jan Ott and Ruut Veenhoven. Edward Elgar Publishing [26,26,26]

Reviewed books for:
- Routledge [22]
- Cambridge U Pr [23]

Served on editorial boards:
- Social Indicators Research: Associate Editor / Co-Editor[2] Jan 2024-??
- Political Behavior Jan-Dec 2019

Reviewed grant proposals:
- Polish National Science Foundation (NCN)
- Israel Science Foundation (ISF)

SERVICE TO STUDENTS:

Publications with students (all students unless indicated: "[not student]"):

[1] As a co-editor of Social Indicators Research, conducted through reviews (printed out, marked up, etc). Note: this is different from regular editing, where I just skim through the paper or read some parts only; and hence this extra work is listed here separately from editing as reviewing [starting oct2025 each such review marked with 2 last digits of year]

[2] This is not just sitting on the board of the journal and having an occasional board meeting, I am editing papers every week spending as of 2026 $1 + hrs/wk$ (earlier as i was less experienced it was $2 + hrs/wk$) on skimming through the submitted manuscripts, inviting reviewers, reading reviews, and making editorial decisions.

10. Golden L [not student], Valente RR [not student], AOK, Mikhaeil E, Do Unpredictable and Unstable Work Hours Reduce Subjective Wellbeing? Social Indicators Research 2026
9. AOK, L Martinez [not student], E Mikhaeil; Happy Colombia, Unhappy Bogota (urban-rural happiness gradient in Colombia) Wellbeing, Space and Society 2026
8. E Mikhaeil, AOK; Well-Being and the authoritarian welfare state: sector of employment and human happiness in Arab countries; Public Organization Review 2025
7. MJ D'Italia, AOK; Constructing General Human Agency Indicators (GHAIs) and a General Personal Agency Scale (GPAS); Social Indicators Research 2025
6. E Mikhaeil, AOK; Public-private job satisfaction differential: The case of Egypt; Public Organization Review 2025
5. AOK, BK Everett, E Mikhaeil; Happiness is In The Air if It Grows (Growing Places are Happier than Shrinking ones); Urban Affairs Review 2024
4. E Mikhaeil, AOK; Subjective well-being and urbanization in Egypt; Cities 2024
3. S Deb, AOK; Exploring the association of urbanization and subjective well-being in India; Cities 2023
2. FA Abarghuie, R Javadi, E Mirabi, AOK; Experience of urban thermal pleasure: A qualitative study to the understanding of thermal-scape quality; Cities 2022
1. R Stansfield [not student], E Aaronson, AOK; Police use of firearms: Exploring citizen, officer, and incident characteristics in a statewide sample; Journal of criminal justice 2021

Supervised dissertations for [year of graduation, first job]:
9. Isabelle Surielow [starting dissertation hours in Fall 2026]
8. Wei Chen [28??]
7. Erick W Udogu [27??]
6. Douglas N Zacher [27??]
5. Marlo Rossi [26??]
4. Michael D'Italia [24, RU-Camden] PDF
3. Shibin Yan [24, Walter Rand Institute for Public Affairs] PDF
2. Aditi Manke [21, Virginia Tech] PDF
1. Shourjya Deb [20, SIPRI] PDF

Supervised dissertations for less than the whole duration:
1. Helena Cabezas [Fall 25 only]

Served on dissertation committees:
10. Jason Schweitzer (Prevention Science) [27??]
9. Daniel Assamah [27??]
8. Monty Sanders [26??]
7. Brian Everett [27??]
6. Kate Cruz [22]
5. Charles Ian Hugh Forsyth (University of Otago NZ) [22]
4. Straso Jovanovski [19]
3. Spencer Clayton [18]
2. Rasheda Weaver [17]
1. Chris Wheeler [17]

Served on dissertation committees for less than the whole duration:
1. Yanan Li [stopped serving just before the proposal in 2023]

Supervised data science master thesis for [year of graduation]:

6. Nalluri Chandrahas [26]
5. Bhavyani Dodda [25]
4. Thanvitha Reddy Munagala [25]
3. Pavan Satya Venigalla [25]
2. Het Rakeshbhai Patel [25]
1. Raman Srinivas Naik [25]

Served on master thesis committees in social science:
2. Ethan Aaronson (crim) [20]
1. Rachel Wagner (psy) [20]

Independent/Directed Study [starting in 2023 counts shown (2014-present)]:
- fa25 x1
- sp25 x4
- fa24 x7
- sp24 x2
- fa23 x1

Researchers/Predocs/Postdocs/Lab Members/Visiting Scholars:
- hired 2 data scientists (Sep 2025-Jul 2026)
- planning to host Visiting Scholar, Professor Leonie C. Steckermeier ........ Fall 2026

DEPARTMENTAL AND CAMPUS-WIDE SERVICE:

At Rutgers-Camden served on the following departmental committees:
- Awards/Events, chair ........ 2013-16, 2021-22; 2023-25
- Awards/Events, member ........ 2012-13, 2019-20, 2023
- Strategic Planning, member ........ 2019-20
- PR/Website, chair ........ 2014-16
- PR/Website, member ........ 2012-13
- PhD, member ........ 2014-21
- Methods Qualifying Exam, member ........ 2012-21

Other Rutgers-Camden departmental service:
- held dissertation seminars for a group of doctoral candidates (in addition to one-on-one) ........ Fall 2025
- led Brown Bag Seminars ........ 2018-19; 2023-24
- As co-editor of a journal organized "Publish or Perish" workshop ........ Oct 2024

At Rutgers-Camden served on the following campus-wide committees:
- AI committee ........ Fall 2025
- Advancement and Promotions (A&P), member/ad hoc chair ........ 2020-25
- Advancement and Promotions (A&P), member ........ 2018-19
- Faculty Senate ........ 2018-25
- Faculty Life, chair ........ 2019-22
- Faculty Life, member ........ 2014-15

Other Rutgers-Camden campus-wide service:
- led creation and maintenance of data science lab and MS in data science and associated certificate programs 2018-22

- served as director for data science certificate: https://graduateschool.camden.rutgers.edu/graduate-programs/#certificates .......... 2022-??
- organized Geographic Information System (GIS) workshop attended by faculty from across the campus https://theaok.github.io/shortGIS/ .......... 2017

COMMUNITY SERVICE

provided data management for:
- Ohio Business Round Table (contracted by Miami University, Ohio; Contract Amount: $7,000) .......... 2022
- 2018 Michigan Economic Competitiveness Study, Michigan Chamber of Commerce (contracted by Northwood University, Michigan; Contract Amount: $5,000) .......... 2018

assisted following Rutgers-Camden students in their community projects as a faculty advisor:
- data science MS Ojobo Agbo Eje: "Camden Health Equity Data Visualization Initiative" project: [click for news item] [hired/supervised 4 students, budget $30k] .......... 2024-26
- public affairs PhD Zach Chengcheng Yue at Walter Rand Institute: Covid-19 in Atlantic County ... Spring-Fall 2025
- public affairs PhD Marlo Rossi at Walter Rand Institute: Abortion in South Jersey; and other projects on abortion: [click for news item] .......... Fall 2025-Spring 2026
- MPA Olorunfunmi Adebajo's internship at the Camden County Community Development Office's Homelessness Assistance Unit: [click for news item], .......... 2024

other community service:
- made GIS maps dashboard (containing work by students in my GIS class): available on campus or via RU VPN only: https://maps-dppa.camden.rutgers.edu/index.php?search=map&title=Special%3ASearch&profile=default&fulltext=1 .......... 2019-23
- assisted Hunterdon Central Regional High School student: happiness paper class project Fall 2025-Spring 2026
- organized Geographic Information System (GIS) workshop for The Walter Rand Institute for Public Affairs https://theaok.github.io/shortGIS/vShortGeoda.pdf .......... 2019

MOST FREQUENT RECENT CONFERENCES

LASA (Latin American Studies Association) 2024

AAG (American Association of Geographers): 2015, 2016, 2017, covid19/none,

ISQOLS (International Society for Quality of Life Studies): 2015, 2017, 2018, covid19/none, 2023, 2024 (webinar), 2026 (regional)

ERSA (European Regional Science Association): 2015, 2017, 2018, covid19/none, 2023

AFFILIATE SCHOLAR
- Australian Centre on Quality of Life (2016-present)
- CURE: Center for Urban Research and Education, Rutgers-Camden (2012-18)
- EHERO: Erasmus Happiness Economics Research Organization, Erasmus, the Netherlands (2013-16)
- Vistula University, Warsaw, Poland (2015-18)

# 3 papers: various social indicators

## 3.1 The Subjective Well-Being Political Paradox: Happy Welfare States and Unhappy Liberals (Okulicz-Kozaryn et al. 2014)

Journal of Applied Psychology
2014, Vol. 99, No. 6, 1300–1308


0021-9010/14/$12.00 http://dx.doi.org/10.1037/a0037654

RESEARCH REPORT

# The Subjective Well-Being Political Paradox: Happy Welfare States and Unhappy Liberals

Adam Okulicz-Kozaryn and Oscar Holmes IV
Rutgers, The State University of New Jersey

Derek R. Avery
Temple University

Political scientists traditionally have analyzed the effect of politics on subjective well-being (SWB) at the collective level, finding that more liberal countries report greater SWB. Conversely, psychologists have focused primarily on SWB at the individual level and shown that being more conservative corresponds in greater SWB. We integrate the theoretical foundations of these 2 literatures (e.g., livability and system justification theories) to compare and contrast the effects of country- and individual-level political orientation on SWB simultaneously. Using a panel of 16 West European countries representative of 1,134,384 individuals from 1970 to 2002, we demonstrated this SWB political paradox: More liberal countries and more conservative individuals had higher levels of SWB. More important, we explored measurement as a moderator of the political orientation–SWB relationship to shed some light on why this paradox exists. When orientation is measured in terms of enacted values (i.e., what the government actually does), liberalism corresponds in higher SWB, but when politics is measured in terms of espoused values (i.e., what individuals believe), greater conservatism coincided in higher SWB.

*Keywords:* subjective well-being, politics, conservatism, liberalism

Happiness or subjective well-being (SWB), defined as “people’s cognitive and affective evaluations of their lives” (Diener, Diener, & Diener, 1995, p. 851) has long been of interest to scholars and philosophers across various disciplines. Henceforth, we use the term *SWB* to encompass the related constructs of happiness and life satisfaction but sometimes use the terms *happiness* and *SWB* interchangeably to be consistent with the literature we review. Within the social psychology literature, for instance, scholars have examined pertinent antecedents of SWB, such as national culture and personality (Diener, Suh, Lucas, & Smith, 1999; Steel & Ones, 2002). Within the political science literature, the idea that governments should promote well-being dates back centuries to Adam Smith, Jeremy Bentham, and even ancient philosophers (Kahneman, Wakker, & Sarin, 1997; Nussbaum, 2005; Rasmussen, 2006). Furthermore, economists have focused on how income impacts citizens’ well-being (Alesina, Di Tella, & MacCulloch, 2004; Clark, Frijters, & Shields, 2008; Easterlin, 1974; Frey & Stutzer, 2000), and organizational scholars’ interest in SWB dates back prior to the Hawthorne studies and extends to topics such as job satisfaction, job engagement, and organizational commitment (Ashkanasy, 2011; Fisher, 2010; Hackman & Oldham, 1980; Hassard, 2012; Judge, Locke, Durham, & Kluger, 1998; Maier & Brunstein, 2001). Despite such widespread attention across disciplines, SWB is usually approached at different levels of analysis without considering how variables may impact SWB differently at each level. As Ashkanasy (2011) pointed out, there can be surprisingly different relationships between SWB and its predictors, depending on the level of analysis.

As an example, the effects of politics on SWB have been studied at both the country and the individual levels. Political scientists have found that societies led by leftist or liberal governments (also referred to as welfare states) report the highest SWB (Bok, 2010; Radcliff, 2001), yet they often overlook the possible connection between individual-level political orientation and SWB. A *welfare state* is a term used to describe a government on the liberal end of the political continuum wherein the state protects and promotes equity and public responsibility and makes broader investments (e.g., health care, education, social services) to provide its citizens with more protections from social, economic, and political hardships (Green-Pedersen, 2004). In contrast, psychologists have found that more conservative individuals report higher SWB than more liberal folks (Napier & Jost, 2008) but largely fail to consider that politics at the country level also could affect SWB. As such, there is little consensus across the literatures on how politics affect

This article was published Online First August 25, 2014.

Adam Okulicz-Kozaryn, Department of Public Policy and Administration, Rutgers, The State University of New Jersey; Oscar Holmes IV, School of Business, Rutgers, The State University of New Jersey; Derek R. Avery, Department of Human Resource Management, Temple University.

We thank the following people for their constructive feedback on previous versions of this article: Herman Aguinis, Sabrina Volpone, Julie Hancock, Samantha Paustian-Underdahl, Allison Rossetti, and Emilija Djurdjevic.

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SWB at both levels of analyses simultaneously. Generating consensus on how politics affect SWB at both levels is important not only in terms of SWB itself but also because variables that lead to lower SWB are related to a number of organizational issues, such as greater absenteeism and turnover (McKee-Ryan & Harvey, 2011; Mullainathan & Shafir, 2013), increased health problems (Dooley et al., 1996; Wilkinson & Pickett, 2006, 2010), IQ decrements (Mullainathan & Shafir, 2013), and lower job satisfaction and organizational commitment (Ensher et al., 2001; McKee-Ryan & Harvey, 2011; Simon et al., 2010).

The purpose of our study is to reconcile the seemingly disparate findings between the literatures on how politics affect SWB at the country and individual level. This research not only responds to the call for multidisciplinary and multilevel examination of SWB (Ashkanasy, 2011; Judge & Kammeyer-Mueller, 2011) but also further illustrates how research implications can be misguided when taken from only one level of analysis (Earley & Mosakowski, 2002; Klein & Kozlowski, 2000). Drawing on livability (Veenhoven, 2008; Veenhoven & Ehrhardt, 1995) and system justification theories (Jost & Banaji, 1994), we use a panel of West European market democracies to shed light on why the relationship between politics and SWB ostensibly has competing relationships at the country and individual levels.

## Background and Theoretical Development

### Livability Theory and Collective Political Orientation

Governments impact the interests of their citizenry. Livability theory posits (Veenhoven & Ehrhardt, 1995; Veenhoven, 2008) and empirical research supports (Appleton & Song, 2008; Bjørnskov, Dreher, & Fischer, 2008; Judge, Piccolo, Podsakoff, Shaw, & Rich, 2010; Okulicz-Kozaryn, 2012) the idea that improvements in one's living conditions can lead to greater well-being. As such, SWB is expected to increase if governmental policies produce collective provisions that create livable conditions satisfying citizen needs and, to some extent, their desires (Radcliff, 2001; Veenhoven & Ehrhardt, 1995). The conundrum for governments is that most consist of differing political parties whose divergent perspectives often lead them to disagree with what the collective provisions should be and how they should be allocated to citizens (Powell & Whitten, 1993; Radcliff, 2001; van Oorschot, 2006). Whereas the collective political orientation of countries falls on a continuum, countries can be and often are populated by a combination of citizens on both extremes of the spectrum. Collectively, liberal or left-wing governments occupy one end and conservative or right-wing governments occupy the other end of the political continuum (Hibbs, 1977).

On the basis of livability theory, political scientists tend to equate liberal or left-wing governments with increased well-being (Álvarez-Díaz, Gonzalez, & Radcliff, 2010; Bok, 2010; Pacek & Radcliff, 2008; Radcliff, 2001), believing that the lack of welfare and over reliance on the free market will result in lower SWB. For example, Esping-Andersen (1990) argued that "the market becomes to the worker a prison within which it is imperative to behave as a commodity in order to survive" (p. 36). In this regard, lower SWB can be expected when workers perceive themselves to be commodities (Easterlin, 2009). Additionally, Lane (2000) contended that because markets are indifferent to the fate of individuals, SWB should be less when there is no safety net protecting citizens from economic hardships. More liberal governments tend to take greater efforts to ensure better health care, public education, and pension systems and do more to protect their citizens from the externalities (e.g., unemployment) of the market economy (Bok, 2010; Scruggs & Allan, 2006). They also actively seek to reduce income inequality that is commonly linked to poor health, crime, and overall misery among citizens (Wilkinson & Pickett, 2006). It is important to note that no government can fully eliminate adverse conditions, but more liberal governments tend to do more to shield citizens by providing a buffer against these hardships and their impact on SWB (Helliwell & Huang, 2008).

Considering the far-reaching negative factors that contribute to low levels of SWB, it is no wonder why many governments have been interested in alleviating these negative factors through the collective provisions and policies that they enact. In fact, the happiest countries in the world, the Scandinavian countries, have lower income inequality and, politically, are considered more liberal countries. On the basis of evidence in support of livability theory, we hypothesize that SWB will be higher in more liberal countries.

> *Hypothesis 1:* Citizens of more liberal countries will report levels of SWB than are higher than those reported in more conservative countries.

### System Justification Theory and Individual Political Orientation

Although livability theory predicts that country-level liberalism enhances SWB, system justification theory predicts a different pattern altogether for individuals. System justification theory posits that people generally tend to (a) support and rationalize the status quo and (b) believe that the current political, social, and economic state of affairs has been realized from relatively fair and legitimate processes (Jost & Banaji, 1994; Jost & Hunyady, 2003; Jost, Pelham, & Carvallo, 2002). The theory also contends that people are hesitant to call attention to and work to disrupt inequities because of their desire to see themselves and the world as generally fair and meritocratic (Benabou & Tirole, 2006; Jost & Banaji, 1994). System-justifying ideology results partly because it provides individuals with the cognitive structure to cope with uncertainty and threats (Cheung, Noel, & Hardin, 2011; Laurin, Kay, & Shepherd, 2011). As such, individuals are motivated to believe in the fairness, legitimacy, and meritocracy of the surrounding sociopolitical system (i.e., maintain structure) because exposing its flaws and contradictions calls into question their own outcomes, reduces their ability to predict cause-and-effect relationships, and detracts from their understanding of the world, all of which introduce instability and erode structure (Jost & Hunyady, 2003; Jost, Kivetz, Rubini, Guermandi, & Mosso, 2005).

In their recent examination of why "conservatives are happier than liberals," Napier and Jost (2008, p. 565) concluded that an important factor related to system justification endorsement is one's individual political orientation. They found that more conservative individuals were more likely to endorse system-justifying beliefs and, more surprisingly, to report higher SWB (Napier & Jost, 2008). This is the case because those who are more politically conservative more readily accept and endorse the legitimacy of their social, political, and economic reality in spite of any adverse conditions within the nation in which they live. This acceptance and endorsement heuristic leads to the belief that individuals are responsible for their outcomes and, ultimately, get what they deserve (i.e., rationalize the status quo). This desire to rationalize the status quo serves as a "*palliative function* of system-justifying ideology" (Jost & Hunyady, 2003; Napier & Jost, 2008, p. 565; italics in original). Thus, Napier and Jost argued that more conservative people report higher SWB because they ignore or rationalize away many of the factors that otherwise might negatively influence their SWB (e.g., unemployment, inequality, disparate health care and education). In support of this notion, evidence (e.g., DiTomaso, 2013) indicates that those with more conservative beliefs were much more likely to attribute unemployment to laziness than to structural issues within the sociopolitical system. Conversely, those holding more liberal views report lower SWB because they benefit less (if at all) from this palliative function because, at least in part, to greater empathy for those experiencing adverse conditions (Graham, Haidt, & Nosek, 2009; Napier & Jost, 2008; Savani & Rattan, 2012). On the basis of system justification theory, we hypothesize the following:

> *Hypothesis 2:* Those with more politically conservative views will report higher SWB.

## Juxtaposing the Two Perspectives: Measurement as a Moderator

So how can it be that political liberalism is tied to greater well-being at the country level (Hypothesis 1) but lesser well-being at the individual level (Hypothesis 2)? We contend that this apparent paradox between the literatures stems from differences in the constructions of political orientation across disciplines and levels of analysis. On the one hand, more macro perspectives on political liberalism (e.g., political science, sociology, public policy) tend to focus on what governments provide or do not provide for their citizens. Countries offering greater state-supported housing, health care, education, and remediation of income inequality are deemed more liberal, whereas those that do not are labeled more conservative (Álvarez-Díaz et al., 2010). Likewise, scholars rate governmental political orientation largely on the basis of interest group ratings of legislators' behavioral support (i.e., voting, advocacy) for liberal and conservative policies (Berry, Fording, Ringquist, Hanson, & Klarner, 2010). On the other hand, more micro approaches (e.g., psychology, behavioral economics) center on individual beliefs about what a government should or should not do for the people it serves (Jost, Federico, & Napier, 2009). Those believing governments should be more involved in the provision of social services are tabbed as more liberal, whereas those who prefer a more hands-off approach to government assistance are considered more conservative.

Although these two methods of defining political orientation might appear similar at first glance, a closer look illustrates a key difference between the two. Namely, the macro approach focuses more on what governments actually do while the micro perspective centers on beliefs about what governments should do. This difference resembles that of the distinction between espoused and enacted values in the organizational literature (Argyris & Schon, 1978). The former pertains to things that an organization (or its membership) purports to endorse whereas the latter describes the ideals embodied by its actions. Applied to the current context, it appears that while scholars with a micro SWB perspective have explored the impact of espoused political orientation, scholars with a macro perspective have devoted their attention to enacted political orientation.

Recognizing this division in the theoretical and empirical construction of political orientation across disciplines provides a potential reconciliation for the seemingly paradoxical findings produced by micro and macro SWB research. Inconsistent results often call for greater attention to prospective moderators that determine the nature of a relationship between two variables. In this case, it appears that measurement type may be such a moderator. When political orientation is determined by what countries actually do (enacted), the tenets of livability theory should apply and the nature of the liberalism–SWB linkage should be positive because the measured government actions (or lack thereof) are influencing individuals' lives. Conversely, when espoused political orientation is the focus, system justification theory's prediction of a negative relationship between liberalism and SWB should hold true because what is being measured is an aggregation of individual worldviews as opposed to actual governmental action.

> *Hypothesis 3:* The country-level political orientation effect on citizen's SWB is moderated by the type of measurement. The effect is positive when enacted political orientation was measured but negative when espoused political orientation was measured.

# Method

## Sample and Data Collection

Archival data served as the sample for this study. All individual-level (Level 1) variables were obtained from the Mannheim Eurobarometer data set (ICPSR 4357), which is a collection of Eurobarometers conducted each year in Europe and commonly used in political science research. This data set contains 1,134,384 individuals nested within Western European countries between the years of 1970 to 2002. About 2,000 people per country were surveyed in earlier years and up to 4,000 people per country were surveyed in more recent years. The 16 European countries included are France, Belgium, the Netherlands, Germany, Italy, Luxembourg, Denmark, Ireland, the United Kingdom, Greece, Spain, Portugal, Finland, Sweden, Austria, and Norway. Country-level (Level 2) variables were obtained from the World Bank and the Organisation for Economic Co-operation and Development (OECD). Two (i.e., decommodification and equality) were produced by political scientists (Scruggs & Allan, 2006) and economists (Deininger & Squire, 1996). Great Britain in the Eurobarometer data is matched with the United Kingdom in the OECD and Scruggs and Allan's data sets.

### Individual-level (Level 1) measures

**SWB.** We measured SWB using the following item from the Mannheim Eurobarometer: "On the whole, are you very satisfied, fairly satisfied, not very satisfied, or not at all satisfied with the life you lead?" The scale ranges from 1 = *not at all satisfied* to 4 =

*very satisfied*. Previous research has found this measure to be reliable and valid (Di Tella & MacCulloch, 2006; Myers, 2000) and closely correlates with objective measures of well-being, such as brain activity measures (Layard, 2005).

**Political orientation.** Political orientation, the degree to which one is more conservative (right) or liberal (left), was measured using the following item: "In political matters people talk of 'the left' and 'the right,' how would you place your views on this scale?" The scale ranges from $1 =$ *left* to $10 =$ *right*. Research has found that self-placement on the right–left self-placement scale has "good test–retest reliability and strong predictive validity" (Napier & Jost, 2008, p. 568; see also Jost, 2006; Jost et al., 2009). Scores on this measure were (a) reversed so that higher values indicate greater liberalism and (b) aggregated to the country level and also included as a Level 2 measure.

### Country-Level (Level 2) Measures

To enhance the robustness of our results, we triangulate the measurement for country-level political orientation using three measures: decommodification, welfare spending, and equality, where the higher the value, the more liberal, or, alternatively speaking, the closer the country to the "welfare state ideal."

**Decommodification.** The decommodification index was retrieved from Scruggs and Allan's (2006) research and captures three dimensions: (a) the ease of access to welfare, (b) welfare value, and (c) breadth of coverage. The key welfare programs captured in this index are pensions, sickness benefits, and unemployment compensation (Cronbach's $\alpha = .55$).

**Welfare spending.** Welfare spending is measured as the total public social expenditure as a percentage of gross domestic product (GDP). This variable, which can range from 0 to 100, was retrieved from the Welfare Expenditure Report produced by the OECD.

**Equality.** The Gini index, which ranges from 0 (*perfect equality*) to 1 (*perfect inequality*), is used commonly in economics and sociology research to capture societal inequity. To make this measure more consistent with the other Level 2 measures, we reverse coded it and multiplied it by 100 so that equality scores ranged from 0 to 100. The Gini index was retrieved from Deininger and Squire's (1996) research.

### Data Analyses

Because the participants in our study were nested within countries and years, their SWB scores are likely not independent and ordinary least squares is inappropriate. We have data about countries over time, but for reasons of parsimony and computational convenience, we model it using a country–year random intercept, as opposed to a three-level model. A one-way analysis of variance (ANOVA) indicated that the variation in SWB attributable to country–year was statistically significant ($ICC = .14$), $F(333, 631834) = 290.18$, $p < .001$. Accordingly, we used multilevel modeling (MLM) with the "xtmixed" command in Stata statistical software (Rabe-Hesketh & Skrondal, 2005). Moreover, because we incorporated variables at both the country (i.e., welfare spending, decommodification, equality) and the individual levels (i.e., SWB and political orientation), we used random intercept models, a specific type of MLM, to test all hypotheses (Kreft & de Leeuw, 1998). Like prior researchers (e.g., Park & DeShon, 2010; Porter, Webb, & Gogus, 2010), we present Rosenthal and Rubin's (2003) $r_{\text{equivalent}}$ as an indicator of effect size for our MLM analyses.

## Results

Descriptive statistics and correlations are presented in Table 1. When modeling SWB, results tend to be similar in ordinal logistic and linear models (Ferrer-i-Carbonell & Frijters, 2004); hence, we used a linear model for ease of interpretation. To avoid multicollinearity, each model includes only one measure of political orientation at the country level (see Table 2 for a summary). Hypothesis 1 proposed that citizens of more liberal countries (i.e., welfare states) would report higher SWB than those in more conservative countries. In support of this hypothesis, we observed moderate to large effects of decommodification (.03, $p < .01$, $r = .60$), welfare spending (.03, $p < .01$, $r = .50$) and equality (.02, $p < .01$, $r = .35$) on SWB.

Table 1
*Individual- and Country-Level Descriptive Statistics and Correlations*

| Variable | *M* | *SD* | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|
| Individual level (Level 1) | | | | | | | |
| 1. Political orientation | 5.69 | 2.09 | — | | | | |
| 2. Subjective well-being | 3.06 | 0.75 | $-0.10^{**}$ | — | | | |
| Country level (Level 2) | | | | | | | |
| 1. Decommodification | 28.51 | 4.29 | — | | | | |
| 2. Welfare spending | 23.71 | 4.61 | $0.72^{**}$ | — | | | |
| 3. Equality | 69.37 | 4.77 | 0.10 | 0.18 | — | | |
| 4. Political orientation | 5.64 | 0.42 | $-0.08$ | $-0.00$ | $-0.26^{*}$ | — | |
| 5. Subjective well-being | 3.07 | 0.28 | $0.54^{**}$ | $0.48^{**}$ | $0.45^{**}$ | $-0.38^{**}$ | — |

*Note.* Political orientation is on a right-to-left continuum ranging from more conservative (1) to more liberal (10). At country level a unit of analysis is a country–year, that is, a given country observed in a given year. In country level (Level 2) panel, subjective well-being scores are aggregated to the country–year. Reported correlations are pairwise with about 300 country–year observations and only 77 in cases of equality because of missing data. At individual level, there are at least 400,000 observations and usually more than 700,000 observations.
$^{*}p < .05$. $^{**}p < .01$.

Table 2
*Summary of Multilevel Models Predicting Subjective Well-Being*

| Variable | Model 1 | Model 2 | Model 3 |
|---|---|---|---|
| Person level (Level 1): | | | |
| Intercept | $3.114^{**}$ (0.011) | $3.082^{**}$ (0.014) | $3.045^{**}$ (0.025) |
| Political orientation | $-0.029^{**}$ (0.001) | $-0.026^{**}$ (0.001) | $-0.033^{***}$ (0.001) |
| $r_{\text{equivalent}}$ | .09 | .08 | .10 |
| Country level (Level 2) | | | |
| Decommodification | $0.030^{**}$ (0.003) | | |
| $r_{\text{equivalent}}$ | .60 | | |
| Welfare spending | | $0.030^{**}$ (0.003) | |
| $r_{\text{equivalent}}$ | | .50 | |
| Equality | | | $0.020^{**}$ (0.006) |
| $r_{\text{equivalent}}$ | | | .35 |
| Political orientation | $-0.301^{**}$ (0.026) | $-0.229^{**}$ (0.036) | $-0.283^{***}$ (0.050) |
| $r_{\text{equivalent}}$ | .60 | .36 | .31 |
| *N* | 407,333 | 430,091 | 114,517 |
| Level 1 residual variance | .46 | .48 | .50 |
| Level 2 residual variance | .03 | .05 | .04 |
| Deviance | 843,007 | 908,862 | 246,296 |

*Note*. Numbers in parentheses are standard errors. Political orientation is on a right-to-left continuum ranging from more conservative (1) to more liberal (10). At the individual level, political orientation is centered within cluster. At the country level, all variables are grand-mean centered. We also calculate country–year mean of political orientation at Level 2. Intraclass correlation (1) for the null model is .14.
$^{**}$ $p < .01$. $^{***}$ $p < .001$.

Hypothesis 2 predicted that more conservative participants would report higher SWB than their more liberal counterparts. Because we centered political orientation within cluster at Level 1, it is noteworthy that its effects at Levels 1 and 2 are independent and the Level 1 variable indicates the participant's relative (to the mean) political orientation within their country. Regressing SWB onto political orientation revealed that both country-level (*b* ranged from $-.30$ to $-.23$, *p*s $< .01$, *r* ranged from .31 to .60) and individual-level (*b* ranged from $-.03$ to $-.03$, *p*s $< .01$, *r* ranged from .08 to .10) political orientation predicted SWB as expected in all three models (i.e., across the different enacted measures). Thus, we find support for Hypothesis 2.

Hypothesis 3 posited that measurement type moderates the effect of political orientation such that the effect is positive when enacted political orientation was measured but negative when espoused political orientation was measured. Because the proposed moderation involves testing differences in the relationships between the various indicators of political orientation and SWB, it cannot be tested by computing a product term between an independent variable and a moderator variable. Rather, the task is to determine whether the coefficients reported for country-level espoused and enacted political orientation differ significantly from one another. Toward this end, we used two tactics. First, Stata provides a post hoc function that uses a Wald test to compare the equivalence of any specified coefficients derived from the multilevel analysis. Second, we (a) computed standardized coefficients with confidence intervals for the models reported in Table 2 and (b) conducted a Monte Carlo simulation of 50,000 estimates of the differences between country-level espoused political orientation coefficient and each enacted political orientation coefficient. All of these results led to identical conclusions: Each of the enacted measures (decommodification, welfare spending, and equality) differed from the espoused measure at the country level (see Table 3 for a summary of these results). As Figure 1 illustrates, the effect of the espoused measure was negative, whereas the effects of the three enacted measures were positive, thereby supporting Hypothesis 3.

## Supplemental Analyses

Although not hypothesized, the multilevel nature of our data presents an opportunity to test for possible interactions among the different measures of political orientation. First, we analyzed cross-level interactions of political orientation at the individual and

Table 3
*Summary of Tests of Moderation by Measurement Type*

| Variable | Decommodification | Welfare spending | Equality |
|---|---|---|---|
| Enacted political orientation | $0.46^{**}$ [0.36, 0.56] | $0.49^{**}$ [0.36, 0.63] | $0.34^{**}$ [0.06, 0.63] |
| Espoused political orientation (Level 2) | $-0.45^{**}$ [$-0.55$, $-0.35$] | $-0.34^{**}$ [$-0.48$, $-0.20$] | $-0.42^{**}$ [$-0.62$, $-0.23$] |
| Difference | $0.91^{**}$ [0.75, 1.07] | $0.83^{**}$ [0.65, 1.03] | $0.76^{**}$ [0.50, 1.03] |
| $\chi^2$ | $164.23^{**}$ | $50.48^{**}$ | $38.02^{**}$ |

*Note*. Coefficients are standardized. Espoused political orientation is on a right-to-left continuum ranging from more conservative (1) to more liberal (10). Numbers in parentheses are 99% confidence intervals. Confidence intervals for differences were derived via Monte Carlo simulation.
$^{**}$ $p < .01$.

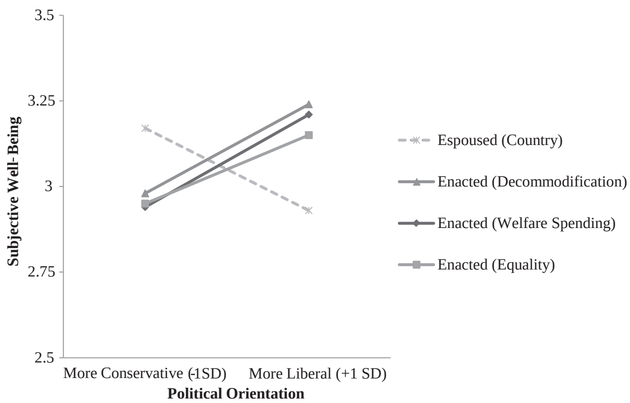


*Figure 1.* The moderating effect of measurement type on the political orientation–subjective well-being relationship.

country levels but found that none were significant with the exception of the interaction of political orientation (right–left) and welfare spending ($\gamma = .001$, $p = .02$). This indicated that the negative association between political orientation and SWB was slightly larger when welfare spending was lower ($b = -.03$, $p < .01$) than when it was higher ($b = -.02$, $p < .01$). Second, we tested interactions among the country-level (Level 2) variables and many were significant. For instance, the interaction between welfare spending and decommodification was positive ($\gamma = .01$, $p < .01$) and simple slopes indicated welfare spending related positively to SWB when decommodification was higher ($\gamma = .02$, $p < .01$) but negatively when decommodification was lower ($\gamma = -.04$, $p < .01$). We also performed a robustness check by including a host of controls commonly used in past research on SWB (age, income, employment status, marital status, per capita gross domestic product, unemployment, and inflation), which did not alter the pattern of results reported.

## Discussion

Integrating livability and system justification theories, we investigated how politics may lead to ostensibly inverse relationships with SWB at the country and individual levels. Consistent with livability theory, more liberal countries had higher SWB. Conversely, more conservative people reported greater SWB in line with system justification theory. To reconcile these paradoxical findings, we proposed and found that measurement moderated the relationship between political orientation and SWB at the country level. When measures capture what a country does (enacted political orientation), greater liberalism corresponds with higher SWB, but when measures tap what citizens believe (espoused political orientation), the pattern is the opposite. We now turn our attention to the implication of these findings.

### Implications for Research and Practice

Perhaps the most significant implication of this work is that it calls attention to a previously unacknowledged paradox within the existent multidisciplinary research on the association between politics and SWB. On the one hand, political scientists measure collective political orientation via enacted values or how countries govern their citizens. In this regard, countries providing greater safety nets are designated more liberal countries (i.e., welfare states). On the other hand, psychologists measure individual political orientation by measuring espoused values, or what individuals believe. In this regard, more conservative values help people rationalize the status quo, thereby buffering their SWB from the negative impact of any existing adverse societal conditions. We found that when individual political orientation is aggregated to the country level, more conservative countries also reported higher SWB. Thus, it is not just the level at which political orientation is measured but rather what the measures reflect that determines the nature of the politics–SWB linkage. As such, our results reconcile the disparate findings and generate consensus regarding how politics affect SWB across the two literatures. It appears that both political scientists and psychologists are correct regarding the relationship between politics and SWB—the culprit of the paradox is measurement differences across the two literatures.

It also appears that the effects of espoused and enacted political orientation on SWB may be largely independent. Individuals living in countries with greater enacted liberalism reported higher SWB than those in countries with less enacted liberalism, irrespective of their own political views or those of their fellow residents. Moreover, more conservative individuals reported higher SWB than their more liberal counterparts irrespective of how their governments actually behaved (i.e., enacted political orientation). This is important because scholars with macro and micro perspectives frequently debate the relative contributions of the person and environment in predicting attitudes and behavior and often settle on a compromise involving interactions between the two (Kristof-Brown & Guay, 2011). In the present study, the effects of political person and environment characteristics on SWB appeared to operate in relative isolation from one another.

To place our effects in perspective, it helps to take a closer look at what the coefficients mean. In our data, decommodification ranged from 18 (Ireland in 1974) to 36 (Sweden in 1994), and such a discrepancy translates into a more than half point difference in SWB [$(36 - 18).03 = .54$] on a 4-point scale. Similarly, the observed ranges in welfare spending (from 11 in Portugal in 1985 to 35 in Sweden in 1994) and equality (from 56 in France in 1975 to 77 in the United Kingdom in 1977) corresponded with SWB differences of .72 and .42, respectively. All else being equal, two individuals on opposite ends of the espoused political orientation spectrum should differ in SWB by 0.3, whereas two countries at the poles would differ by the entire range of the scale. This clearly indicates that one's political environment and perspective contribute to SWB (albeit in different ways), which is important because SWB influences workplace productivity (Judge, Ilies, & Dimotakis, 2010; Judge & Kammeyer-Mueller, 2011; Judge & Locke, 1993; Judge et al., 1998; Judge, Piccolo, et al., 2010), team cohesion, organizational citizenship behavior, counterproductive work behavior, and turnover (Chiaburu, Peng, Oh, Banks, & Lomeli, 2013; Eatough, Chang, Miloslavic, & Johnson, 2011; Simon et al., 2010).

Governments and organizations depend on each other to provide for the citizenry. Jobs, education, and security, which contribute considerably to individual well-being, are just a few issues requiring the two to work together effectively (Diener et al., 1995; Diener & Fujita, 1995; Diener, Sandvik, Seidlitz, & Diener, 1993; Judge & Kammeyer-Mueller, 2011). This should be especially important to policy makers and managers because a growing body of literature is uncovering how factors that occur outside of organizations can affect employee attitudes and behaviors within them (Brief et al., 2005; Martins, Eddleston, & Veiga, 2002; Pugh, Dietz, Brief, & Wiley, 2008). Because politics is omnipresent, managers and policy makers should be aware of how these factors contribute to SWB so that they can make better decisions. For instance, our findings contribute to the current debate about the effect of welfare (or its components, e.g., health care or education) on SWB. Sometimes this debate is implicit—current health care reform in the United States is assumed to increase people's well-being (Blanchflower & Oswald, 2011). In other cases, the debate is more explicit. In fact, economics Nobel Prize laureates Joseph Stiglitz and Amartya Sen explicitly argued, in a recent report by the Commission on the Measurement of Economic Performance and Social Progress, that welfare increases SWB (Stiglitz, Sen, & Fitoussi, 2009). Our findings provide a measure of empirical support for their argument.

## Study Limitations, Future Research Directions, and Conclusions

Our results should be interpreted in light of the study's limitations. First, although our measures for SWB and political orientation have seen use in previous research and have been shown to be reliable and valid (Di Tella & MacCulloch, 2006; Myers, 2000; Napier & Jost, 2008), using a single item indicator does not allow us to easily estimate the internal consistency of scores on these measures. Future researchers should use multiple-item measures when possible. Second, the low internal consistency of scores on the decommodification composite suggests a fairly high degree of measurement error, which typically attenuates relationships between constructs. Although the Spearman–Brown prophecy formula indicates that doubling the number of indicators (from three to six) would achieve a conventionally acceptable level of internal consistency (i.e., .70), we urge researchers to continue developing measures of this construct. Third, despite the large individual-level sample size, our study was limited to 16 Western European countries. Although this relatively small number of countries may make our results conservative, it is also possible that there may be a different relationship between politics and SWB in other parts of the world (e.g., Asia) because of different values or culture. Scholars generalizing our results beyond a Western European context should do so with appropriate caution and may wish to consider country-level moderators.

As with any new findings, we suggest that future researchers should attempt to replicate our findings using additional measures and data sets such as the International Social Survey Programme, newer Eurobarometers that have more recently become available, the World Values Survey, and the Database of Political Institutions from the World Bank. Future researchers also might further explore potential interactions involving political variables. For instance, we observed several significant interactions between most country-level (Level 2) variables as briefly mentioned in the results. Moreover, subsequent inquiry could focus on refining the concept of political ideology (right–left) by perhaps creating subdimensions and exploring how they relate to SWB. Notably, views on fiscal and social issues are often misaligned. Accordingly, political views (e.g., welfare spending, taxation, preferences for size of government, preferences for redistribution, public good provision, health care, and education) could be used to define conservatism and liberalism. The goal of this exercise would be to find out what it is specifically about the broad concept of conservatism that contributes to SWB. Finally, our study found an inverse relationship between enacted and espoused values. Previous research (House, Hanges, Javidan, Dorfman, & Gupta, 2004; Taras, Steel, & Kirkman, 2010) also uncovered several examples where values and practices have been found to relate negatively without identifying the specific reason for each case. We suggest future researchers investigate the specific reasons these inverse relationships occur to contribute to the theoretical understanding of the values-to-practice phenomenon.

Politics are everywhere—individuals have political views and countries have political cultures. In this study, we showed how politics at the individual and collective levels simultaneously and differentially influence SWB. Our results explained the SWB political paradox by uncovering that how scholars measure politics matters. It is interesting that both enacted left-wing politics (liberalism) and espoused right-wing politics (conservatism) relate positively to SWB.

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Received July 15, 2013
Revision received July 11, 2014
Accepted July 14, 2014 ■

### 3.2 The Happiness-Energy Paradox: Energy Use is Unrelated to Subjective Well-Being (Okulicz-Kozaryn and Altman 2019)

Applied Research in Quality of Life (2020) 15:1055–1067
https://doi.org/10.1007/s11482-019-09719-y

# The Happiness-Energy Paradox: Energy Use is Unrelated to Subjective Well-Being

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Received: 3 April 2018 / Accepted: 27 February 2019 / Published online: 12 March 2019



## Abstract

Earth's per capita energy use continues to grow, despite technological advances and widespread calls for reduction in energy consumption. The negative environmental consequences are well known: resource depletion, pollution, and global warming. However many remain reluctant to cut energy consumption because of the widespread, although, implicit, belief that a nation's well being depends on its energy consumption. This article systematically examines the evidential support for the relationship between energy use and subjective well-being at the societal level, by integrating data from multiple sources, collected at multiple levels of government, and spanning four decades. This analysis reveals, surprisingly, that the most common measure of subjective well-being, life satisfaction, is unrelated to energy use – whether measured at the national, state or county level. The nil relationship between happiness and energy use is reminiscent of the well-known Easterlin Paradox, however the causal mechanisms responsible to each remain in question. We discuss the possible causes for the Happiness-Energy paradox and potential policy implications.


**Keywords** Energy use · Energy consumption · Energy intensity of economy · Sustainability · Happiness · Life satisfaction · Subjective well-being (SWB)

**Electronic supplementary material** The online version of this article (https://doi.org/10.1007/s11482-019-09719-y) contains supplementary material, which is available to authorized users.

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## Introduction

The environmental consequences of human energy consumption are one of the world's biggest problems. In particular, energy consumption is a key driver of climate change (MacKay 2008; Pachauri et al. 2014). For instance, 98% of US carbon dioxide emissions are due to energy consumption (Energy Information Administration 2008). Despite technological advances, Earth's per-capita energy use has increased about 40% over the past four decades, and continues to grow. An essential policy and ethical challenge is to find ways to be environmentally efficient at production of human well-being, to achieve good life for all within planetary boundaries (Dietz et al. 2009; O'Neill et al. 2018).

Energy is a strategic resource. Countries wage wars over energy sources, and much of politics is driven by energy. Many countries rely heavily on energy production and many use it as a political tool. Virtually all countries seek to obtain more energy sources. A recent example is fracking (Inman 2014). Yet, we need to consume less–for at least two obvious reasons: First, worldwide, most of the energy consumed is non-renewable, and this pattern of use will persist for some time (MacKay 2008). Second, energy consumption causes pollution, and pollution harms the environment, other species, and ourselves (MacKerron and Mourato 2009; Gandelman et al. 2012; Ferreira et al. 2013). Thus, natural resource depletion and pollution potentially cancel out the benefits of energy consumption.

Many argue that consumption must be reduced dramatically. A recent report by an Intergovernmental Panel on Climate Change is alarming (Pachauri et al. 2014 and http://www.ipcc.ch).

A major review estimated that global climate change could reduce global GDP by 20 %. (Stern et al. 2006) Indeed, the threat is serious enough that claims for not growing the economy anymore or even "degrowing" are being strongly argued (Kallis 2011; Kallis et al. 2012). Some argue that reduction as high as a factor of ten is needed in affluent societies (Pretty 2013).

Moreover, it is generally believed that there is a fundamental tradeoff between societal energy consumption and self-interest at the individual and national level. It is believed that substantially reducing energy consumption requires individual sacrifices: It is widely (although not universally) believed that increased energy consumption is necessary both for overall national economic growth and to increase the directly consumed amenities (such as cooling, travel, lighting) that increase well-being (Jorgenson et al. 2014; Stern and Kander 2012; Dietz 2015; Carter 1977; Smil 2005). However recent research in the field of subjective well-being (SWB) offers theoretical reason for hope – the key point from SWB literature relevant here is that there is a consumption threshold beyond which SWB become satiated (Lamb and Steinberger 2017). In what follows we bridge the study of energy consumption and SWB and examine the empirical relationship between these at a global scale.

While energy consumption is useful, most energy consumption both pollutes and depletes natural resources (Arrow et al. 2004; Soytas et al. 2007). How energy use affects human well-being on the whole remains an open question, which we examine in this research. Traditional approaches to examining the effects of development have measured societal benefits using the Gross Domestic Product (GDP) and its adjustments – per capita and purchasing power parity (Jorgenson 2014). Relying solely on

GDP as a measure of societal benefit has a broad range of shortcomings which are now becoming broadly recognized – and does not generally capture individual's evaluation of their quality of life (Giannetti et al. 2015). We evaluate the benefits and problems of energy use using both GDP and gold-standard measures of subjective well-being (SWB), life satisfaction.[1] Combining these measures reveals surprising relationships among energy consumption and societal benefit.

By combining data on energy consumption and SWB we find that the people in economically developed areas consuming more energy are not happier. This finding is consistent across time and multiple levels of spatial aggregation–it applies to patterns of energy consumption at the local, national, and global scale. Further, across both developed and developing countries with normal economies we observe a strong inverse relationship between SWB and the energy-intensiveness (the amount of energy per unit of GDP). This suggests that a national increase in energy consumption does not contribute to SWB (likely the opposite) except where such consumption leads to unusually great increases in GDP. By combining data on energy consumption and SWB we find that the people in economically developed areas that consume more energy are not happier. This finding is consistent across time and multiple levels of spatial aggregation–it applies to patterns of energy consumption at the local, national, and global scale.[2]

## Analysis and Results

### Selection of Measures and Data

There is a consensus that the survey items that we use are good measures of subjective well-being (Diener 2009; Oswald and Wu 2009; Stiglitz et al. 2009). All SWB measures come from surveys representative of given areas. Such measures are reasonably valid and reliable (Diener et al. 2013). One caveat is limited cross-cultural comparability (Diener and Suh 2003). Our cross-country results showed strong relationships and it is unlikely that the whole effect is due to measurement error, and we mostly used data within the US.

Unlike SWB, energy consumption measures used here are not very similar or directly comparable and differ across studies due to availability of measures. For the purposes of determining broad and durable patterns, we identified the existing data sources that have been systematically and continuously measured over periods of at least five years; have coverage of units within the target population (states, counties, nations) that is as complete as possible; and are of high quality.

[1] The concept of subjective well being (SWB) has been well-studied, for recent reviews see Diener (2009) and Diener et al. (2013). It is now widely recognized in this field of study that SWB is a broad multidimensional concept that unifies such popular concepts as (affective) happiness, and life satisfaction. Because it is a multidimensional concept, a complete measurement of all aspects of SWB requires collecting a large set of measures from each subject. Most broad-scale research in this field aims to measure the cognitive dimension of SWB, i.e., life satisfaction, because of its general domain (necessary for cross-comparisons), and established validity and reliability (see, Diener et al. 2013). We follow the standard practice in this area, and make use only empirical data and use life-satisfaction as our measure of SWB.

[2] As argued later, only developing nations could improve SWB through greater energy consumption, but across the developed world, the relationship between energy consumption and SWB is nil.

**Global Patterns** We started by examining the relationship between energy use and SWB across the world, by country. We used data from the World Database of Happiness (http://www1.eur.nl/fsw/happiness/hap_nat/nat_fp.php?mode=8). The SWB scale is based on multiple data sources and is for the most part based on responses to questions of the form:" All things considered, how satisfied are you with your life as a whole these days?" on a scale from 0 = "dissatisfied" to 10 = " satisfied." The measure of total per capita energy consumption (kg of oil equivalent per capita) comes from the World Bank (http://data.worldbank.org/indicator/EG.USE.PCAP.KG.OE). All data were averaged over the years 2000–2009.

Data are plotted in Fig. 1 . The basis for the received wisdom that increasing well-being requires increased energy consumption is illustrated by the figure on the left. Across countries there is a clear positive association between energy use and SWB. SWB generally increases alongside energy consumption, but the variance across countries is large and there are many outliers. For example, some countries that have high energy use, such as Russia (RU), are unhappy. In contrast, many countries such as Costa Rica (CR) or Mexico (MX) are able to reach the highest level of SWB while maintaining very low energy use. As energy use rises, its relationship with SWB flattens out,[3] especially after the threshold of 5000 kg of oil equivalent per capita. In general, developed countries have a greater SWB (Mazur 2011; Jorgenson 2014).

How does the relationship between energy use and SWB change if we take economic development into account? It is well known that the relationship between energy use and SWB is affected by level of development (e.g., Jorgenson and Givens 2015; Knight and Rosa 2011). As recently highlighted by Dietz et al. (2009), it is important to examine how efficient a nation is in producing human well-being. Surprisingly, we find that when we measure energy as a function of economic efficiency, the relationship reverses. This is shown in the second panel of Fig. 1, which displays on the x axis the energy intensity of gross domestic product (energy/GDP). Countries that consume less energy per unit of wealth are happier. Some of the happiest countries are the highly-developed Nordic countries (DK, FI, NO, SE). Their energy consumption is high (Fig. 1, panel 1), but relative to income, their energy consumption is among the lowest in the world (Fig. 1, panel 2). In a descriptive sense, high energy intensity means that a country requires a high cost to convert energy into GDP. Another way to put this is that some countries are more efficient (in the economic, not technical sense) at converting energy into wealth.

There are a number of plausible causal explanations for this observed pattern. One hypothesis consistent with the data is that energy-use is largely unrelated to SWB once basic needs are met, but is driven up by other factors (some of which are explored below). A second plausible hypothesis is that countries with lower energy-intensity of GDP employ more energy-efficient technologies: These technologies reduce energy input for the same output of goods or services, so that consumption can remain high at low levels of pollution. A third plausible hypothesis is that countries with lower energy-intensity of GDP have economies that are rely less on inherently energy-intensive industries. A fourth plausible hypothesis is that country-specific characteristics (e.g. as industrial development, financial development, air temperature, and population) simultaneously drive energy use up and well-being down.

[3] This is known as saturation, first mentioned with respect to human development by Martinez and Ebenhack (2008).

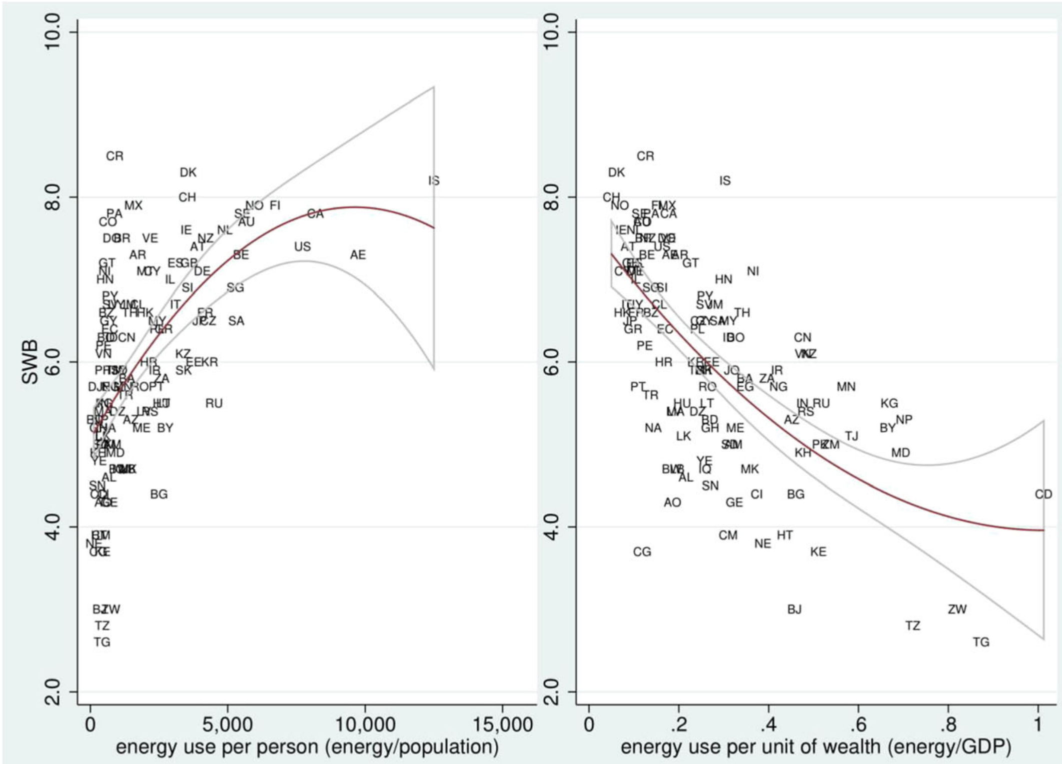


**Fig. 1** Energy is measured as kg of oil equivalent. Countries that consume more energy per person are happier (first panel). The positive relationship in first panel is spurious–taking into account the level of development, the relationship reverses–the more energy per unit of wealth, the less SWB (second panel). Quadratic fit shown with 95% confidence intervals. Energy use refers to use of primary energy before transformation to other end-use fuels (kg of oil equivalent). All data were averaged over the 2000–2009 period. Country codes are in Table S1 in supplementary material. Several outliers on energy, GDP, or energy/GDP were dropped: Ukraine, Mozambique, Turkmenistan, Uzbekistan, Ethiopia, Luxemburg, and Qatar. And two additional outliers Kuwait, and Trinidad and Tobago–they would change L-shaped relationship in the second panel to U-shaped because they have extreme leverage–compare with Fig. S2 using full sample in supplementary material. Also, the relationships between income, energy and SWB are further explored in Figs. S3-S8–examining these interrelationships is a topic for the future research–present study focuses on documenting the main relationship only

These hypotheses cannot be definitely tested with the available data, however examining individual countries favors the first hypothesis: The Netherlands (NL) is rich, energy efficient, and happy. But there are also outliers. For instance, Colombia (CO) is happy and energy efficient, but poor. In general, Latin America poses a puzzle for SWB researchers. Latin Americans are relatively poor but happy. They also use very little energy and have similar energy intensity to that of the US. Notably, all Latin American countries cluster at the top left in the first panel. Great SWB is possible using little energy. East European post-Soviet countries, on the other hand, cluster at the bottom and some at the left. Some countries are relatively unhappy despite low energy intensity of GDP such as Congo (CG). Some countries, on the other hand, are relatively happy despite high energy intensity such as Trinidad and Tobago (TT) and Kazakhstan (KZ).

**US States and Counties** To further probe the hypotheses discussed above, and to answer old question of whether more energy is needed to increase well-being if there is already a great deal of energy being consumed (Mazur and Rosa 1974), we turned to the US, since it is among the countries that use the most energy per capita. State and county level SWB data come from the Behavioral Risk Factor Surveillance System (http://www.cdc.gov/brfss) using a similar measure of SWB: “In general, how satisfied are you with your life?” on a scale from 1 = “very dissatisfied” to 4 = “very satisfied.” State energy data came from the US Energy Information Administration (http://www.eia.gov/state) and is measured as total energy consumption per capita in the residential sector.[4] California’s residential electricity consumption per capita came from the Energy Consumption Data Management System (http://www.ecdms.energy.ca.gov/elecbycounty.aspx).[5] State level data were averaged over 2005–2010; and California data were averaged over 2006–2010. Results are shown in Fig. 2.

Energy-hungry states are not happier. There are two outliers, Hawaii and California, consuming much less energy in the residential sector than others. We zoom in on California counties in the second panel of the same figure. There is also a great deal of variation in energy use across California counties, and the relationship with SWB is also nil. We have experimented with energy intensity of GDP as we did earlier across countries, but in the case of the US subregions the results are not different.

**Over Time Movement** It is well-known that SWB is related to income in a cross-section, but not over time (Easterlin 1974; Easterlin et al. 2012). To account for this we supplemented our cross-sectional results with an exploration of SWB and energy consumption over time. We used the General Social Survey (http://gss.norc.org), which measures SWB as follows: “Taken all together, how would you say things are these days–would you say that you are very happy, pretty happy, or not too happy?” 1= “not to happy”, 2=“pretty happy”, 3=“very happy”. Energy consumption is measured as total energy use per capita in the residential sector, the same measure as used for states. Figure 3 shows SWB and energy use over time by census division. There is not much co-movement: the two series correlate at .2 only. Energy consumption is only weakly related to SWB.

High residential energy use in the US is often linked to energy inefficient suburbanization: sprawling car-dependent large housing (so called McMansions) (Duany et al. 2001). Energy use and pollution per capita are higher in sprawling suburbs than in compact cities (Meyer 2013). Such suburbs typically crop up in affordable and spacious South and indeed much of the country, and least in increasingly compact and expensive North East and West Coast[6]–Figure 3 agrees: New England and Pacific have their energy use per capita declining. Another problem for climate change is that the rest of the World may want to mimic this unsustainable lifestyle.[7]

Americans continue to consume large amounts of energy as compared to other countries. With the notable exceptions of New England and the Pacific region, energy

[4] The series code is TERPB, and further information is available at https://www.eia.gov/state/seds/sep_use/notes/use_technotes.pdf

[5] Total residential electricity consumption has two components: (A) validated Quarterly Fuel and Energy Report (QFER) retail sales/delivery data and (B) self-generation. Electricity Consumption = retail sales + self-generation. Retail sales: electricity amounts delivered from the grid and consumed by the customer. Self-generation: electricity produced and consumed on-site by the customer (for example, parking lot roofs covered with PV panels). Self-generation only contributes to total consumption and not to total sales.

[6] Parts of California, notably Los Angeles, are often argued as examples of wasteful suburbanization, yet Los Angeles new homes are much smaller than average (https://www.laweekly.com/news/how-the-size-of-the-typical-la-home-has-grown-over-the-years-7364336).

[7] Notably, Central and Eastern Europe have been suburbanizing recently (Stanilov and Sýkora 2014).

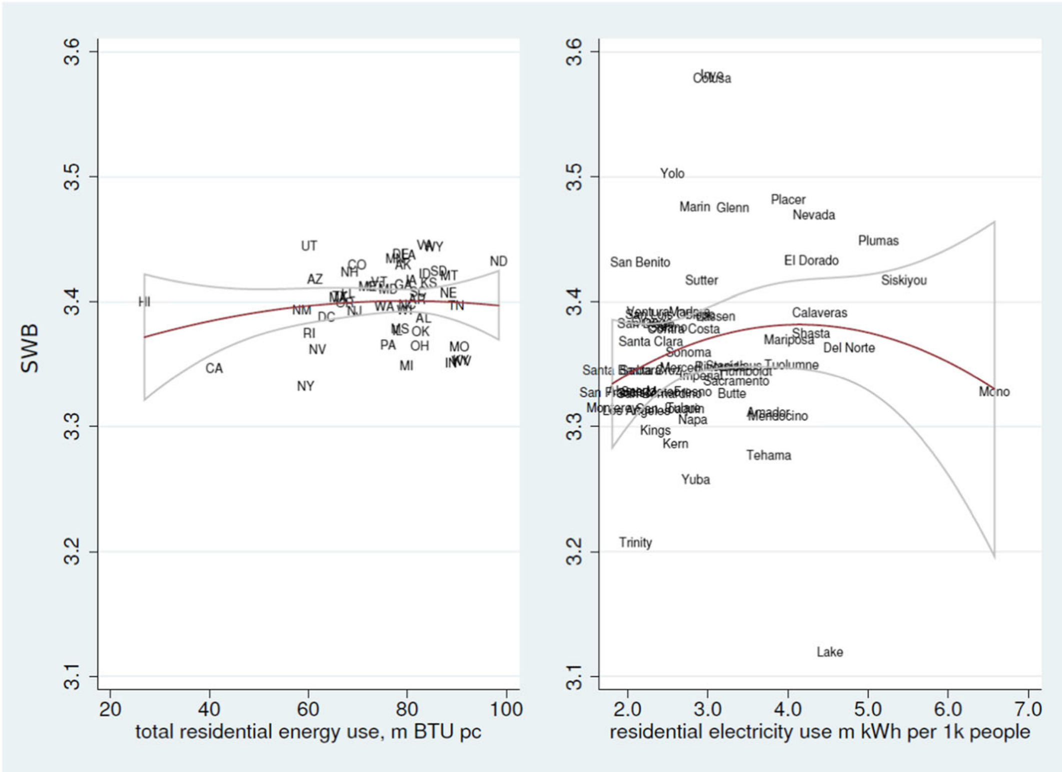


**Fig. 2** SWB and total residential energy use across US states and residential electricity use across California counties. The relationship was also quite flat in terms of total energy consumption and its GDP intensity. State level data were averaged over 2005–2010, and California data were averaged over 2006–2010

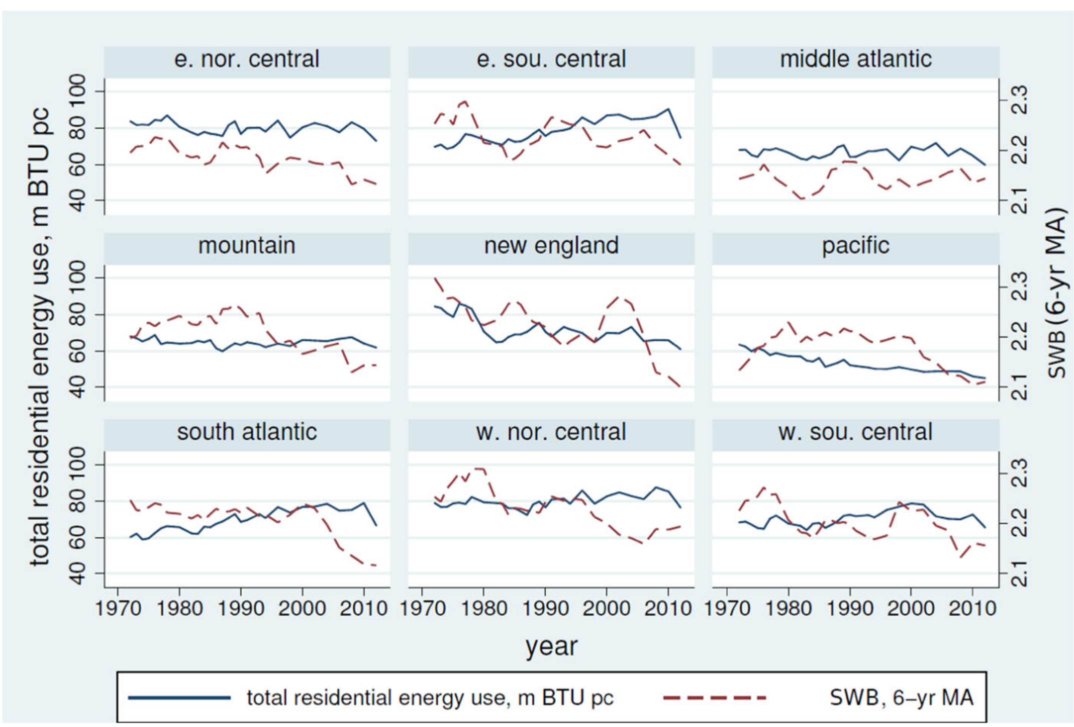


**Fig. 3** SWB (6-yr moving average) and total energy consumption in residential sector per capita. Correlation is .2 only (for unsmoothened series)

use is not decreasing, and sometimes it is increasing. Americans do not spend any less on energy either: 5–10% of personal expenditure over past 50 years were on energy (BEA 2014).

## Discussion

The analysis above shows that energy use is apparently unrelated to SWB, especially in developed nations such as the US. Why? There are a number of potential causal explanations rooted in on existing theories of individual behavior, social behavior, and economics. Existing observed information does not allow us to definitively test each of these causal theories here. The discussion below aims to suggest productive areas for future by outlining four possible behavioral explanations, and several additional economic/industrial explanations.

One possibility is that bounded rationality plays a strong role: Decisions on how much to produce and consume based on expected SWB are tricky, humans are often predictably wrong and experience much less SWB than expected (Kahneman et al. 1997), and are wrong in their perceptions of energy consumption and savings (Attari et al. 2010; Dietz 2014). Thus it is perhaps not surprising that there is a discrepancy between expectation (more energy use, more SWB) and experience (more energy use, no more SWB) is not surprising.[8]

Second explanation is that energy consumption is a positional good: Consumption buys position, but because everyone is in competition with everyone else, the race cannot be won (Frank 2012; Kasser 2003). A related explanation is a “hedonic treadmill” (Brickman et al. 1978): It is possible that consumption does not increase well-being because of hedonic adaptation. More consumption would not make people happy if their basic needs are already satisfied.

A third potential explanation builds on Veblen’s concept of conspicuous or wasteful consumption (Veblen 2005a, b). Such consumption does not satisfy needs but simply aims to demonstrate that one is better than others. Much of such consumption wastes energy without satisfying human needs, and does not make us any happier (Csikszentmihalyi 1999; Frank 2004, 2005, 2012).

A fourth potential explanation based on the livability theory (Veenhoven 2014), is that consumption result in more well-being only up to the point that is used to satisfy basic human needs. Hence, energy used to alleviate extreme temperatures, provide food, shelter, basic transportation, etc, will increase human SWB (if it out-weighs the costs such as pollution). But energy consumed on non-essential human needs (arguably most energy consumed in developed nations) will result in little SWB, if any, and our findings are consistent with such an explanation.

In parallel with these potential behavioral explanations, industrial and economic factors could contribute to the relationship between energy use and SWB. One possibility that any direct positive effect of energy use on SWB is offset by an increase in pollution. (See, for example, Li et al. 2018 for a review of the effects of air pollution on

[8] While we believe the the behavioral explanations are pivotal, there are other potential explanations. For instance, access to technology may affect the threshold energy level needed to meet core human needs.

SWB). If such an indirect effect exists, the relative dependence of a nation's economy on industries that are energy intensive and/or highly polluting, and the availability and deployment of energy efficient technologies in these industries, would be likely to mediate together the relationship between SWB and energy consumption (Stern 2011). These mediators in turn may be causally affected by country-specific characteristics such as economic history, population, and climate. Future research can focus on testing specific explanations–here we have focused on documenting the overall relationship.

## Conclusions

Again, while for simplicity we use term "energy", it is important to keep in mind that we have used different measures of energy. At the country level, among developed countries, there is no relationship between energy use and development. Further, for all countries with normal economies, the less energy-intensive is a country's economy, the happier are its people. (Fig. 1, Panel 2). Across US states and California counties, energy consumption and SWB have no apparent relationship. Likewise, the changes over time in energy use are almost unrelated to changes in SWB across US Census regions. With this correlational study we aim to bring the relationship between energy use and well-being to wider audiences, and encourage more research in this area.

While correlation is not sufficient for causality, it is typically necessary – especially across relatively homogeneous units of observation (Mazur 2011). Although this is an observational study and does not demonstrate causality, the patterns observed suggest that in the US, and across the developed World, change in energy use will not cause significant change in SWB. Further it suggests that, even in developing countries, increases in energy consumption do not contribute to SWB except where such consumption leads to unusually great increases in GDP. These patterns contrasts with widely held assumption about the relationship between SWB and energy consumption. There are many possible explanations–and while the available data does not support a causal analysis, we offer some conjectures for future research.

We found that energy consumption is neither necessary for SWB, nor linked directly to it, and we speculate the relation between energy use and SWB is very similar to the relation between economic growth and SWB, i.e., the Easterlin Paradox (Easterlin et al. 2010): common wisdom has it that over time income should increase SWB, and so common wisdom has it that energy use should increase SWB, but neither does. Specifically, Easterlin Paradox states that at a point in time SWB varies directly with income, but over the long run (10 years or more) SWB does not increase when a country's income increases. Two countries that dramatically increased both GDP and energy use are Japan and more recently China. From 1960 till 1990 Japanese energy use per capita increased more than three fold; from 1990 to 2007 Chinese energy use per capita more than doubled, but SWB remained flat in both China and Japan.[9]

[9] Primary energy use (before transformation to other end-use fuels) in kilograms of oil equivalent, per capita from World Development Indicators accessed through https://www.google.com/publicdata. SWB data come from Frey and Stutzer (2002) and Easterlin et al. (2012)–note that Chinese SWB slightly increased in last several years. Also note that in the US both energy use per capita and SWB remain flat over past four decades (using GSS data and above measure of energy use). We provide additional analyses in supplementary material.

It is striking that there were only several attempts to relate energy consumption to SWB. In a small sample of 55 countries, one early study used a set of 27 indicators to measure quality of life (Mazur and Rosa 1974), which was later extended over time (Mazur 2011). Mazur (2013) confirmed his earlier findings: among industrial nations, already high in their energy and electricity consumption, further increases in per capita energy or electricity consumption are unrelated to changes in SWB. Another study is a recent article that analyzes cross national data (Winfrey 2013). But neither study explores energy intensity of GDP, nor variation at finer geographic representation than a country. This is an important contribution of this study: there is more heterogeneity across units at more aggregated levels (e.g., countries are less comparable than US states), and there is less precision (e.g., state level SWB and energy averages are less precise descriptions of individuals than county level averages). And it is well known that there may be different relationships at different levels of aggregation (e.g., Ashkanasy 2011), i.e. earlier cross-national findings may not hold at state or county levels. Furthermore, cross-sectional relationships often differ from time-series relationships (Easterlin et al. 2012), i.e. earlier cross-sectional findings needed confirmation in time-series analysis in present study.

We do not claim causality in this correlational study. Yet, as persuasively pointed out by labor economist Andrew Oswald (e.g., Blanchflower and Oswald 2011; Oswald 2014), correlational studies are not without merit despite what many economists think–many scientific breakthroughs were first discovered in observational studies, for instance, that smoking is related to cancer. It is often overlooked that experiments tend to suffer from many problems that are not inherent in observational studies such as lack of external validity, small sample size, artificial laboratory setting, and so forth, for discussion see Pawson and Tilley (1997). At the same time, we encourage research into the causal relationships between energy use and SWB.

There is a need for future research in this important area. There have been many calls to systematically collect SWB data, and we should collect energy use data for the same subjects. Such data would allow the exploration of the relationship at the individual level.

We conjecture that in the developed world, we can decrease energy consumption without much loss in SWB, if any. While people from India and China may need to consume more energy, it is possible that Americans could consume consume considerably less without reducing SWB. For example, Texans might reduce energy consumption by half, following the example of California, and remain content. We agree with Dietz et al. (2009) that we should strive for environmentally efficient well-being: achievement of human well-being at low level of environmental stress; and we agree with Steinberger and Roberts (2010) that high human development can be achieved at moderate energy and carbon levels.

Based on the our analysis, we suggest an interventions to decrease energy consumption. First, we may simply need to increase awareness of the empirical relationship between SWB and energy consumption: there is no evidence that increasing already substantial consumption does buy substantial SWB. Just as increasing income beyond a point does not result in much SWB (Kahneman and Deaton 2010), increasing energy consumption beyond a point does not result in much SWB either. In short, SWB can be achieved at low levels of energy consumption–human flourishing requires energy to satisfy basic needs only. We are hopeful that awareness and education can change behavior. Many other ways to curb consumption have been suggested (Dietz 2014, 2015; Asensio and Delmas 2015; Dumas 1987; Attari et al. 2010). Proposals to consume less energy are urgently needed as market economy is often incentivizing people to consume more energy, for instance when new sources are discovered as with current natural gas bonanza in the US (Hopkins 2017).

**Acknowledgements** We thank anonymous reviewer for pointing out that the relation between energy use and SWB is very similar to the relation between economic growth and SWB (i.e., the Happiness Paradox).

**Compliance with Ethical Standards** This material has not been published in whole or in part elsewhere; the manuscript is not currently being considered for publication in another journal; all authors have been personally and actively involved in substantive work leading to the manuscript, and will hold themselves jointly and individually responsible for its content.

**Conflict of Interest** No conflict of interest declared.

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### 3.3 Constructing General Human Agency Indicators (GHAIs) and a General Personal Agency Scale (GPAS) (D'Italia and Okulicz-Kozaryn 2025)

Social Indicators Research (2025) 176:337–411
https://doi.org/10.1007/s11205-024-03458-w

ORIGINAL RESEARCH

# Constructing General Human Agency Indicators (GHAIs) and a General Personal Agency Scale (GPAS)

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Accepted: 3 October 2024 / Published online: 11 November 2024


**Abstract**
Despite its importance for the social sciences, *human agency* remains an ambiguous and underoperationalized construct. After engaging prior research to articulate clear criteria for defining agency and synthesize a multidimensional conceptual framework for human agency, this study develops and validates preliminary General Human Agency Indicators (GHAIs) to measure subconstructs within that framework. Utilizing the Midlife in the United States (MIDUS) dataset, we aggregated a list of 30 survey items previously used in agency research and conducted an iterative process of exploratory factor analysis (EFA) and item elimination to reduce that list to a set of 9–13 items with a strong, consistent factorial structure. Using confirmatory factor analysis (CFA), we identified two bifactor models that demonstrated good fit: a nine-item General Personal Agency scale (GPAS) and a nine-item GHAIs tool combining six items from the GPAS with three measuring agency achievement. Initial evidence for the construct validity of the tools was produced through tests of internal consistency and correlational analysis, indicating that the proposed GPAS and GHAIs effectively measure personal agency, intrinsic agency, instrumental agency, and agency achievement.

**Keywords** Human agency · Personal agency · Capabilities approach · Self-determination Theory · MIDUS

## 1 Introduction

### 1.1 Human Agency, Underoperationalization, and Mismeasurement

*Human agency* (or *agency*) refers to a person's capacity to enact control over their lives and engage their physical and social environments to pursue self-determined goals (Ahearn, 2001; Kabeer, 1999; Kotan, 2010; Sen, 1985). A fundamental mechanism through which the social world is constructed and navigated; agency is a critical topic for the social

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sciences. A literature review by Cavazzoni et al. (2022) provides an overview of how agency has been explored in fields as diverse as psychology, sociology, social geography, global development, women's empowerment, childhood studies, and research on morality and sexuality.

Agency has particular importance for fields that engage human well-being. For example, Bhattacharyya (1995) defines the discipline of community development as the cultivation of social relations characterized by solidarity and agency and, in the Capabilities Approach to development, it is discussed as the enactment of human freedom, the expansion of which is the ultimate purpose of development activities (Sen, 1985, 1999). Human agency is a topic of interest to the study of subjective well-being, as well-being is produced when individuals achieve self-determined goals and functionings (Comim, 2005; Kotan, 2010). Research demonstrates a strong positive relationship between agency and life satisfaction (Graham & Nikolova, 2013; Hojman & Miranda, 2018; Wang, 2015), experienced across cultures as a consistent sequence: (1) as socioeconomic opportunities expand, people place greater value on freedom, (2) as valuation of freedom increases, so does the influence of agency on life satisfaction, (3) life satisfaction increases commensurate to the increased effect of agency (Welzel & Inglehart, 2010). Agency has also been discussed in the context of its relationships with empowerment (Chavis & Wanderman, 1990; Kieffer, 1984), prosocial behaviors (Christoph et al., 2014), community engagement (Peterson et al., 2008), volunteerism (Cicognani et al., 2015), and democratic participation (Sen, 1999).

Despite and, in some sense, because of its significance across disciplines, definitions of human agency abound, with no clear consensus regarding how the construct should be operationalized (Cavazzoni et al., 2022; Ibrahim & Alkire, 2007). Agency has been conceptualized as an internal attitude (Alsop et al., 2006 Pleeging et al., 2021), one's capacity for action (Cavazzoni et al., 2022; Giddens, 1984; Graham & Nikolova, 2013; Onyx & Bullen, 2000; Smith et al., 2000), or some combination thereof (Bryan et al., 2014; Hitlin & Elder, 2006; Kabeer, 1999; Kotan, 2010; Sen, 1985; Williams & Merten, 2014). Some extend the term to explicitly include interactions with one's environment (Bentley-Edwards, 2016; Christensen & Hooker, 2000; Horvath, 1998; Krauss et al., 2014) and relationships with others, either alongside (Alsop et al., 2006; Bandura, 2018; Bhattacharyya, 1995; Narayan et al., 2007; Yount et al., 2020) or independently from them (Lautamo et al., 2021; Salem et al., 2020; Steckermeier, 2019). Agency has also been used to describe both individual and collective capacity to participate in and transform existing sociocultural structures and norms (Bhattacharyya, 1995; Harvey, 2002; Lautamo et al., 2021; Veronese et al., 2019b; Zimmerman et al., 2019). A table expanding on Cavazzoni et al.'s (2022) original survey of previous agency definitions is included as Appendix A.

Disagreement over how agency is defined inevitably leads to challenges related to its measurement (Cavazzoni et al., 2022). A critical example of this is how agency is operationalized alongside communion as one of the "Big Two" traits in personality research (Gebauer et al., 2014). In personality research, agency refers to one's extraversion and openness to experience (DeYoung, 2006; Digman, 1997; Erdle et al., 2009; Paulhus & John, 1998) and their desire for independence, individuation, and "agentic contrast" (p. 454)—that is, how one articulates their identity by differentiating themselves from others. The agency personality construct has been useful for exploring personality (Gebauer et al., 2014) and for organizing psychological characteristics related to social behaviors (Wiggins, 1991), social values (Trapnell & Paulhus, 2012), self-enhancement strategies (Campbell et al., 2002), and developmental goals (Charles & Carstensen, 2010). However, while the agency personality construct shares some similarities with human agency as it has been engaged elsewhere in the social sciences, in that it reflects a mode of enacting control and over oneself and one's environment, none of the characteristics attributed to the agency personality trait are necessary for the self-determination, pursuit, or achievement of goals per se, which is an essential component of human agency. Rather, the agency personality trait is better understood as something akin to individualism, complementing the collectivistic orientation of the communion trait.

Agency has also been assessed using proxy measures like income, education, employment opportunities, and access to resources (Alkire, 2008; Alsop et al., 2006; Bhattacharyya, 1995; Kabeer, 1999). However, this approach has been criticized on the grounds that proxy measures are poor indicators of what people value most (Anand et al., 2009; Helliwell & Barrington-Leigh, 2010). Socioeconomic proxies are merely means to the greater end of advancing human freedom, and one's ability to convert them into desired outcomes is influenced by personal, social, political, cultural, and environmental factors (Alkire, 2008; Sen, 1999). Proxy measures also risk creating unobserved or confounding variable bias, as multiple proxies might produce similar effects on agency; this risk is compounded in fields like poverty analysis where proxies are already in regular use and limits the capacity for research to probe interactions between agency and other topics of interest (Alkire, 2008). Similarly, tools like Vallacher and Wegner's (1989) Behavior Identification Form (BIF) and Yount et al.'s (2020) Women's Agency Scale 61 (WAS-61) use behaviors and measures for perceived influence over domains and activities to assess agency. While these measures were useful for the authors' specific purposes of assessing action identity and women's agency, respectively, they do not allow for subjective valuation of the relevant activities, which is important in agency research (Alkire, 2005). This effectively causes them to function as proxy indicators, subject to the limitations thereof.

Several attempts have been made to develop direct, subjective indicators for human agency, examples of which are shown in Table 1 alongside sample items from the BIF. Some researchers have deployed single-item indicators (e.g., Graham & Nikolova, 2013; Hojman & Miranda, 2018), eliminating opportunities to the dimensionality of the construct. Even when they consist of multiple items, direct measures of agency are often unidimensional and conflated with concepts like autonomy, freedom, internal locus of control, purposeful choice, and self-efficacy (c.f. Alsop et al., 2006; Bandura, 2018; Cavazzoni et al., 2022; Graham & Nikolova, 2013; Hojman & Miranda, 2018; Inglehart et al., 2008; Veenhoven, 2000; Verme, 2009; Welzel & Inglehart, 2010). While these concepts are likely important for informing our understanding of human agency and how it functions, they are previously defined psychosocial constructs; used in isolation as unidimensional indicators, they too function as proxy measures. Unidimensional assessment of agency also fails to address a preponderance of literature that conceptualizes it as multidimensional (c.f. Alkire, 2008; Bandura, 2001; Kotan, 2010; Sen, 1999; Smith et al., 2000; Yount et al., 2020). Scholars like Lautamo et al. (2021), Smith et al. (2000), and Yount et al. (2020) have successfully developed direct, subjective, multidimensional indicators to measure aspects of human agency, but none of these measures on their own are intended to measure general human agency but, rather, subconstructs defined by the authors that do not necessarily engage conceptualizations or latent structures proposed by others.

## 2 Conceptualizing Human Agency

There is presently an "urgent need" to develop a shared understanding of human agency and common tools for its measurement (Cavazzoni et al., 2022, p. 1148). Exploration of previous agency research produces several insights regarding its common themes and characteristics. In their analysis, Cavazzoni et al. (2022) define agency as "people's ability to

**Table 1** Sample measures from previous agency research

| | |
|---|---|
| Single-item measures | I feel free to decide for myself how to lead my life. (Hojman & Miranda, 2018) |
| | How satisfied/dissatisfied are you with "Your freedom to choose what you do with your life". (Graham & Nikolova, 2013) |
| | Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Please use this scale where 1 means "none at all" and 10 means "a great deal" to indicate how much freedom of choice and control you feel you have over the way your life turns out. (Inglehart et al., 2008; Welzel & Inglehart, 2010) |
| | I feel free to decide for myself how to lead my life. (Hojman & Miranda, 2018) |
| | How satisfied/dissatisfied are you with "Your freedom to choose what you do with your life". (Graham & Nikolova, 2013) |
| Behavior Identification Form (BIF) (Vallacher & Wegner, 1989) | *Your task is to choose the identification, a or b, that best describes the behavior for you. Simply place a check mark in the space beside the identification statement that you pick* |
| | Making a list |
| | *a. Geting organized** |
| | *b. Writing things down* |
| | Reading |
| | *a. Following lines of print* |
| | *b. Gaining knowledge** |
| | Joining the Army |
| | *a. Helping the Nation's defense** |
| | *b. Signing up* |
| | *Higher-level alternative |
| Personal Agency Scale Items (Smith et al., 2000) | I get what I want or need by relying on my own efforts and ability |
| | I control what happens to me by making choices in my best interest |
| | Using the right resources or tools helps me to achieve my goals |
| Assessment tool for perceived agency (ATPA-22) (Lautamo et al., 2021) | I feel that different areas of my daily life are balanced |
| | I am satisfied with the amount of daily activities I manage to do |
| | I feel that I have a suitable amount to do on a daily basis |
| Measurement model of agency (Hitlin & Elder, 2006) | |
| *Planfulness* | When you have a problem to solve, one of the first things you do is get as many facts about the problem as possible |
| *Optimism* | How likely is it that you will go to college? |
| *Self-efficacy* | When get what you want, it's usually because you worked hard for it |

**Table 1** (continued)

| | |
|---|---|
| Interpersonal agency scale items (Smith et al., 2000) | I achieve my goals by knowing when to ask others for help |
| | I accomplish my goals by letting others know my needs and wants |
| | I get what I want or need by seeking the advice of others |
| Collective agency (Yount et al., 2020) | I feel like I have a pretty good understanding of the important issues that face my community |
| | I am often a leader in groups |
| | I can usually organize people to get thigs done |

exert control over one's life and pursue goals" (p. 1126). Similarly, Kabeer (1999) defines it as "the ability to define one's goals and act upon them." (p. 438). Broadly, these definitions align with a preponderance of those used in other studies, as shown in Appendix A, and reflect two aspects of agency, control and goal pursuit, that are commonly encountered in discussions on the topic.

Other scholars have sought to articulate the fundamental characteristics of human agency. Alkire (2008) identifies five key features of Sen's (1999) conceptualization of human agency:

> *"(i) agency is exercised with respect to goals the person values; (ii) agency includes effective power as well as direct control; (iii) agency may advance wellbeing or may address other-regarding goals; (iv) to identify agency also entails an assessment of the value of the agent's goals; (v) the agent's responsibility for a state of affairs should be incorporated into his or her evaluation of it" (p. 6).*

Kotan (2010) describes agency as involving "(a) action, power and causality, (b) purposiveness and (c) the determination of objectives" (p. 369); these three characteristics are further reduced to two: "The ability to act to influence or affect the state of the world" and "The ability to judge and reflect upon goals and situations and to determine one's own goals and objectives as reasons for action" (Kotan, 2010, p. 370). Burger and Walk (2016) also identify three elements of agency—perceived control, commitment to self-determined goals, and self-efficacy—that parallel Kotan's. This pattern that emerges when these sets of characteristics are compared suggests that any operationalization of human agency should, at a minimum, engage one's enactment of *control* over one's life through the *self-determination* and pursuit of values, goals, and desired outcomes, their capacity to *influence* circumstances to achieve those goals and outcomes, and their *ability* to perform behaviors and leverage resources to exert the influence necessary to achieve desired ends.

Many scholars have also explored various dimensions of human agency. Sen (1999) asserts that agency necessarily involves two elements: *agency freedom*, the capacity or potential to pursue goals, and *agency achievement*, one's success in achieving desired outcomes. Agency freedom is a prerequisite for agency achievement, as it reflects the conditions under which one strives to attain their goals (Sen, 1999). In practice, research typically focuses only on agency freedom, although some scholars have utilized well-being indicators to approximate agency achievement (Hojman & Miranda, 2018; Okulicz-Kozaryn, 2015; Veenhoven, 2000). Agency freedom is itself comprised of both personal and social competencies (Cavazzoni et al., 2022) and involves one's ability to advance personal goals either on their own or in cooperation with others (Narayan & Petesch, 2007). Smith et al. (2000) articulates these domains as *personal agency*, one's capacity to achieve goals through individual efforts, and *interpersonal* agency, the ability to engage others to cooperatively pursue desired outcomes. As with agency freedom, scholarship tends to focus on personal agency, rather than interpersonal agency.

Bandura (2001) describes personal agency as the ability to make choices, plan actions, and perform those actions effectively, which he distinguishes from what could be considered two subdimensions of interpersonal agency: *proxy agency*, one's ability to leverage social relationships to pursue personal goals, and *collective agency*, the capability of groups to produce, pursue, and attain shared goals. Similarly, Yount et al. (2020) engage the concept of collective agency and articulate alongside it two dimensions subsumable under personal agency: *intrinsic agency*, one's internal motivations, perceptions, and attitudes related to the determination and pursuit of goals, and *instrumental agency*, the array of strategies one has available to enact their freedom, achieve goals, and establish control over their life. Reciprocally, proxy and collective agency (Bandura, 2001) could be considered subdimensions of Smith et al.'s interpersonal agency. These various categorizations of agency are mutually complementary and can be organized into a cohesive conceptual framework, shown in Fig. 1. Agency achievement is produced by agency freedom, which is comprised of both personal and interpersonal dimensions. Personal agency is made up of intrinsic and instrumental subdimensions, while interpersonal agency represents one's capacity to engage others to achieve personal (proxy agency) or common goals (collective agency).

The proposed framework for agency aligns with the basic human needs defined by Self-Determination Theory (SDT) (Deci & Ryan, 1985). SDT posits that all people possess inherent psychological needs for autonomy, competence, and relatedness; pursuing fulfillment of these needs is fundamental to human experience, necessary for well-being (Ryan & Deci, 2001) and contributes to personal growth, intrinsic motivation, vitality and "aliveness" (Ryan & Frederick, 1997, p. 530), and concordance of goals with personal interests and values (Sheldon & Elliot, 1999). Autonomy is reflected in both one's intrinsic and instrumental agency, competence is represented by instrumental and proxy agency, and both proxy and collective agency engage one's relatedness to others. Alkire (2005, 2008) has previously discussed theoretical connections between human agency and SDT in her exploration of subjective quantitative agency measures, noting that indicators for both autonomy and competence are important for the study of agency and that agency necessarily interacts with one's relatedness, as one may possess motivations and goals that involve other individuals and their well-being. Alkire (2005) also makes connections between agency and well-being, specifically, Ryff's (1989) multidimensional model of Psychological Well-Being (PWB); this supports Sen's position that agency freedom and achievement precede and contribute to well-being freedom and achievement.

The present framework also conforms to the elements of control, self-determination, influence, and ability that are common across discussions of the characteristics of agency. One's capacity for self-determination is manifested through both intrinsic and collective agency, and their ability to pursue self-determined ends is exercised through both instrumental and proxy agency. Subsumed under agency freedom, the dimensions of personal and interpersonal agency represent the totality of one's capability to enact control by influencing their circumstances to achieve those ends. As it was produced by interweaving strands of previous agency

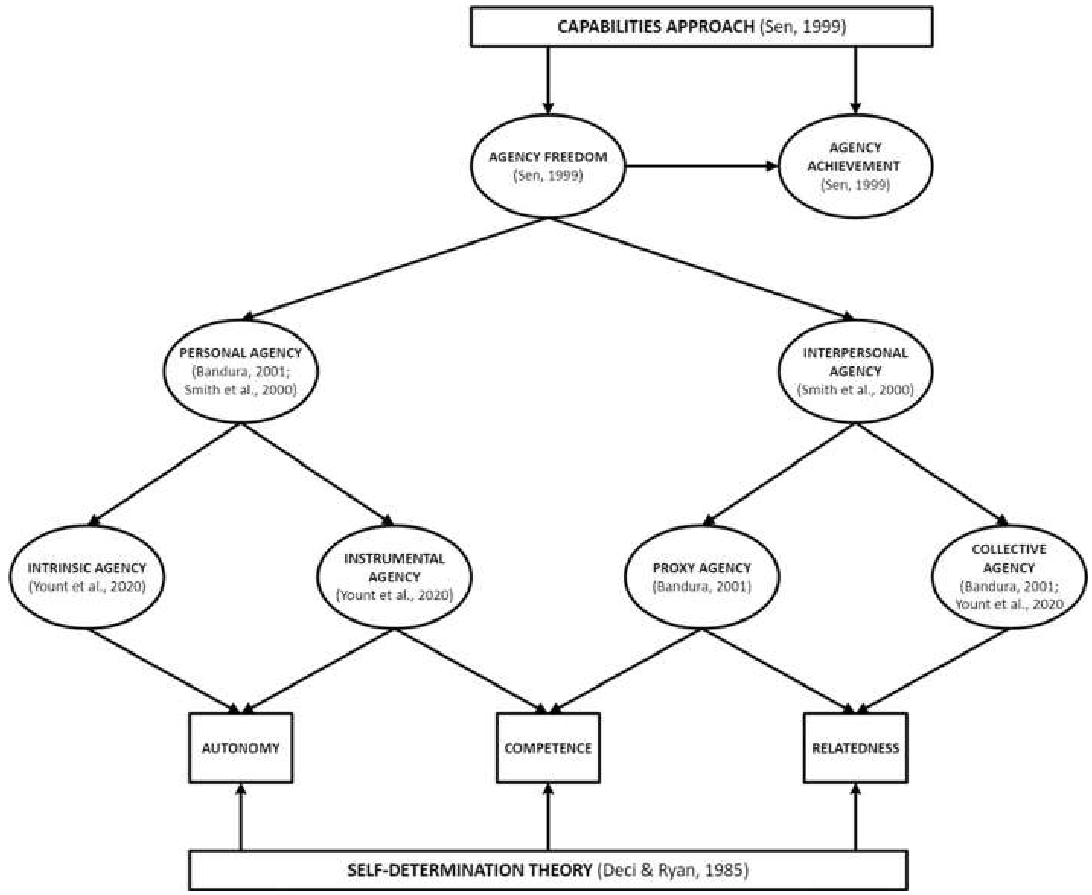



**Fig. 1** Conceptual framework of human agency

research, aligns with existing theorizations of how agency functions, and engages the essential agency characteristics of control, self-determination, influence, and ability, we assert that the proposed conceptual framework for human agency is a useful starting point for developing a common operationalization of the construct.

However, without effective indicators for measurement, even the best conceptual framework remains confined to the realm of abstract theory. Now that essential characteristics of human agency have been identified and the concept has been organized into a multidimensional framework that synthesizes existing scholarship on the topic, we now turn our attention to the development of preliminary General Human Agency Indicators (GHAIs) sufficient for assessing agency and its subdimensions. Such measures may contribute to a deeper understanding of agency, its effects on human functioning, and how individuals interact with their environments, circumstances and resources to pursue well-being. This, in turn, could have significant implications for global and community development, the formulation of public policy, implementation of empowerment activities, and social science research (Cavazzoni et al., 2022; Sen, 1999).

## 3 Methods

### 3.1 Study Design, Data, and Item Selection

We utilized secondary data analysis with waves II (MIDUS II) and III (MIDUS III) of The Midlife in the United States (MIDUS) dataset, which were collected from 2004 to 2006 and 2013 to 2014, respectively (Ryff et al., 2017, 2019). MIDUS is a nationally representative dataset that includes information about a variety of psychological and socioeconomic subjects. Data was collected through an initial phone interview, with respondents selected by random digit dialing, followed by respondents' completion of a self-administered questionnaire. MIDUS II contains 4,032 observations of "non-institutionalized, English-speaking adults in the coterminous United States, aged 35 to 86" (MIDUS II documentation) who completed both the phone interview and questionnaire; MIDUS III includes 2,732 adults, now aged 40 to 94, who completed the interview and questionnaire and had participated in the MIDUS II study.

Using Cavazzoni et al.'s (2022) analysis as a starting point, we aggregated a list of survey tools and items used in previous scholarship on human agency or one of its subdimensions. We then searched the MIDUS codebook for items that were a) identical to an item used in prior research, b) similar to a previously used item, or c) the reverse of such an item. Next, we evaluated items to ensure they reflected at least one of the agency characteristics of control, self-determination, influence, or ability and conformed to the characteristics of subjectivity discussed by Alkire's (2005): subjective measures should a) represent the perceptions and valuations of the subject, b) allow for both positive and negative assessments, and c) emphasize overarching or enduring perceptions and valuations instead of "fleeting emotional states" (p. 222). We also assessed items for alignment with Alkire's (2008) categorization of agency measures, which include complementary global and multidimensional measures, measures of effective power or direct control, measures of the advancement of well-being and other valued outcomes, and measures of autonomy and ability. Items that were not reflective of agency characteristics, did not meet subjectivity criteria, or fit into at least one category of existing agency measure were removed from the study. This selection method allowed us to establish preliminary face and content validity for the selected measures as, instead of developing novel items, which inherently involves some level of subjective bias, we instead relied on those that had already been validated in peer-reviewed research and conformed to existing standards of subjectivity and established categories of agency measures. The selection process also allowed us to assess items from multiple prior agency measures simultaneously, enabling us to probe for potential relationships between items previously used to measure similar but heretofore disjoint concepts like perceived, personal, interpersonal, and intrinsic agency.

### 3.2 Data Analysis

Multiple steps were taken to ensure the appropriateness of selected items for factor analysis. We calculated Variance Inflation Factor scores and item tolerance to test for multicollinearity, and employed tests for univariate and bivariate normality with Mardia's (1970) tests for multivariate skewness and kurtosis, Henze and Zirkler's (1990) consistent test, and the Doornik and Hansen (2008) omnibus test to assess distribution of data. We also examined Pearson's $r$ and Spearman rank correlations between items, calculated the Kaiser–Meyer–Olkin Measure of Sampling Adequacy (KMO) (Kaiser, 1970), and conducted Bartlett's (1950) Test of Sphericity. As a final preparation for factor analysis, data was then subdivided into two samples for use exploratory (EFA) and confirmatory factor analysis (CFA).

As data was non-normal, we were unable to utilize maximum likelihood (ML) modeling, the typical precursor to CFA (Costello & Osborne, 2005), as it relies on assumptions of normality (Fabrigar et al., 1999). Instead, iterated principal axis factoring (IPF) was used. IPF is a robust estimation method that produces more accurate estimates than principal axis factoring (StataCorp, 2021a, 2021b) and is not affected by non-normal data (Costello & Osborne, 2005; Fabrigar et al., 1999). IPF requires that the number of factors to extract is specified in advance (StataCorp, 2021a, 2021b); therefore, Horn's (1965) Parallel Analysis (PA) and Velicer's (1976) minimum average partial method (MAP) were used to determine the appropriate number of factors to extract and retain. PA has consistently been shown to be one of the most accurate methods for determining latent structure (Hayton et al., 2004); however, a combination of decision-making rules produces more reliable interpretations than relying on a single method (Slocum-Gori & Zumbo, 2011) and, because PA sometimes recommends retaining too many factors, MAP is particularly complementary due to its tendency to underreport the number of factors to retain (Hayton et al., 2004).

In the social sciences, it is generally assumed that factors are correlated (Costello & Osborne, 2005). Therefore, after evaluating the dimensionality of data, we rotated factors using direct oblimin, which allows for both orthogonal and oblique solutions to be produced. Rotated solutions were examined for latent factorial structure. Following criteria asserted by Comrey and Lee (2013) and Tabachnick and Fidell (2013), we considered items with "fair" or better loadings ($\lambda \geq 0.45$) to be salient on a given factor and those with "poor" or better loadings ($\lambda \geq 0.32$) to be cross-loaded. We removed items from the dataset that were not consistently salient or cross-loaded, then re-estimated the dimensionality of remaining items with PA and MAP. This process of extraction, rotation, and data reduction was repeated until all remaining items demonstrated a consistent, interpretable latent factorial structure. To identify potential alternative models, we then reviewed the solutions generated using relaxed thresholds of $\lambda \geq 0.4$ for salience (Gorsuch, 1983; Hinkin, 1995, 1998; Stevens, 1992) and a difference between loadings of $< 0.2$ (Hinkin, 1998) to indicate cross-loading.

We assessed the fit of selected EFA models through CFA using asymptotic distribution free (ADF) estimation, which does not require normal data (StataCorp, 2021a, 2021b). We examined model fit by calculating two absolute fit indices, the root mean square error of approximation (RMSEA) and standardized root mean square residual (SRMR), alongside the Comparative Fit Index (CFI) and Tucker-Lewis Index (TLI), which evaluate relative fit. Results of model chi-square tests were ignored because both non-normality of data (McIntosh, 2007) and large sample size made it likely that the test would reject the model (Bentler & Bonnet, 1980). We interpreted absolute fit indices using cutoffs of $\leq 0.05$ for good fit (Byrne, 2013; Fabrigar et al., 1999) and $\leq 0.08$ to indicate acceptable fit (Fabrigar et al., 1999; Hu & Bentler, 1999), and relative fit indices using cutoffs of $\geq 0.95$ for good fit and Byrne's (1994) less conservative threshold of $\geq 0.9$ for acceptable fit. Because agency freedom and, therefore, personal agency is a precondition of agency achievement, a final model using the best-fitting GHAIs was estimated to test the predictive effects of personal agency on agency achievement.

As the final step in our analysis, we performed several tests of preliminary construct validity on the best-fitting models. We assessed internal consistency and content, substantive, and structural validity by calculating Cronbach's α, average interitem correlations using both unstandardized and standardized items for the GPAS and all proposed GHAI scales and subscales, and analysis of pairwise Pearson's *r* correlations between individual scale items. Next, we tested convergent and concurrent validity by calculating pairwise Pearson's *r* correlations between GPAS and GHAIs scales, correlates of agency, and other associated concepts: life satisfaction (Comim, 2005; Hojman & Miranda, 2018; Wang, 2015), self-acceptance and self-esteem (Azizli et al., 2015; Serdiuk et al., 2018; Skinner et al., 1996), purpose in life and autonomy (Alkire, 2005; Serdiuk et al., 2018), and positive relations with others and social integration (Christoph et al., 2014; Veenhoven, 2004). As agency is central to the human experience and can arguably be influenced by a diverse variety of factors, it was challenging to identify variables to test for discriminant validity; therefore, we compared coefficients between the GHAIs and agency correlates with those between GHAIs scales and the "Big Two" agency trait (Gebauer et al., 2014), given the demonstrable theoretical distinctions between the two constructs described above. Finally, to evaluate the predictive validity of the proposed tools, we calculated pairwise Pearson's *r* correlations between GPAS and GHAIs scores in MIDUS II with those of agency correlates and associated concepts in MIDUS III.

## 4 Results

### 4.1 Item Selection

30 items were selected that matched or closely resembled those previously deployed in tools measuring human agency. 25 items were worded differently than, but conceptually aligned with, previous items; seven of these similar items were reverse measures of items used by Black (2016), Lautamo et al. (2021), and in the Basic Psychological Need Satisfaction Scale (Deci & Ryan, 2000; Gagné, 2003). The other five items selected were exact matches for items deployed by Nestadt, et al. (2022). No items were identified that were representative of those used previously to assess interpersonal, proxy, or collective agency (Smith et al., 2000; Yount et al., 2020); therefore, the remainder of the study focused predominantly on identifying measures for personal agency, its subdimensions, and agency achievement.

All selected items met Alkire's (2005) criteria for subjectivity and conformed to at least one established category of agency measures (2008). Most items were global, rather than pertaining to a specific life domain; however, indicators for direct control, effective power, autonomy, ability, and the advancement of well-being or other goals were identified, indicating that the initial list of selected measures engaged the agency construct in ways reflective of the breadth of prior research. All items also reflected at least one of the agency characteristics of control, self-determination, influence, or ability. Relationships between previously used agency items and those selected for the current study are summarized in Table 2.

Most items selected were included in the MIDUS dataset as scale items measuring other constructs. Twelve items were selected from three scales representing Ryff's (1989) PWB constructs of autonomy (5 of 7), environmental mastery (6 of 7, with the final item relating to fitting in with one's community), and purpose in life (1), although the single item selected from this latter construct was included in data collection but excluded from the final scale. All 12 items from Lachman and Weaver's (1998) perceived control scale were

**Table 2** Selected agency items and associated MIDUS constructs (bold items included in final GPAS and GHAIs)

PSYCHOLOGICAL WELL-BEING (RYFF, 1989)

| Previously used agency item | Used in | MIDUS item | MIDUS Construct |
|---|---|---|---|
| I generally feel free to express my ideas and opinions | Gagné, (2003) | I AM NOT AFRAID TO VOICE MY OPINIONS, EVEN WHEN THEY ARE IN OPPOSITION TO THE OPINIONS OF MOST PEOPLE | Autonomy |
| I can easily express my thoughts and opinions to other people | Lautamo et al., (2021) | | |
| When I disagree with others, I tell them | Beyers et al., (2003) | | |
| When I act against the will of others, I usually get nervous | Beyers et al., (2003) | MY DECISIONS ARE NOT USUALLY INFLUENCED BY WHAT EVERYONE ELSE IS DOING | |
| I have a strong tendency to comply with the wishes of others | | | |
| I make up my own mind about doing good or bad things | Black, (2016) | | |
| No one can make me do something I know to be wrong | | | |
| My actions in most situations are based on what other people tell me is the right thing to do | | | |
| In most cases, I can make my own decisions about what is right or wrong in a situation | | | |
| How true would it be to say that your actions with respect to [the domain] are motivated by and reflect your own values and/or interests? | Ibrahim and Alkire, (2007) | I JUDGE MYSELF BY WHAT I THINK IS IMPORTANT, NOT BY THE VALUES OF WHAT OTHERS THINK IS IMPORTANT | |
| I often change my mind after listening to others | Beyers et al., (2003) | I TEND TO BE INFLUENCED BY PEOPLE WITH STRONG OPINIONS | |
| I often agree with others, even if I'm not sure | | | |
| I generally feel free to express my ideas and opinions | Gagné, (2003) | IT'S DIFFICULT FOR ME TO VOICE MY OWN OPINIONS ON CONTROVERSIAL MATTERS | |
| I can easily express my thoughts and opinions to other people | Lautamo et al., (2021) | | |

**Table 2** (continued)

| PSYCHOLOGICAL WELL-BEING (RYFF, 1989) | | | |
|---|---|---|---|
| Previously used agency item | Used in | MIDUS item | MIDUS Construct |
| I feel like I am free to decide for myself how to live my life | Gagné, (2003) | IN GENERAL, I FEEL I AM IN CHARGE OF THE SITUATION IN WHICH I LIVE | Environmental Mastery |
| How true is the following statement for you?: I feel free to decide for myself how to lead my life | Hojman and Miranda, (2018) | | |
| I make active choices about what to do daily | Lautamo et al., (2021) | | |
| I can influence my living situation in satisfying ways | Lautamo et al., (2021) | **I HAVE DIFFICULTY ARRANGING MY LIFE IN A WAY THAT IS SATISFYING TO ME** | |
| I do tasks that give me a feeling of competence or satisfaction | | | |
| I can take care of my everyday tasks independently | Lautamo et al., (2021) | **THE DEMANDS OF EVERYDAY LIFE OFTEN GET ME DOWN** | |
| I can influence my living situation in satisfying ways | Lautamo et al., (2021) | **I HAVE BEEN ABLE TO BUILD A LIVING ENVIRONMENT AND A LIFESTYLE FOR MYSELF THAT IS MUCH TO MY LIKING** | |
| I manage to perform in my environment | Lautamo et al., (2021) | I AM QUITE GOOD AT MANAGING THE MANY RESPONSIBILITIES OF MY DAILY LIFE | |
| I am satisfied with the amount of daily activities I manage to do | | | |
| I feel that I have a suitable amount to do on a daily basis | | | |
| I control things by managing my affairs properly | Smith et al., (2000) | | |
| I do tasks that I feel challenge me appropriately | Lautamo et al., (2021) | **I OFTEN FEEL OVERWHELMED BY MY RESPONSIBILITIES** | |
| I energetically pursue my goals | Snyder (1991) | **I AM AN ACTIVE PERSON IN CARRYING OUT THE PLANS I SET FOR MYSELF** | Purpose in Life (removed from final scale) |
| I meet the goals that I set for myself | | | |

**Table 2** (continued)

| Perceived control (Lachman & Weaver, 1998) | | | |
|---|---|---|---|
| Previously used agency item | Used in | MIDUS item | MIDUS Construct |
| There is little I can do to change many of the important things in my life | Nestadt et al. (2022) | **THERE IS LITTLE I CAN DO TO CHANGE THE IMPORTANT** | Perceived Constraints |
| I often feel helpless in dealing with the problems of life | Nestadt et al. (2022) | **I OFTEN FEEL HELPLESS IN DEALING WITH THE PROBLEMS OF LIFE** | |
| In my daily life, I frequently have to do what I am told | Gagné (2003) | **OTHER PEOPLE DETERMINE MOST OF WHAT I CAN AND CANNOT DO** | |
| Luck, more than what you do, is responsible for whether things turn out for the best | Black (2016) | **WHAT HAPPENS IN MY LIFE IS OFTEN BEYOND MY CONTROL** | |
| I am the one responsible for my own behavior, good or bad | | | |
| I feel responsible for the consequences of my actions | | | |
| There is not much opportunity for me to decide for myself how to do things in my daily life | Gagné (2003) | THERE ARE MANY THINGS THAT INTERFERE WITH WHAT I WANT TO DO | |
| I do tasks that I feel are important to me | Lautamo et al., (2021) | | |
| I have little control over the things that happen to me | Nestadt et al. (2022) | **I HAVE LITTLE CONTROL OVER THE THINGS THAT HAPPEN TO ME** | |
| When things don't turn out as I expected, it seems like someone else took control of things | Black (2016) | | |

**Table 2** (continued)

Perceived control (Lachman & Weaver, 1998)

| Previously used agency item | Used in | MIDUS item | MIDUS Construct |
|---|---|---|---|
| There is really no way I can solve some of the problems I have | Nestadt et al. (2022) | **THERE IS REALLY NO WAY I CAN SOLVE THE PROBLEMS I HAVE** | |
| I can handle problems and pressure (by discussing them or taking actions) | Lautamo et al., (2021) | | |
| I can solve daily challenges in a reasonable way | | | |
| I can think of many ways to get out of a jam | Snyder (1991) | | |
| There are lots of ways around any problem | | | |
| Even when others get discouraged, I know I can find a way to solve the problem | | | |
| Even when others want to quit, I know that I can find ways to solve the problem | Poteat et al. (2018); Veronese et al. (2019a), (b), (2020a), (b) | | |
| When I have a problem, I can come up with lots of ways to solve it | | | |
| Sometimes I feel that I am being pushed around in life | Nestadt et al. (2022) | I SOMETIMES FEEL I AM BEING PUSHED AROUND IN MY LIFE | |
| I feel pressured in my life | Gagné, (2003) | | |

**Table 2** (continued)

| Perceived control (Lachman & Weaver, 1998) | | | |
|---|---|---|---|
| Previously used agency item | Used in | MIDUS item | MIDUS Construct |
| Once I decide on a goal, I do whatever I can to achieve it | Smith et al., (2000) | **I CAN DO JUST ABOUT ANYTHING I REALLY SET MY MIND TO** | Personal Mastery |
| I can think of many ways to get the things in life that are important to me | Poteat et al. (2018); Snyder (1991); Veronese et al. (2019a), (b), (2020a), (b) | **WHEN I REALLY WANT TO DO SOMETHING, I USUALLY FIND A WAY TO SUCCEED AT IT** | |
| When you get what you want, it's usually because you worked hard for it | Hitlin & Elder (2006); Williams and Merton (2014) | WHETHER OR NOT I AM ABLE TO GET WHAT I WANT IS IN MY OWN HANDS | |
| I get what I want or need by relying on my own efforts and ability | Smith et al., (2000) | | |
| I control what happens to me by making choices in my best interest | | | |
| Some people believe they can decide their own destiny, while others think they do not have control over their destiny. Please, to what extent do you believe you can decide your own destiny? | Victor et al. (2013) | WHAT HAPPENS TO ME IN THE FUTURE MOSTLY DEPENDS ON ME | |
| Otheritems | | | |
| Previously used agency item | Used in | MIDUS item | MIDUS Construct |
| Once I decide on a goal, I do whatever I can to achieve it | Smith et al., (2000) | WHEN THINGS DON'T GO ACCORD-ING TO MY PLANS, MY MOTTO IS, "WHERE THERE'S A WILL, THERE'S A WAY." | Selective Primary Control (Heckhausen & Schulz, 1993; Wrosch, Heckhausen, & Lachman, 2000) |
| When I have a problem, I can come up with lots of ways to solve it | Poteat et al. (2018); Veronese et al. (2019a), (b), (2020a), (b) | WHEN FACED WITH A BAD SITUATION, I DO WHAT I CAN TO CHANGE IT FOR THE BETTER | |
| There are lots of ways around any problem | Snyder (1991) | | |

**Table 2** (continued)

| Otheritems | | | |
|---|---|---|---|
| Previously used agency item | Used in | MIDUS item | MIDUS Construct |
| I can solve daily challenges in a reasonable way | Lautamo et al., (2021) | EVEN WHEN I FEEL I HAVE TOO MUCH TO DO, I FIND A WAY TO GET IT ALL DONE | |
| I find it difficult to decide what I want<br>When people ask me what I want, I immediately know the answer | Beyers et al. (2003) | I KNOW WHAT I WANT OUT OF LIFE | Self-Directedness and Planning (Prenda & Lachman, 2001) |
| How much control do you feel you have in making personal decisions that affect your everyday activities? | Ibrahim and Alkire, (2007) | At present, how much control do you have over your LIFE IN GENERAL? A lot, some, a little, or none at all? | None |
| "Please use this scale where '1' means "none at all" and '10' means "a great deal" to indicate how much freedom of choice and control you feel you have over the way your life turns out." | Inglehart et al., (2008); Veenhoven, (2000); Verme, (2009) | Using a 0 to 10 scale where 0 means "no control at all" and 10 means "very much control," how would you rate the amount of control you have over your life overall these days? | |

selected, reflecting the subscales of perceived constraints and personal mastery. Three of five items were selected from a scale representing selective primary control (Heckhausen & Schulz, 1993; Wrosch, Heckhausen, & Lachman, 2000) and one of three items was selected from Prenda and Lachman's (2001) self-directedness and planning scale. Only two selected items, both related to one's level of control over their life, were not part of an existing scale.

Altogether, findings from the item selection process produced initial evidence of the content validity of selected items. All items met multiple established criteria for agency measures and were selected from scales measuring constructs that align conceptually with agency (Alkire, 2005, 2008; Burger & Walk, 2016; Kotan, 2010). However, some items demonstrated questionable content validity; specifically, items from the PWB autonomy scale appeared to emphasize assertiveness, individuation, and agentic contrast. As previously discussed, these characteristics are typically unique to discussions of agency as a personality trait (Gebauer et al., 2014) and are not requisite characteristics for the self-determination and pursuit of goals.

### 4.2 Data Management and Descriptive Statistics

Descriptive statistics for the full sample and subsamples for MIDUS II and MIDUS III datasets are summarized in Appendix B alongside findings from tests for multivariate normality described below. Values for seven items were reversed so that larger values consistently reflected stronger agentic perceptions. VIF scores were less than 2.5 and tolerance values were greater than 0.4 for all items in both the full sample and subsamples, indicating a low risk of multicollinearity among items.

Data in all samples violated assumptions of normality. Mean scores and standard deviations indicated that, on average, respondents reported moderate to high levels scores for each item, and cursory examination of item skewness and kurtosis suggested that all items demonstrated a left-tailed distribution. Tests of univariate and bivariate normality for all items were significant at the level of $p < 0.0001$, confirming that items were non-normally distributed. Similarly, Mardia's tests for multivariate skewness and kurtosis, Henze-Zirkler's consistent test, and the Doornik-Hansen omnibus test were all significant at the level of $p < 0.0001$, leading to a rejection of the null hypothesis that items possess multivariate normality.

### 4.3 Suitability of Data for Factor Analysis and Determination of Factors to Retain

The average interitem correlation was $r = 0.26$ in both MIDUS II and III. Item-rest Pearson's correlations ranged from $r = 0.31$-$0.69$ and $r = 0.3$-$0.67$ for unstandardized and standardized items, respectively, in MIDUS II, and from $r = 0.26$-$0.7$ and $r = 0.27$-$0.68$ in MIDUS III, indicating positive relationships between individual items and remaining items in the series. All pairwise Spearman rank correlations between items were significant ($p < 0.05$) and positive, although relationships varied in strength from negligible to strong in both MIDUS II ($\rho = 0.08$-$0.64$) and MIDUS III ($\rho = 0.09$-$0.63$). Overall, correlational analyses demonstrated relationships between selected items, providing preliminary evidence of their suitability for factor analysis.

Five items were frequently associated with coefficients less than 0.2 in both datasets, although they also demonstrated moderate to strong relationships with some items. Two were related to selective primary control ("When things don't go according to plan, my motto is 'Where there's a will there's a way'", "Even when feel I have too much to do, I find a way to get it all done"), two were from the PWB autonomy scale ("I am not afraid to voice my opinions, even when they are in opposition to the opinions of most people", "I tend to be influenced by people with strong opinions"), and the final item, "At present, how much control do you feel you have over your life in general?", was independent from any scale.

The KMO index was 0.944 for both waves of data, exceeding the suggested cutoff of 0.6 (Kaiser, 1970). Bartlett's Test of Sphericity (Bartlett, 1954) was significant in both MIDUS II ($\chi^2$=38,616.06, $p<0.001$) and III ($\chi^2$=27,390.5, $p<0.001$). These findings corroborated those from correlational analyses, indicating study data was suitable for factor analysis.

## 4.4 Exploratory Factor Analysis

For both MIDUS II and III, PA indicated that five components be retained while MAP recommended that two factors be extracted from the data; to fully explore the factorial structure of the selected items, initial extractions included solutions within this range. Five-factor rotations produced four factors with at least two salient loadings ($\lambda \geq 0.45$) in both waves, which were interpretable as the constructs from which several items were derived: perceived constraints and personal mastery (Lachman & Weaver, 1998), selective primary control (Heckhausen & Schulz, 1993; Wrosch, Heckhausen, & Lachman, 2000), and the PWB autonomy construct (Ryff, 1989). However, seven items cross-loaded on factors in at least one wave of data, and seven were never salient on any factor in either sample, indicating a possible overextraction of factors. Four-factor solutions resulted in three salient factors consistent across waves that could be understood to represent perceived constraints, the PWB autonomy construct, and a combination of selective primary control and personal mastery, with five items cross-loading and six never being salient on a factor.

Three-factor models were the first to demonstrate a clear structure of two consistent factors, the first of which reflected a combination of perceived constraints and environmental mastery (Ryff, 1989) and the second a combination of selective primary control and personal mastery. While only one item from the PWB autonomy construct cross-loaded in MIDUS II, it was not salient on any factor in MIDUS III; in total, ten items were never salient on any factor in either sample. This latent structure was maintained in two-factor models, with single items selected from the constructs of self-directedness and planning (Prenda & Lachman, 2001), purpose in life (Ryff, 1989), and environmental mastery loading onto the second factor, one item cross-loading in MIDUS III only, and nine items never demonstrating salience. However, in the two-factor solution, factors also appeared to be interpretable as personal agency, one's attitudes and perceptions regarding their ability to enact control over their life and pursue self-determined goals, and agency achievement, one's ability to succeed in these capacities.

Multifactor solutions consistently produced a ratio of first-to-second eigenvalues greater than five, the only exception being the MIDUS III five-factor extraction (1:2 $\lambda = 4.92$). This suggested the presence of a general dominant factor which explained between 69 and 85 percent of variance in MIDUS II solutions and 67 to 84 percent of variance in MIDUS III (in five- and two-factor rotations, respectively). Therefore, a one-factor solution was also extracted despite not being recommended by PA or MAP. 20 items were consistently salient on this general factor, with items from Lachman and Weaver's (1998) perceived constraints scale and Ryff's (1989) environmental mastery scale typically loadings that were good ($\lambda \geq 0.55$) or better (Tabachnick & Fidell, 2013). No items from the selective primary control or PWB autonomy scales were consistently salient, neither was one item from the personal mastery scale ("What happens in the future mostly depends on me") and the non-scale item "At present, how much control do you have over your life in general?". Factor loadings and other information for initial extractions with one through three factors are provided in Appendix C.

Because of the quantity of non-salient and cross-loaded items, extracted latent factors were difficult to explain beyond the scales that items represented; however, two- and three-factor models indicated the presence of an alternative factorial structure. To improve interpretability of these factors, items were eliminated if they were not consistently salient ($\lambda \geq 0.45$) or cross-loaded on the two factors that emerged from two- and three-factor solutions, as were those that were not salient on the single-factor model in both waves. A total of 10 items were removed, which included all items from the selective primary control and PWB autonomy scales. We then conducted IPF extractions on the remaining items; this process of reduction and re-estimation was repeated until all remaining items were consistently salient or cross-loaded in both multifactor and single-factor solutions. Six iterations of extraction, rotation, and reduction were conducted in total; findings from the process are summarized in Table 3.

In the third and fourth iterations of extractions, clear latent structures began to emerge. Two-factor solutions produced factors interpretable as personal agency and agency achievement. Agency achievement persisted as a latent variable in three-factor extractions, while items reflecting personal agency subdivided into factors interpreted as intrinsic and instrumental agency. However, strict application of reduction criteria eliminated the agency achievement factor at the conclusion of the fourth round of extractions. The nine items that remained at the end of the data reduction process comprised a General Personal Agency Scale (GPAS) explainable as a single latent variable, personal agency, or as a combination of intrinsic and instrumental agency subfactors. Factor loadings for this final round of extractions are found in Table 4.

While the purpose of data reduction was to simplify the latent structure of the selected items to produce a measure for personal agency, the loss of agency achievement indicators resulted in a substantive reduction of the potential explanatory power of the emerging GHAIs. Further, the agency achievement factor demonstrated moderate to strong correlations with personal, intrinsic, and instrumental agency in the third and fourth iterations of extraction, suggesting the presence of positive relationships between the latent variables. Therefore, solutions from the fourth round of extractions, the last that included at least three items consistently salient on agency achievement, were reexamined using relaxed factor loading cutoffs of $\lambda \geq 0.4$ for salience and $\lambda \geq 0.2$ for cross-loading items, which are commonly encountered in EFA (Gorsuch, 1983; Hinkin, 1995, 1998; Stevens, 2002). Based on these less conservative cutoffs, all items were consistently salient on a single factor in one- and two-factor models. In the three-factor solutions, the item "I often feel helpless in dealing with the problems of life" cross-loaded onto intrinsic and instrumental agency, and "I have been able to build a living environment and a lifestyle for myself that is much to my liking" cross-loaded onto instrumental agency and agency achievement in MIDUS II and III. Therefore, both the nine-item GPAS and the 13-item GHAIs measuring personal agency and agency achievement scale from the fourth iteration of extractions were retained for CFA. Two alternative models based on the 13-item GHAIs were also developed: an 11-item version that included cross-loaded items and only the three highest loading items for intrinsic agency, and a nine-item tool that eliminated cross-loaded items, leaving only the three highest loading items for intrinsic agency, instrumental agency, and agency achievement.

**Table 3** Summary of data reduction process

| Iteration | 1 | 2 | 3 | 4* | 5 | 6* |
|---|---|---|---|---|---|---|
| # of items | 30 | 20 | 17 | 13 | 11 | 9 |
| PA factors | 5 | 3 | 3 | 3 | 2 | 2 |
| MAP factors | 2 | 2 | 2 | 1 | 1 | 1 |
| Items removed | Control over life in general at present | Know what I want out of life | In charge of situation in which I live | Actively carry out plans I set for self | Able to build lifestyle to my liking | |
| | Where there's a will there's a way | Whether I get what want is in own hands | Good managing daily responsibilities | When really want something, find way | Do just about anything I set my mind to | |
| | Do what can to change for better | Rate control over life | Many things interfere with what I want do | | | |
| | Even when feel too much, get it all done | | Feel pushed around in life | | | |
| | Not afraid to voice opinions in opposition | | | | | |
| | Decisions not influenced by others doing | | | | | |
| | Judge self by what I think is important | | | | | |
| | Influenced by people with strong opinions | | | | | |
| | Difficult voice opinion on controversial | | | | | |
| | Happens to me in future depends on me | | | | | |

*Solutions were retained for CFA

**Table 4** Factor loadings for 9-item personal agency scale (GPAS)

| | Single factor | | | | 2-Factor | | | | | | Loadings on individual factors | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Personal agency | | Uniqueness | | Intrinsic agency | | Instrumental agency | | Uniqueness | | Intrinsic agency | | Instrumental agency | | Uniqueness | |
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Little can do to change important things | **.634** | **.671** | .599 | .549 | **.604** | **.694** | | | .577 | .509 | **.659** | **.710** | | | .566 | .496 |
| Others determine what I can and cannot do | **.631** | **.637** | .602 | .594 | **.555** | **.623** | | | .594 | .570 | **.634** | **.660** | | | .598 | .564 |
| What happens in life is beyond my control | **.636** | **.680** | .596 | .538 | **.715** | **.768** | | | .528 | .467 | **.683** | **.719** | | | .533 | .483 |
| Little control over things happen to me | **.676** | **.708** | .543 | .499 | **.747** | **.785** | | | .471 | .428 | **.718** | **.744** | | | .485 | .446 |
| Really no way I can solve problems I have | **.698** | **.748** | .513 | .441 | **.760** | **.689** | | | .442 | .420 | **.743** | **.766** | | | .447 | .413 |
| Helpless dealing with problems of life | **.756** | **.801** | .429 | .359 | .430 | .421 | .392 | .444 | .438 | .361 | **.694** | **.742** | **.686** | **.744** | .519 | .446 |
| Difficult arranging life in satisfying way | **.643** | **.668** | .586 | .553 | | | **.595** | **.639** | .514 | .471 | | | **.707** | **.735** | .500 | .461 |
| Demands of everyday life often get me down | **.599** | **.629** | .641 | .604 | | | **.761** | **.780** | .459 | .434 | | | **.722** | **.744** | .479 | .447 |
| Overwhelmed by my responsibilities | **.594** | **.585** | .647 | .658 | | | **.738** | **.721** | .480 | .517 | | | **.709** | **.687** | .497 | .528 |
| Eigenvalue | 3.845 | 4.206 | | | 3.913 | 4.268 | .585 | .555 | | | 2.852 | 3.148 | 1.995 | 2.118 | | |
| Proportion of variance | 1.000 | 1.000 | | | .870 | .885 | .130 | .115 | | | 1.000 | 1.000 | 1.000 | 1.000 | | |

Additional extractions and rotations were conducted for each alternative model and for each subscale in all models. The factorial structure and loadings for the 11- and 9-item GHAIs corresponded with the 13-item model; the only difference being that the item "I have been able to build a living environment and a lifestyle for myself that is much to my liking" cross-loaded on personal agency and agency achievement in two-factor rotations of the 11-item agency models. Rotations of individual factors conformed to findings produced by full-model rotations, with cross-loaded variables from full models consistently salient on each of the latent factors they had previously cross-loaded onto, indicating that these items might influence the relationships between factors. Factor loadings for the GPAS and 9-item GHAIs, which demonstrated the best model fit during CFA, are found in Table 5 alongside loadings for their individual scales; loadings for 13- and 11-item agency models and individual scales are included in Appendix D.

Of the 13 items included in final models, five were similar to items used by Lautamo et al. (2021), four were exact matches to those employed by Nestadt et al. (2022), and three were like items included in the agentic pathways section of Snyder's (1991) Hope Scale. Other items were like those deployed by Black (2016), Gagné (2003), Poteat et al. (2018), and Veronese, et al. (2019a, b), 2020a, b). Final items included four from Ryff's (1989) environmental mastery scale and the single item excluded from her purpose in life scale; the remaining items were from Lachman and Weaver's (1998) perceived constraints and personal mastery scales. The intrinsic agency factor was comprised entirely of perceived constraints items and instrumental agency of environmental mastery items; agency achievement included a combination of items reflecting environmental mastery, personal mastery, and purpose in life. This distribution of items supports the face, content, and structural validity of the selected models, as agency represents the interaction of one's capacities to enact control over life through the pursuit and achievement of self-determined goals—that is, to overcome perceived constraints through mastery of one's environment to shape one's circumstances according to one's purpose and desired ends.

### 4.5 Confirmatory Factor Analysis

Based on findings from EFA, a total of sixteen path models were specified and estimated. Five models failed to converge without modification due to issues related to sample size and the number of fitted parameters in the model. When this occurred, it was resolved by removing paths for items that cross-loaded during EFA; covariance between intrinsic and instrumental agency was also eliminated from models when it was found to be insignificant. Relationships between all remaining variables and constructs were statistically significant with practically significant effect sizes. Fit index values for all final, converged CFA models are shown in Table 6.

CD values indicated that models accounted for between 86 and 98 percent of variance in MIDUS II models and between 88 and 99 percent of variance in MIDUS III models. Among single factor models, only the GPAS consistently demonstrated acceptable absolute fit (RMSEA and SRMR $\leq 0.08$), suggesting that personal agency and agency achievement are distinct yet interrelated constructs. All two-factor models were shown to have acceptable absolute fit or better, and absolute indices for all three-factor and bifactor models indicated good fit. Findings from relative fit indices were less consistent. Only the bifactor 9-item personal agency and three-factor and bifactor 9-item agency models consistently demonstrated acceptable fit ($\geq 0.9$) or better, although 13- and 11-item bifactor agency models produced some evidence of acceptable relative fit in MIDUS II. CFI

**Table 5** Factor loadings for 9-item personal agency and agency achievement scales (GHAIs)

Full scale factor loadings

| Item | Single factor | | | | 2-Factor | | | | | | 3-Factor | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Agency | | Uniqueness | | Personal agency | | Agency achievement | | Uniqueness | | Intrinsic agency | | Instrumental agency | | Agency achievement | | Uniqueness | |
| | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| What happens in life is beyond my control | **.588** | **.641** | .654 | .589 | **.569** | **.621** | | | .645 | .584 | **.658** | **.763** | | | | | .549 | .447 |
| Little control over things happen to me | **.657** | **.675** | .569 | .545 | **.641** | **.669** | | | .554 | .533 | **.847** | **.802** | | | | | .340 | .381 |
| Really no way I can solve problems I have | **.660** | **.710** | .565 | .495 | **.584** | **.668** | | | .570 | .497 | **.602** | **.586** | | | | | .510 | .466 |
| Difficult arranging life in satisfying way | **.651** | **.692** | .576 | .521 | **.604** | **.668** | | | .571 | .515 | | | **.569** | **.660** | | | .526 | .452 |
| Demands of everyday life often get me down | **.594** | **.629** | .647 | .605 | **.685** | **.703** | | | .580 | .548 | | | **.751** | **.756** | | | .458 | .442 |
| Overwhelmed by my responsibilities | **.589** | **.587** | .653 | .655 | **.695** | **.677** | | | .577 | .593 | | | **.753** | **.699** | | | .458 | .515 |
| Actively carry out plans I set for self | .479 | .442 | .771 | .804 | | | .403 | .399 | .724 | .734 | | | | | .421 | .444 | .721 | .703 |
| Do just about anything I set my mind to | .488 | .476 | .762 | .774 | | | **.665** | .544 | .546 | .612 | | | | | **.667** | **.554** | .547 | .609 |
| When really want something, find way | .536 | .453 | .712 | .795 | | | **.794** | **.848** | .378 | .309 | | | | | **.800** | **.829** | .375 | .336 |
| Eigenvalue | 3.092 | 3.217 | | | 3.166 | 3.286 | .690 | .790 | | | 3.248 | 3.359 | .727 | .802 | .541 | .489 | | |
| Proportion of variance | 1.000 | 1.000 | | | .821 | .806 | .179 | .194 | | | .719 | .722 | .161 | .172 | .120 | .105 | | |

Loadings on individual factors

| Item | Personalagency | | Uniqueness | | Intrinsic agency | | Instrumental agency | | Agency achievement | | Uniqueness | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| What happens in life is beyond my control | **.609** | **.657** | .630 | .568 | **.673** | **.743** | | | | | .547 | .449 |
| Little control over things happen to me | **.682** | **.695** | .535 | .517 | **.814** | **.789** | | | | | .338 | .378 |
| Really no way I can solve problems I have | **.657** | **.714** | .569 | .490 | **.684** | **.716** | | | | | .532 | .488 |
| Difficult arranging life in satisfying way | **.637** | **.682** | .594 | .535 | | | **.655** | **.702** | | | .571 | .507 |

**Table 5** (continued)

Loadings on individual factors

| Item | Personalagency | | Uniqueness | | Intrinsic agency | | Instrumental agency | | Agency achievement | | Uniqueness | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Demands of everyday life often get me down | **.630** | **.655** | .603 | .572 | | | **.741** | **.754** | | | .451 | .432 |
| Overwhelmed by my responsibilities | **.629** | **.624** | .604 | .611 | | | **.742** | **.710** | | | .450 | .496 |
| Actively carry out plans I set for self | | | | | | | | | .490 | .503 | .760 | .747 |
| Do just about anything I set my mind to | | | | | | | | | **.670** | **.642** | .551 | .589 |
| When really want something, find way | | | | | | | | | **.797** | **.785** | .366 | .384 |
| Eigenvalue | 2.46556 | 2.70721 | | | 1.58276 | 1.68515 | 1.52809 | 1.56467 | 1.3236 | 1.28093 | | |

Bold numbersindicate a loading of good or better ($\lambda \geq .55$); table does not include loadings < .32

**Table 6** Fit indices for CFA models

9-Item personal agency models (GPAS)

| | Single factor | | 2-Factor | | Bifactor | |
|---|---|---|---|---|---|---|
| | M2 | M3 | M2 | M3 | M2* | M3* |
| RMSEA | ***.073*** | ***.072*** | **.043** | **.047** | **.041** | **.036** |
| PCLOSE | .000 | .000 | .934 | .675 | **.949** | **.977** |
| CFI | .779 | .784 | ***.930*** | ***.915*** | **.951** | **.962** |
| TLI | .705 | .713 | ***.899*** | .878 | ***.907*** | ***.928*** |
| SRMR | ***.072*** | ***.063*** | **.030** | **.033** | **.025** | **.024** |
| CD | .893 | .904 | .950 | .954 | .974 | .974 |

9-Item personal agency and agency achievement models (GHAIs)

| | Single Factor | | 2-Factor | | 3-Factor | | Bifactor (covariance) | | Bifactor (directional) | |
|---|---|---|---|---|---|---|---|---|---|---|
| | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3* | M2 | M3 |
| RMSEA | .088 | .084 | ***.069*** | ***.070*** | **.040** | **.041** | **.030** | **.035** | **.030** | **.035** |
| PCLOSE | .000 | .000 | .000 | .000 | **.971** | .936 | **1.000** | **.988** | **1.000** | **.988** |
| CFI | .679 | .732 | .821 | .828 | ***.943*** | **.946** | **.973** | **.966** | **.973** | **.966** |
| TLI | .572 | .643 | .743 | .752 | ***.911*** | ***.916*** | **.949** | ***.938*** | **.949** | ***.938*** |
| SRMR | .110 | .108 | ***.060*** | ***.060*** | **.030** | **.035** | **.025** | **.038** | **.025** | **.038** |
| CD | .858 | .875 | .948 | .956 | .975 | .979 | .975 | .982 | .974 | .964 |

13-Item personal agency and agency achievement models (GHAIs)

| | Single Factor | | 2-Factor | | 3-Factor | | Bifactor | |
|---|---|---|---|---|---|---|---|---|
| | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3* |
| RMSEA | ***.067*** | ***.064*** | **.058** | **.058** | **.042** | **.043** | **.036** | **.042** |
| PCLOSE | .000 | .000 | .002 | .012 | **.996** | **.960** | **1.000** | **.974** |
| CFI | .624 | .669 | .718 | .733 | .866 | .859 | ***.911*** | .874 |
| TLI | .549 | .602 | .657 | .674 | .826 | .817 | .867 | .825 |
| SRMR | .106 | .100 | ***.076*** | ***.077*** | **.047** | ***.055*** | **.035** | ***.055*** |
| CD | .909 | .915 | .963 | .961 | .983 | .984 | .985 | .989 |

11-Item personal agency and agency achievement models (GHAIs)

| | Single Factor | | 2-Factor | | 3-Factor | | Bifactor | |
|---|---|---|---|---|---|---|---|---|
| | M2 | M3 | M2 | M3 | M2 | M3 | M2* | M3 |
| RMSEA | ***.075*** | ***.073*** | ***.063*** | ***.065*** | **.045** | **.048** | **.036** | **.047** |
| PCLOSE | .000 | .000 | .000 | .000 | .885 | .630 | **.999** | .717 |
| CFI | .654 | .690 | .760 | .766 | .888 | .881 | ***.942*** | .893 |
| TLI | .568 | .612 | .693 | .694 | .842 | .832 | ***.900*** | .840 |
| SRMR | .105 | .106 | **.050** | ***.075*** | **.046** | ***.055*** | **.032** | ***.058*** |
| CD | .893 | .900 | .955 | .958 | .978 | .981 | .983 | .988 |

*Reduced from full EFA model

Bold statistics indicate good fit; italics indicate acceptable fit and TLI values are influenced by correlations between items and the number of estimated parameters, which may explain why relative fit of models are not significantly different from independent ones. Further, non-normal data can inflate absolute fit index values and underestimate relative fit index values (Finney & DiStefano, 2006), so models may fit better than values indicate.

Overall, the bifactor models for the GPAS and 9-item GHAIs fit best, consistently demonstrating RMSEA and SRMR values below $\leq 0.05$, CFI and TLI indices above 0.9, and explaining between 96 and 98 percent of variance in the data. As agency achievement is theorized to be produced from agency freedom (Sen, 1999), of which personal agency is a component, a final 9-item GHAIs model was fitted that included a direct regression path from personal agency to agency achievement; this model consistently indicated that the predictive effect of personal agency on agency achievement was statistically and practically significant. Path diagrams for 9-item personal agency and 9-item agency models are displayed in Fig. 2, and diagrams for 13- and 11-item agency models are found in Appendix D.

### 4.6 Internal Consistency and Construct Validity

Unstandardized and standardized Cronbach's alpha coefficients for most modeled scales and subscales were greater than 0.7; only the three-item agency achievement scale included in the 9-item GHAIs showed questionable consistency in MIDUS II ($\alpha = 0.672$) and III ($\alpha = 0.666$). All final scales were comprised of items with more than five ordered categories and were therefore treated as ordinal approximations of continuous variables; pairwise Pearson's $r$ correlations were calculated to test strength and directionality of relationships, rather than Spearman rank coefficients. Average interitem correlations for all scales were of moderate or greater effect size ($r \geq 0.3$), and all pairwise Pearson's $r$ correlations among the 13 items included across measures were positive and significant at the level of $p \leq 0.05$, ranging from $r = 0.2$ to 0.55 in MIDUS II and $r = 0.2$-0.58 in MIDUS III. Correlations between personal agency and agency achievement items were typically weaker than those between items within individual factors. Correlations between factors in each identified model consistently demonstrated significant, positive relationships of moderate to strong effect ($r = 0.44$-0.67 in MIDUS II and $r = 0.38$-0.71 in MIDUS III); in the GHAIs, the relationship between intrinsic and instrumental agency factors was consistently stronger than that of either factor with agency achievement, mirroring the latent structure identified during factor analysis. Together, these findings provide evidence for the content, substantive, and structural validity of the proposed scales. Internal consistency statistics and correlations between agency items and factors are summarized in Appendix E.

Pairwise correlations between the GPAS and GHAIs and correlates of agency showed weak to strong positive relationships between all measures in MIDUS II and III. Again, relationships between personal, intrinsic, and instrumental agency were consistently stronger than those between agency achievement and personal agency, further supporting the structural validity of the proposed indicators. Correlations among the GPAS and GHAIs scales were typically larger than those between agency measures and agency correlates. Of agency correlates, the GPAS and GHAIs consistently demonstrated the most substantive relationships with self-acceptance, self-esteem, and purpose in life. Overall, agency measures showed mostly moderate to strong significant positive relationships with agency correlates, supporting both the convergent and concurrent validity of the GHAIs.

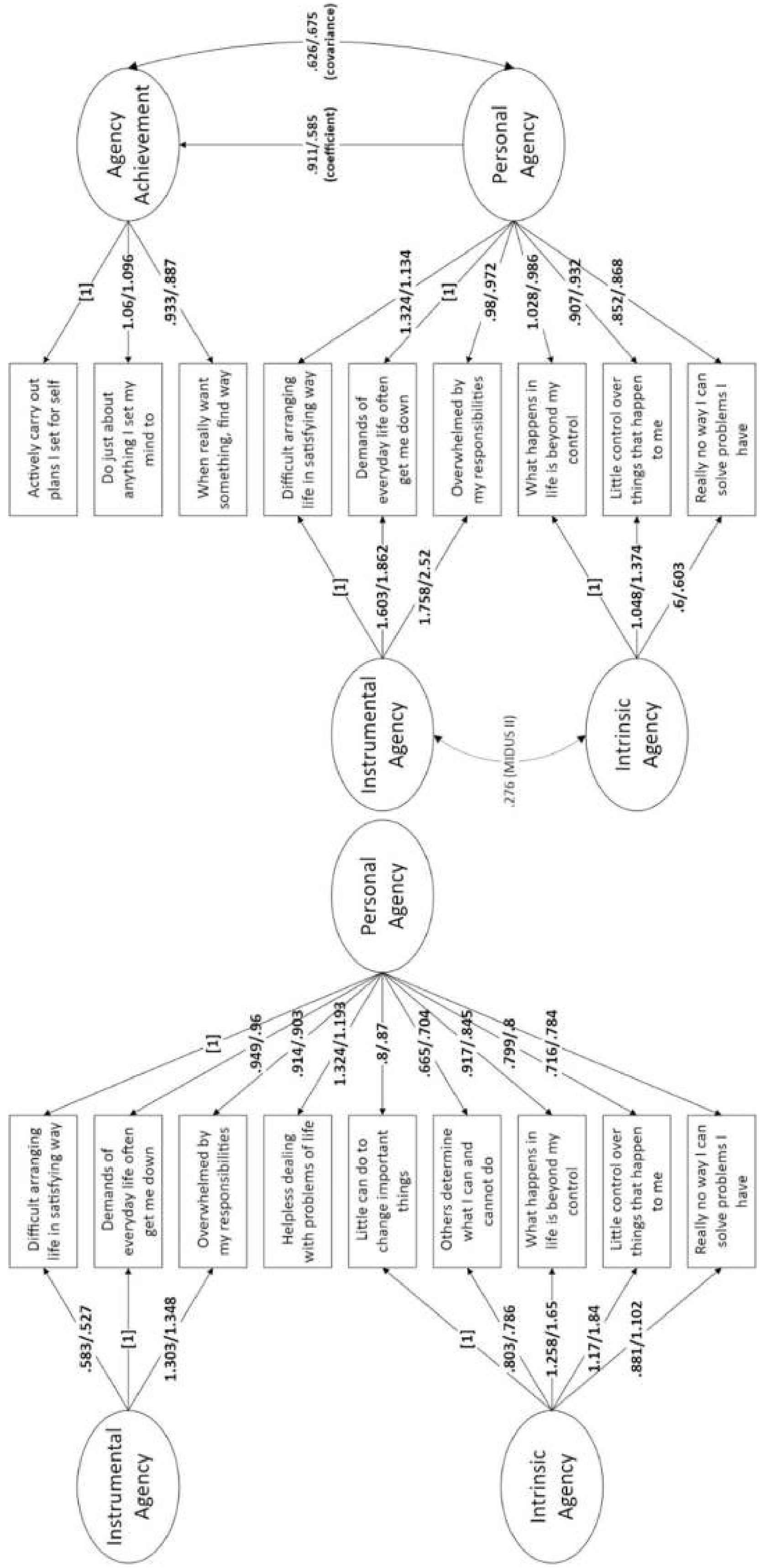



**Fig. 2** Path diagrams for GPAS and 9-item GHAIs

The weakest correlations were frequently encountered between agency measures and the "Big Five" agency trait, buttressing the argument that the personality construct is distinct from human agency as it is defined in other areas of social science research and offering some indication that the GPAS and GHAIs possess discriminant validity. Correlations between MIDUS II GHAI scores and MIDUS III agency correlates were consistent with those in individual waves, albeit with reduced coefficients. The stability of the relationships between measures over time provides preliminary evidence of the GPAS an GHAIs' predictive validity and further supports the criterion validity of the proposed tools. Correlations between the GPAS and 9-item GHAIs and agency correlates are found in Tables 7 and 8, respectively, and those between the 13- and 11-item GHAIs and agency correlates are included in Appendix E.

## 5 Discussion

Through a rigorous process of item selection and data analysis, this study produced robust initial evidence of construct validity for both a nine-item General Personal Agency Scale (GPAS) and a nine-item General Human Agency Indicators (GHAIs) tool measuring personal agency and its subdimensions alongside agency achievement. Face and content validity for the tools were established through a multi-stage item selection process, aggregating items that matched or were similar to those previously used in agency research and ensuring they were representative of agency characteristics, met subjectivity criteria, and fit within existing categories of agency measures. Factor loadings from EFA and fit indices from CFA indicate the content, substantive, and structural validity of the GPAS and 9-item GHAIs, demonstrating good-fitting bifactorial structures that align with our proposed conceptual framework for human agency. Although less consistent across MIDUS II and III, some evidence was also produced indicating the factorial and construct validity of expanded 13- and 11-item GHAIs, potentially enabling exploration of how one's ability to solve problems and build a lifestyle to one's liking may influence relationships between agency subdimensions. Only single-factor GHAIs models did not consistently

**Table 7** Pairwise correlations between 9-item personal agency (GPAS) factors and agency correlates

| | MIDUS II | | | MIDUS III | | | TWO-WAVE | | |
|---|---|---|---|---|---|---|---|---|---|
| Construct | (1) | (2) | (3) | (1) | (2) | (3) | (1) | (2) | (3) |
| (1) Personal Agency | 1.00 | | | 1.00 | | | 1.00 | | |
| (2) Intrinsic Agency | .98* | 1.00 | | .98* | 1.00 | | .98* | 1.00 | |
| (3) Instrumental Agency | .90* | .79* | 1.00 | .92* | .81* | 1.00 | .90* | .79* | 1.00 |
| (4) Life Satisfaction | .49* | .44* | .50* | .50* | .46* | .49* | .37* | .34* | .37* |
| (5) Self-Acceptance (PWB) | .68* | .63* | .68* | .69* | .63* | .70* | .53* | .49* | .52* |
| (6) Self-Esteem | .68* | .63* | .68* | .67* | .62* | .66* | .51* | .47* | .50* |
| (7) Purpose in Life (PWB) | .65* | .62* | .60* | .66* | .63* | .63* | .49* | .48* | .44* |
| (8) Positive Relations w/Others (PWB) | .53* | .50* | .51* | .52* | .48* | .51* | .41* | .39* | .40* |
| (9) Social Integration | .35* | .33* | .33* | .36* | .34* | .34* | .31* | .30* | .28* |
| (10) Agency (Big 5) | .30* | .29* | .28* | .28* | .26* | .26* | .24* | .23* | .23* |
| (11) Autonomy (PWB) | .47* | .43* | .48* | .47* | .43* | .47* | .38* | .35* | .36* |

*$p < 0.05$

**Table 8** Pairwise correlations between 9-item personal agency and agency achievement (GHAIs) factors and agency correlates

| | MIDUS II | | | | | | MIDUS III | | | | | | TWO-WAVE | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Construct | (1) | (2) | (3) | (4) | (5) | (6) | (1) | (2) | (3) | (4) | (5) | (6) | (1) | (2) | (3) | (4) | (5) | (6) |
| (1) Agency | 1.00 | | | | | | 1.00 | | | | | | 1.00 | | | | | |
| (2) Personal Agency | .98* | 1.00 | | | | | .99* | 1.00 | | | | | .98* | 1.00 | | | | |
| (3) Agency Achievement (2-factor) | .79* | .67* | 1.00 | | | | .71* | .60* | 1.00 | | | | .79* | .67* | 1.00 | | | |
| (4) Intrinsic Agency | .90* | .91* | .61* | 1.00 | | | .93* | .94* | .57* | 1.00 | | | .90* | .91* | .61* | 1.00 | | |
| (5) Instrumental Agency | .90* | .94* | .57* | .73* | 1.00 | | .92* | .95* | .51* | .78* | 1.00 | | .90* | .94* | .57* | .73* | 1.00 | |
| (6) Agency Achievement (3-factor) | .80* | .67* | 1.00* | .61* | .58* | 1.00 | .72* | .61* | 1.00* | .57* | .53* | 1.00 | .80* | .67* | 1.00* | .61* | .58* | 1.00 |
| (7) Life Satisfaction | .52* | .50* | .43* | .41* | .50* | .43* | .51* | .51* | .35* | .44* | .49* | .36* | .38* | .38* | .29* | .31* | .36* | .30* |
| (8) Self-Acceptance (PWB) | .72* | .69* | .61* | .58* | .68* | .62* | .72* | .70* | .55* | .60* | .70* | .56* | .56* | .53* | .47* | .44* | .51* | .48* |
| (9) Self-Esteem | .71* | .69* | .58* | .58* | .66* | .58* | .69* | .67* | .53* | .59* | .65* | .54* | .53* | .51* | .43* | .43* | .49* | .44* |
| (10) Purpose in Life (PWB) | .66* | .63* | .53* | .57* | .59* | .54* | .67* | .65* | .49* | .59* | .62* | .50* | .50* | .48* | .41* | .43* | .43* | .41* |
| (11) Positive Relations (PWB) | .55* | .52* | .46* | .45* | .51* | .46* | .54* | .52* | .40* | .45* | .51* | .41* | .42* | .40* | .35* | .34* | .38* | .35* |
| (12) Social Integration | .36* | .34* | .29* | .30* | .32* | .29* | .37* | .36* | .27* | .32* | .34* | .27* | .31* | .30* | .25* | .27* | .27* | .25* |
| (13) Agency (Big 5) | .36* | .32* | .40* | .28* | .29* | .40* | .33* | .29* | .36* | .26* | .27* | .36* | .30* | .26* | .35* | .23* | .24* | .35* |
| (14) Autonomy (PWB) | .50* | .48* | .44* | .40* | .47* | .44* | .49* | .47* | .39* | .41* | .47* | .40* | .39* | .38* | .34* | .32* | .36* | .34* |

*$p < 0.05$

meet standards for acceptable absolute fit, further supporting conceptualizations of agency as multidimensional. While not all models consistently demonstrated good relative fit, this can be at least partially explained by a combination of convergence issues and challenges related to both intercorrelations among agency items and non-normality of data.

Calculation of Cronbach's alpha coefficients and pairwise Pearson's *r* correlations between both individual scale items and latent factors indicated that both the GPAS and GHAIs were internally consistent, providing further evidence of their content, substantive, and structural validity, and the significance, direction, and effect sizes of relationships between agency indicators and correlates of agency present preliminary proof that both the GPAS and GHAIs possess convergent, discriminant, and criterion validity. Conducting correlational analyses across two waves of data demonstrated the stability of results over time, and correlations between MIDUS II agency measures and MIDUS III agency correlates indicated the predictive validity of the GPAS and GHAIs. Altogether, our findings make a strong case for the construct validity of the GPAS and 9-item GPAIs and provide moderate evidence for the expanded 13- and 11-item GHAIs.

### 5.1 Contribution

Because human agency is so frequently a misunderstood or misoperationalized construct, the first substantive contributions of this study were to identify common themes and characteristics of agency and organize prior conceptualizations of the construct into a cohesive framework that been implied, but never specified, in previous research. This framework builds on Sen's (1999) dichotomy of agency freedom and agency achievement, and further reifies agency's multidimensionality by clarifying its personal and interpersonal components, under which are situated the subconstructs of intrinsic, instrumental, proxy, and collective agency. The proposed human agency framework also identifies how these components of personal agency align with the essential human needs for autonomy, competence, and relatedness described by Deci and Ryan (1985) and other proponents of SDT, positioning agency as both a critical expression of and mechanism for human thriving and navigation of the social world.

Our item selection process also makes a novel methodological contribution to agency research and scale design. Constructing a composite tool from items identical to or like those used in previous research and validating them through rigorous examination using characteristics and other criteria related to the construct in question reduces the inherent bias of subjective item selection and establishes initial face and content validity for the tool produced. It also for synthesis of new measures from existing tools which, in the present context, was helpful given the lack of consensus regarding the agency construct. This approach should be considered in similar situations where there is disagreement over how a concept should be defined and operationalized.

This study successfully produced global, multidimensional indicators for personal agency and agency achievement that conform to Sen's (1985, 2006) definitions of agency freedom and achievement and align with characteristics of agency established by Alkire (2005, 2008), Burger and Walk (2016), and Kotan (2010). Previous research has explored the effects of agency using proxy measures (Alkire, 2008; Bhattacharyya, 1995), single-item measures (e.g., Graham & Nikolova, 2013; Hojman & Miranda, 2018; Okulicz-Kozaryn, 2015) and unidimensional scales (Kesavayuth et al., 2022; Pleeging et al., 2021; Serdiuk et al., 2018). Rather than relying on purportedly objective proxies or unidimensional indicators, the GPAS and GHAIs utilize items that capture respondents' subjective experience of perceptions and outcomes and capture multiple interrelated dimensions of agency. Although research on SDT (Reis et al., 2018) and psychological well-being (Margolis et al., 2021) has explored agency by combining measures approximating autonomy and competence, it has not integrated these concepts into a broader conceptual framework for human agency like the one articulated in the present study.

To our knowledge, the GPAS an GHAIs are the first tools synthesized from existing conceptualizations and measures of human agency. The development of and validation of these multidimensional agency measures has critical implications for the social sciences. It addresses a critical gap in the literature, as there is presently no established measure for operationalizing personal agency (Cavazzoni et al., 2022; Ibrahim & Alkire, 2007). The GPAS and GHAIs have utility and value for a variety of disciplines, as personal agency is a fundamental mechanism through which the social world is engaged. They can be used to explore how agency may influence diverse outcomes like prosocial behaviors (Christoph et al., 2014), group membership and volunteerism (Cicognani et al., 2015), community and civic engagement, democratic governance, and the development of other resources and capabilities that support well-being (Alkire, 2005; Ibrahim & Alkire, 2007; Peterson et al., 2008). The GPAS and GHAIs may also contribute to the advancement of empowerment theory and the processes by which individuals leverage assets and capabilities to assert control over their lives and participate in change processes (Chan & Mak, 2020; Kieffer, 1984; Rappaport, 1987; Zimmerman, 2000).

The GPAS and GHAIs have particular value for the study of community development (Bhattacharyya, 1995), the Capabilities Approach to global development (Sen, 1985, 1999), and subjective well-being (Comim, 2005; Kotan, 2010), as each field understands agency to be essential for human well-being and thriving. If the expansion of freedom, the primary end of development activities, is enacted through agency (Sen, 1999) and the "development" in community development refers to the production of agency among group members (Bhattacharyya, 1995) then, up to this point, these fields have lacked robust indicators for one of their most critical outcomes. This has severely inhibited our ability to probe, among other things, how personal agency might predict desired outcomes like well-being, solidarity, and community and civic engagement, as well as how personal agency might be predicted by environmental factors, policies, and interventions intended to contribute to development initiatives. Now that such indicators have been created and validated, the GPAS and GHAIs allow us to assess what is most essential to development—that is, whether the work of the field is advancing freedom by empowering individuals and communities to enact control over their lives and pursue self-determined goals that contribute to outcomes that they value (Alkire, 2008; Bhattacharyya, 1995; Burger & Walk, 2016; Cavazzoni et al., 2022; Kabeer, 1999; Kotan, 2010; Sen, 1999).

The GPAS and GHAIs also offer several practical benefits. Because they rely on subjectively determined, global indicators, the GPAS and GHAIs are useful for analysis of a general population and can be complemented by supplementary measures or adapted for subpopulations of interest like women (Yount et al., 2020) and children (Poteat et al., 2018; Veronese et al. 2019a, b, 2020a, b). As recommended by Alkire (2008), the tools include items that measure autonomy, ability, direct control, and effective power, and address outcomes phrased so that they can refer to both well-being and other-regarding goals. The latent structures of the GPAS and GHAIs also allow for researchers to select from a variety of scales to study agency subconstructs of interest. The GPAS and best-fitting GHAIs are only nine items in length, making them easier to deploy alongside other measures without contributing to response attrition, which may be more likely to be experienced when using longer tools like the BIF (Vallacher & Wegner, 1989), ATPA-22 (Lautamo, et al., 2021), or WAS-61 (Yount et al., 2020). While tools like Hitlin and Elder's (2006) agency model, Snyder's (1991) Agentic Pathways subscale, and Smith and et al.'s (2000) Personal Agency Scale are comparably brief, they conceptualize agency in ways that deviate from how it is commonly discussed in literature, either conflating it with or subsuming it under other constructs like hope, optimism, or planfulness. The GPAS and GHAIs do not incorporate proxy measures for agency or indicators for resources that may contribute to agency; however, the brevity of the proposed tools allows them to be deployed alongside measures, potentially enabling analysis of how those resources influence personal agency and agency achievement.

Different versions of the tools also provide distinct benefits: the GPAS provides the most nuanced representation of personal agency and its subfactors, which is typically the construct engaged in agency research. nine-item GHAIs provides the best-fitting and most elegant measures of personal agency, its subfactors, and agency achievement, and the 13- and 11-item GHAIs enable exploration of the possible influence of one's ability to solve problems and achieve a lifestyle they value on the relationships between intrinsic and instrumental agency and between both personal and instrumental agency and agency achievement, respectively. Finally, because of the nature of their construction and validation we assert that, with further testing, the GPAS and GHAIs may emerge as exemplar tools against which other agency indicators might be compared to assess their construct validity. This would be a substantive contribution to the field, as it could serve as a starting point for the development and validation of more bespoke and effective agency indicators.

### 5.2 Limitations

Despite their potential for informing social science research, the GPAS and GHAIs have several limitations related to the use of the MIDUS dataset. Our study was not able to identify items previously used to assess interpersonal, proxy, or collective agency. This contributed to a significant gap in our findings as, based on our conceptual framework for human agency, these constructs comprise one half of agency freedom and represent the relatedness component of self-determination theory. While the GHAIs are a promising start, without appropriate measures for interpersonal, proxy, and collective agency, our proposed measures are a conspicuously incomplete attempt at developing a full general human agency index.

Due to issues related to sample size and the number of fitted parameters, it was not possible to converge several of the full models recommended by EFA. To test the fit of these models, a larger sample size is required. As this study used a national dataset, the GPAS and GHAIs have not been tested for generalizability beyond the United States population, nor have they been shown to be generalizable to subgroups within that population; this is particularly true of children and youth, who were not included in sampling. Because MIDUS does not include established human agency indicators it was not possible to test criterion validity for the proposed tools through direct comparison with exemplars, and further testing of concurrent and predictive validity using regression analysis is needed before the criterion validity of the GPAS and GHAIs can be confirmed.

Finally, the GPAS and GHAIs are only useful for interpreting agency through a quantitative lens. While helpful for identify general patterns among a given population, such analyses should be complemented by qualitative research to develop a richer and more nuanced understanding of how personal agency is experienced phenomenologically across socioeconomic and cultural contexts. This need for accompanying qualitative research is compounded by the subjectively determined nature of human agency—as only respondents can discern whether and how they are experiencing agency, qualitative input is necessary to ensure that research on human agency accurately reflects their articulated lived experience.

### 5.3 Future Directions

These limitations offer several directions for future research on human agency. An immediate next step would be to leverage correlational and regression analyses to continue to probe relationships between the GPAS and GHAI and concepts previously associated with personal agency like well-being indicators and participatory behaviors. Here, the identification of significant relationships would reinforce the convergent, discriminant, and predictive validity of the tools and help us understand the effects of agency on outcomes of interest. To test their generalizability, fixed-effects models should be employed to assess whether there are differences in how agency is experienced across subpopulations categorized by socioeconomic indicators like age, race, biological sex, income, and education, and time series models would allow further testing of scales' predictive validity.

Analyses could be performed using the MIDUS dataset; however, collection of primary data may offer additional benefits. International data could be collected to examine agency in a global context, compare how it is experienced across cultures, and better assess the generalizability of the GPAS and GHAIs. Primary data incorporating other validated agency measures like the BIF (Vallacher & Wegner, 1989), ATPA-22 (Lautamo, et al. 2020), or Hitlin and Elder's (2006) agency model would enable confirmatory analysis of the GPAS' and GHAIs' criterion validity. Further, data collection with a tool comprised of multiple agency measures could expand on the present study by generating GHAIs selected from a wider variety of items representing all agency dimensions articulated in our conceptual framework. A larger sample would likely resolve the convergence issues encountered during CFA, potentially clarifying the roles of cross-loaded variables in the 13- and 11-item GHAIs. CFA could also test for predictive relationships between agency dimensions and exogenous variables, contributing to our understanding of how agency is experienced, nurtured, and enacted.

Future research should also address the opportunity to use aspects of our item selection process to create measures for interpersonal, proxy, and collective agency to complement and complete the GHAIs, either by selecting from existing or developing novel indicators based on established criteria for and characteristics of agency. The GPAS and GHAIs are important steps forward; however, on their own they are insufficient for fully operationalizing human agency because they do not capture its social dimensions. Finally, findings that emerge from quantitative analyses utilizing the GPAS and GPAIs must be informed by a rich, phenomenological understanding of agency that can only be gained through qualitative research. Should suitable interpersonal agency measures prove difficult to identify, qualitative methods may also be useful for developing novel indicators.

### 5.4 Conclusion

Human agency is an essential mechanism through which the social world is constructed and navigated, but underoperationalization has limited our ability to assess agency and its relationships with social interaction, well-being, and the thriving of individuals and communities. This study advances agency research by engaging past scholarship to identify critical characteristics of agency and constructing a cogent conceptual framework representing its latent structure. Components of that framework were reified and through the creation and initial validation of the GPAS and GPAIs, contributing a set of tools that assess subjectively determined, multidimensional aspects of personal agency and agency achievement. At a minimum, we hope that these tools will generate further discussion on how to best measure human agency and expand ongoing efforts to come to a unified operationalization of the construct.

## Appendix A: Definitions of agency (adapted from Cavazzoni et al., 2022)

| Article | Agency definition |
|---|---|
| Ahearn (2001), p. 112 | ''Agency refers to the socioculturally mediated capacity to act." |
| Alkire (2008), p. 6 | "i) Agency is exercised with respect to goals the person values; ii) agency includes effective power as well as direct control; iii) agency may advance wellbeing or may address other-regarding goals; iv) to identify agency also entails an assessment of the value of the agent's goals." |
| Alsop et al., (2006), p. 11 | "Agency is defined as an actor's or group's ability to make purposeful choices—that is, the actor is able to envisage and purposively choose options." |
| Bandura (2001), p. 8 | "Agency thus involves not only the deliberative ability to make choices and action plans, but the ability to give shape to appropriate courses of action and to motivate and regulate their execution." |
| Barandiaran et al. (2009), p. 369 | "A system doing something by itself according to certain goals or norms within a specific environment." |
| Barker (2005), p. 632 | "The socially determined capability to act and make a difference." |
| Beer (1995), p. 173 | "Any embodied system [that pursues] internal or external goals by its own actions while incontinuous long-term interaction with the environment in which it is situated." |
| Bentley-Edwards (2016), p. 78 | "The perception of what one is able to do to control their environment or circumstance." |
| Berhane et al. (2019), p. S53 | The "ability to define goals, and act on them" |
| Beyers et al. (2003), P. 360 | "AGENCY reflects the possibility of self-directed behavior." |
| Bhattacharyya (1995), p. 61 | "The capacity of a people to order their world… to create, reproduce, change, and live according to their own meaning systems, the powers effectively to define themselves as opposed to being defined by others." |
| Black, (2016), p. 296 | "Moral agency refers to the ability of individuals to determine their behavior when it affects others' well-being." |
| Bryan et al., (2014), p. 242 | "The sense that one is in control of one's life, and is the initiator of one's own actions." |
| Burger and Walk (2016) | "Individuals' capacity to gain control over their lives largely independently of social structure (Chin and Phillips 2004)," |

| Article | Agency definition |
|---|---|
| Cadenas et al., (2021), pp. 93–94 | "Critical agency can be conceptualized as a component of critical consciousness that combines motivation and beliefs of self-efficacy to address societal injustices or it is identified as ones perceived ability to make a difference for social change." |
| Cavazzoni et al., (2022), p. 1126 | "People's ability to exert control over one's life and pursue goals." |
| Cheong et al., (2017), p. 25 | "The 'ability to define one's goals and act upon them'." (citing Kabeer, 1999, p. 438) |
| Christensen and Hooker (2000), p. 133 | "AGENTS are entities which engage in normatively constrained, goal-directed, interaction with their environment." |
| Franklin and Graesser, (1996), p. 25 | "An autonomous agent is a system situated within and a part of an environment that senses that environment and acts on it, over time, in pursuit of its own agenda and so as to effect what it senses in the future." |
| Giddens (1984), p. 14 | "To be able to intervene in the world, or to refrain from such intervention, with the effect of influencing a specific process or state of affairs." |
| Graham and Nikolova (2013), p. 4 | "The capacity to pursue a purposeful and fulfilling life." |
| Grower & Ward, (2018), p. 139 | "Sexual agency is a multidimensional construct that reflects the awareness of self as a sexual being; the ability to identify, negotiate, and communicate one's sexual needs; and the successful initiation of behaviors that allow for the satisfaction of these needs." |
| Habashi & Worley, (2009), p. 44 | "The ability of the agent to reinvent the local resources that are produced by global/local discourse while responding to the same global hegemony." |
| Harvey (2002), p.173 | "The capacity of persons to transform existing states of affairs," |
| Hitlin and Elder, (2006), p. 38 | "An individual capacity for meaningful and sustained action." the sense of having the capacity for meaningful and successful action, something related, but not equivalent, to the perception of having structural opportunities to exercise such capacities." (p. 40) "Agency represents a human capacity to influence one's own life within socially structured opportunities." (pp. 56–57) "Agency, in this model, represents an individual capacity, one that is both the result of individual differences (planfulness) as well as achieved successes (self-efficacy) and a sense of temporal, self-reflective understanding about one's life chances (optimism)." (p. 60) |
| Horvath (1998), p. 139 | "A mode of human functioning that involves self-concern, self-protection, self-determination, self-efficacy, and an instrumental approach to the environment" |
| Kabeer (1999), p. 438 | "The ability to define one's goals and act upon them." |
| Kauffman, (2000), p. 8 | "A system that can act on its own behalf in an environment." |
| Klein et al., (2018) | "Sexual agency is commonly defined as the ability to act according to one's own wishes and have control of one's own sexual life." (quoting Fahs and McClelland, 2016, p. 396) |
| Kotan (2010), p. 370 | "The ability to exert power so as to influence the state of the world, do so in a purposeful way and in line with self-established objectives." |

| Article | Agency definition |
|---|---|
| Krauss et al., (2014), p. 1552 | "Psychological agency refers to beliefs about one's abilities in nonsocial environments, such as intellectual or artistic skills (Zimmerman and Zahniser 1991), and the ability to set goals and organize one's actions to achieve them (Bandura 2006; Larson and Angus 2011)." |
| Lautamo et al., (2021) | "The capacity of individuals to act independently and to make their own free choices." (citing Barker, 2005) |
| Maes, (1994), p. 136 | "A system that tries to fulfill a set of goals in a complex, dynamic environment." |
| McWhirter & McWhirter, (2016), p. 553 | "Critical agency combines commitment to and efficacy for taking action against racism and discrimination." |
| Moore et al., (2016), p. 890 | "Belief in one's ability to affect change," "Agency refers to the ability to intentionally influence one's life circumstances," (p. 891, citing Bandura, 2006) |
| Narayan and Petesch, (2007), p. 15 | "People's ability to act individually or collectively to further *their own interests*." |
| Nestadt et al. (2022), pp. NP8819-NP8820 | "Agency is the ability to define one's goals and take action to realize them (Kabeer, 1999). Practically, it is the ability to make choices and act in accordance with what one desires to do without impediment (Blanchard et al., 2013; Kabeer, 1999; Mosedale, 2005)." |
| Onyx and Bullen (2000), p. 29 | "The capacity of the individual to plan and initiate action." |
| Pleeging et al., (2021), p. 1025 | "The belief that we are able to achieve our goals." |
| Poteat et al., (2018) | "A global belief in one's ability to make and attain goals in general; Snyder et al., (1996)" |
| Reeve & Tseng, (2011), p. 258 | "We define agentic engagement as students' constructive contribution into the flow of the instruction they receive." |
| Richardson et al., (2019), p. 3 | "Agency is the ability to identify one's goals and act upon them (Kabeer, 1999)." |
| Richardson, (2018), p. 541 | "The ability to make choices and act upon those choices" (citing Malhotra & Schuler, 2005 and Kabeer, 1999) |
| Russell and Norvig, (1995), p. 33 | "An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through effectors." |
| Salem et al., (2020), p. 653 | "As women's exercise of choice in [decision-making, freedom of movement, and gender attitudes]," |
| Samari, (2017), p. 562 | "The ability to define life choices in an evolving historic and social context… Agency includes the ability to formulate one's own strategic choices, to control resources, and to make attitudinal changes under evolving constraints (Crandall et al. 2016; Dyson and Moore 1983; Yount et al. 2016)." (citing Kabeer, 1999) |
| Sen (1985), p. 203 | "What a person is free to do and achieve in pursuit of whatever goals or values he or she regards as important." |
| Mortimer & Shanahan, (2007) | ''The ability to exert influence on one's life.'' |
| Smith et al., (2000), p. 458 | Personal agency involves "achieving desired outcomes on one's own behalf (e.g., through ability, choices, perseverance, or planning)," |
| Smithers, 1995, p. 97 | "Agent systems are systems that can initiate, sustain, and maintain an ongoing and continuous interaction with their environment as an essential part of their normal functioning." |

| Article | Agency definition |
|---|---|
| Stattin et al., (2017), p. 309 | "Political agency is defined as person's intentional attempts to affect other peoples' minds about political and issues." |
| Steckermeier (2019), p. 31 | "Agency combines two different aspects: The ability to act independently from others—comparable to the process aspect of freedom in the capabilities approach; and the ability to choose from different opportunities—denoted as the opportunity aspect in the capabilities approach (Sen 2007, p. 10; Archard 2015, p. 5)." |
| Thoits (2003), p. 190 | "The ability to initiate self-change (e.g., Kiecolt, 1994; Lazarus & Folkman, 1984)." |
| Veronese et al. (2018), p. 863 | "Agency can be defined as a creative and dynamic act of resistance to oppose the oppressor and/or occupier (Peteet,1994)." |
| Veronese et al., (2019b), p. 2 | "The capacity to act positively across space and time with respect to oppressive structures in one's environment (Jeffrey, 2012)." |
| Veronese et al. (2020a), p. 243 | "The transformational and generative operations by which cognitive models are translated into proficient action… as well as the changes that occur in multilevel regulation of skills as they are perfected." (quoting Bandura, 1991, p. 61) |
| Victor et al. (2013), p. 32 | "Kotan (2010, p. 370) defines agency as 'the ability to exert power so as to influence the state of the world, do so in a purposeful way and in line with self-established objectives.' " |
| Ward et al. (2018), p. 30 | "Although scholars define sexual agency in many ways, in general it includes the acknowledgment of self as a sexual being; the ability to identify, communicate, and negotiate one's sexual needs; and the successful initiation of behaviors that allow for the satisfaction of these desires (Fetterolf & Sanchez, 2015; Froyum, 2010; Horne & Zimmer-Gembeck, 2005)." |
| Williams and Merten (2014), p. 1565 | "Agency is more than independence or autonomy; the construct refers to a person's capacity, willingness, and ability to actively construct their life course (Elder and Hitlin 2006)." |
| Yount et al. (2016) | "Women's agency refers to their ability to make strategic life choices under historically evolving constraints (Kabeer, 1999; VanderEnde et al., n.d.)." |
| Yount et al., (2020), p. 6 | "Ability to make strategic choices under constraints." (citing Kabeer, 1999) |
| Zimmerman et al., (2019), p. 1 | "'The capacity to make purposeful choices' (Kabeer, 1999)" |

## Appendix B: Descriptive Statistics and Tests of Multivariate Normality

Descriptive statistics—MIDUS II

| Agency items | Min | Max | Full sample *N* | Mean | SD | Skew | Kurt | EFA sample *N* | Mean | SD | Skew | Kurt | CFA sample *N* | Mean | SD | Skew | Kurt |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Control over life in general at present* | 1 | 4 | 4955 | 3.69 | 0.6 | −1.96 | 6.8 | 2478 | 3.68 | 0.6 | −1.9 | 6.5 | 2477 | 3.7 | 0.6 | −2.01 | 7.2 |
| Where there's a will there's a way* | 1 | 4 | 4014 | 3.21 | 0.8 | −0.71 | 2.9 | 2011 | 3.2 | 0.8 | −0.68 | 2.9 | 2003 | 3.22 | 0.8 | −0.73 | 3 |
| Do what can to change for better* | 1 | 4 | 4017 | 3.44 | 0.6 | −0.79 | 3 | 2014 | 3.44 | 0.7 | −0.8 | 3 | 2003 | 3.45 | 0.6 | −0.78 | 2.9 |
| Even when feel too much, get it all done* | 1 | 4 | 4005 | 3.22 | 0.8 | −0.73 | 2.9 | 2007 | 3.21 | 0.8 | −0.69 | 2.7 | 1998 | 3.24 | 0.8 | −0.78 | 3 |
| Know what I want out of life* | 1 | 4 | 4010 | 3.14 | 0.8 | −0.68 | 2.8 | 2009 | 3.14 | 0.8 | −0.68 | 2.9 | 2001 | 3.14 | 0.8 | −0.69 | 2.8 |
| Not afraid to voice opinions in opposition* | 1 | 7 | 4020 | 5.56 | 1.6 | −1.28 | 3.8 | 2015 | 5.58 | 1.5 | −1.28 | 3.8 | 2005 | 5.53 | 1.6 | −1.27 | 3.8 |
| In charge of situation in which I live* | 1 | 7 | 4021 | 5.96 | 1.3 | −1.73 | 6 | 2016 | 5.95 | 1.2 | −1.69 | 5.8 | 2005 | 5.97 | 1.3 | −1.77 | 6.2 |
| **Difficult arranging life in satisfying way** | **1** | **7** | **4018** | **5** | **1.9** | **−0.48** | **1.9** | **2018** | **4.97** | **1.8** | **−0.44** | **1.9** | **2000** | **5.03** | **1.9** | **−0.52** | **1.9** |

Descriptive statistics—MIDUS II

| Agency items | Min | Max | Full sample *N* | Mean | SD | Skew | Kurt | EFA sample *N* | Mean | SD | Skew | Kurt | CFA sample *N* | Mean | SD | Skew | Kurt |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Decisions not influenced by others doing* | 1 | 7 | 4021 | 5.49 | 1.6 | −1.08 | 3.3 | 2018 | 5.39 | 1.6 | −0.99 | 3.1 | 2003 | 5.59 | 1.5 | −1.18 | 3.6 |
| **Demands of everyday life often get me down** | **1** | **7** | **4020** | **4.76** | **1.9** | **−0.25** | **1.7** | **2016** | **4.77** | **1.9** | **−0.23** | **1.7** | **2004** | **4.74** | **1.9** | **−0.28** | **1.7** |
| **Actively carry out plans I set for self*** | **1** | **7** | **4023** | **5.59** | **1.5** | **−1.22** | **3.9** | **2018** | **5.59** | **1.5** | **−1.18** | **3.8** | **2005** | **5.6** | **1.5** | **−1.26** | **4** |
| Judge self by what I think is important* | 1 | 7 | 4024 | 6.09 | 1.2 | −1.74 | 6.2 | 2019 | 6.08 | 1.2 | −1.74 | 6.1 | 2005 | 6.1 | 1.2 | −1.74 | 6.3 |
| **Able to build lifestyle to my liking*** | **1** | **7** | **4022** | **5.9** | **1.3** | **−1.71** | **5.9** | **2019** | **5.86** | **1.4** | **−1.65** | **5.6** | **2003** | **5.95** | **1.3** | **−1.78** | **6.3** |
| Influenced by people with strong opinions | 1 | 7 | 4022 | 4.74 | 1.8 | −0.16 | 1.7 | 2017 | 4.72 | 1.8 | −0.15 | 1.7 | 2005 | 4.77 | 1.8 | −0.17 | 1.7 |
| Good managing daily responsibilities* | 1 | 7 | 4020 | 6.09 | 1.3 | −1.97 | 7.1 | 2016 | 6.08 | 1.3 | −1.89 | 6.8 | 2004 | 6.09 | 1.2 | −2.06 | 7.5 |
| Difficult voice opinion on controversial | 1 | 7 | 4017 | 4.97 | 2 | −0.51 | 1.8 | 2016 | 4.96 | 2 | −0.51 | 1.8 | 2001 | 4.98 | 2 | −0.51 | 1.8 |
| **Overwhelmed by my responsibilities** | **1** | **7** | **4019** | **4.85** | **1.9** | **−0.36** | **1.7** | **2016** | **4.83** | **1.9** | **−0.34** | **1.7** | **2003** | **4.87** | **1.9** | **−0.38** | **1.7** |

Descriptive statistics—MIDUS II

| Agency items | Min | Max | Full sample *N* | Mean | SD | Skew | Kurt | EFA sample *N* | Mean | SD | Skew | Kurt | CFA sample *N* | Mean | SD | Skew | Kurt |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Little can do to change important things** | **1** | **7** | **4007** | **5.61** | **1.7** | **−1.23** | **3.4** | **2009** | **5.58** | **1.7** | **−1.18** | **3.3** | **1998** | **5.65** | **1.7** | **−1.29** | **3.6** |
| **Helpless dealing with problems of life** | **1** | **7** | **4008** | **5.42** | **1.8** | **−0.88** | **2.5** | **2009** | **5.38** | **1.8** | **−0.86** | **2.5** | **1999** | **5.47** | **1.7** | **−0.91** | **2.5** |
| **Do just about anything I set my mind to*** | **1** | **7** | **4014** | **5.8** | **1.4** | **−1.55** | **5.1** | **2011** | **5.78** | **1.4** | **−1.53** | **5** | **2003** | **5.82** | **1.4** | **−1.57** | **5.1** |
| **Others determine what I can and cannot do** | **1** | **7** | **4012** | **5.88** | **1.5** | **−1.37** | **4** | **2012** | **5.83** | **1.5** | **−1.31** | **3.7** | **2000** | **5.92** | **1.5** | **−1.44** | **4.2** |
| **What happens in life is beyond my control** | **1** | **7** | **4009** | **5.19** | **1.8** | **−0.72** | **2.2** | **2009** | **5.2** | **1.8** | **−0.73** | **2.3** | **2000** | **5.17** | **1.8** | **−0.7** | **2.2** |
| **When really want something, find way*** | **1** | **7** | **4007** | **6** | **1.2** | **−1.75** | **6.7** | **2013** | **5.97** | **1.2** | **−1.76** | **6.6** | **1994** | **6.03** | **1.1** | **−1.72** | **6.8** |
| Many things interfere with what I want do | 1 | 7 | 4009 | 4.3 | 1.8 | −0.02 | 1.7 | 2011 | 4.27 | 1.8 | 0.01 | 1.7 | 1998 | 4.33 | 1.8 | −0.05 | 1.7 |
| Whether I get what want is in own hands* | 1 | 7 | 4009 | 5.25 | 1.5 | −0.95 | 3.1 | 2012 | 5.24 | 1.5 | −0.91 | 3 | 1997 | 5.26 | 1.6 | −0.98 | 3.2 |

Descriptive statistics—MIDUS II

| Agency items | Min | Max | Full sample | | | | | EFA sample | | | | | CFA sample | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | *N* | Mean | SD | Skew | Kurt | *N* | Mean | SD | Skew | Kurt | *N* | Mean | SD | Skew | Kurt |
| **Little control over things happen to me** | **1** | **7** | **4008** | **5.42** | **1.6** | **−1** | **3** | **2011** | **5.41** | **1.6** | **−0.97** | **2.9** | **1997** | **5.44** | **1.6** | **−1.04** | **3.1** |
| **Really no way I can solve problems I have** | **1** | **7** | **4008** | **5.98** | **1.4** | **−1.66** | **5.3** | **2012** | **5.97** | **1.4** | **−1.63** | **5.2** | **1996** | **6** | **1.4** | **−1.68** | **5.4** |
| Feel pushed around in life | 1 | 7 | 4012 | 5.54 | 1.7 | −0.9 | 2.5 | 2012 | 5.49 | 1.8 | −0.85 | 2.4 | 2000 | 5.59 | 1.7 | −0.95 | 2.6 |
| Happens to me in future depends on me* | 1 | 7 | 4013 | 5.89 | 1.4 | −1.67 | 5.5 | 2014 | 5.89 | 1.4 | −1.68 | 5.6 | 1999 | 5.89 | 1.4 | −1.66 | 5.5 |
| Rate control over life | 0 | 10 | 4002 | 7.91 | 1.8 | −1.24 | 4.8 | 2004 | 7.86 | 1.9 | −1.23 | 4.7 | 1998 | 7.96 | 1.8 | −1.25 | 4.9 |

*Item reversed

Tests of multivariate normality—MIDUS II

| Test | Value | $\chi^2$ | DF | *p* |
|---|---|---|---|---|
| Mardia—skewness | 75.31973 | 48,119.183 | 4960 | 0.0000 |
| Mardia—kurtosis | 1207.884 | 30,643.314 | 1 | 0.0000 |
| Henze-Zirkler | 2.113454 | $4.98e^{9}$ | 1 | 0.0000 |
| Doornik-Hansen | | | 60 | 0.0000 |

Descriptive statistics—MIDUS III

| Agency items | Min | Max | Full sample *N* | Full sample Mean | Full sample SD | Full sample Skew | Full sample Kurt | EFA sample *N* | EFA sample Mean | EFA sample SD | EFA sample Skew | EFA sample Kurt | CFA sample *N* | CFA sample Mean | CFA sample SD | CFA sample Skew | CFA sample Kurt |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Control over life in general at present* | 1 | 4 | 3288 | 3.63 | 0.6 | − 1.76 | 5.8 | 1640 | 3.62 | 0.7 | − 1.77 | 5.9 | 1648 | 3.63 | 0.6 | − 1.75 | 5.8 |
| Where there's a will there's a way* | 1 | 4 | 2888 | 3.22 | 0.8 | − 0.65 | 2.9 | 1439 | 3.2 | 0.8 | − 0.68 | 3 | 1449 | 3.23 | 0.7 | − 0.62 | 2.8 |
| Do what can to change for better* | 1 | 4 | 2897 | 3.45 | 0.6 | − 0.75 | 2.8 | 1441 | 3.45 | 0.6 | − 0.75 | 3 | 1456 | 3.46 | 0.6 | − 0.75 | 2.6 |
| Even when feel too much, get it all done* | 1 | 4 | 2887 | 3.18 | 0.8 | − 0.66 | 2.8 | 1439 | 3.14 | 0.8 | − 0.6 | 2.6 | 1448 | 3.23 | 0.8 | − 0.71 | 3 |
| Know what I want out of life* | 1 | 4 | 2863 | 3.16 | 0.8 | − 0.68 | 2.9 | 1427 | 3.16 | 0.8 | − 0.68 | 2.9 | 1436 | 3.16 | 0.8 | − 0.67 | 2.8 |
| Not afraid to voice opinions in opposition* | 1 | 7 | 2907 | 5.61 | 1.5 | − 1.31 | 4.1 | 1447 | 5.57 | 1.5 | − 1.26 | 3.9 | 1460 | 5.64 | 1.5 | − 1.35 | 4.3 |
| In charge of situation in which I live* | 1 | 7 | 2914 | 5.96 | 1.3 | − 1.77 | 6.3 | 1452 | 5.93 | 1.3 | − 1.74 | 6.2 | 1462 | 5.98 | 1.3 | − 1.79 | 6.3 |
| **Difficult arranging life in satisfying way** | **1** | **7** | **2906** | **5.11** | **1.8** | **− 0.6** | **2.1** | **1451** | **5.08** | **1.8** | **− 0.58** | **2.1** | **1455** | **5.14** | **1.8** | **− 0.61** | **2.1** |
| Decisions not influenced by others doing* | 1 | 7 | 2911 | 5.56 | 1.5 | − 1.14 | 3.6 | 1450 | 5.56 | 1.5 | − 1.19 | 3.7 | 1461 | 5.56 | 1.5 | − 1.1 | 3.4 |

Descriptive statistics—MIDUS III

| | | | Full sample | | | | | EFA sample | | | | | CFA sample | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Agency items | Min | Max | *N* | Mean | SD | Skew | Kurt | *N* | Mean | SD | Skew | Kurt | *N* | Mean | SD | Skew | Kurt |
| **Demands of everyday life often get me down** | **1** | **7** | **2913** | **4.88** | **1.8** | **−0.4** | **1.9** | **1451** | **4.91** | **1.8** | **−0.43** | **1.9** | **1462** | **4.85** | **1.9** | **−0.37** | **1.8** |
| **Actively carry out plans I set for self*** | **1** | **7** | **2910** | **5.59** | **1.5** | **−1.22** | **3.9** | **1450** | **5.56** | **1.5** | **−1.14** | **3.7** | **1460** | **5.62** | **1.5** | **−1.29** | **4.2** |
| Judge self by what I think is important* | 1 | 7 | 2911 | 5.89 | 1.3 | −1.66 | 5.8 | 1451 | 5.9 | 1.4 | −1.67 | 5.8 | 1460 | 5.89 | 1.3 | −1.66 | 5.9 |
| **Able to build lifestyle to my liking*** | **1** | **7** | **2913** | **5.84** | **1.4** | **−1.62** | **5.3** | **1451** | **5.81** | **1.4** | **−1.58** | **5.2** | **1462** | **5.88** | **1.4** | **−1.65** | **5.5** |
| Influenced by people with strong opinions | 1 | 7 | 2912 | 4.77 | 1.7 | −0.18 | 1.8 | 1452 | 4.73 | 1.7 | −0.14 | 1.8 | 1460 | 4.81 | 1.7 | −0.23 | 1.9 |
| Good managing daily responsibilities* | 1 | 7 | 2910 | 6.08 | 1.2 | −1.86 | 6.8 | 1451 | 6.07 | 1.2 | −1.81 | 6.7 | 1459 | 6.1 | 1.2 | −1.92 | 7 |
| Difficult voice opinion on controversial | 1 | 7 | 2907 | 5.06 | 1.9 | −0.61 | 2 | 1450 | 5.06 | 1.9 | −0.61 | 2 | 1457 | 5.06 | 1.9 | −0.61 | 2 |
| **Overwhelmed by my responsibilities** | **1** | **7** | **2907** | **4.99** | **1.9** | **−0.5** | **1.9** | **1450** | **4.95** | **1.9** | **−0.46** | **1.9** | **1457** | **5.02** | **1.9** | **−0.55** | **2** |
| **Little can do to change important things** | **1** | **7** | **2897** | **5.42** | **1.7** | **−1.03** | **2.9** | **1444** | **5.4** | **1.7** | **−1.04** | **2.9** | **1453** | **5.43** | **1.7** | **−1.02** | **2.9** |

Descriptive statistics—MIDUS III

| | | | Full sample | | | | | EFA sample | | | | | CFA sample | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Agency items | Min | Max | *N* | Mean | SD | Skew | Kurt | *N* | Mean | SD | Skew | Kurt | *N* | Mean | SD | Skew | Kurt |
| **Helpless dealing with problems of life** | **1** | **7** | **2898** | **5.44** | **1.7** | **−0.91** | **2.6** | **1443** | **5.45** | **1.7** | **−0.93** | **2.7** | **1455** | **5.43** | **1.7** | **−0.89** | **2.6** |
| **Do just about anything I set my mind to*** | **1** | **7** | **2897** | **5.64** | **1.5** | **−1.37** | **4.4** | **1442** | **5.6** | **1.5** | **−1.34** | **4.2** | **1455** | **5.68** | **1.4** | **−1.4** | **4.6** |
| **Others determine what I can and cannot do** | **1** | **7** | **2899** | **5.84** | **1.5** | **−1.36** | **4.1** | **1443** | **5.82** | **1.5** | **−1.33** | **4** | **1456** | **5.86** | **1.5** | **−1.39** | **4.2** |
| **What happens in life is beyond my control** | **1** | **7** | **2893** | **5.14** | **1.8** | **−0.67** | **2.2** | **1441** | **5.15** | **1.7** | **−0.69** | **2.3** | **1452** | **5.13** | **1.8** | **−0.66** | **2.2** |
| **When really want something, find way*** | **1** | **7** | **2905** | **5.91** | **1.2** | **−1.59** | **6.3** | **1447** | **5.87** | **1.2** | **−1.54** | **5.9** | **1458** | **5.96** | **1.2** | **−1.65** | **6.7** |
| Many things interfere with what I want do | 1 | 7 | 2902 | 4.34 | 1.8 | −0.06 | 1.8 | 1445 | 4.35 | 1.8 | −0.08 | 1.8 | 1457 | 4.32 | 1.8 | −0.03 | 1.8 |
| Whether I get what want is in own hands* | 1 | 7 | 2900 | 5.17 | 1.5 | −0.9 | 3.1 | 1445 | 5.11 | 1.5 | −0.9 | 3.1 | 1455 | 5.22 | 1.5 | −0.91 | 3.2 |
| **Little control over things happen to me** | **1** | **7** | **2904** | **5.31** | **1.6** | **−0.89** | **2.8** | **1447** | **5.32** | **1.6** | **−0.94** | **2.9** | **1457** | **5.29** | **1.6** | **−0.84** | **2.7** |

Descriptive statistics—MIDUS III

| Agency items | Min | Max | Full sample | | | | | EFA sample | | | | | CFA sample | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | *N* | Mean | SD | Skew | Kurt | *N* | Mean | SD | Skew | Kurt | *N* | Mean | SD | Skew | Kurt |
| **Really no way I can solve problems I have** | **1** | **7** | **2895** | **5.86** | **1.4** | **−1.44** | **4.4** | **1443** | **5.83** | **1.4** | **−1.4** | **4.4** | **1452** | **5.88** | **1.4** | **−1.48** | **4.5** |
| Feel pushed around in life | 1 | 7 | 2893 | 5.61 | 1.7 | −0.98 | 2.7 | 1442 | 5.61 | 1.7 | −0.97 | 2.7 | 1451 | 5.61 | 1.7 | −0.99 | 2.7 |
| Happens to me in future depends on me* | 1 | 7 | 2889 | 5.69 | 1.4 | −1.43 | 4.8 | 1441 | 5.67 | 1.4 | −1.45 | 4.8 | 1448 | 5.71 | 1.4 | −1.42 | 4.7 |
| Rate control over life | 0 | 10 | 2851 | 7.71 | 2 | −1.17 | 4.5 | 1422 | 7.65 | 2 | −1.17 | 4.6 | 1429 | 7.76 | 1.9 | −1.18 | 4.5 |

*Item reversed

Tests of multivariate normality—MIDUS III

| Test | Value | $\chi^2$ | DF | *p* |
|---|---|---|---|---|
| Mardia—skewness | 78.0815 | 34,579.611 | 4960 | 0.0000 |
| Mardia—kurtosis | 1222.27 | 23,770.41 | 1 | 0.0000 |
| Henze-Zirkler | 1.730598 | $2.08e^{9}$ | 1 | 0.0000 |
| Doornik-Hansen | | 30,822.486 | 60 | 0.0000 |

## Appendix C: Factor Loadings for Initial 3-, 2-, and Single-Factor IPF Extractions w/Direct Oblimin Rotation

| | Three-factor | | | | | | | | Two-factor | | | | | | Single factor | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Factor 1 | | Factor 2 | | Factor 3 | | Uniqueness | | Factor 1 | | Factor 2 | | Uniqueness | | Factor 1 | | Uniqueness | |
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Control over life in general at present | | | | | | | 0.825 | 0.828 | | | | | 0.824 | 0.829 | 0.420 | 0.414 | 0.823 | 0.828 |
| Where there's a will there's a way | | | **0.700** | 0.545 | | | 0.610 | 0.744 | | | **0.697** | **0.573** | 0.625 | 0.742 | 0.444 | 0.356 | 0.803 | 0.873 |
| Do what can to change for better | | | **0.618** | 0.512 | | | 0.640 | 0.721 | | | **0.631** | **0.574** | 0.639 | 0.719 | 0.492 | 0.402 | 0.758 | 0.839 |
| Even when feel too much, get it all done | | | 0.512 | 0.433 | | | 0.756 | 0.803 | | | 0.505 | 0.486 | 0.767 | 0.801 | 0.399 | 0.336 | 0.841 | 0.887 |
| Know what I want out of life | | | 0.443 | 0.364 | | | 0.685 | 0.690 | | | 0.475 | 0.450 | 0.697 | 0.723 | 0.514 | 0.484 | 0.736 | 0.765 |
| Not afraid to voice opinions in opposition | | | 0.359 | | 0.334 | 0.358 | 0.733 | 0.777 | | | 0.411 | 0.385 | 0.823 | 0.864 | 0.370 | | 0.863 | 0.912 |
| In charge of situation in which I live | | | 0.423 | 0.419 | | | 0.592 | 0.603 | | | 0.449 | 0.443 | 0.606 | 0.602 | **0.615** | **0.614** | 0.622 | 0.623 |

| | Three-factor | | | | | | | | Two-factor | | | | | | Single factor | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Factor 1 | | Factor 2 | | Factor 3 | | Uniqueness | | Factor 1 | | Factor 2 | | Uniqueness | | Factor 1 | | Uniqueness | |
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Difficult arranging life in satisfying way | **0.619** | **0.617** | | | | | 0.497 | 0.483 | **0.611** | **0.602** | | | 0.537 | 0.516 | **0.655** | **0.682** | 0.571 | 0.535 |
| Decisions not influenced by others doing | | | 0.343 | | | 0.340 | 0.793 | 0.783 | | | 0.386 | 0.376 | 0.839 | 0.859 | 0.355 | | 0.874 | 0.901 |
| Demands of everyday life often get me down | **0.651** | **0.689** | | | | | 0.593 | 0.549 | **0.651** | **0.681** | | | 0.616 | 0.585 | 0.550 | **0.576** | 0.698 | 0.668 |
| Actively carry out plans I set for self | | | 0.538 | 0.513 | | | 0.638 | 0.649 | | | **0.557** | **0.573** | 0.636 | 0.650 | 0.546 | 0.507 | 0.702 | 0.744 |
| Judge self by what I think is important | | | 0.430 | 0.342 | | | 0.793 | 0.838 | | | 0.472 | 0.417 | 0.817 | 0.862 | 0.331 | | 0.890 | 0.929 |
| Able to build lifestyle to my liking | | | 0.449 | 0.393 | | | 0.635 | 0.636 | | | 0.474 | 0.439 | 0.641 | 0.638 | **0.575** | **0.581** | 0.670 | 0.662 |
| Influenced by people with strong opinions | | | | | | 0.364 | 0.825 | 0.795 | | | | | 0.914 | 0.923 | | | 0.913 | 0.924 |

| | Three-factor | | | | | | | | Two-factor | | | | | | Single factor | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Factor 1 | | Factor 2 | | Factor 3 | | Uniqueness | | Factor 1 | | Factor 2 | | Uniqueness | | Factor 1 | | Uniqueness | |
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Good managing daily responsibilities | | | 0.452 | 0.405 | | | 0.700 | 0.719 | | | 0.479 | 0.467 | 0.704 | 0.727 | 0.504 | 0.471 | 0.746 | 0.778 |
| Difficult voice opinion on controversial | | | | | | | 0.738 | 0.767 | | | | | 0.812 | 0.857 | 0.435 | 0.377 | 0.811 | 0.858 |
| Overwhelmed by my responsibilities | **0.642** | **0.615** | | | | | 0.570 | 0.598 | **0.633** | **0.604** | | | 0.619 | 0.636 | **0.558** | **0.561** | 0.688 | 0.685 |
| Little can do to change important things | **0.569** | **0.626** | | | | | 0.625 | 0.568 | **0.583** | **0.642** | | | 0.644 | 0.602 | **0.555** | **0.580** | 0.692 | 0.663 |
| Helpless dealing with problems of life | **0.698** | **0.767** | | | | | 0.445 | 0.378 | **0.710** | **0.774** | | | 0.442 | 0.377 | **0.702** | **0.739** | 0.507 | 0.453 |
| Do just about anything I set my mind to | | | **0.554** | 0.545 | | | 0.629 | 0.602 | | | 0.522 | 0.472 | 0.666 | 0.676 | 0.529 | 0.530 | 0.720 | 0.719 |
| Others determine what I can and cannot do | 0.538 | **0.596** | | | | | 0.603 | 0.594 | 0.545 | **0.605** | | | 0.601 | 0.592 | **0.615** | **0.609** | 0.622 | 0.629 |

| | Three-factor | | | | | | | | Two-factor | | | | | | Single factor | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Factor 1 | | Factor 2 | | Factor 3 | | Uniqueness | | Factor 1 | | Factor 2 | | Uniqueness | | Factor 1 | | Uniqueness | |
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| What happens in life is beyond my control | **0.626** | **0.654** | | | | | 0.561 | 0.521 | **0.638** | **0.670** | | | 0.615 | 0.554 | **0.560** | **0.619** | 0.687 | 0.617 |
| When really want something, find way | | | **0.646** | **0.722** | | | 0.518 | 0.514 | | | **0.615** | **0.687** | 0.556 | 0.563 | **0.601** | 0.526 | 0.639 | 0.723 |
| Many things interfere with what I want do | **0.668** | **0.656** | | | | | 0.614 | 0.574 | **0.684** | **0.666** | | | 0.611 | 0.572 | 0.531 | **0.602** | 0.719 | 0.638 |
| Whether I get what want is in own hands | | | 0.387 | 0.544 | | | 0.682 | 0.624 | | | 0.336 | 0.442 | 0.740 | 0.738 | 0.505 | 0.470 | 0.745 | 0.779 |
| Little control over things happen to me | **0.649** | **0.726** | | | | | 0.508 | 0.465 | **0.660** | **0.743** | | | 0.567 | 0.503 | **0.603** | **0.631** | 0.636 | 0.602 |
| Really no way I can solve problems I have | **0.639** | **0.702** | | | | | 0.519 | 0.450 | **0.653** | **0.716** | | | 0.550 | 0.461 | **0.625** | **0.691** | 0.609 | 0.523 |
| Feel pushed around in life | **0.636** | **0.617** | | | | | 0.555 | 0.581 | **0.637** | **0.618** | | | 0.575 | 0.587 | **0.607** | **0.610** | 0.632 | 0.628 |

| Item | Three-factor | | | | | | | | Two-factor | | | | | | Single factor | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Factor 1 | | Factor 2 | | Factor 3 | | Uniqueness | | Factor 1 | | Factor 2 | | Uniqueness | | Factor 1 | | Uniqueness | |
| | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Happens to me in future depends on me | | | 0.354 | 0.497 | | | 0.730 | 0.651 | | | | 0.412 | 0.822 | 0.737 | 0.416 | 0.483 | 0.827 | 0.767 |
| Rate control over life | 0.371 | | | 0.371 | | | 0.668 | 0.637 | 0.379 | 0.320 | | 0.328 | 0.690 | 0.667 | **0.558** | **0.576** | 0.688 | 0.669 |
| Eigenvalue | 8.348 | 8.268 | 1.507 | 1.632 | 0.866 | 0.962 | | | 8.322 | 8.236 | 1.483 | 1.605 | | | 8.269 | 8.180 | | |
| Proportion of variance | 0.779 | 0.761 | 0.141 | 0.150 | 0.081 | 0.089 | | | 0.849 | 0.837 | 0.151 | 0.163 | | | 1.000 | 1.000 | | |

Bold numbers indicate a loading of good or better ($\lambda \geq 0.55$); table does not include loadings $< 0.32$

## Appendix D: Factor Loadings and Path Diagrams for 13- and 11-Item Agency Scales

### Factor Loadings for 13-Item Personal Agency And Agency Achievement Scales (GHAIs)

Full Scale Factor Loadings

| | SINGLE FACTOR | | | | 2-FACTOR | | | | | | 3-FACTOR | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Agency | | Uniqueness | | Personal agency | | Agency achievement | | Uniqueness | | Intrinsic agency | | Instrumental agency | | Agency achievement | | Uniqueness | |
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Little can do to change important things | **0.620** | **0.650** | 0.616 | 0.578 | **0.623** | **0.703** | | | 0.599 | 0.545 | **0.570** | **0.686** | | | | | 0.580 | 0.508 |
| Others determine what I can and cannot do | **0.637** | **0.634** | 0.594 | 0.598 | **0.572** | **0.624** | | | 0.593 | 0.589 | 0.495 | **0.579** | | | | | 0.583 | 0.569 |
| What happens in life is beyond my control | **0.605** | **0.660** | 0.634 | 0.564 | **0.701** | **0.714** | | | 0.581 | 0.530 | **0.723** | **0.759** | | | | | 0.523 | 0.461 |
| Little control over things happen to me | **0.648** | **0.684** | 0.580 | 0.532 | **0.739** | **0.754** | | | 0.524 | 0.488 | **0.757** | **0.772** | | | | | 0.460 | 0.426 |
| Really no way I can solve problems I have | **0.680** | **0.735** | 0.538 | 0.460 | **0.711** | **0.745** | | | 0.506 | 0.440 | **0.726** | **0.655** | | | | | 0.446 | 0.423 |

Full Scale Factor Loadings

| | SINGLE FACTOR | | | | 2-FACTOR | | | | | | 3-FACTOR | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Agency | | Uniqueness | | Personal agency | | Agency achievement | | Uniqueness | | Intrinsic agency | | Instrumental agency | | Agency achievement | | Uniqueness | |
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Helpless dealing with problems of life | **0.752** | **0.798** | 0.435 | 0.363 | **0.692** | **0.764** | | | 0.430 | 0.356 | 0.374 | 0.404 | 0.383 | 0.453 | | | 0.434 | 0.353 |
| Difficult arranging life in satisfying way | **0.663** | **0.693** | 0.560 | 0.520 | 0.524 | **0.558** | | | 0.567 | 0.525 | | | **0.595** | **0.686** | | | 0.480 | 0.415 |
| Demands of everyday life often get me down | **0.585** | **0.621** | 0.658 | 0.615 | **0.574** | **0.604** | | | 0.648 | 0.609 | | | **0.730** | **0.709** | | | 0.477 | 0.470 |
| Overwhelmed by my responsibilities | **0.578** | **0.571** | 0.666 | 0.674 | **0.576** | **0.571** | | | 0.654 | 0.666 | | | **0.718** | **0.664** | | | 0.490 | 0.544 |
| Able to build lifestyle to my liking | 0.514 | 0.510 | 0.736 | 0.740 | | | 0.485 | 0.460 | 0.657 | 0.653 | | | 0.316 | 0.389 | 0.453 | 0.380 | 0.635 | 0.637 |
| Actively carry out plans I set for self | 0.486 | 0.435 | 0.764 | 0.811 | | | 0.518 | **0.591** | 0.665 | 0.632 | | | | | 0.494 | 0.498 | 0.670 | 0.646 |

Full Scale Factor Loadings

| | SINGLE FACTOR | | | | 2-FACTOR | | | | | | 3-FACTOR | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Agency | | Uniqueness | | Personal agency | | Agency achievement | | Uniqueness | | Intrinsic agency | | Instrumental agency | | Agency achievement | | Uniqueness | |
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Do just about anything I set my mind to | 0.481 | 0.458 | 0.769 | 0.790 | | | **0.604** | **0.530** | 0.617 | 0.655 | | | | | **0.612** | 0.515 | 0.595 | 0.638 |
| When really want something, find way | 0.541 | 0.442 | 0.707 | 0.805 | | | **0.777** | **0.734** | 0.430 | 0.506 | | | | | **0.782** | **0.781** | 0.401 | 0.392 |
| Eigenvalue | 4.742 | 4.952 | | | 4.792 | 4.996 | 0.737 | 0.811 | | | 4.846 | 5.050 | 0.772 | 0.867 | 0.609 | 0.602 | | |
| Proportion of variance | 1.000 | 1.000 | | | 0.867 | 0.860 | 0.133 | 0.140 | | | 0.778 | 0.775 | 0.124 | 0.133 | 0.098 | 0.092 | | |

Bold numbers indicate a loading of good or better (λ ≥ 0.55); table does not include loadings < 0.2

Loadings on Individual Factors

| | Personal agency | | Uniqueness | | Intrinsic agency | | Instrumental agency | | Agency achievement | | Uniqueness | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Little can do to change important things | **0.634** | **0.671** | 0.599 | 0.549 | **0.659** | **0.710** | | | | | 0.566 | 0.496 |
| Others determine what I can and cannot do | **0.631** | **0.637** | 0.602 | 0.594 | **0.634** | **0.660** | | | | | 0.598 | 0.564 |
| What happens in life is beyond my control | **0.636** | **0.680** | 0.596 | 0.538 | **0.683** | **0.719** | | | | | 0.533 | 0.483 |

| Loadings on Individual Factors | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Personal agency | | Uniqueness | | Intrinsic agency | | Instrumental agency | | Agency achievement | | Uniqueness | |
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Little control over things happen to me | **0.676** | **0.708** | 0.543 | 0.499 | **0.718** | **0.744** | | | | | 0.485 | 0.446 |
| Really no way I can solve problems I have | **0.698** | **0.748** | 0.513 | 0.441 | **0.743** | **0.766** | | | | | 0.447 | 0.413 |
| Helpless dealing with problems of life | **0.756** | **0.801** | 0.429 | 0.359 | **0.694** | **0.742** | **0.703** | **0.755** | | | 0.506 | 0.430 |
| Difficult arranging life in satisfying way | **0.643** | **0.668** | 0.586 | 0.553 | | | **0.735** | **0.768** | | | 0.460 | 0.411 |
| Demands of everyday life often get me down | **0.599** | **0.629** | 0.641 | 0.604 | | | **0.699** | **0.720** | | | 0.511 | 0.482 |
| Overwhelmed by my responsibilities | **0.594** | **0.585** | 0.647 | 0.658 | | | **0.686** | **0.665** | | | 0.529 | 0.557 |
| Able to build lifestyle to my liking | | | | | | | 0.491 | 0.499 | **0.554** | 0.535 | 0.693 | 0.713 |
| Actively carry out plans I set for self | | | | | | | | | **0.580** | **0.591** | 0.664 | 0.651 |
| Do just about anything I set my mind to | | | | | | | | | **0.618** | **0.605** | 0.618 | 0.634 |
| When really want something, find way | | | | | | | | | **0.755** | **0.726** | 0.431 | 0.473 |
| Eigenvalue | 3.845 | 4.206 | | | 2.852 | 3.148 | 2.235 | 2.368 | 1.595 | 1.529 | | |

**Factor loadings for 11-item personal agency and agency achievement scales (GHAIs)**

Full Scale Factor Loadings

| | SINGLE FACTOR | | | | 2-FACTOR | | | | | | 3-FACTOR | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Agency | | Uniqueness | | Personal agency | | Agency achievement | | Uniqueness | | Instrumental agency | | Intrinsic agency | | Agency achievement | | Uniqueness | |
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| What happens in life is beyond my control | **0.577** | **0.627** | 0.667 | 0.607 | **0.610** | **0.625** | | | 0.640 | 0.595 | | | **0.685** | **0.753** | | | 0.528 | 0.439 |
| Little control over things happen to me | **0.632** | **0.651** | 0.601 | 0.576 | **0.663** | **0.661** | | | 0.569 | 0.558 | | | **0.803** | **0.747** | | | 0.387 | 0.415 |
| Really no way I can solve problems I have | **0.647** | **0.709** | 0.582 | 0.497 | **0.617** | **0.690** | | | 0.572 | 0.488 | | | **0.625** | **0.588** | | | 0.494 | 0.441 |
| Helpless dealing with problems of life | **0.750** | **0.784** | 0.438 | 0.386 | **0.713** | **0.784** | | | 0.424 | 0.364 | 0.418 | 0.522 | 0.321 | 0.309 | | | 0.444 | 0.379 |
| Difficult arranging life in satisfyingway | **0.674** | **0.713** | 0.546 | 0.492 | **0.603** | **0.662** | | | 0.545 | 0.492 | **0.614** | **0.711** | | | | | 0.483 | 0.415 |

Full Scale Factor Loadings

| | SINGLE FACTOR | | | | 2-FACTOR | | | | | | 3-FACTOR | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Agency | | Uniqueness | | Personal agency | | Agency achievement | | Uniqueness | | Instrumental agency | | Intrinsic agency | | Agency achievement | | Uniqueness | |
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Demands of everyday life often get me down | **0.601** | **0.639** | 0.639 | 0.591 | **0.682** | **0.713** | | | 0.586 | 0.541 | **0.734** | **0.721** | | | | | 0.477 | 0.469 |
| Overwhelmed by my responsi-bilities | **0.593** | **0.590** | 0.649 | 0.652 | **0.683** | **0.678** | | | 0.592 | 0.598 | **0.722** | **0.661** | | | | | 0.491 | 0.545 |
| Able to build lifestyle to my liking | 0.534 | 0.534 | 0.714 | 0.715 | 0.228 | 0.301 | 0.407 | 0.356 | 0.674 | 0.675 | 0.315 | 0.402 | | | 0.443 | 0.370 | 0.637 | 0.637 |
| Actively carry out plans I set for self | 0.498 | 0.459 | 0.752 | 0.790 | | | 0.476 | 0.483 | 0.677 | 0.677 | | | | | 0.491 | 0.492 | 0.672 | 0.644 |
| Do just about anything I set my mind to | 0.484 | 0.470 | 0.766 | 0.779 | | | **0.622** | 0.531 | 0.598 | 0.636 | | | | | **0.617** | 0.521 | 0.597 | 0.632 |
| When really want some-thing, find way | 0.548 | 0.454 | 0.699 | 0.794 | | | **0.804** | **0.808** | 0.386 | 0.390 | | | | | **0.784** | **0.778** | 0.400 | 0.397 |
| Eigenvalue | 3.947 | 4.122 | | | 4.009 | 4.177 | 0.727 | 0.810 | | | 4.070 | 4.235 | 0.757 | 0.831 | 0.563 | 0.521 | | |

Full Scale Factor Loadings

| | SINGLE FACTOR | | | | 2-FACTOR | | | | | | 3-FACTOR | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Agency | | Uniqueness | | Personal agency | | Agency achievement | | Uniqueness | | Instrumental agency | | Intrinsic agency | | Agency achievement | | Uniqueness | |
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Proportion of variance | 1.000 | 1.000 | | | 0.846 | 0.838 | 0.154 | 0.162 | | | 0.755 | 0.758 | 0.141 | 0.149 | 0.105 | 0.093 | | |

Bold numbers indicate a loading of good or better ($\lambda \geq 0.55$); table does not include loadings < 0.2

Loadings on Individual Factors

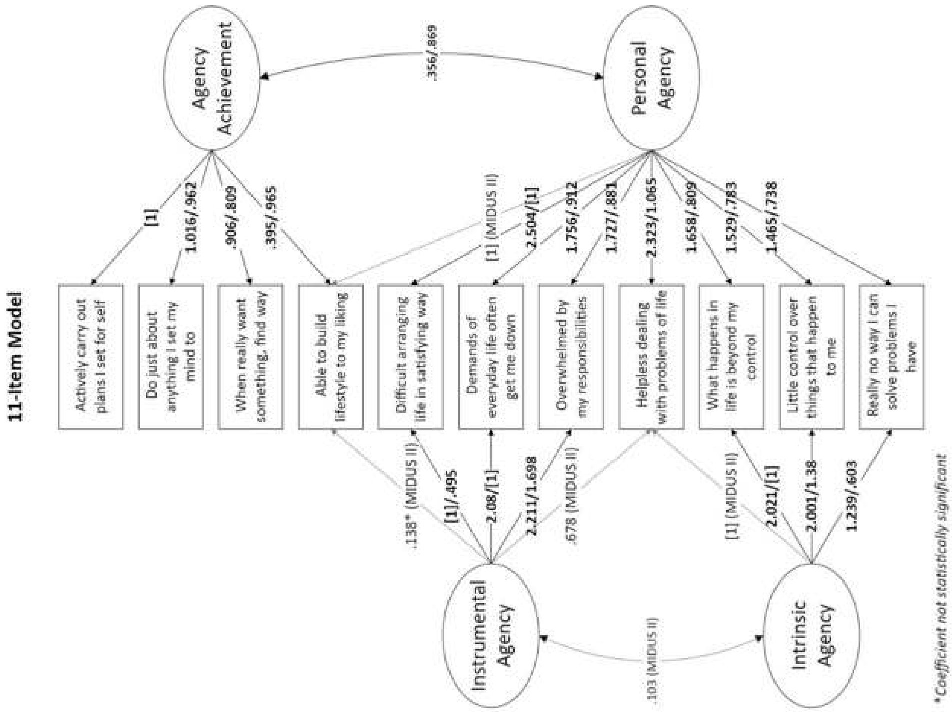
| | Personal agency | | Uniqueness | | Intrinsic agency | | Instrumental agency | | Agency achievement | | Uniqueness | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| What happens in life is beyond my control | **0.597** | **0.639** | 0.643 | 0.591 | **0.690** | **0.726** | | | | | 0.524 | 0.473 |
| Little control over things happen to me | **0.651** | **0.666** | 0.576 | 0.557 | **0.759** | **0.748** | | | | | 0.425 | 0.440 |
| Really no way I can solve problems I have | **0.650** | **0.719** | 0.578 | 0.483 | **0.723** | **0.775** | | | | | 0.478 | 0.400 |
| Helpless dealing with problems of life | **0.763** | **0.797** | 0.417 | 0.365 | **0.661** | **0.705** | **0.703** | **0.755** | | | 0.506 | 0.430 |
| Difficult arranging life in satisfying way | **0.676** | **0.713** | 0.543 | 0.492 | | | **0.735** | **0.768** | | | 0.460 | 0.411 |
| Demands of everyday life often get me down | **0.635** | **0.666** | 0.596 | 0.557 | | | **0.699** | **0.720** | | | 0.511 | 0.482 |
| Overwhelmed by my responsibilities | **0.628** | **0.622** | 0.606 | 0.613 | | | **0.686** | **0.665** | | | 0.529 | 0.557 |

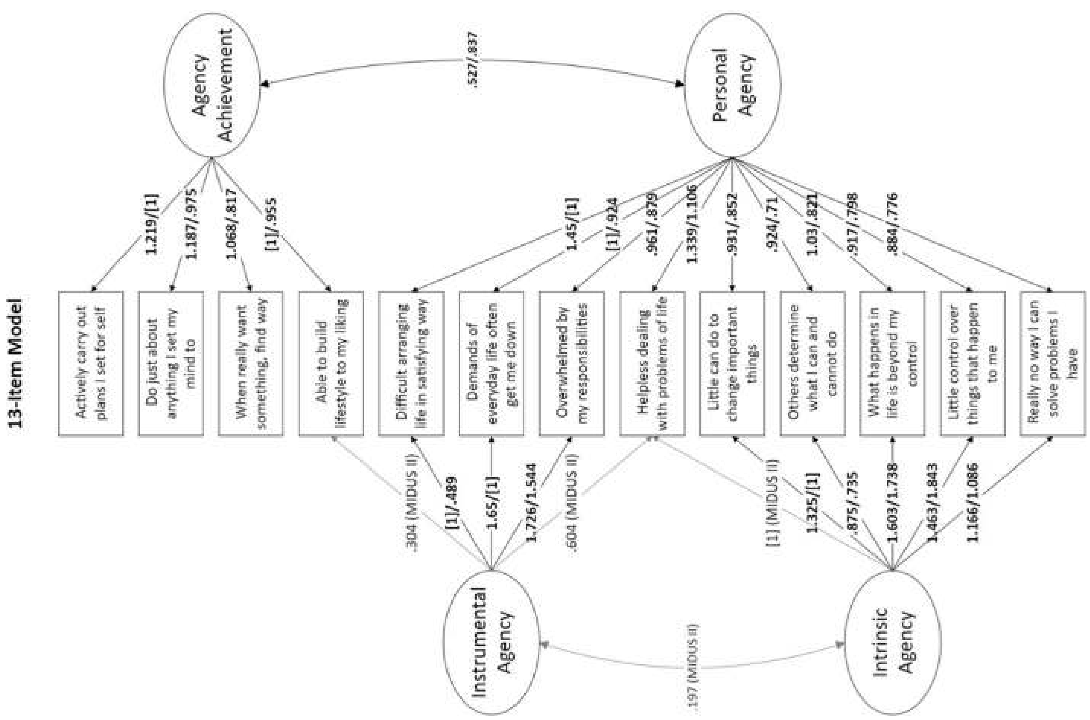
Loadings on Individual Factors

| | Personal agency | | Uniqueness | | Intrinsic agency | | Instrumental agency | | Agency achievement | | Uniqueness | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Item | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Able to build lifestyle to my liking | 0.469 | 0.478 | 0.780 | 0.772 | | | 0.491 | 0.499 | **0.554** | 0.535 | 0.693 | 0.713 |
| Actively carry out plans I set for self | | | | | | | | | **0.580** | **0.591** | 0.664 | 0.651 |
| Do just about anything I set my mind to | | | | | | | | | **0.618** | **0.605** | 0.618 | 0.634 |
| When really want something, find way | | | | | | | | | **0.755** | **0.726** | 0.431 | 0.473 |
| Eigenvalue | 3.261 | 3.570 | | | 2.010 | 2.185 | 2.235 | 2.368 | 1.595 | 1.529 | | |

For cross-loaded items in 3-factor solutions, uniqueness reported is for factor on which items had highest loading

## Path Diagrams for 13- and 11-Item GHAIs


11-Item Model
Actively carry out plans I set for self
Do just about anything I set my mind to
When really want something, find way
Able to build lifestyle to my liking
Difficult arranging life in satisfying way
Demands of everyday life often get me down
Overwhelmed by my responsibilities
Helpless dealing with problems of life
What happens in life is beyond my control
Little control over things that happen to me
Really no way I can solve problems I have
Agency Achievement
Personal Agency
Instrumental Agency
Intrinsic Agency
.356/.869
[1]
1.016/.962
.906/.809
.395/.965
[1] (MIDUS II)
2.504/[1]
1.756/.912
1.727/.881
2.323/1.065
1.658/.809
1.529/.783
1.465/.738
.138* (MIDUS II)
[1]/.495
2.08/[1]
2.211/1.698
.678 (MIDUS II)
[1] (MIDUS II)
2.021/[1]
2.001/1.38
1.239/.603
.103 (MIDUS II)
*Coefficient not statistically significant


13-Item Model
Actively carry out plans I set for self
Do just about anything I set my mind to
When really want something, find way
Able to build lifestyle to my liking
Difficult arranging life in satisfying way
Demands of everyday life often get me down
Overwhelmed by my responsibilities
Helpless dealing with problems of life
Little can do to change important things
Others determine what I can and cannot do
What happens in life is beyond my control
Little control over things that happen to me
Really no way I can solve problems I have
Agency Achievement
Personal Agency
Instrumental Agency
Intrinsic Agency
.527/.837
1.219/[1]
1.187/.975
1.068/.817
[1]/.955
1.45/[1]
[1]/.924
.961/.879
1.339/1.106
.931/.852
.924/.71
1.03/.821
.917/.798
.884/.776
.304 (MIDUS II)
[1]/.489
1.65/[1]
1.726/1.544
.604 (MIDUS II)
[1] (MIDUS II)
1.325/[1]
.875/.735
1.603/1.738
1.463/1.843
1.166/1.086
.197 (MIDUS II)

## Appendix E: Internal Consistency Statistics and Construct Validity

### Internal Consistency Statistics for Selected Agency Items and Final GPAS and GHAIs Scales

| Original agency items | | | Intrinsic agency scales | | | |
|---|---|---|---|---|---|---|
| | 9-ITEM | | 5-ITEM | | 3-ITEM | |
| | M2 | M3 | M2 | M3 | M2 | M3 |
| Alpha | 0.910 | 0.911 | 0.809 | 0.833 | 0.753 | 0.782 |
| Std. Alpha | 0.914 | 0.913 | 0.812 | 0.835 | 0.760 | 0.786 |
| Avg. Interitem covariance | 0.556 | 0.561 | 1.177 | 1.294 | 1.312 | 1.411 |
| Avg. interitem correlation | 0.261 | 0.260 | 0.464 | 0.503 | 0.513 | 0.550 |

| Final agency scales | | | | | | | Instrumental agency scale | |
|---|---|---|---|---|---|---|---|---|
| | 13-ITEM | | 11-ITEM | | 9-ITEM | | 3-ITEM | |
| | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Alpha | 0.876 | 0.884 | 0.855 | 0.864 | 0.817 | 0.829 | 0.724 | 0.738 |
| Std. Alpha | 0.877 | 0.884 | 0.857 | 0.864 | 0.820 | 0.830 | 0.717 | 0.731 |
| Avg. Interitem covariance | 0.912 | 0.966 | 0.913 | 0.956 | 0.882 | 0.924 | 1.265 | 1.283 |
| Avg. interitem correlation | 0.354 | 0.370 | 0.353 | 0.366 | 0.337 | 0.351 | 0.388 | 0.405 |

| Personal agency scales | | | | | | | Agency achievement scales | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 9-ITEM | | 7-ITEM | | 6-ITEM | | 4-ITEM | | 3-ITEM | |
| | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 | M2 | M3 |
| Alpha | 0.864 | 0.881 | 0.839 | 0.858 | 0.801 | 0.826 | 0.715 | 0.703 | 0.672 | 0.666 |
| Std. Alpha | 0.867 | 0.884 | 0.842 | 0.861 | 0.805 | 0.830 | 0.722 | 0.709 | 0.685 | 0.678 |
| Avg. Interitem covariance | 1.216 | 1.316 | 1.307 | 1.392 | 1.228 | 1.328 | 0.706 | 0.715 | 0.754 | 0.760 |
| Avg. interitem correlation | 0.420 | 0.458 | 0.433 | 0.470 | 0.408 | 0.448 | 0.393 | 0.379 | 0.420 | 0.412 |

**Pairwise Pearson's *r* correlations between final agency items**

| MIDUS II | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Item | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) | (13) |
| (1) Little can do to change important things | 1 | | | | | | | | | | | | |
| (2) Others determine what I can and cannot do | 0.41* | 1 | | | | | | | | | | | |
| (3) What happens in life is beyond my control | 0.43* | 0.45* | 1 | | | | | | | | | | |
| (4) Little control over things happen to me | 0.44* | 0.40* | 0.55* | 1 | | | | | | | | | |
| (5) Really no way I can solve problems I have | 0.50* | 0.46* | 0.45* | 0.54* | 1 | | | | | | | | |
| (6) Helpless dealing with problems of life | 0.51* | 0.44* | 0.48* | 0.47* | 0.52* | 1 | | | | | | | |
| (7) Difficult arranging life in satisfying way | 0.34* | 0.39* | 0.35* | 0.35* | 0.37* | 0.53* | 1 | | | | | | |
| (8) Demands of everyday life often get me down | 0.31* | 0.31* | 0.32* | 0.35* | 0.34* | 0.49* | 0.48* | 1 | | | | | |

| MIDUS II | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Item | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) | (13) |
| (9) Overwhelmed by my responsibilities | 0.30* | 0.32* | 0.32* | 0.34* | 0.34* | 0.47* | 0.48* | 0.54* | 1 | | | | |
| (10) Able to build lifestyle to my liking | 0.23* | 0.28* | 0.22* | 0.25* | 0.27* | 0.39* | 0.44* | 0.31* | 0.30* | 1 | | | |
| (11) Actively carry out plans I set for self | 0.23* | 0.26* | 0.22* | 0.25* | 0.28* | 0.34* | 0.33* | 0.26* | 0.23* | 0.41* | 1 | | |
| (12) Do just about anything I set my mind to | 0.27* | 0.29* | 0.26* | 0.25* | 0.29* | 0.32* | 0.29* | 0.21* | 0.20* | 0.31* | 0.34* | 1 | |
| (13) When really want something, find way | 0.29* | 0.30* | 0.26* | 0.28* | 0.31* | 0.35* | 0.32* | 0.24* | 0.24* | 0.39* | 0.39* | 0.54* | 1 |
| MIDUS III | | | | | | | | | | | | | |
| (1) Little can do to change important things | 1 | | | | | | | | | | | | |
| (2) Others determine what I can and cannot do | 0.42* | 1 | | | | | | | | | | | |
| (3) What happens in life is beyond my control | 0.46* | 0.48* | 1 | | | | | | | | | | |
| (4) Little control over things happen to me | 0.51* | 0.48* | 0.58* | 1 | | | | | | | | | |

MIDUS II

| Item | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) | (11) | (12) | (13) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (5) Really no way I can solve problems I have | 0.54* | 0.49* | 0.51* | 0.57* | 1 | | | | | | | | |
| (6) Helpless dealing with problems of life | 0.56* | 0.49* | 0.49* | 0.50* | 0.57* | 1 | | | | | | | |
| (7) Difficult arranging life in satisfying way | 0.39* | 0.42* | 0.39* | 0.42* | 0.45* | 0.56* | 1 | | | | | | |
| (8) Demands of everyday life often get me down | 0.36* | 0.36* | 0.35* | 0.38* | 0.41* | 0.54* | 0.52* | 1 | | | | | |
| (9) Overwhelmed by my responsibilities | 0.31* | 0.35* | 0.36* | 0.35* | 0.37* | 0.49* | 0.51* | 0.55* | 1 | | | | |
| (10) Able to build lifestyle to my liking | 0.25* | 0.29* | 0.26* | 0.28* | 0.32* | 0.40* | 0.47* | 0.32* | 0.30* | 1 | | | |
| (11) Actively carry out plans I set for self | 0.23* | 0.27* | 0.23* | 0.25* | 0.29* | 0.32* | 0.36* | 0.26* | 0.23* | 0.41* | 1 | | |
| (12) Do just about anything I set my mind to | 0.27* | 0.27* | 0.28* | 0.27* | 0.31* | 0.33* | 0.28* | 0.25* | 0.21* | 0.29* | 0.33* | 1 | |
| (13) When really want something, find way | 0.26* | 0.25* | 0.25* | 0.24* | 0.28* | 0.29* | 0.29* | 0.21* | 0.20* | 0.34* | 0.37* | 0.53* | 1 |

$^*p < 0.05$

## Pearson's *r* correlations between agency factors

| 2-Factor Personal Agency Models | | | | |
|---|---|---|---|---|
| | MIDUS II | | MIDUS III | |
| | (1) | (2) | (1) | (2) |
| (1) Intrinsic agency | 1 | | 1 | |
| (2) Instrumental agency | 0.67 | 1 | 0.71 | 1 |

All correlations significant ($p \leq 0.05$)

| 2-Factor Personal Agency/Agency Achievement Models | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 13-Item | | | | 11-Item | | | | 9-Item | | | |
| | MIDUS II | | MIDUS III | | MIDUS II | | MIDUS III | | MIDUS II | | MIDUS III | |
| MIDUS II | (1) | (2) | (1) | (2) | (1) | (2) | (1) | (2) | (1) | (2) | (1) | (2) |
| (1) Personal agency | 1 | | 1 | | 1 | | 1 | | 1 | | 1 | |
| (2) Agency achievement | 0.61 | 1 | 0.56 | 1 | 0.59 | 1 | 0.50 | 1 | 0.54 | 1 | 0.45 | 1 |

All correlations significant ($p \leq 0.05$)

| 3-Factor Personal Agency/Agency Achievement Models | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 13-Item | | | | | | 11-Item | | | | | | 9-item | | | | | |
| | MIDUS II | | | MIDUS III | | | MIDUS II | | | MIDUS III | | | MIDUS II | | | MIDUS III | | |
| MIDUS II | (1) | (2) | (3) | (1) | (2) | (3) | (1) | (2) | (3) | (1) | (2) | (3) | (1) | (2) | (3) | (1) | (2) | (3) |
| (1) Intrinsic agency | 1 | | | 1 | | | 1 | | | 1 | | | 1 | | | 1 | | |
| (2) Instrumental agency | 0.64 | 1 | | 0.67 | 1 | | 0.62 | 1 | | 0.65 | 1 | | 0.62 | 1 | | 0.67 | 1 | |
| (3) Agency achievement | 0.54 | 0.48 | 1 | 0.43 | 0.39 | 1 | 0.50 | 0.48 | 1 | 0.40 | 0.40 | 1 | 0.50 | 0.45 | 1 | 0.44 | 0.38 | 1 |

All correlations significant ($p \leq 0.05$)

## Pairwise correlations between 13-item human agency factors and correlates of agency

| | MIDUS II | | | | | | MIDUS III | | | | | | TWO-WAVE | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Construct | (1) | (2) | (3) | (4) | (5) | (6) | (1) | (2) | (3) | (4) | (5) | (6) | (1) | (2) | (3) | (4) | (5) | (6) |
| (1) Agency | 1.00 | | | | | | 1.00 | | | | | | 1.00 | | | | | |
| (2) Personal Agency | 0.98* | 1.00 | | | | | 0.99* | 1.00 | | | | | 0.98* | 1.00 | | | | |
| (3) Agency Achievement (2-factor) | 0.86* | 0.74* | 1.00 | | | | 0.77* | 0.68* | 1.00 | | | | 0.86* | 0.74* | 1.00 | | | |
| (4) Intrinsic Agency | 0.94* | 0.98* | 0.68* | 1.00 | | | 0.95* | 0.97* | 0.62* | 1.00 | | | 0.94* | 0.98* | 0.68* | 1.00 | | |

| | MIDUS II | | | | | | MIDUS III | | | | | | TWO-WAVE | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Construct | (1) | (2) | (3) | (4) | (5) | (6) | (1) | (2) | (3) | (4) | (5) | (6) | (1) | (2) | (3) | (4) | (5) | (6) |
| (5) Instrumental Agency | 0.90* | 0.87* | 0.76* | 0.76* | 1.00 | | 0.92* | 0.91* | 0.67* | 0.80* | 1.00 | | 0.90* | 0.87* | 0.76* | 0.76* | 1.00 | |
| (6) Agency Achievement (3-factor) | 0.79* | 0.66* | 0.97* | 0.64* | 0.60* | 1.00 | 0.71* | 0.61* | 0.99* | 0.58* | 0.56* | 1.00 | 0.79* | 0.66* | 0.97* | 0.64* | 0.60* | 1.00 |
| (7) Life Satisfaction | 0.53* | 0.49* | 0.53* | 0.44* | 0.53* | 0.47* | 0.53* | 0.51* | 0.46* | 0.46* | 0.53* | 0.41* | 0.39* | 0.37* | 0.38* | 0.33* | 0.39* | 0.33* |
| (8) Self-Acceptance (PWB) | 0.75* | 0.69* | 0.74* | 0.62* | 0.72* | 0.67* | 0.74* | 0.71* | 0.68* | 0.62* | 0.75* | 0.61* | 0.57* | 0.53* | 0.57* | 0.48* | 0.54* | 0.51* |
| (9) Self-Esteem | 0.73* | 0.68* | 0.69* | 0.63* | 0.70* | 0.62* | 0.71* | 0.69* | 0.63* | 0.62* | 0.70* | 0.58* | 0.54* | 0.51* | 0.52* | 0.46* | 0.52* | 0.46* |
| (10) Purpose in Life (PWB) | 0.69* | 0.65* | 0.63* | 0.62* | 0.62* | 0.58* | 0.70* | 0.68* | 0.59* | 0.63* | 0.66* | 0.54* | 0.52* | 0.50* | 0.48* | 0.47* | 0.46* | 0.44* |
| (11) Positive Relations (PWB) | 0.58* | 0.53* | 0.57* | 0.49* | 0.55* | 0.51* | 0.56* | 0.53* | 0.50* | 0.48* | 0.55* | 0.45* | 0.45* | 0.41* | 0.43* | 0.38* | 0.42* | 0.39* |
| (12) Social Integration | 0.38* | 0.35* | 0.36* | 0.33* | 0.35* | 0.33* | 0.39* | 0.37* | 0.34* | 0.34* | 0.37* | 0.31* | 0.33* | 0.31* | 0.31* | 0.29* | 0.30* | 0.28* |
| (13) Agency (Big 5) | 0.36* | 0.31* | 0.42* | 0.29* | 0.30* | 0.41* | 0.32* | 0.29* | 0.39* | 0.26* | 0.29* | 0.38* | 0.30* | 0.25* | 0.35* | 0.24* | 0.24* | 0.35* |
| (14) Autonomy (PWB) | 0.52* | 0.47* | 0.51* | 0.43* | 0.49* | 0.47* | 0.50* | 0.48* | 0.45* | 0.43* | 0.49* | 0.41* | 0.40* | 0.38* | 0.39* | 0.35* | 0.37* | 0.36* |

*$p < 0.05$

**Pairwise correlations between 11-item human agency factors and correlates of agency**

| | MIDUS II | | | | | | MIDUS III | | | | | | TWO-WAVE | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Construct | (1) | (2) | (3) | (4) | (5) | (6) | (1) | (2) | (3) | (4) | (5) | (6) | (1) | (2) | (3) | (4) | (5) | (6) |
| (1) Agency | 1.00 | | | | | | 1.00 | | | | | | 1.00 | | | | | |
| (2) Personal Agency | 0.98* | 1.00 | | | | | 0.99* | 1.00 | | | | | 0.98* | 1.00 | | | | |
| (3) Agency Achievement (2-factor) | 0.82* | 0.71* | 1.00 | | | | 0.75* | 0.64* | 1.00 | | | | 0.82* | 0.71* | 1.00 | | | |
| (4) Instrumental Agency | 0.93* | 0.94* | 0.65* | 1.00 | | | 0.95* | 0.96* | 0.59* | 1.00 | | | 0.93* | 0.94* | 0.65* | 1.00 | | |
| (5) Intrinsic Agency | 0.89* | 0.92* | 0.61* | 0.75* | 1.00 | | 0.91* | 0.92* | 0.57* | 0.79* | 1.00 | | 0.89* | 0.92* | 0.61* | 0.75* | 1.00 | |
| (6) Agency Achievement (3-factor) | 0.80* | 0.68* | 1.00* | 0.62* | 0.60* | 1.00 | 0.73* | 0.62* | 1.00* | 0.57* | 0.56* | 1.00 | 0.80* | 0.68* | 1.00* | 0.62* | 0.60* | 1.00 |
| (7) Life Satisfaction | 0.55* | 0.52* | 0.48* | 0.53* | 0.42* | 0.47* | 0.54* | 0.53* | 0.41* | 0.53* | 0.44* | 0.41* | 0.40* | 0.39* | 0.34* | 0.39* | 0.32* | 0.33* |
| (8) Self-Acceptance (PWB) | 0.76* | 0.72* | 0.68* | 0.73* | 0.59* | 0.66* | 0.76* | 0.73* | 0.62* | 0.75* | 0.60* | 0.61* | 0.58* | 0.55* | 0.52* | 0.55* | 0.46* | 0.51* |
| (9) Self-Esteem | 0.74* | 0.71* | 0.63* | 0.70* | 0.60* | 0.61* | 0.72* | 0.70* | 0.58* | 0.69* | 0.60* | 0.57* | 0.55* | 0.53* | 0.47* | 0.52* | 0.44* | 0.46* |
| (10) Purpose in Life (PWB) | 0.68* | 0.66* | 0.58* | 0.63* | 0.58* | 0.57* | 0.69* | 0.67* | 0.54* | 0.66* | 0.59* | 0.54* | 0.51* | 0.49* | 0.44* | 0.46* | 0.44* | 0.44* |

| Construct | MIDUS II | | | | | | MIDUS III | | | | | | TWO-WAVE | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (1) | (2) | (3) | (4) | (5) | (6) | (1) | (2) | (3) | (4) | (5) | (6) |
| (11) Positive Relations (PWB) | 0.58* | 0.55* | 0.52* | 0.55* | 0.45* | 0.50* | 0.56* | 0.54* | 0.45* | 0.55* | 0.45* | 0.45* | 0.44* | 0.42* | 0.39* | 0.42* | 0.35* | 0.39* |
| (12) Social Integration | 0.38* | 0.36* | 0.33* | 0.35* | 0.31* | 0.32* | 0.39* | 0.38* | 0.31* | 0.37* | 0.32* | 0.31* | 0.33* | 0.31* | 0.29* | 0.30* | 0.27* | 0.28* |
| (13) Agency (Big 5) | 0.37* | 0.32* | 0.42* | 0.30* | 0.28* | 0.41* | 0.33* | 0.30* | 0.38* | 0.29* | 0.26* | 0.38* | 0.30* | 0.26* | 0.36* | 0.24* | 0.23* | 0.35* |
| (14) Autonomy (PWB) | 0.51* | 0.49* | 0.47* | 0.49* | 0.40* | 0.46* | 0.50* | 0.48* | 0.42* | 0.49* | 0.41* | 0.41* | 0.40* | 0.38* | 0.36* | 0.37* | 0.33* | 0.35* |

$*p < 0.05$

## Declarations

**Conflict of interest** The authors have no competing interests to declare that are relevant to the content of this article. No funding has been received to support the development of this study, nor is the author affiliated with or involved in any organization or entity with any financial or non-financial interest in the subject matter or materials discussed in this manuscript. As of January 1, 2024, one of the authors began serving as a co-editor of Social Indicators Research; to avoid any conflict of interest, an alternate editor was involved in the review and editing of this manuscript.



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# 4 papers: urban-rural happiness gradient

## 4.1 Urbanism and happiness: A test of Wirths theory of urban life (Okulicz-Kozaryn and Mazelis 2018)

*Article*

Urban Studies

*Urban Studies*
2018, Vol. 55(2) 349–364

DOI: 10.1177/0042098016645470
journals.sagepub.com/home/usj
SAGE

# Urbanism and happiness: A test of Wirth's theory of urban life

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## Abstract

Social scientists have long studied the effects of cities on human wellbeing and happiness. This article demonstrates that people in cities are less happy, confirming a long-standing argument in the literature. But it had not yet been tested whether it is urbanism that negatively affects happiness, or if urban problems such as crime and poverty are to blame. Wirth posited that urbanism itself led to negative effects, but Fischer noted the necessity of empirical tests of Wirth's ideas. This study uses a happiness measure to provide a new look at the old question of urban unhappiness. Using the Behavioral Risk Factor Surveillance System from the US Centers for Disease Control and Prevention, we aim to untangle the effects of the city itself and urban problems on happiness in the USA. We find that the core characteristics of urban life (in particular size and density) contribute to urban unhappiness, controlling for urban problems. Urban unhappiness persists regardless of urban characteristics.

## Keywords

city, happiness, life satisfaction, subjective wellbeing, urbanism

## 摘要

社会科学研究者一直在研究城市对人类福祉和幸福感的影响。本文证明城市里的人们确实不那么幸福，印证了研究文献中长期存在的一个论点。但到底是都市生活对幸福感产生了负面影响，还是应归咎于犯罪和贫困等贫困问题，这一点尚未得到验证。沃斯 (Wirth) 认为都市生活本身导致了负面影响，但费歇尔 (Fischer) 指出有必要对沃斯思想进行实证检验。本研究采用了一项幸福度指标来重新审视都市生活之不幸感的旧问题。我们使用美国疾病控制和预防中心的行为风险因素监测系统，旨在厘清城市本身和城市问题对美国幸福感的影响。我们发现，在假定城市问题不产生影响的情况下，城市生活的核心特征（特别是规模和密度）促成了城市中的不幸福感。不论城市特征如何，城市中的不幸福感都会持续下去。

## 关键词

城市、幸福、生活满意度、主观幸福感、都市生活

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Received August 2015; accepted March 2016

This article addresses a fundamental social science question: how does the modern world affect those who live in it? Modernisation and industrialisation brought urbanisation, rife with complex problems. Tönnies ([1887] 2002) and Durkheim ([1893] 1997) challenged us to investigate what living in cities does to us. The first American sociologists, Louis Wirth among them, were urban sociologists, similarly concerned with city characteristics, urban problems and happiness (Wirth, 1938). They did not use the term happiness, but other terms to denote the negative side of the urban way of life, such as 'malaise'.[1] Wirth's theory of urban life focused on how urbanism led to various negative consequences, including (1) cognitively, in terms of alienation, (2) behaviourally, in terms of deviance, and (3) structurally, in terms of anomie (normlessness), all of which would lead to unhappiness.

There is a long-standing debate about whether people in cities are unhappy. The academic debate started in sociology (Durkheim, [1893] 1997; Park, 1915; Weber et al., [1921] 1962; Wirth, 1938), but was abandoned by sociologists in the 1970s after Fischer's major works were published. Recently, the topic was taken up by other disciplines, yet it has not addressed sociological research, and has used other approaches and foci. Regardless of the fact that urban life has much to offer residents, there is a consensus emerging among scholars that city living makes us, on average, unhappy and unhealthy (Adams, 1992; Adams and Serpe, 2000; Amato and Zuo, 1992; Balducci and Checchi, 2009; Berry and Okulicz-Kozaryn, 2009, 2011; Evans, 2009; Lederbogen et al., 2011). Those authors, however, used small data sets with few variables, typically operationalised city simply as the size of settlement, or used a dichotomous urban/rural variable. No study has controlled for city-level variables such as per capita income; no study so far has used a model that accounts for both personal and city-level factors that affect happiness. A fundamental question as framed by Fischer (1972, 1973, 1975) remains unanswered: is it cities themselves, in terms of size, density, and heterogeneity, that lead to unhappiness, or is it problems people typically associate with cities – notably poverty, crime and lack of support – that make people unhappy?

The first urban sociologists proposed that urbanisation created malaise (unhappiness) because of the core characteristics of cities: increased population size created anonymity and impersonality, density created sensory overload and withdrawal from social life, and heterogeneity led to anomie and deviance (see Park et al., [1925] 1984; Simmel, 1903; Tönnies [1887] 2002; Wirth, 1938). Some might assert that cities are defined by their problems, and argue that measures can be used interchangeably to examine both urbanism and urban problems. However, we follow Park, Wirth, and others to define urbanism by using population size and density. We seek to advance the understanding of urban unhappiness by better measuring urbanism using several size and density variables, and controlling for urban characteristics and problems to test whether urban unhappiness persists regardless of them. Recent large-scale survey data representative of a wide range of cities, as well as counties that are not cities as a comparison group, make this test possible.

## Theoretical background

There are many advantages to cities. Labour specialisation, economies of scale, invention and creativity are all made possible by high-density living (e.g. Florida, 2008; Glaeser, 2011; O'Sullivan, 2009). Human civilisation as we know it was made possible by cities and the labour specialisation that took place in them (O'Sullivan, 2009); this made organic solidarity possible (Durkheim,

[1893] 1997). Average standard of living has improved tremendously, even with high levels of inequality: by one estimate, those in the bottom decile today have a better standard of living than all but the top 1% did 100 years ago (Bok, 2010). For an overview of the progress that Americans have made since colonial times, see Fischer (2010). Progress in standard of living and urbanisation are closely related, leading some scholars to conclude that we are happier in cities than we are elsewhere (Glaeser, 2011); this is not so. In fact, Americans have not become any happier over time (Easterlin, 1974), and we are, on average, the least happy in cities (Berry and Okulicz-Kozaryn, 2011). The most celebrated thinkers in America's intellectual history almost universally expressed ambivalence or animosity toward the city, among them: Jefferson, Emerson and Thoreau (White and White, 1977). Wirth (1938) famously summarised the negatives of city life, building on the worries of earlier European sociological theories (Tönnies, [1887] 2002). Cities are collections of heterogeneous people with different specialisations; there is a lot of variability among city dwellers. Relations between them are anonymous, superficial, impersonal, transitory, unstable and insecure (Wirth, 1938). Early theorists lamented what they saw as a lack of norms, which led to moral deviance such as crime (Wirth, 1938). Cities are fast-paced and crowded, creating a psychological overload because of the intensification of stimuli for urban dwellers and occasioning detachment, and a retreat to social isolation (Simmel, 1903). Simmel argued that people necessarily adapted to urban life by developing a blasé attitude: with the rapid pace and all the people we come across, we would not be able to handle it if we had to deeply engage with everyone and everything, so we become detached. Simmel noted that this reserve provides more personal freedom and privacy to urban dwellers, but it also necessitates impersonal interactions with those we encounter (Simmel, 1903). Cities generate moral *disorder*: misunderstandings, tension and conflict (Smelser and Alexander, 1999). For more elaboration see Fischer and Merton (1976).

Recent evidence raises other concerns about cities: density and heterogeneity predict lower social capital (Helliwell and Wang, 2010) and low social capital is associated with poor health and higher mortality (Kawachi et al., 1997; Subramanian et al., 2002). Poverty is concentrated in cities, and the more concentrated the poverty, the worse it is for people (Jargowsky, 1997). There is an abundance of crime (Bettencourt et al., 2007, 2010) and segregation (Jargowsky, 1997; Massey and Denton, 1993). There is more crime in cities than there is elsewhere, and high crime predicts poor health (Lynch et al., 2004; Zimmerman and Bell, 2006).

High density predicts low happiness (Fassio et al., 2013; Lawless and Lucas, 2011). One specific reason why there may be less happiness in cities is that 'the pecuniary nexus tends to displace personal relations' (Wirth, 1938: 1). In addition, 'fashion tends to take the place of custom' (Park, 1915: 605), and consumption does not have a lasting impact on life satisfaction because people adapt to material goods (Easterlin, 2003). Recent neurological evidence confirms early sociological theory: great cities actually overstimulate our brains to the point where it is not healthy (Lederbogen et al., 2011). Schwartz (2004) shows that the abundance of choice, exemplified in cities, often leads to depression, loneliness, anxiety and stress. Americans do not merely prefer smaller areas as documented earlier by Fuguitt and Zuiches (1975) and Fuguitt and Brown (1990), they are also happier in smaller areas (Lawless and Lucas, 2011).

There have been many attempts at discerning the effect of city living on happiness. But the limitation of even the most recent research is that it does not control for city-level variables such as per capita income, and city disadvantages such as crime and poverty, something that Fischer suggested researchers should do over 40 years ago. Fischer (1973: 222) acknowledged that people are less happy in the biggest cities, but he was not sure whether it was urbanism or other confounding variables that make residents unhappy in cities:

> (1) Such preferences [for rural living] may be a function of idealized images founded on popular conceptions of urban and rural life. (2) They may indicate utopian hopes of maintaining urban opportunities in small communities. [...] (3) These evaluations may result from the contemporary state of American cities rather than from the nature of cities per se.

The remainder of this article will try to answer item (3) above. We aim to discover whether city residents are less happy, controlling for problems people typically associate with contemporary cities. If so, these problems cannot be the cause of lower happiness in cities. Rather, urbanism's core characteristics – large numbers of people and high density – along with other factors, make city dwellers unhappy:

H1: people in cities are less happy than are those in other areas if cities are defined by:
H1a: population size
H1b: density
H2: urban unhappiness will persist after controlling for problems people typically associate with cities.

In testing the above hypotheses we control for city characteristics that may affect happiness and for known person-level predictors of happiness.

## Data and measurement

We analyse the 2005 Behavioral Risk Factor Surveillance System (BRFSS) from the United States Centers for Disease Control and Prevention. The BRFSS is representative of the USA, with a total sample size of over 100,000 people per year, well suited for the study of cities. The regular BRFSS data set, however, is not representative of most counties, so we use the SMART (Selected Metropolitan/Micropolitan Area Risk Trends) version of the BRFSS that is representative of counties; for simplicity we will refer to it as the BRFSS. The downside of the SMART version is a smaller number of counties, 232, but it still provides a good sample of the USA with enough variation.[2] See the appendix (available online) for a listing of counties. In addition to the 2005 BRFSS, as a robustness check, we include models using data for 2006, 2007 and 2008 in the appendix (available online), because the more recent waves have bigger samples, and to account for potential lag effects from variables measuring urban problems at the county level. In the main text we report the 2005 BRFSS results, because the county-level data we use are either for 2003, 2004 or 2005. All county-level data come from the Inter-university Consortium for Political and Social Research: County Characteristics, 2000–2007.

We use data at person and county levels. We do not have data at the city level. City attributes are measured at the county level; many metropolitan areas cover the whole county. Even where a metropolitan area covers part of a county, it seldom ends abruptly and changes into a non-urban area or open land. Where a metropolitan area abruptly ends, then it is due to a natural boundary such as an ocean in the case of coastal cities, but in that case the county ends as well. Furthermore, this limitation is ameliorated by another operationalisation of cities: the urban–rural continuum, which differentiates between metropolitan areas, cities, towns of various sizes and smaller areas. Another triangulation of measurement is performed using a person-level location variable: metropolitan status code.

## The person-level dependent variable: Happiness

As noted above, we treat malaise as unhappiness, following research by Fischer (1972, 1973), who pioneered using happiness data to study urban malaise. Yet data and statistical software did not enable a proper test in the early 1970s. We use large-scale data and a model controlling for both personal and ecological characteristics.

Happiness is a multidimensional concept (e.g. Hills and Argyle, 2001), though for simplicity we use the terms happiness, life satisfaction and subjective wellbeing interchangeably (the dimensions overlap). The BRFSS has only one survey item measuring happiness.[3]

The survey item reads 'In general, how satisfied are you with your life?'. For simplicity, we recoded answers so that a higher numeric value means more happiness: 1 is 'very dissatisfied', 2 'dissatisfied', 3 'satisfied' and 4 'very satisfied'. Over 90% of respondents were either satisfied or very satisfied with their lives. The same variable has been used in recent happiness studies by Oswald and Wu (2010, 2011), Glaeser et al. (2014) and Winters and Li (2015).

The happiness measure, even though self-reported and subjective, is reliable, valid (Myers, 2000), and closely correlates with similar, more objective measures such as brain waves (Layard, 2005). Unhappiness strongly correlates with suicide incidence and mental health problems (Bray and Gunnell, 2006). Finally, to be clear, we study here general/overall happiness, not a domain-specific happiness such as neighbourhood or community satisfaction.

## The main person- and county-level independent variables: City size/city population

In this study, we sacrifice breadth of measurement of urbanism at the cost of precision and use the most precise measures available: population size and density. Population size is used in two ways: as an urban–rural continuum with nine categories defined by the USDA (US Department of Agriculture) and as a metropolitan status code based on a survey item from the BRFSS. We chose continuous/ordinal measures because they convey more information than do binary or nominal ones (see Meyer, 2013; Wirth, 1938). The county-level urban–rural continuum ranges from 1 ('metro>1 million') to 9 ('nonmetro<2500, not adjacent to metro').[4]

The metropolitan status code differs within a county; it is a person-level variable and reflects the urban dweller status more precisely than does the county-level urban–rural continuum. The metropolitan status code is broken down into the following dummy variables: 'not in an MSA', 'inside a suburban county of the MSA' and 'outside centre OR in MSA without centre'. 'Centre city of MSA' is the base case, and MSA stands for Metropolitan Statistical Area, a geographical area associated with an urban area of 50,000 people or more, as defined by the United States Office of Management and Budget.

## The main county-level independent variable: Population density

We measure density as population (in thousands) per square mile, and there are staggering differences; density ranges from four people per square mile in Sweetwater County, WY to 70,000 people per square mile in New York County, NY.

The measures of urbanism we use do not capture heterogeneity directly. We do not include heterogeneity for three reasons. First, it is not obvious which type of heterogeneity is to be measured. In modern times we think of racial, ethnic and class diversity, but in Wirth's conception, influenced by Durkheim's focus on the increasing division of labour and occupational specialisation that accompanied urbanisation, heterogeneity may have been more about occupational diversity; Wirth writes of heterogeneity in 'social groups' (1938). It also may have been more about diversity in nation of origin, which while related to contemporary understandings of ethnicity, is not synonymous with the latter. As Wilson (2012) and Finney and Simpson (2005) note, sometimes the focus on race and ethnicity is misplaced. So we might consider measures of ethnicity or national origin, race, religion, income or occupation, for example. Second, there is precedent in recent literature to omit heterogeneity as part of a definition of urbanism, though literature consistently considers size and density (e.g. Meyer, 2013). Third, we looked at various measures of racial diversity, but did not find a consistent and robust relationship.

### *County-level controls: Measures of urban characteristics*

As we have argued above, this paper's key contribution is to disentangle the effects of the city (size, density) and urban problems on the happiness of city residents. To achieve that, we control for the following city characteristics: crime, housing stress, low education, low employment, persistent poverty, population loss, personal income (USD 1000)/cap, and percent Black. This set of controls addresses a fundamental confounding factor and an alternative explanation for city unhappiness: there are many poor people either stuck because they cannot afford to move, or who come to cities seeking a better life. Poverty and crime are seen as fundamental urban problems (Jargowsky, 1997).[5] Classical theorists posited that lack of support from others is an issue in cities (Park, 1915; Wirth, 1938), though Fischer (1982, 1995) argues that this urban problem is a myth.[6] However, we control for lack of support with a person-level variable: social/emotional support.

We control for eight city characteristics, but there are countless characteristics one could hypothetically include. Therefore we use county fixed effects (we can measure city size at person level, because there is a person-level survey item asking respondents about the location of their residence). Finally, we focus on several of the happiest and least happy counties. We focus on issues thought to be major urban problems, such as poverty and crime, as defined in the literature (e.g. Jargowsky, 1997), as well as other variables that might have an effect, such as personal income and percent Black. 'Problems' such as congestion or high cost of living are arguably very closely related to population density, and hence are characteristics of urbanism as defined here.

### *Person-level controls: Predictors of happiness*

We control for several variables because they predict happiness as shown in the literature. Exploring the interactions of these variables with city size and density may be an interesting topic for future research, but it is beyond the scope of this study.

Increased income boosts happiness and unemployment depresses it (e.g. Di Tella and MacCulloch, 2006; Di Tella et al., 2001a, 2001b). Married people are happier than those who are unmarried (e.g. Diener and Seligman, 2004; Myers, 2000). African Americans are less happy than are Whites in the USA (e.g. Berry and Okulicz-Kozaryn, 2009, 2011), and they are concentrated in cities, so it is important to control for race in the context of urban unhappiness. Happiness is U-shaped with age, that is, both the young and the old are happier than are people during midlife (e.g. Sanfey and Teksoz, 2005).

Health is consistently an important predictor of happiness, so we control for health status, and we also control for education, which some authors find to predict happiness (see Dolan et al., 2008). There are substantial urban–rural differences in education. In general, urban relationships are impersonal and transitory (e.g. Wirth, 1938) and some posit that there is less support in cities (White and White, 1977) and less trust in cities (Berry and Okulicz-Kozaryn, 2011). We measure lack of support using the following question: 'How often do you get the social and emotional support you need?'. It ranges from 1 (never) to 5 (always).[7] Together with several county-level variables this accounts for urban problems in testing whether it is such urban problems – or cities themselves – that create urban unhappiness.

The appendix (available online) lists all the variables used in this study along with their definitions, and includes summary statistics as well.

## Model and results

Our modelling approach is similar to Fernandez and Kulik (1981) who also tested the effect of place on happiness and found that people are happier in smaller areas. By comparison with their work, the sample in the present study is over ten times larger, uses additional robustness checks and controls for what are typically characterised as urban problems.

We use a standard OLS regression with clustered standard errors by county. We also use BRFSS sampling weights to account for oversampling. We treat the four-step happiness variable as continuous (see Blanchflower and Oswald, 2011; Ferrer-i-Carbonell and Frijters, 2004).[8]

Results are shown in Table 1. We first estimate a base model (A) that includes known predictors of happiness. Column (B) adds two basic proxies for urban problems: crime and poverty. We also include percent Black, not as an urban problem but because many US central cities include a large African American population, African Americans score lower on happiness measures than do white Americans, and therefore the lack of this variable in a model could bias results because of different levels of happiness between racial groups. Finally, column (C) includes several additional controls for city characteristics and possible urban problems: low education, housing stress, low employment, population loss and personal income per capita.

We start with a person-level location variable. City size dummies are listed first in the table and indicate difference relative to the centre city. The most significant, both statistically and practically, is the 'not in MSA' dummy. As hypothesised, people are happiest outside of cities. Estimates are stable across specifications; it does not matter whether the city has stereotypical urban problems. People living outside of metropolitan areas as compared with central cities are happier by 0.05 on a scale from 1 to 4. This difference may appear to be small in practical terms, but it is still important. First, it contradicts claims that people are happier in cities, notably those by Glaeser (2011). Second, happiness is largely due to person-level influences such as unemployment, and ecology is secondary. Across 232 counties in our sample, happiness ranges between 3.18 and 3.56, with the vast majority of counties, 209 of the 232, in the 3.30 and 3.40 ranges. Third, an ecological difference of 0.05 (coefficient on 'not in an MSA' from Table 1) on a happiness scale from 1 to 4 can translate to large effects: for example, all else being equal, if one-third of 1% of the US population, about 1 million people, who live in

**Table 1.** Survey weighted regressions of happiness.

| | A | B | C |
|---|---|---|---|
| Outside centre OR in MSA without centre | −0.0001 | 0.0002 | −0.0017 |
| Inside suburban county of MSA | −0.0024 | 0.0098 | 0.0079 |
| Not in an MSA | 0.0467*** | 0.0445** | 0.0451** |
| *County-level controls* | | | |
| Crime rate index | | 7.90e−06** | 7.90e−06* |
| Persistent poverty | | 0.0061 | 0.0175 |
| % Black | | −0.0012** | −0.0009* |
| Low education | | | 0.017 |
| Housing stress | | | −0.0019 |
| Low employment | | | −0.028 |
| Population loss | | | −0.02 |
| Pers. inc. (USD 1000)/cap | | | 0.0001 |
| *Person-level controls* | | | |
| Income | 0.0255*** | 0.0260*** | 0.0259*** |
| Married or member of an unmarried couple | 0.1406*** | 0.1410*** | 0.1409*** |
| Unemployed | −0.1687*** | −0.1676*** | −0.1672*** |
| Age | −0.0078*** | −0.0081*** | −0.0081*** |
| Age squared | 0.0001*** | 0.0001*** | 0.0001*** |
| White | −0.0558*** | −0.0581*** | −0.0558*** |
| Education level | −0.0085+ | −0.0110* | −0.0110* |
| Soc/emo support | 0.1847*** | 0.1842*** | 0.1844*** |
| General health | 0.1342*** | 0.1357*** | 0.1356*** |
| *N* | 138,453 | 132,677 | 131,657 |

*Note*: +$p < 0.10$, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

central cities instead lived in non-urban areas, it would mean about 50,000 (0.05*1 million) people would be satisfied with their lives rather than dissatisfied.

All person-level variables are significant, while most county-level variables are not significant. It could be that the location variable is significant because of the large sample size at the person level. Yet, in Table 2 we use a county-level size variable, the urban–rural continuum. It ranges from 1 (‘metro>1 million’) to 9 (‘non-metro<2500, not adjacent to metro’). This triangulation of measurement serves as a robustness check; the smaller the place, the more happiness, no matter how the city is measured.

The coefficient estimates are 0.01, and there are eight categories on the urban–rural continuum, so a change from 1 (‘metro>1 million’) to 9 (‘non-metro<2500, not adjacent to metro’) is associated with a happiness difference of about 0.1. What is striking about this second set of results is the persistence of statistical significance at the $p < 0.01$ level on the urban–rural continuum variable; the significance of all county-level control variables is much less stable. Urban unhappiness exists, and hence we find support for hypotheses H1a and H1b. To account for possible lag effects from contextual, county-level variables on happiness, we reran these models using the 2006, 2007 and 2008 BRFSS data (the results are in the appendix, available online).

Finally, Table 3 shows that the denser the county, the less happy people are there. Adding controls attenuates the effect. Crime and poverty explain away some of the city effect. Percent Black also explains some of the city effect. Yet, density remains negative

**Table 2.** Survey weighted regressions of happiness.

| | A | B | C |
|---|---|---|---|
| Urban–rural continuum | 0.0121*** | 0.0083** | 0.0094** |
| *County-level controls* | | | |
| Crime rate index | | 7.198e−06** | 7.366e−06* |
| Persistent poverty | | 0.0024 | 0.0144 |
| % Black | | −0.0011** | −0.0008+ |
| Low education | | | 0.0174 |
| Housing stress | | | −0.0013 |
| Low employment | | | −0.0282 |
| Population loss | | | −0.0191 |
| Pers. inc. (USD 1000)/cap | | | 0.0002 |
| *Person-level controls* | | | |
| Income | 0.0256*** | 0.0261*** | 0.0259*** |
| Married or member of an unmarried couple | 0.1401*** | 0.1409*** | 0.1408*** |
| Unemployed | −0.1683*** | −0.1674*** | −0.1670*** |
| Age | −0.0077*** | −0.0080*** | −0.0081*** |
| Age squared | 0.0001*** | 0.0001*** | 0.0001*** |
| White | −0.0573*** | −0.0582*** | −0.0561*** |
| Education level | −0.0085+ | −0.0111* | −0.0111* |
| Soc/emo support | 0.1846*** | 0.1841*** | 0.1843*** |
| General health | 0.1342*** | 0.1357*** | 0.1356*** |
| *N* | 138,453 | 132,677 | 131,657 |

*Note*: $+p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

and significant when controlling for a full set of covariates.

Results from all models are consistent. Defining the city in multiple ways and controlling for commonly considered urban problems and other characteristics changes the results little, which suggests that it is the defining features of cities – size and density – that are responsible for the urban–rural happiness gradient, indicating support for hypothesis H2. Solving urban problems, if it were possible, would not likely make people in cities as happy as people are elsewhere. Urbanism – in terms of large population size and high density – is associated with lower happiness.

We use a county fixed effects model to account for potential omitted variable bias, any unobserved heterogeneity, and anything specific about the area that may affect results. US cities differ from each other substantially. It is not possible to control for all urban problems (or all confounding variables), but any such factors would be picked up in the fixed effects model. Lower happiness in urban areas persists in the fixed effects model.

Finally, we focus on several specific cases and make some illustrative comparisons. A ranking of happy counties (table 8 in the appendix, available online) shows that the happiest counties are not cities, and the least happy counties are large cities. The three happiest counties (happiness > 3.5) in this sample are rural, yet close to large cities. They therefore have the amenities nearby that cities offer, but they escape potential downfalls of larger size and higher density. Douglas County, CO, borders Denver, but remains largely rural and has a population density of 300/sq. mile. Shelby County, TN, is partially urban; it houses Memphis, yet

**Table 3.** Survey weighted regressions of happiness.

| | A | B | C |
|---|---|---|---|
| Population density per sq mile, 05–09 * 1,000,000 | −1.5241 + | −0.9695 | −1.3860* |
| *County-level controls* | | | |
| Crime rate index | | 5.560e−06* | 5.825e−06* |
| Persistent poverty | | 0.0050 | 0.0171 |
| % Black | | −0.0009* | −0.0006 |
| Low education | | | 0.0165 |
| Housing stress | | | 0.001 |
| Low employment | | | −0.0101 |
| Population loss | | | −0.0241* |
| Pers. inc. (USD 1000)/cap | | | 0.0007 |
| *Person-level controls* | | | |
| Income | 0.0254*** | 0.0258*** | 0.0255*** |
| Married or member of an unmarried couple | 0.1388*** | 0.1405*** | 0.1406*** |
| Unemployed | −0.1676*** | −0.1671*** | −0.1670*** |
| Age | −0.0077*** | −0.0080*** | −0.0081*** |
| White | −0.0598*** | −0.0591*** | −0.0559*** |
| Education level | −0.0078 | −0.0107* | −0.0109* |
| Soc/emo support | 0.1846*** | 0.1842*** | 0.1845*** |
| General health | 0.1341*** | 0.1356*** | 0.1354*** |
| Age squared | 0.0001*** | 0.0001*** | 0.0001*** |
| *N* | 138,453 | 132,677 | 131,657 |

*Note*: $+p < 0.10$, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

most of the county is rural and the population density is relatively low at 1200/sq. mile. Johnson County, KS, is mostly rural, yet close to Kansas City and has a population density of 1100/sq. mile. On the other side of the scale, the counties with the lowest happiness scores (happiness < 3.2) in this sample are urban: the first one, St Louis city, MO, is all urban; it covers the central city of St Louis, and has about five times the population density of two of the three happiest counties, at 5700/sq. mile (and 19 times the density of the happiest county). The second and third least happy counties, Bronx, NY and Kings, NY (Brooklyn) are parts of the American city with the largest population, New York City; both counties have a density of over 30,000/sq. mile. The happiness differences between urban and non-urban counties, even though they appear small, are substantial because most people (>90%) are either satisfied '3' or very satisfied '4' with their lives and the standard deviation across counties in this sample is only 0.06. Hence, the urban–rural happiness gradient is quite substantial relative to the overall variability.

## Discussion and future research

Fischer asked the right question about urban unhappiness: is it the city itself – in terms of its core characteristics of size and density – or is it what are typically thought of as urban problems that generate unhappiness? We have excluded alternative explanations (our controls and fixed effects at the county level), that urban problems such as poverty or crime are solely responsible for urban unhappiness. We have confirmed that people

**Table 4.** Fixed effects survey weighted regressions of happiness.

| | FE |
|---|---|
| Outside centre OR in MSA without centre | −0.003 |
| Inside suburban county of MSA | 0.0265 |
| Not in an MSA | 0.0927* |
| Income | 0.0261*** |
| Married or member of an unmarried couple | 0.1361*** |
| Unemployed | −0.1650*** |
| Age | −0.0076*** |
| Age squared | 0.0001*** |
| White | −0.0570*** |
| Education level | −0.0082 + |
| Soc/emo support | 0.1843*** |
| General health | 0.1337*** |
| Constant | 2.1122*** |
| *N* | 138,453 |

are less happy in cities, in part because of population size and density, the defining features of the city. Other variables also have an effect, but size and density matter, controlling for urban problems.

There is clearly a 'city paradox': cities provide sought-after resources, but we pay a price: the relative unhappiness of city life. High-density living fosters specialisation, information exchange, labour pooling, skills matching, knowledge spillovers, and more. On the other hand it can also foster crime, spread of disease, stress, cognitive and sensory overload, and other problems. Cities act like a magnifying glass, bringing out the best and the worst in us. This synergy has been estimated at the rate of 1.15 relative to the population – doubling population size increases everything (say crime or creativity) that an average person does by 15% (Bettencourt et al., 2007, 2010). Cities do not make people happier as Glaeser (2011) argues. People prefer to live outside of cities (Fuguitt and Brown, 1990; Fuguitt and Zuiches, 1975; Palmer, 2012) and, as we document here, people are happier outside of them.

Yet Americans are moving to metropolitan areas. This may point to an explanation of the Easterlin Paradox (1974) that economic growth does not result in happiness growth; perhaps this is due to people moving to less happy places.[9] Moving to a city will not necessarily make one less happy, nor will moving to a small town or village make one more happy. How moving to an area affects personal happiness is beyond the scope of this cross-sectional study. Yet, there is an urban–rural happiness gradient and it persists, regardless of urban problems such as poverty or crime. By pointing to urban unhappiness, we do not argue that urbanisation should be countered. There are many benefits to cities, and of course, some people are happier in cities than they would be elsewhere.

This study is about the USA and its results should not be generalised to other countries. At best these results can be generalised to other developed countries. However, in developing countries, there is likely to be a negative relationship between happiness and the city (Berry and Okulicz-Kozaryn, 2009).

As with any study, there needs to be more work done in this area; even though we continue a long-standing line of research, this is the first study to use county-level data to investigate the urban unhappiness hypothesis empirically. Future research may study how neighbourhoods (for example, census tracts) affect the wellbeing of residents. Cities are collections of neighbourhoods, and all characteristics, including happiness and urban problems, are distributed unevenly within a city and across its neighbourhoods. There are notable spatial inequalities (Ballas, 2013), so segregation and inequality are key fields for future work. Finer geographic resolution also has disadvantages; there are no surveys representative of neighbourhoods across a wide range of cities, so a neighbourhood study is not an alternative or an improvement over the present study but a complement to it. While neighbourhoods often seem independent or self-contained ('a mosaic of little worlds which touch but do not interpenetrate' (Park, 1915: 608)), there are spillovers as well; for example, poverty and crime in one neighbourhood does indirectly affect people and their happiness in other neighbourhoods. While people in public space may not engage in focused interaction, effectively being 'alone, together', people do encounter others in public transportation and in public spaces, especially in large cities (Morrill et al., 2005; Putnam, 2001).

Future research would do well to investigate why people choose to live in cities, given that urbanites are less happy, using person-level panel data. Future work might also explore the effect of the city on the happiness of different social groups. Additional research could use case studies of the happiest cities as opposed to the least happy cities and investigate more deeply than is possible with currently available data what makes residents happy. As more data become available, future research can focus on the exploration of multiple advantages and disadvantages of cities. We have striven to include the most important and easily measurable controls, but it is important to extend this research further by including an even larger set of city characteristics in the future. Notably, segregation, inequality, various types of pollution, and natural amenities are topics for future research in this area (Ballas, 2013; Brereton et al., 2008).

It may be possible that 'deviant' people choose to live in cities – and they may be less happy than are others and so that may depress overall city happiness. Indeed, Wirth (1938) argued that deviance would be higher in cities; Fischer (1995) referred to this as unconventionality, a more positive framing, but with the same notion that people with unusual interests or members of small subcultures would be more common in cities. These qualities may be to some degree picked up by person-level controls: income, education and unemployment, and by the county-level in the crime rate index. But otherwise, there is room for improvement, and future research could help in this area. In general, though, deviance and 'qualities that make people unhappy' are difficult to measure. There is also a possibility that unhappy people are pulled to cities; people who feel they do not belong in smaller areas and people who are unhappy where they live migrate to cities, but it is possible that even after urban migration their happiness remains. LGBTQ people (e.g. Doderer, 2011) and immigrants (e.g. Damm, 2009) prefer cities. But it is equally possible that happy and energetic extroverts move to cities to realise their full potential.

While there may be an element of self-selection, we think that it is possible that cities have the ability to change people in such a way that they become less happy. For instance, cities may intensify the pecuniary and consumerist orientation in people, and make them more stressed and overworked. Indeed:

> To say that self-selection explains some of the differences between town and city is not to say that urbanism is irrelevant to those very same differences. It remains an indirect cause by stimulating selective migration. Although, part of urban/nonurban self-selection is coincidental, most of it results from processes originating in the nature of urbanism itself. (Fischer, 1982: 256)

And this is our interpretation and conclusion: urbanism is at least partially responsible for lower happiness in cities, regardless of urban problems.

## Acknowledgement

We thank Michael Timberlake, the North American Editor of *Urban Studies*, whose comments on earlier writings by one of us

(Okulicz-Kozaryn) on this subject inspired this study – to search for the effects of city size per se as opposed to the effects of compositional differences of bigger cities. We thank Paul Jargowsky and Arielle Kuperberg for their advice and comments, and our fellow affiliated scholars of the Center for Urban Research and Education at Rutgers-Camden for pushing us to be clearer in our argument. We are also grateful to the students in our courses, who have helped us think and rethink ideas about cities, but we take full responsibility for weaknesses in our arguments.

## Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

## Notes

1. We use happiness, life satisfaction and (subjective) wellbeing interchangeably, as in much of the literature; happiness is defined in the 'Data and measurement' section. Wirth, who used the term 'malaise' in his influential article, 'Urbanism as a way of life', referred to general dissatisfaction and unhappiness, and hence we use a measure of happiness to study urban malaise, as Fischer (1972, 1973) did. In other words, we interpret the early 20th century focus on 'urban malaise' as 21st century 'urban unhappiness or dissatisfaction'.
2. As a robustness check, we reran our analyses using the regular BRFSS and found similar results, which are available from the corresponding author upon request.
3. The happiness measure used here is quite comprehensive and also popular in the existing literature. This does not preclude the possibility that some aspects of happiness/life satisfaction/wellbeing may be actually improved among city dwellers. This is a topic for future research, but would require a different type of data with a very detailed questionnaire about happiness, or an in-depth qualitative study which would necessarily reduce the size and representativeness of such a data set. For future research, it may be useful to use data sets measuring distinct cognitive and affective aspects of subjective wellbeing. As of 2015, we are unaware of a data set representative of a large number of US counties that includes such measures.
4. For all categories of both size variables, see Table 7 in the appendix, available online.
5. Only very recently have crime and poverty been suburbanising. This study, however, uses 2005 data.
6. It is better to overcontrol than undercontrol, especially in a large data set as used here (Gujarati, 2002).
7. This survey item does not capture levels of trust or social capital, which may also be important, but it is the best measure available in the BRFSS to measure support, which is related to trust and social capital. Future research should try to use other measures in order to capture trust and social capital.
8. We used the following Stata command: regress <happiness> <person-level variables> <county-level variables> [pw = _cntywt], robust cluster(<county>). We also tried a multilevel logit model, and the results were similar. The bivariate relationship between size of a place and happiness is negative. Stata code and additional results are available upon request from the corresponding author.
9. The Easterlin Paradox shows that Americans did not become happier over the past several decades despite enormous economic progress over that time. The fact that over the same time America (sub)urbanised substantially, and that people are less happy in metropolitan areas as shown here, may help explain the Easterlin Paradox.

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4.2 Livability and Subjective Well-Being Across European Cities (Okulicz-Kozaryn and Valente 2019)

Applied Research Quality Life (2019) 14:197–220
https://doi.org/10.1007/s11482-017-9587-7

# Livability and Subjective Well-Being Across European Cities

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Received: 20 July 2017 / Accepted: 19 December 2017 / Published online: 19 January 2018


**Abstract** This study documents for the first time the correlation between livability and subjective well-being (SWB) across European cities. Livability is measured with the popular Mercer Quality of Living Survey and correlates considerably with SWB, measured as place and life satisfactions. There are outliers, for instance: the "unlivable" but "happy" Belfast (fool's paradise) and the "livable," but "unhappy" Paris (fool's hell). In addition, we find geographic patterns: while the Mercer index ranks higher Western cities, subjective well-being is higher in Northern cities. Smaller cities score higher on both livability and SWB, confirming thus the urban sociological theory of urban malaise while contradicting urban economic theory of city triumph.

**Keywords** Satisfaction · Happiness · Subjective well-being · Quality of life · Urban quality of life · Cities · City rankings · Livability · Best places to live · Mercer · Economic theory · Utility

Livability refers to the quality of life, standard of living, or general well-being of a population in a specific region, area, or city. It is the sum of factors that add

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up to a community's quality of life (economic prosperity, social equity and stability, educational opportunities, recreation and cultural possibilities, etc.) According to the Mercer Quality of Living Survey,[1] the cities with the highest levels of livability are predominantly European (Okulicz-Kozaryn 2013; Mercer 1999). Vienna, Zurich, Geneva, and Copenhagen, among others, rank as the most livable cities. This paper aims to explore for the first time whether urban livability is correlated to happiness in Europe. Does living in a city with high levels of livability increase happiness? Or is individual happiness independent from the livability of a city? Similarly, is the relationship between livability and happiness consistent throughout Europe or are there regional differences?

To address these questions, it is important to recognize that Europe is largely divided into two main areas: East and West. These regions are distinct in how their institutions are organized and on the characteristics of their residents. The East tends to be more traditional and some might even consider it "backward," while poorly governed and with low opportunities for economic advance, yet residents tend to be cheerful and spontaneous. The East is a region experiencing a new post-communist era, and is less advanced in terms of democracy and capitalism, although some argue that these differences are disappearing (Hudabiunigg 2004). Unlike the East, the West has institutions that are well governed and organized governments are more progressive and well-endowed, providing residents with opportunities for financial success. Yet, people in the West are known to be "grim and stiff." This duality seems to be applicable to the South and North regions of Europe as well, where Eastern qualities can be also applied to the South, and Western qualities to the North.

These contrasting differences should be of concern to policymakers and European citizens alike, given that one of the founding principles of the European Union is to promote economic, social and territorial cohesion among member states (Union 2004). In addition, with ever increasing urbanization[2] and the free movement of EU citizens within the European Union, understanding the relationship between livability and subjective well-being may help explain residency decisions, and provide measurement to promote territorial cohesion among member states.

## The Social Indicators Literature: Objective and Subjective

According to the Webster dictionary, livability is defined as "suitability for human living." Some scholars, define livability as (objective) quality of life, welfare, 'level of living,' or habitability (Veenhoven 2000). Another definition for livability is quality of place (Burton 2014) and its synonyms: environmental quality or urban quality, defined as the "the physical characteristics of community, the way it is planned,

[1]The Mercer Survey, also referred to as the Mercer Index, evaluates cities based on 39 factors including political, economic, environmental, personal safety, health, education, transportation, and other public service factors. We discuss the index later in depth.

[2]Europe, as the rest of the world, is urbanizing: in 1950 about half of Europeans lived in cities, now it is about 74% and the urban proportion will increase by another 10 percentage points to about 84% by 2050 (http://esa.un.org/unup/).

designed, developed, and maintained" (Burton 2014, p. 5312). The Mercer Index used in our analysis, mostly measures material standards or levels. Thus, livability is tangible and objective. Perhaps the term 'standard of living,' or 'level of living,' are actually the best terms to describe how livability is measured. A shortcoming of the current measurement of livability is that it fails to account for intangible qualities of place such as vibrancy, authenticity, and distinctiveness. Nevertheless, it is much more comprehensive than traditional economic approaches that tend to equate development, or progress, with income or consumption.

Subjective well-being (SWB) is one of the most comprehensive measurements available. Diener and Lucas (quoted in Steel et al. 2008, p.142) define it as people's evaluations of their lives, which include "both cognitive judgments of one's life satisfaction in addition to affective evaluations of mood and emotions," which is virtually the same as Veenhoven's (2008, p. 2) definition: "overall judgment of life that draws on two sources of information: cognitive comparison with standards of the good life (contentment) and affective information from how one feels most of the time (hedonic level of affect)." In this paper, we use these overall SWB definitions when referring to subjective well-being, and we use the terms "happiness" and "life satisfaction" interchangeably.[3]

The relationship between livability and subjective well-being should be positive: if livability is high, human needs are satisfied and as a result happiness follows (Diener et al. 1993; Veenhoven 1991; Veenhoven and Ehrhardt 1995). Figure 1 visualizes this relationship: Livability is illustrated as Florida's (2008) pyramid of place in Panel a) (also see Burton 2014) and the pyramid's bottom is similar to Maslow's pyramid of a person's needs (Maslow [1954] 1987) in Panel b). The foundation of both pyramids are basic needs. The top of Maslow's pyramid is made of psychological and self-fulfillment needs, whereas Florida places higher dimensions of livability at the top. The Mercer Index is relatively similar to Florida (2008)'s pyramid of place shown in panel a) in Fig. 1, especially its bottom.

As shown in Panel c), subjective well-being is a function of basic needs first (the foundation of the pyramids), but once they are satisfied, SWB depends on the higher dimensions prescribed in both pyramids. Panel c) is also a reformulation of the well-being and income graph from Inglehart (1997), (also see discussion in Inglehart 1997, p. 1849), and it illustrates the "affluence paradox" (Pacione 2003)—the more income, economic development or affluence, the less these matter for SWB. At higher level of economic development, what matters for SWB are the characteristics described at the higher dimensions of both pyramids. This phenomenon is similar to the diminishing marginal returns from income on subjective well-being observed at country, region, and person levels (Okulicz-Kozaryn 2012). Note that the definitions used in Fig. 1 are not definitive, and were used for the purpose of illustration

[3]Some scholars make a distinction between happiness and life satisfaction—life satisfaction refers to cognition and happiness refers to affect. Life satisfaction is a cognitive aspect of happiness (Dorahy et al. 1998) We cannot differentiate between the two as we have only one measurement, hence, we mostly measure 'life satisfaction, not 'happiness.' But as described above, there is an overlap between the two. In addition, we will use a place satisfaction measure.

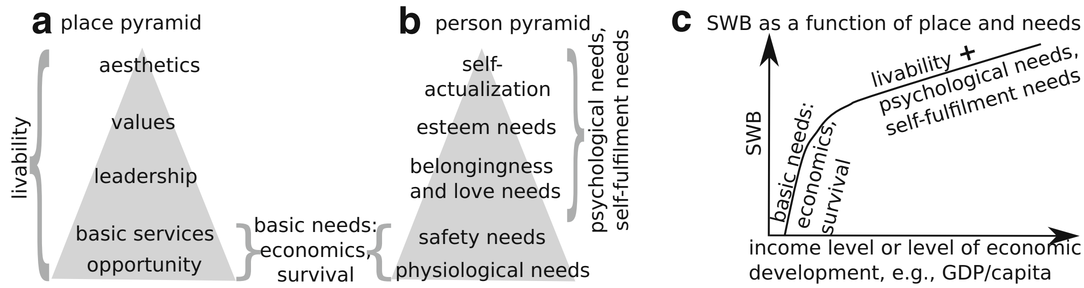


**Fig. 1** **a** Livability as place pyramid; **b** Maslow's hierarchy of needs; **c** SWB as a function of (**a**) and (**b**)

only. Livability can be defined in a much broader way than what is illustrated in Florida (2008)'s pyramid—for instance, it can include welfare provision or broadly understood income redistribution (Okulicz-Kozaryn et al. 2014). The Mercer Index, used in this study, captures most of the characteristics in the bottom of the pyramids (basic needs: economics and survival) and to a lesser degree some of the aspects in the higher dimensions. There has been debate about whether SWB would increase linearly with income in Panel c), and some scholars even transform the x axis into a log scale and use cross-section as opposed to time series when illustrating this relationship (e.g., Stevenson and Wolfers 2013), however, the literature has shown that usually in the long term SWB has nil relationship with income (Easterlin et al. 2010). There are cases, however, when income and SWB go hand in hand over time (Veenhoven and Vergunst 2013).

There is an extensive literature on the livability-SWB nexus (Veenhoven 2000; Cummins 2000; Diener and Suh 1997; Schneider 2005; Pacione 2003). From these, we've learned that objective and subjective indicators provide different perspectives, each has strengths and weaknesses and both should be used as complements, and most importantly, it is critical to use subjective indicators as they add greatly to economic measures such as income.

There are several major happiness theories that can explain the relationship between livability and SWB. One of the main theories is Veenhoven's Livability Theory (Veenhoven and Ehrhardt 1995; Veenhoven 2000, 2014a). Livability depends on ecology, but also on some fundamental human and social needs such as those at the bottom of Maslow's pyramid of needs (Maslow [1954] 1987). In other words, there are some universal human needs that need to be satisfied (Veenhoven and Ehrhardt 1995).

Another theory proposed by Michalos links livability and SWB (Michalos 2014), refer to Table 1. Michalos' classification is somewhat similar to Veenhoven's four qualities of life (Veenhoven 2000) discussed later in Table 2. Concurrently, there is also the overall Quality Of Life (QOL) theory subsuming objective with subjective indicators (Veenhoven 2000; Michalos 2014; Giannias and Sfakianaki 2014; Burckhardt et al. 2003). Quality of life can be defined as "a global measure based on an aggregation of well-being across several life domains (e.g., recreational, social activities, finances), usually assessed using a combination of objective and subjective indicators" (Steel et al. 2008). This is very similar to the QOLS scale (Burckhardt

**Table 1** Michalos' two variable theory: a fool's paradise and a fool's hell (Michalos 2014)

| | lo livability | hi livability |
|---|---|---|
| lo SWB | Real hell [deprivation, unhappy poor] | Fool's hell [dissonance, unhappy rich] |
| hi SWB | Fool's paradise [adaptation, happy poor] | Real paradise [well-being, happy rich] |

Cummins' classification is shown in the square brackets (Sirgy 2002, p. 61)

et al. 2003). However, such aggregation may not be the best idea—there is a conceptual and empirical difference between the two—it is better to explore the relationship between objective and subjective indices instead.

There is a handful of empirical studies linking objective with subjective indicators. Senlier et al. (2009) and Wkeziak-Bialowolska (2016) find a significant link between most domain perceptions, or satisfactions, and overall place satisfaction. Balducci and Checchi (2009) on the other hand, find a significant link only between some domain perceptions and overall happiness for specific cities separately and after controlling for person level characteristics. Liao (2009) uses both objective and subjective measures and finds that in most domains the correlation between objective and subjective is insignificant. In general, livability and SWB have been found to be poorly or moderately correlated (Schneider 2005; von Wirth et al. 2015; Cummins 2000; Okulicz-Kozaryn 2013).

Oswald and Wu (2009) mostly contradicts the literature by finding moderately high correlation (.6) between subjective and objective indicators. Several explanations for this incongruity are possible: a large representative sample of state level data and adjustment of life satisfaction measure with person level predictors. Perhaps the key is their usage of a comprehensive SWB measure, life satisfaction, and comprehensive index of QOL containing multiple domains. Our study will also use the life satisfaction measure of SWB and the comprehensive QOL Mercer Index finding concurrent results.

We know how livability relates to place satisfaction (Okulicz-Kozaryn 2013). But, what is the relationship of livability with overall life satisfaction? This is the first study linking overall livability, defined as objective quality of life or standard of living, with subjective well-being (SWB) measured as life satisfaction across European cities. Before turning to data analysis, we add one more angle to our study—that of urbanicity or size of place.

**Table 2** Veenhoven's four qualities of life (Veenhoven 2000)

| | Outer qualities | Inner qualities |
|---|---|---|
| Life chances | Livability of environment [Mercer rank, pop size] | Person's life-ability [N/A] |
| Life results | Utility of life [NA] | Appreciation of life [place, life satisfaction] |

The measures used in this study are in brackets. Note that place also affects life-ability to some degree, for instance, urban living is unhealthy to the human brain (Lederbogen et al. 2011)

## Urban Economic v Sociological Literatures: Size of a Place as Livability

With respect to cities, economists and sociologists are less focused on the differences and relationships between objective and subjective measures. One of their interests however, is urbanicity[4]—the degree to which a place is urban, often measured as population size. These urban economic and sociological literatures are largely separate from the social indicators literature, and both mainstream economists[5] and sociologists (Veenhoven 2014b) still tend to dismiss SWB. It is important to connect the literatures, because economic theory argues in favor of large cities, while sociological theory is more ambivalent, but tends to argue in favor of smaller places.

Classic sociologists point to the problems of city life: over stimulation, withdrawal, vice, impersonality, and shallowness (Park et al. [1925] 1984; Simmel 1903; Tönnies [1887] 2002; Wirth 1938). Popular urbanists such as Florida, Jacobs, and Zukin are more ambivalent (Zukin 2009; Florida 2014, 2016a, b, c; Jacobs [1961] 1993)—surely they appreciate cities—yet, they are also critical, and value the small-town feel (e.g., see famous appreciation of small-town feel of Greenwich Village by Jacobs ([1961] 1993)). Economists invariably point to the economic benefits that emerge in cities such as labor specialization, productivity, agglomeration economies, economies of scale, invention and creativity (Florida 2008; Glaeser 2011b; O'Sullivan 2009). According to the Central Place Theory (e.g., O'Sullivan 2009) consumption in large scale can only take place in the largest cities: large museums, opera houses, symphonies, etc.

Thus, we propose to explicitly test the size of a place as a measure of livability. We are not aware of any other research where livability is explicitly defined as the size of a place, although some studies have implicitly suggested it, particularly those using economic theory (e.g. Glaeser 2011a, b).[6] In general, economists main goal is to maximize utility or welfare,[7] measured as income or consumption (Autor 2010). Geographically, the greatest income or consumption per capita is always found in the

[4]There has been many studies linking urbanity to SWB, for instance, see the World Database of Happiness (http://worlddatabaseofhappiness.eur.nl), subject section Eb02 'Urbanity'.

[5]There are many "maverick economists" studying the so called, "economics of happiness." Some such as Richard Easterlin and Andrew Oswald have significantly contributed to the social indicators literature. Notwithstanding, there are a few skeptics who do SWB research, but sneer it at the same time (e.g., Angus Deaton (e.g., Deaton 2013) and Ed Glaeser (Glaeser et al. (2014, 2016)). Many, if not the vast majority of economists do not consider SWB, social indicators or any social science outside of economics as worthwhile (Economist 2014, 2016; Naim 2016; Fourcade et al. 2015).

[6]There are also studies by economists using "hedonic pricing" and "compensating differentials," see for instance Oswald and Wu (2009), Giannias and Sfakianaki (2014), Glaeser et al. (2016), and Albouy (2008). We do not dwell into this economic literature as it is based on revealed preferences and rationality assumption—and we know that humans are not rational (Shiller 2015; Zafirovski 2014; Akerlof and Shiller 2010; Ariely 2009; Kahneman 1994; Sen 1977). There are also economists using broader measures such as crime and pollution—for review see Lambiri et al. (2007).

[7]Veenhoven (2000, p. 6) also confirms that "economists sometimes use the term 'welfare' for livability of environment.

largest cities, hence, the bigger the city, the more utility or welfare. Thus indirectly, size of place refers to economist's notion of livability:

$$livability \approx size\ of\ a\ place \tag{1}$$

If defining livability as size of place seems far-fetched, see the conceptualization by sociologist Veenhoven (2000) in Table 2, where livability of environment is defined as the intersection of outer qualities (place, environment) and life chances. Clearly, the city epitomizes the apex of life chances. Nowhere else there is so much variety and opportunity (Tönnies [1887] 2002; Milgram 1970; Fischer 1995; Glaeser 2011b; O'Sullivan 2009; Campbell 1981).

Although non-intuitive, there is support in the economic literature for the $livability \approx size\ of\ a\ place$ equation. As one economist explains, "more populous cities offer a higher QOL that is implied by wages and costs alone: if two cities offer the same wages and costs, the more populated city is deemed the one more amenable to the average individual" (Albouy 2008, p. 20). Of course, this is not the only measurement economists use, as shown in (Albouy 2008; Lambiri et al. 2007; Myers 1988), nor would they use it explicitly. However, a careful review of the most recent economic literature on the topic (Glaeser et al. 2014, 2016; Glaeser 2011a, b, 2014; Albouy 2008) reveals that this definition is implicit. Albouy (2008) provides a specific treatment of QOL and size of a place, and concludes that the bigger the place, the higher the quality of life (see Table 1 in Albouy 2008). In some ways, the economic approach to livability seems to be similar to how development and progress used to be measured solely as income (e.g., per capita gross domestic product). It was not until recently that some progressive economists acknowledge this shortcoming (e.g., Stiglitz et al. 2009), while non-economists knew it for decades (e.g., Campbell et al. 1976).

Another popular measure of QOL among economists is $\frac{cost\ of\ living}{wage}$ (Albouy 2008), which is a reasonably good proxy for size of a place—cost rises much faster than wage with population size (Okulicz-Kozaryn 2015b). Similarly, economists like Glaeser et al. (2016) for example, tend to assume that city growth is a consequence of people's rational preference for the city, and hence, they must be happy there. Therefore, city growth is a result of utility or SWB maximization. Here, for simplicity, we just focus on city size, not city growth. There are also some dissenters among economists arguing that there is higher utility in smaller places, as reviewed, for instance, in Albouy (2008) or Pines (1972).

In general, economists would predict that more money means more SWB (Autor 2010), and since wages and consumption are greatest in the largest cities, the argument follows that this will yield the highest happiness levels. Alternatively, an urban economist would predict that SWB will be constant across space: the more money, the more utility, but "urban dis-utilities" like commute distance and time have to be compensated by higher wages, and net utility is constant. According to the axiom of spatial equilibrium—one of the most important founding principles of urban economics—in equilibrium, individuals cannot improve their overall utility levels via migration (Glaeser et al. 2016), making SWB constant across cities. Our results contradict economic theory and indicate the opposite: larger cities do not have the highest SWB levels, nor is SWB constant across cities. We find that the highest levels of SWB are found in the smallest places.

## Data

We use the 2015 Mercer Index, and complement the analysis by using the 2012 data as a robustness check in the Appendix. The SWB data come from the Eurostat database at http://ec.europa.eu/eurostat/web/cities/data/database (urb_percep database). The Eurostat data are city-level aggregates from the Flash Eurobarometer Survey 419 (Quality of Life in European Cities 2015) and the Flash Eurobarometer Survey 366 (Quality of Life in European Cities 2012).[8] The Flash Eurobarometer Surveys interviewed European urbanites aged 15+ via telephone in their mother tongue on behalf of the European Commission. The basic sample design applied in all countries is multi-stage random (probability). In each household, the respondent was drawn at random following the "last birthday rule" (for more information see https://doi.org/10.4232/1.12516). SWB is measured using two variables:

LIFE SATISFACTION measured with "Q3.3 On the whole, are you very satisfied, fairly satisfied, not very satisfied or not at all satisfied with ...? - The life you lead."

PLACE SATISFACTION measured with "Q3.4 On the whole, are you very satisfied, fairly satisfied, not very satisfied or not at all satisfied with ...? - The place where you live."

Eurostat provides the percentage of respondents in each category as separate variables. We created one variable by using the following transformation:

$$\begin{aligned} \textit{variable used here} = &(\textit{'strongly/very YES'} * 1) + (\textit{'somewhat/rather YES'} * .5) \\ &+ (\textit{'somewhat/rather NO'} * -.5) + (\textit{'strongly/very NO'} * -1) \end{aligned}$$

The new variable ranges between a theoretical $-100$ where everybody "strongly/very" disagrees to +100 where everybody "strongly/very" agrees. Okulicz-Kozaryn (2013) used a synthetic index, $\frac{('strongly/veryYES' + 'somewhat/ratherYES')}{('strongly/veryNO' + 'somewhat/ratherNO')}$ which is very similar to our variable—they both correlate at about .95, and the transformation we use is slightly better as it uses ordinal scale information as opposed to treating 'strongly/very' and 'somewhat/rather' as the same. Livability is also measured with two variables: the Mercer Index Ranking and population size.

The MERCER is a city ranking survey based on the Mercer Index and can be downloaded from https://www.imercer.com/uploads/GM/qol2015/h5478qol2015/index.html. The Mercer is probably the most popular survey used to rank cities in terms of their livability or standard of living. Other "best places to live" rankings appear to follow the Mercer. For instance, the Economist and Forbes base their rankings primarily on data from the Mercer ranking (e.g., http://www.livablecities.org/blog/value-rankings-and-meaning-livability). Kotkin (2011) claims

[8] We were unable to find a direct statement per the source of the 2015 Data in the Eurostat Metadata. The 2012 Data is directly referenced in Eurostat Metadata to "Quality of life in cities - Perception survey in 79 European cities - European Commission, Flash Eurobarometer 366, October 2013".

that the Economist ranking is ''remarkably similar'' to the Mercer Ranking. The ranking calculates livability based on 39 factors, grouped in 10 different categories. The Mercer survey questioned expatriates on the importance of each of the 39 issues. The weights assigned to each category are as follows (most heavily weighted items are in bold):[9]

23 **Political and social environment** (political stability, crime, law enforcement, etc)
4 Economic environment (currency exchange regulations, banking services, etc)
6 Socio-cultural environment (censorship, limitations on personal freedom, etc)
19 **Health and sanitation** (medical supplies and services, infectious diseases, sewage, waste disposal, air pollution, etc)
3 Schools and education (standard and availability of international schools, etc)
13 **Public services and transportation** (electricity, water, public transport, traffic congestion, etc)
9 **Recreation** (restaurants, theaters, cinemas, sports and leisure, etc)
11 **Consumer goods** (availability of food/daily consumption items, cars, etc)
5 Housing (housing, household appliances, furniture, maintenance services, etc)
6 Natural environment (climate, record of natural disasters)

The POPULATION size of a city is the second measurement of livability. As argued earlier, it is a problematic measure, but it is derived from economic theory, which implicitly argues that the "larger the place, the better." Population size is a good proxy for opportunity, and livability can be defined as an intersection of life chances and outer qualities (Veenhoven 2000). In general, urbanicity can be measured as population size, density, and heterogeneity (Wirth 1938). For simplicity, we just use population size as a measure. Furthermore, there is a large variability in population size. In the sample used here, some places are small towns of less than 100k people, and some places are large cities with a million or more people. City populations were still largely unavailable for 2015, hence, we extrapolate population from previous years using linear interpolation with Stata command `ipolate` with the `epolate` option.

## Results

Table 3 shows the correlations. We begin by examining how each of the two variables measuring SWB and livability correlate with each other. As expected, the two measures of SWB, PLACE and LIFE SATISFACTIONS correlate strongly at .75.

On the other hand, the MERCER ranking actually correlates with POPULATION (.24) in the opposite direction to what economic theory would argue: the larger the place, the lower the Mercer ranking.[10] Second, we turn to the correlations of livability

[9] We obtained the weights by contacting Mercer in 2011. We have contacted them again to see if there was any change and were told that it has not changed. Morais et al. (2013) reports the same weights. A full list containing the 39 factors can be found in Okulicz-Kozaryn (2013).
[10] Higher value in rankings denotes lower rank, of course.

**Table 3** Pairwise correlations

| Variables | Mercer | Population | Place satisfaction | Life satisfaction |
|---|---|---|---|---|
| Mercer | 1.00 | | | |
| Population | 0.24 | 1.00 | | |
| Place satisfaction | − 0.57* | − 0.38* | 1.00 | |
| Life satisfaction | − 0.63* | − 0.15 | 0.75* | 1.00 |

* $p < .05$

measures with SWB measures. As predicted by sociological theory, and opposite to economic theory predictions, POPULATION correlates negatively with PLACE SATISFACTION at $-.38$ and with LIFE SATISFACTION at $-.15$. The MERCER, on the other hand, indicates that the higher the place in livability ranking, the higher the SWB. The MERCER ranking correlates quite strongly with PLACE and LIFE SATISFACTIONS at about .6 indicating a considerable overlap between livability and SWB. Interestingly, (Oswald and Wu 2009) also found a correlation of .6 between SWB and objective measures of quality of life across US states, which they argue to be very high:

> A correlation coefficient of 0.6 is unusual by the standards of behavioral science. It is high by the cut-offs suggested by Cohen's rules-of-thumb (which argued that in human data an r value over 0.5 should be seen as a large association, and 0.3 a medium one). An r = 0.6 is the same degree of correlation, for example, as has been found for one's own life-satisfaction readings taken 2 weeks apart.

While the correlation found in our study is almost identical with Oswald and Wu (2009)'s results, it is considerably higher than the .36 correlation found in Okulicz-Kozaryn (2013). Several explanations for this discrepancy are possible. Although the Mercer Survey has not changed measurement and measures livability in the same way, it uses a bigger and more representative sample. Likewise, the Eurobarometer measurement of SWB may have improved. Okulicz-Kozaryn (2013) used one of the first data collections on the European urban satisfactions survey. Our study uses a slightly different measurement from Okulicz-Kozaryn (2013) who used a synthetic index to measure place satisfaction. Furthermore, Okulicz-Kozaryn (2013) used the Mercer Index, and our study, like Oswald and Wu (2009)'s, uses the ranking (although both correlate at about .95). One difference that helps explain the discrepancy is that Okulicz-Kozaryn (2013) used survey means over the years of 2004, 2006 and 2009. Using single year, as opposed to multi-year averages would increase correlations by about .05 to 0.1.

Hence, the .36 correlation found in Okulicz-Kozaryn (2013) would predictably go up to as high as .46, which is not very different from the .57 correlation found here. Furthermore, the scatterplot linking place satisfaction with the Mercer index in Okulicz-Kozaryn (2013) is similar to the one reported here.

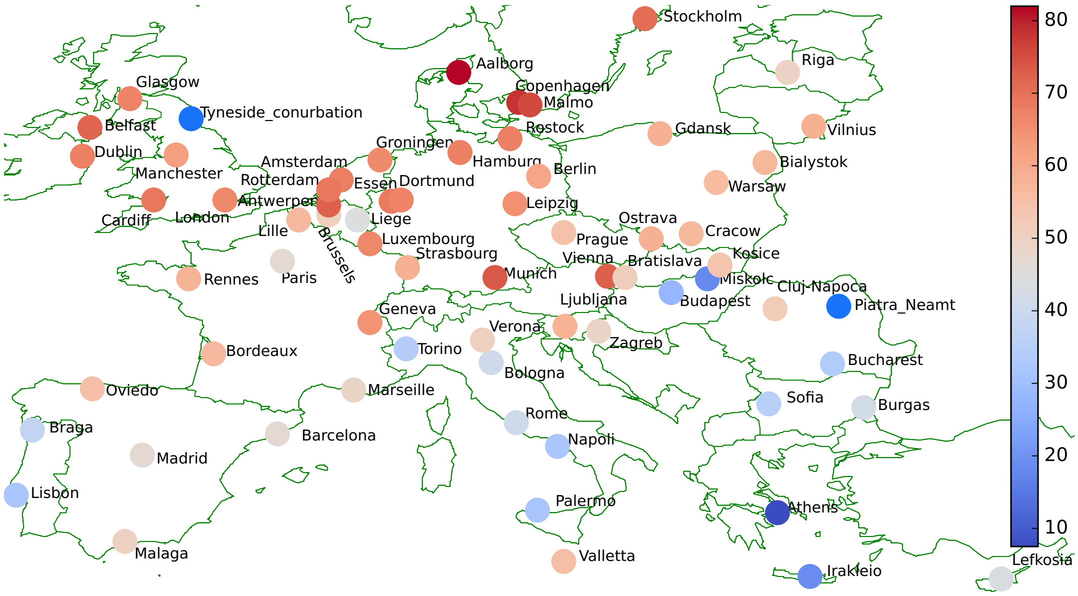


**Fig. 2** Life satisfaction ranges from low (blue) to high (red)

Next, we turn to visual explorations. LIFE SATISFACTION is shown in the map in Fig. 2.

Cities in the North have higher levels of happiness than cities in the South region (including France). This result is concurrent with Okulicz-Kozaryn (2011)'s findings across European regions in 1996: there are clusters spanning national boundaries. The North Western part of Europe, or more specifically Germany, the Netherlands, the UK, and Scandinavia constitute a large cluster of cities with high levels of happiness.

Studies have found that being close to the coast improves SWB (White et al. 2013; Wheeler et al. 2012), however, we do not find this pattern here: cities with the lowest level (e.g., Athens) and with the highest level (e.g., Copenhagen) of subjective well-being are both coastal. Similarly, our results do not necessarily support research indicating that people prefer warmer temperature in the winter and colder temperature in the summer (Rehdanz and Maddison 2005)

Figure 3 plots SWB against the Mercer ranking. A clear pattern emerges: the Mercer ranks Western cities higher than Eastern, thence livability ranges in the East-West dimension. All Western cities are ranked at 65 or higher on the Mercer ranking, and all Eastern cities are ranked below it. Hence, objective quality of life has a clear East-West dimension, which is understandable to some degree: the post communist East still suffers from lower income and other disadvantages such as lower civic engagement and lower subjective feeling of freedom (Okulicz-Kozaryn 2008, 2015a). As the results show in Fig. 3, there is a cluster of cities which ranked high on both SWB and the MERCER circled as West, and a cluster with lower PLACE SATISFACTION and lower LIFE SATISFACTION circled as East. The West-East dimension is clearer than the North-South, where there is a little mixing, but it is still clear that


**Fig. 3** SWB (place and life satisfactions) against the Mercer ranking. Linear fit shown: the higher the livability (Mercer rank), the higher the SWB. "Western" and "Eastern" clusters of cities circled

the South has considerably lower SWB than the North. In fact, if SWB is set at a low level such as 44, we can observe that all cities below this threshold are Southern: Athens, Bucharest, Budapest, Istanbul, Lisbon, Rome, Sofia, Ankara, Athens (greater

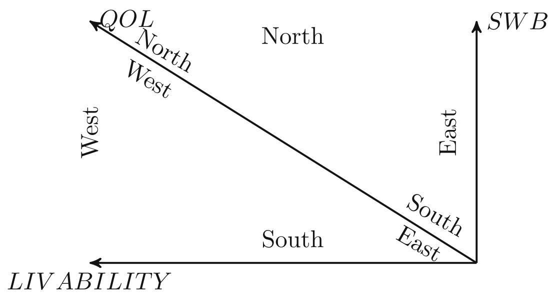


**Fig. 4** Four Europes: the unlivable East and the livable West region, and the unhappy South and the happy North region. Hence, overall, in terms of QOL: a "supreme" North-West region and an "inferior" South-East region. Note: the graph is drawn in such a way as to correspond directionally with Fig. 3: Northern cities are located at the top, and Eastern cities to the left

city), Bologna, Braga, Burgas, Diyarbakir, Irakleio, Lefkosia, Lisbon (greater city), Miskolc, Napoli, Palermo, Piatra Neamt, Torino. Many of these places are omitted from the scatterplots because they are not ranked in the Mercer. They are displayed, however, in the map in Fig. 2.

Given that SWB stretches from the "unhappy" region of the South to the "happy" region of the North and livability stretches from "substandard" East to "preeminent" West, the highest ranking region on both dimensions is the North-West and the lowest ranking is in the South-East as visualized in Fig. 4.

There are some outliers: livable but unhappy, or unlivable but happy places. Outliers can be described in terms of Michalos (2014) terminology: Belfast and Ljubljana are a fool's paradise: higher SWB than expected from livability; while Paris and Athens are a fool's hell: lower SWB than expected from livability. Although, in the case of Athens, the Greek crisis might have affected the results. The cities that fall outside of the circled West and East regions, have lower, in some cases much lower, SWB than expected from the Mercer ranking (refer to Fig. 3).

In general, places ranked high on life satisfaction (top panel) are also ranked high on place satisfaction (bottom panel), but a few places do not correspond on both dimensions, for instance, Istanbulites are quite satisfied with their lives, but not so much with the place they live in.

The higher a city is ranked on the Mercer, the greater the SWB—Western cities score higher on both dimensions and South-Eastern cities score lowest on both. Outliers are instructive—in several places, people were more satisfied with their cities than expected from the Mercer: all are relatively smaller cities such as Stockholm, Glasgow, and Belfast in the North-West, Leipzig in the Central region and Ljubljana and Zagreb in the Central South. The few places where people were less satisfied than expected from the Mercer ranking were rather large cities: Brussels, Paris and Lisbon. This suggests that even though the Mercer ranks smaller places higher, it still under-ranks small places and over-ranks large cities.

Figure 5 repeats the exercise from Fig. 3, except that it now plots population on the x axis. The bigger the place, the lower the PLACE SATISFACTION ($r = -.38$) and LIFE SATISFACTION ($r = -.15$; insignificant). Scatterplots exclude cities with

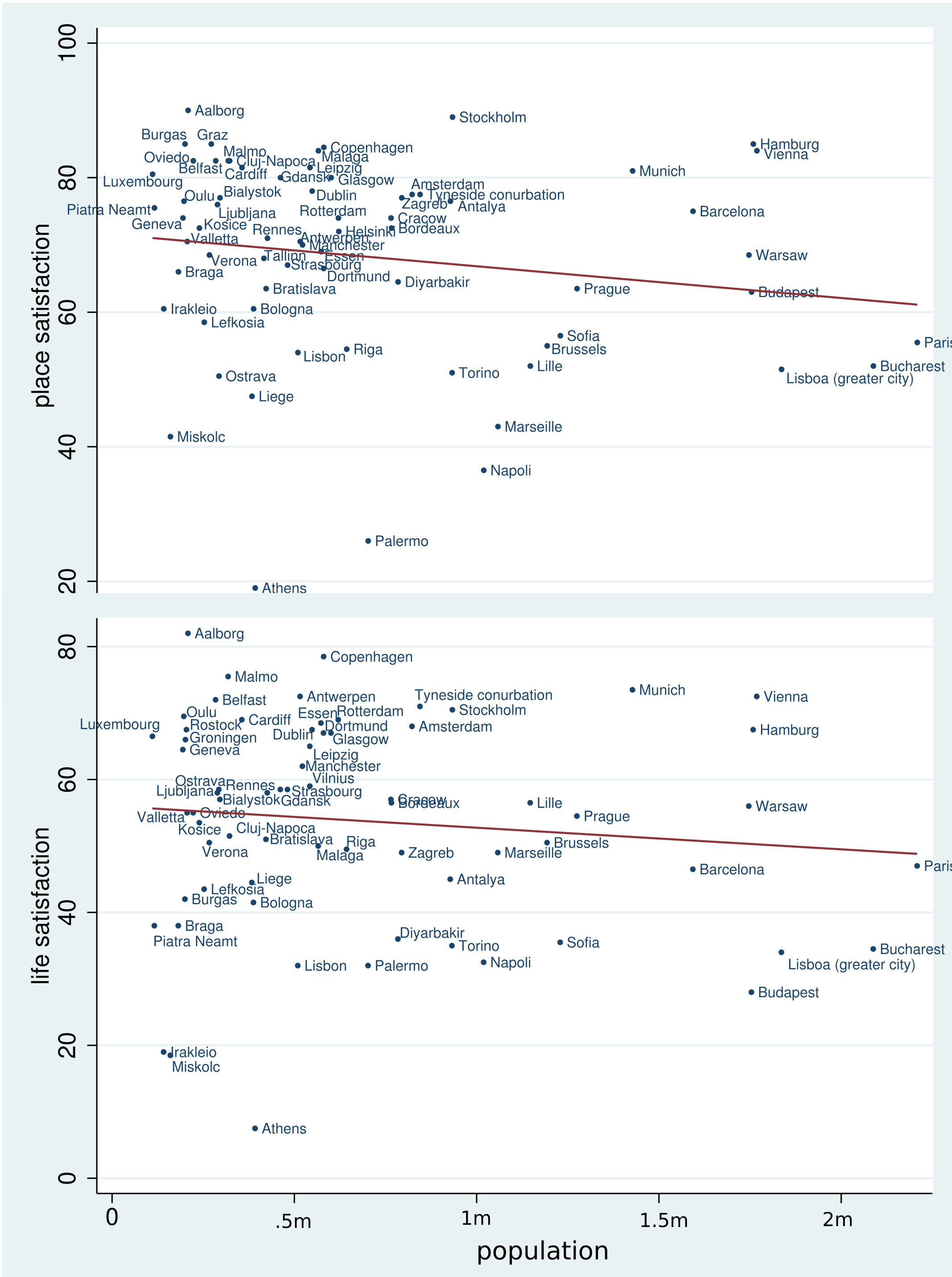


**Fig. 5** SWB (place and life satisfactions) against population size. Linear fit is shown. Cities with population > 2.5 m were dropped so that city labels are readable, but graphs using all available observations are similar and shown in the Appendix. The population data was not available for the year of 2015 and were extrapolated from previous years using Stata 'ipolate' command with option 'epolate.' The relationship for the year 2012, when only a few observations were interpolated/extrapolated, are similar and can be found in the Appendix

a population > 2.5 m so that city names are readable (all data are shown in the Appendix). The relationships are less clear than in Fig. 3, and cities are not easily grouped. It is however important to show this weakly negative relationship because it runs counter to economic theory.

## Conclusion

This study started with the hypothesis that Europe can be divided into regions: East vs. West and South vs. North. Our results confirm this hypothesis. Livability rises from the East to the West, and SWB rises from the South to the North. When combining both dimensions, the overall quality of life is lowest in the South-East and highest in the North-West.

These results contradict the urban economic paradigm of spatial equilibrium that claims that people cannot improve their SWB by migrating to other place (Glaeser et al. 2016). Clearly, both SWB and livability differ widely across European cities, and the free movement of European citizens allow for easy migration. Likewise, our results challenges the economic proposition that people are happiest in the largest cities (e.g., Glaeser 2011b; Glaeser et al. 2016).

The city embodies life chances. It is a person's habitat, where she works, lives, and uses various amenities. This study links these life chances, embodied in these European cities, to life results (SWB). More is sometimes better as we found with livability measured with the Mercer index, but more is not always better (Schwartz 2004), and this is what we found in terms of the size of a place. Our results show that SWB correlates with livability at about 0.6, concurrent with Oswald and Wu (2009)'s findings. The correlation of SWB with objective measures is in some way a confirmation of subjective measures, and it can work both ways: subjective measures can also be used to confirm objective measures. We found confirmation for the Mercer Index Ranking (though it still under-ranks small places), but not for the size of a place.

Subjective indicators do not replace, but complement objective indicators (Stiglitz et al. 2009). At the same time subjective indicators are in some ways more useful. Only SWB can be measured completely, while livability and QOL consist of innumerable items that cannot be measured fully (Veenhoven 2000), although economists recently tried in vain (Benjamin et al. 2014, 2015, 2017). The key advantage of using the SWB yardstick is that it overcomes the difficulty of measuring utility in social welfare, for instance, it helps answer the question of whether or not we should invest limited resources in parks, bike lanes, or waterfronts—for discussion see Diener (2009) and Okulicz-Kozaryn (2016).

This is the first study linking SWB with livability at the city level in Europe. Future studies, might explore this relationship in other regions[11] and continue to explore the complementary nature of using objective and subjective measures in happiness

[11]Research examining size of place and happiness in Latin America did not find a significant difference in happiness levels based on city size (Valente and Berry 2016).

studies. Also, future research might gain more insight by comparing changes over time, especially as more waves of data become available.

## Discussion

We have found the Mercer Index to be substantially correlated with SWB as in Oswald and Wu (2009). Still, we argue that it is not the objective quality of cities but how people perceive them that ultimately matters as argued by Okulicz-Kozaryn (2013). It is the city on one's mind, and not the actual city on the ground that counts. As John Milton said, "the mind is its own place and in itself, can make a heaven of hell, a hell of heaven" (John Milton cited in Campbell 1981, p. 1). Using this metaphor, 'livable' (as per the Mercer Index) Paris is a "hell" (relatively low SWB), and 'unlivable' Belfast is "heaven" (high SWB).

What determines the actual (experienced) livability is what people feel, and not what exists in the real world. Subjective indicators directly tap quality of life as experienced (Schneider 2005). And there is a systematic difference between what we think to influence our quality of life and what actually does. Psychologists call this phenomenon expected versus experienced utility (Kahneman et al. 1997; Schkade and Kahneman 1998; Kahneman 2000; Kahneman and Krueger 2006). This is related to the cognitive theory of happiness, holding that SWB depends on perception of how life is (and in particular on the difference of that perception with how-life-should-be). In effect, the theory of happiness conscious perception is not always required. Needs can be gratified without knowing and still affect how well we feel, as is evidently the case with newborns. In the same vein, pollution can lower our happiness without a perception of pollution. Likewise, the availability of more choice options in a big city can affect happiness positively.[12]

Livability rankings of "best places" actually merely measure standard of living, and not the broadly understood quality of life.[13] There are city-level qualities, such as trust, tolerance, creativity, and so forth, that determine the overall quality of life and are yet left out from livability measurement.[14] These city-level qualities become more important in more developed areas such as in the majority of European cities studied here. There is also a "paradox of affluence" (Pacione 2003) illustrated earlier in Fig. 1—the more economic growth, the less economics matter.

Can selection explain our results? Perhaps. Big cities may attract relatively many unhappy people, such as singles, working age people, drop-outs and people with mental problems, who may live happier in the city than in the province (and moved for that reasons) but still decrease the average level of SWB in big cities.

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[12] We are grateful to an anonymous reviewer for this point.

[13] The Mercer Index disclaimer states that: "One may live in the highest ranked city in terms of quality of living [standards] and still have a very bad quality of life because of unfortunate personal circumstances (illness, unemployment or loneliness, etc.").

[14] One could add them, but still there are virtually uncountable factors—and there is no point in attempting to include them all as some have attempted without success (Benjamin et al. 2014, 2015, 2017). And this is the advantage of using SWB, which captures everything that affects one's well-being.

But likewise, cities do arguably attract the best and the brightest, the most motivated and talented persons. Furthermore, since the vast majority of the population live in cities, the large majority of people are born and raised in cities. Thus, urban unhappiness is not due to sorting or selection, but rather due to the effect of city life—cities make people unhappy. For instance, in cities people are increasingly exposed to light, air, and information (marketing and advertising) pollution. Arguably, cities can also increase stress, overwork, and can magnify pecuniary and consumerist orientation. Selection or sorting would be easy to test, but not with data used here, therefore, this is left to future research.

Our results contradict economic theory, which by using solely a monetary yardstick of income or consumption, would argue that the place with the greatest income or consumption should have the greatest livability and subjective well-being. The richest or most expensive cities like London and Paris are not the most livable (as per the Mercer index), and do not have high levels of SWB. In fact, it is the other way around: London has the lowest SWB when compared to other cities in the UK (Office for National Statistics 2011; Chatterji 2013), similarly Helsinki in Finland (Morrison 2015), and Bucharest in Romania also have the lowest levels of SWB, and so forth—virtually all of the largest cities across Europe and the developed world, have the lowest subjective well-being levels in any given country (Okulicz-Kozaryn 2015b).

While unquestionably important, income (utility or economic welfare) is but one indicator in a list of many social and human factors that combine to explain one's overall happiness and choice of residency. Thus, given the intrinsic complexity of human choices and rationality (Shiller 2015; Zafirovski 2014; Akerlof and Shiller 2010; Ariely 2009; Kahneman 1994; Sen 1977; Thaler 2012; Thaler and Sunstein 2008; Kahneman and Thaler 2006), SWB research can gain significant insight by broadening its scope of analysis to multi-disciplinary approaches that are not solely theory-driven (Krugman 2012; Finance and Economics 2013). If livability is defined, for example, only as consumption or income per capita, then we would expect a positive correlation between livability and city size: the bigger the city, the higher the livability. Such definition is problematic, however, because it excludes other variables that determine livability. Incongruent results in the literature might be the product of studies with restrictive approaches and much can be gained by widening the scope of SWB analysis.[15]

## Appendix

As a robustness check this appendix mostly repeats the analyzes in the body of the paper, but for the year of 2012 instead of 2015. The relationships are similar, however, life satisfaction values differ widely from 2012 to 2015—we have double checked with sources and this is indeed the case. We do not have an explanation for the large

[15] For example, while some researchers using multi disciplinary approach argue that high consumption and high urbanization reduce SWB (Kasser 2003; Frank 2012; Wirth 1938; Okulicz-Kozaryn 2015b), others using a more limited approach claim the opposite Glaeser (2011b) and Stevenson and Wolfers (2013).

**Table 4** Pairwise correlations

| Variables | Mercer | Population | Place satisfaction | Life satisfaction |
|---|---|---|---|---|
| Mercer | 1.00 | | | |
| Population | 0.21 | 1.00 | | |
| Place satisfaction | − 0.54* | − 0.24* | 1.00 | |
| Life satisfaction | − 0.67* | − 0.09 | 0.61* | 1.00 |

*$p < .05$

changes, and such discrepancy in such a short period of time (3 years) is troubling. However, we are confident in the analysis since the relationships are similar. For a discussion of some of the 2012-15 changes see O'Sullivan (2016).

Note that "London" here means the "Greater London Area"—this is probably what most people understand by "London." Other places are delineated by city limits. For a discussion refer to http://ec.europa.eu/eurostat/web/cities/spatial-units.

Table 4 repeats the correlation from the body of the paper for the year of 2012.


**Fig. 6** SWB against the Mercer. Linear fit shown. Data for 2012

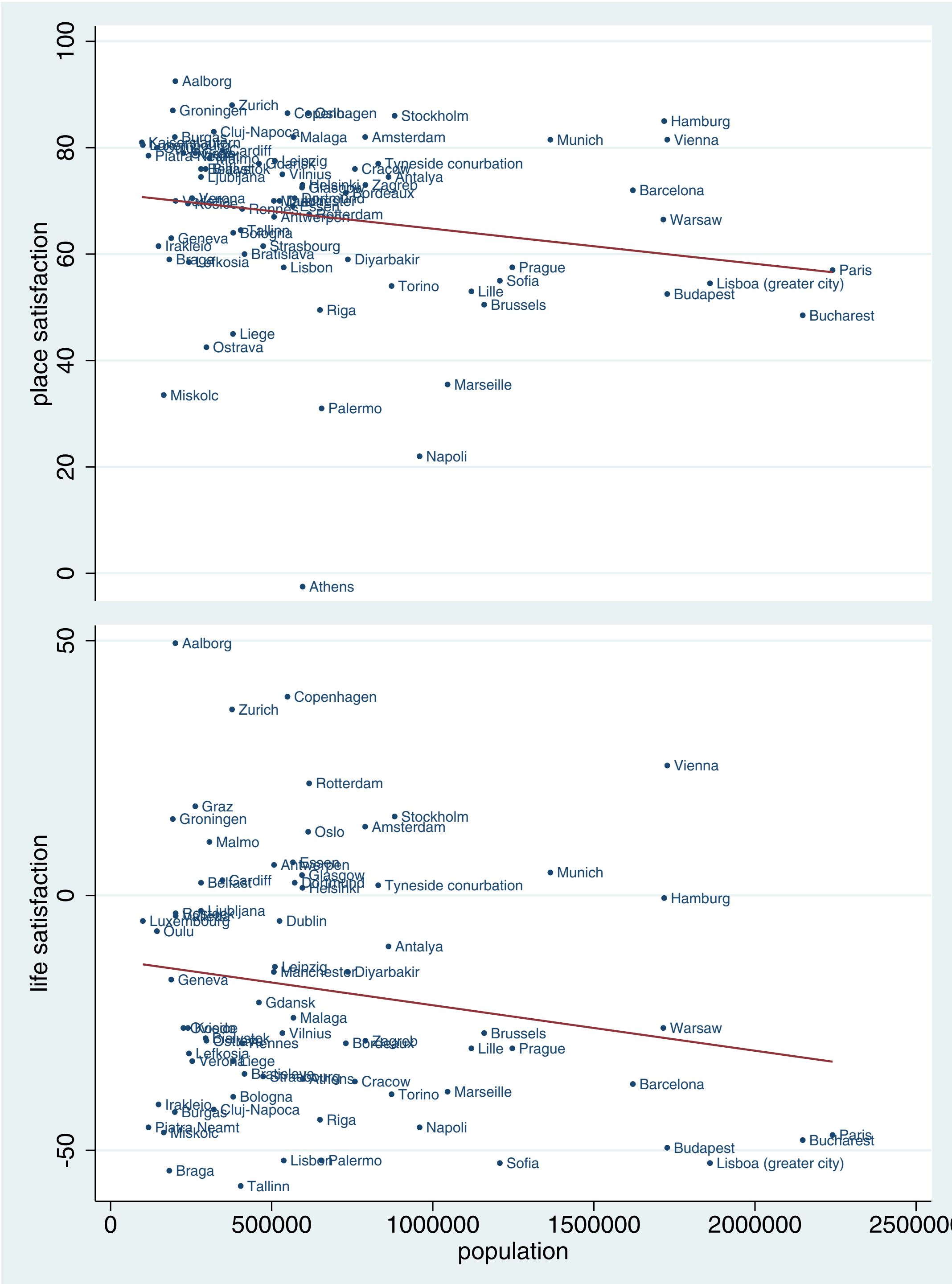


**Fig. 7** SWB against population size. Linear fit shown. Data for 2012. Cities > 2.5 m are dropped so that city labels are readable; graphs using all available observations are similar and shown below. Some (13) population values were not available for 2012 and were extrapolated from previous years using Stata 'ipolate' command with option 'epolate.'

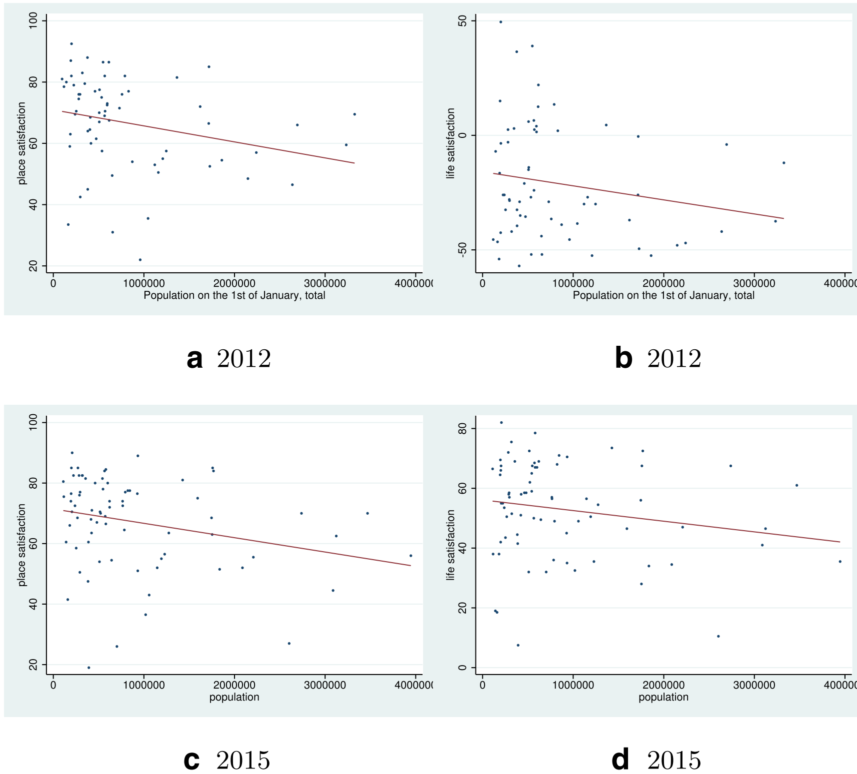


**Fig. 8** SWB against population size. Linear fit shown. Cutoff at 4 m

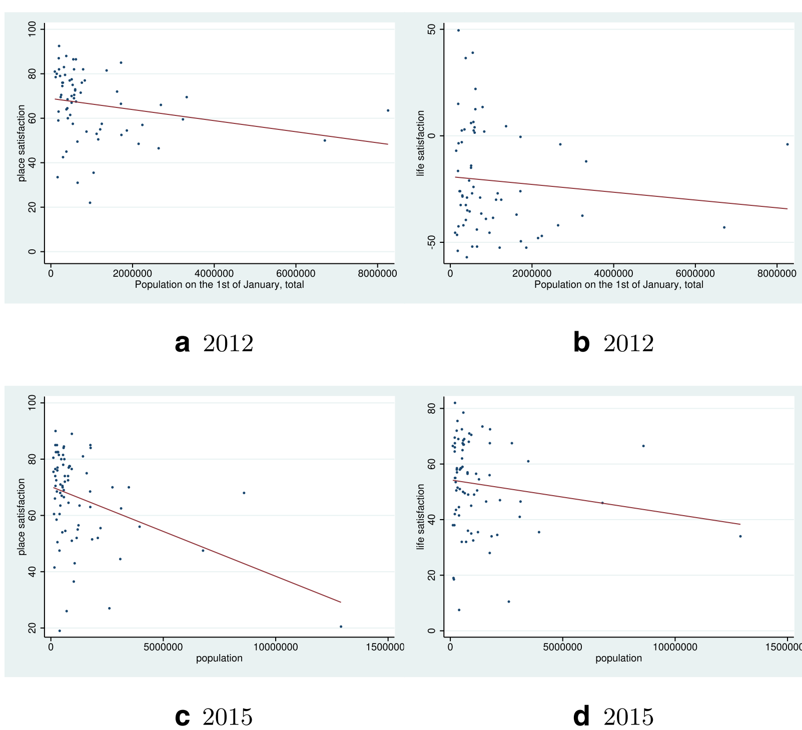


**Fig. 9** SWB against population size. Linear fit shown. All data shown

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4.3 Urban unhappiness is common (Okulicz-Kozaryn and Valente 2021)

Cities 118 (2021) 103368

Contents lists available at ScienceDirect

Cities

journal homepage: www.elsevier.com/locate/cities

# Urban unhappiness is common

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ARTICLE INFO

*Keywords:*
Happiness
Life satisfaction
Subjective wellbeing (SWB)
Cities
Urbanicity
Urban-rural
Urban-rural happiness gradient
World Values Survey (WVS)

ABSTRACT

This study shows, for the first time, that city unhappiness is common across the world. We use the World Values Survey cumulative dataset 1981–2020 from www.worldvaluessurvey.org. In all developed countries, without exception, we find that city dwellers are not happier than rural residents.

Research by Berry and Okulicz-Kozaryn (2011) provided preliminary evidence of an "urban-rural happiness gradient" in many countries, where happiness levels rise from lowest in the largest cities to highest in the smallest places. The gradient is non-linear—the very largest cities are markedly less happy than all other areas in a country, e.g., New York City (Okulicz-Kozaryn & Mazelis, 2018; Senior, 2006), London (Chatterji, 2013; Office for National Statistics, 2011), Helsinki (Morrison, 2015), Bucharest (Lenzi & Perucca, 2016), and Sydney (cited in Morrison, 2011). The goal of this paper is to test the gradient across countries using one dataset with a uniform set of measures. This study shows, for the first time, that city unhappiness is common across the world.

Most research on the urbanicity–happiness relationship points to an urban-rural happiness gradient, where happiness raises from its lowest level in the largest cities, to the highest in the smallest rural areas (e.g., Campbell et al., 1976; Okulicz-Kozaryn & Valente, 2020). Yet, most research has been conducted in the US or Western Europe, and there are only three cross-country investigations using a common dataset: Berry and Okulicz-Kozaryn (2009), Easterlin et al. (2010) and Burger et al. (2020).

There are multiple problems with the Gallup data used by Easterlin et al. (2010) and Burger et al. (2020). And Berry and Okulicz-Kozaryn (2011) have failed to analyze the data individually for each country. Hence, there is a literature gap and a need to test the urban-rural happiness gradient hypothesis across countries using an appropriate dataset. The present study is the first to report the urban–rural happiness gradient across countries showing the gradient for each country separately. We use a suitable and accurate dataset and control for relevant predictors of happiness.

## 1. Data and model

We use the World Values Survey cumulative file 1981-2020 from www.worldvaluessurvey.org, which is representative of about 90% of the world population,[1] and is much better suited for the study than an inadequate and poorly designed Gallup survey.[2]

The variables are listed in Table 1. Country codes and descriptive

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[1] While the WVS is conducted in about 100 countries that represent about 90% of the world population, due to missing data for the particular variables of interest, the present's study coverage is slightly smaller, covering about 70 countries (depending on the model and specification).

[2] Easterlin et al. (2010) acknowledge Gallup's limitations and attempts to address them. Burger et al. (2020), on the other hand, do not. First, the Gallup data is not meant for research but for commerce—Gallup charges $30,000 (per year) for data access (authors' inquiry). Second, the urbanicity classification is twice less precise than that in the World Values Survey (WVS) used in the present study: 4 versus 8 categories. Third, while the WVS uses precise population size with numeric cutoffs, Gallup uses fuzzy concepts such as "rural area," "small town or village," and "large city." Fourth, Gallup uses self-reports of urbanicity, which is highly subjective and problematic in this case—many, if not most people, would likely classify themselves completely arbitrarily into "rural area" vs "village" and so forth. The WVS uses interviewer's information about place of residence. Fifth, apparently much of the data are missing—Easterlin et al. (2010) notes that in 14 countries "rural area" responses were exceptionally low. About half of the world population is rural, but in Burger et al. (2020) only about a quarter of respondents are rural residents.

https://doi.org/10.1016/j.cities.2021.103368
Received 25 May 2021; Received in revised form 2 July 2021; Accepted 21 July 2021
Available online 17 August 2021

statistics are in the Supplementary Online Material (SOM).

Subjective wellbeing (SWB) is an umbrella term for various subjective measures of wellbeing, notably positive and negative effects, happiness, and life satisfaction. Most of the SWB research, including this study, uses the life satisfaction measure, which is a global self-evaluation of one's life as a whole. Life satisfaction is mostly cognitive and not affective—a respondent evaluates her life as a whole globally (professional, personal, family, community, etc.) (Diener, 2009). Following usual practice, for simplicity, we use these terms interchangeably: SWB, happiness, and life satisfaction, but we mostly refer to life satisfaction as defined above.[3]

The SWB question reads, “All things considered, how satisfied are you with your life as a whole these days? Using this card on which 1 means you are ‘completely dissatisfied’ and 10 means you are ‘completely satisfied’ where would you put your satisfaction with your life as a whole?”

Urbanicity is operationalized with the WVS variable “X049,” objective and recorded by the interviewer, not the respondent. There are eight categories ranging from ‘<2k’ to ‘>500k.’ This is an important advantage, because urbanicity or urbanness is a continuum, not a binary urban versus rural dichotomy. We conduct the analysis using a set of dummy variables for all eight categories (leaving out the base case) in the SOM. For simplicity and ease of exposition, however, we present simplified results in the body of the paper using three categories only. Thus, please refer to the SOM for the results of all categories.

Because in many countries, there are either no observations or few observations in the first two bottom categories, ‘<2k’ and ‘2-5k,’ we combined them together for the analyses in the main body of the paper. These two categories together proxy a city-free natural environment most closely resembling the natural human habitat where we have evolved, and it includes: wilderness, open country, and small villages. The other critical category that must be measured is large cities. There is likely to be a threshold at several hundred thousand, hence we use the top category on the WVS variable “X049,” which is ‘>500k,’ as a proxy of large cities. Such places least resemble natural human habitat and are mostly consisting of man–made objects such as asphalt, concrete, glass, etc., and accordingly are likely to be the least happy.

The classification into large cities versus natural areas produces a third category in between for places with a population size of 5-500k. The two cutoffs are driven by theory. It would be a gross oversimplification to use an urban–rural dichotomy with one cutoff, for example, ‘<100k’ vs. ‘>100k’ (or any other value). A place never changes abruptly from rural to urban at some cutoff point—it is a continuum—although it can be simplified to carefully chosen extreme categories, one must always start with the continuum. Since this aggregation or simplification into 3 categories is still somewhat arbitrary, we present alternative aggregations in the SOM in addition to the full 8-step urbanness gradient.

In the choice of controls we generally follow Okulicz-Kozaryn & Valente, 2020. Table 1 lists the control variables used in the body of the paper and there are specific controls worth discussing. Young, single, childless persons and young men with tertiary education are relatively more satisfied with urban areas as a place of residence (Carlsen & Leknes, 2019). Income, class, and education not only predict greater SWB, but are also confounded with and higher in cities.[4]

One great advantage of city life that is often forgotten is freedom (Park et al., 1925 1984), hence we control for freedom. Likewise, trust is important, as it predicts SWB, and it is lower in cities (Milgram, 1970). Health is a key predictor of SWB, and the subjective health measure used here is a reasonable measure of actual health (Subramanian et al., 2009).

We use a standard OLS regression with robust standard errors. We treat the 10-step happiness variable as continuous since research has shown that an ordinal variable can be treated as continuous (Ferrer-i-Carbonell & Frijters, 2004). It is important to underscore that OLS has become the default method in happiness research (Blanchflower & Oswald, 2011). Theoretically, while there is still debate about the cardinality of SWB, there are strong arguments to treat it as a cardinal variable (Ng, 1996, 1997).

## 2. Results and discussion

We only present results from one model for each country[5]–it includes all necessary and some additional controls (yet, not over-saturated where too many controls would result in collinearity and many missing observations). We use here models with controls listed in Table 1. The model presented here uses 3 urbanicity categories, ‘<5k (base)’, ‘5k-500k’, and ‘>500k’. Results are set in Table 2 and also visualized in Fig. 1. We are interested in the comparison between ‘<5k’ vs. ‘>500k’ because places larger than several hundred thousand are the most unnatural environment for humans as compared to natural environments, where our species has evolved.

The results in Table 2 show that in 80% of countries with significant happiness differences across urbanicity, people are less happy in cities than in smaller areas. The only exceptions are in the East European Post Soviet countries (ALB, ROU, RUS), and in South-Asian countries (BGD and IND). Notably, these are all poor or developing countries. In all developed countries, people are happier in smaller places than in large places—without exception, we find that city dwellers are not happier than rural residents.[6] This finding is important because it contradicts a

[3] The SWB measure is at least adequately reliable, valid, and considered acceptable for public policy making and public administration (Diener, 2009; Stiglitz et al., 2009). It is also used frequently in urban research (e.g., Chen et al., 2015; Ma, Dong, Chen, & Zhang, 2018; Moeinaddini et al., 2020; Mouratidis, 2018; Mouratidis, 2019; Valente & Berry, 2016; Wang et al., 2019; Wkeziak-Bialowolska, 2016). There are cross-cultural comparability caveats, however, and SWB may not be adequately comparable across countries (Diener, 2009; Kahneman et al., 1999). This limitation should be kept in mind when comparing results across countries in the present study. More focus should be on within-country differences, and this is what this study is mostly about—the difference between smaller and larger places in terms of SWB within different countries. We treat each country separately and do not pull the data together. In short, one should focus on within-country differences across urbanicity and exercise caution when comparing effects across countries.

[4] Simply comparing unadjusted means may result in oversimplified or biased research claiming that people are happier in cities (e.g., Burger et al., 2020)—there is confounding of urbanicity with higher income, education and class—see SOM for tables with and without controls.

[5] There is a tradeoff in this study between ease of presentation and elaboration as there are dozens of countries and presenting elaborated specifications would result in unwieldy presentation—additional specifications are in the SOM.

[6] In all developed countries studied here, AUS, CAN, DEU, ESP, ITA, NLD (only in SOM), NZL, SWE, and the USA, people in the largest areas have lower happiness levels than those in smaller areas. At least in less elaborate specifications shown in the SOM, but even in the most elaborate specifications, even when the coefficient on larger places is insignificant, it is still negative.The urban-rural gradient is the greatest in EGY, VEN, and VNM where the effect sizes are larger than 1, while the effect sizes for most other places are small to moderate, around 0.3 to 0.5 (on a 1–10 SWB scale).Yet, as indicated earlier, because of the limited cross-cultural comparability of the SWB measure, when interpreting our results, the focus should be on within–country SWB differences across urbanicity, and not on comparing cross–country effect sizes.It is worth noting that in the first column (5-500k), the majority of the results are negative with only 5 countries yielding a positive result: GHA, MDA, PER, RUS, and ZAF—reiteratively, what is remarkable is that none of these countries are considered developed.Note: the result for VEN should be interpreted with caution—this is the main difference with table exT4-3 in SOM and probably has to do with the fact that there are only 60 obs on the base case category. Other results are similar between the two tables.In the vast majority of countries, the results show a negative effect, and are only positive in East European Post Soviet (ALB, ROU, and RUS), and South-Asian countries (BGD and IND). East European Post Soviet countries are still quite centralized where power, opportunity, and resources are located in the large cities. India and Bangladesh are curious outliers (for some discussion see Deb, 2020).

**Table 1**
Variable definitions.

| Name | Description |
|---|---|
| Happiness | 'All things considered, how satisfied are you with your life as a whole these days?' 1='dissatisfied' to 10='satisfied' |
| Place size | 'OBSERVATIONS BY THE INTERVIEWER; Code size of town where interview was conducted' |
| Year survey | Year of survey |
| Age | Age |
| Age2 | Age squared |
| Male | Male |
| Married | Married or living together as married |
| Divorced/separated/ widowed | divorced/separated/widowed |
| Education | 'Highest educational level attained' |
| Income | 'Scale of incomes' |
| Class | 'Social class (subjective)' |
| Health | 'State of health (subjective)' |
| Postmaterialist | 'Post materialist index' |
| God important | 'How important is God in your life? Please use this scale to indicate- 10 means very important and 1 means not at all important.' |
| Religion important | 'For each of the following aspects, indicate how important it is in your life. Would you say it is: Religion' |
| Autonomy | 'Autonomy index' |
| Freedom | 'Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Please use this scale where 1 means 'none at all' and 10 means 'a great deal' to indicate how much freedom of choice and control you feel you have over the way your life turns out.' |
| Trust | 'Most people can be trusted' |

common belief that emerged recently, possibly due to ideological reasons (e.g., Burger et al., 2020; Glaeser, 2011; Glaeser et al., 2016), claiming that urban areas are happier. The effort to contravene the findings that cities tend to be less happy than smaller areas is arguably due to economics axioms: money is centered in cities (production, productivity, income, and consumption increase with population size), and therefore, cities have greater utility, so they must be happier. Yet, empirical evidence says otherwise.

Also note that in about a third or even half of the countries (depending on the model), there is no SWB difference across urbanicity. This is also a finding worth reporting as it runs counter to the common pro-urbanism and city triumphalism claim (e.g., Glaeser, 2011). One would think that cities are the best places to live as people flock there in doves. Thus, a finding showing no difference for many cases is already surprising.

Even though coefficient estimates are small to moderate, the practical significance of the results is very strong because of the sheer size of urbanization. Even a minuscule negative effect of 0.1 (on a scale of 1-10) on a large place versus a smaller place for a small country of 10 million people translates into an effect equivalent of making 100 thousand people go from the most miserable to the most happy level on the SWB scale of 1-10. Globally, for billions of people living in cities, there is a massive amount of human misery produced.

Why are people less happy in large cities in the developed world, yet happier in some developing countries? There is at least one reason. In many developing countries, life is simply unbearable outside of the city lacking basic necessities such as shelter, food, water, sanitation, and healthcare. In developed countries, even the smallest places have reasonable access to necessities, and small places do not suffer from urban disamenities. As per Maslow's pyramid of needs (Maslow, [1954] 1987), survival and opportunity come first, and this arguably can explain much of the paradox found in this paper—despite the city being biologically, neurologically, and socially negative for humans (Lederbogen et al., 2011; Simmel, 1903; Wirth, 1938), cities can be useful for human wellbeing at the early stage of a country's economic development.

**Table 2**
OLS regressions of SWB on place size for each country separately controlling for predictors of SWB listed in Table 1.

| | 5-500k | 500k- | N |
|---|---|---|---|
| ALB | -0.4* | 0.4+ | 1582 |
| ARG | -0.2 | -0.0 | 855 |
| AUS | -0.0 | -0.1 | 3728 |
| AZE | -0.1 | 0.3 | 964 |
| BFA | 0.3 | 0.0 | 567 |
| BGD | 0.0 | 0.7* | 2104 |
| BGR | -0.0 | -0.5* | 1229 |
| BLR | -0.1 | -0.1 | 2815 |
| BRA | -0.2 | -0.4* | 3576 |
| CAN | -0.1+ | -0.3* | 3177 |
| CHL | -0.7* | -0.7* | 3527 |
| CHN | 0.0 | -0.4* | 2005 |
| COL | 0.0 | -0.1 | 1376 |
| DEU | -0.1 | 0.0 | 4795 |
| DZA | -0.4* | -0.6 | 1596 |
| ECU | -0.9* | -0.7* | 1182 |
| EGY | -0.4* | -1.1* | 3428 |
| ESP | -0.1 | -0.1 | 1487 |
| ETH | 0.3 | 0.4 | 1017 |
| GEO | 0.1 | 0.1 | 2401 |
| GHA | 0.3* | -0.0 | 2572 |
| HUN | 0.0 | -0.4* | 887 |
| IDN | 0.1 | -0.0 | 2056 |
| IND | -0.0 | 0.3* | 5857 |
| IRN | -0.3* | -0.0 | 2119 |
| IRQ | -0.1 | -0.0 | 1123 |
| ITA | -0.1 | 0.2 | 585 |
| JOR | 0.1 | -0.2 | 2089 |
| KAZ | -0.0 | -0.3* | 1497 |
| KGZ | -0.1 | -0.3* | 2293 |
| LBN | 0.1 | 0.2 | 731 |
| LTU | 0.3 | 0.3 | 750 |
| LVA | -0.1 | -0.6* | 963 |
| MAR | 0.0 | -0.2 | 845 |
| MDA | 0.2* | 0.2 | 2478 |
| MEX | -0.1 | -0.2+ | 3544 |
| MKD | -0.2 | -0.1 | 1385 |
| MYS | 0.1 | -0.4* | 1541 |
| NGA | -0.1 | -0.1 | 4488 |
| NZL | -0.1 | | 417 |
| PAK | 0.4+ | 0.3 | 900 |
| PER | 0.3* | -0.5 | 1026 |
| PHL | 0.4 | 0.5 | 2294 |
| POL | -0.1 | -0.1 | 1533 |
| ROU | -0.2* | 0.3* | 3568 |
| RUS | 0.2* | 0.2* | 3253 |
| RWA | -0.7* | -0.4+ | 2398 |
| SRB | 0.1 | -0.4* | 2539 |
| SVN | 0.2+ | -0.2 | 1620 |
| SWE | 0.2 | 0.2 | 1769 |
| THA | 0.1 | 0.1 | 2178 |
| TUN | 0.1 | | 826 |
| UKR | 0.0 | -0.1 | 2985 |
| URY | 0.2 | 0.1 | 2017 |
| USA | -0.1 | -0.2* | 3372 |
| UZB | 0.0 | -0.3* | 1247 |
| VEN | -1.7* | -1.2* | 1034 |
| VNM | 0.1 | -1.5* | 2039 |
| ZAF | 0.2* | 0.0 | 5330 |
| ZWE | 0.1 | 0.2 | 1487 |

* $p < 0.05$.
+ $p < 0.1$; robust std. err.

The most important policy implication from these results is that while low-density non-urban living for most people is not viable in the short run (e.g., Meyer, 2013), more consideration should be given to smaller areas that have been left behind, as lamented by some (e.g., Atkins & Allred, 2021; Fuller, 2017; Hanson, 2015), but not heard by most. Redirecting resources away from smaller places should be given more thought and consideration. A bolder policy proposal would be simply to promote rural living through tax breaks or subsidies. Yet, we


**Fig. 1.** Coeficients from Table 2 on cities larger than 500,000. Only significant results shown (white indicates no significant results). Note that the map is zoomed to show the detail and any cut off countries have no significant results.

hesitate to go that far in our recommendations here—a thorough discussion of pros and cons of various policy implications is beyond the scope of this brief communications paper.

## CRediT authorship contribution statement

Adam Okulicz-Kozaryn, Rubia Valente: Conceptualization, Methodology, Writing - Review & Editing.

Adam Okulicz-Kozaryn: Software, Formal analysis, Data Curation, Writing - Original Draft, Visualization, Investigation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cities.2021.103368.

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## 4.4 Subjective well-being and urbanization in Egypt (Mikhaeil et al. 2024)

Cities 147 (2024) 104804

Contents lists available at ScienceDirect

Cities

journal homepage: www.elsevier.com/locate/cities

# Subjective well-being and urbanization in Egypt

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ARTICLE INFO

*Keywords:*
Subjective well-being
Life satisfaction
Happiness
Urbanization
Rural
Urban
Egypt
Cairo

ABSTRACT

World Values Survey data is used to test the association between urbanization, operationalized as population size, and subjective well-being, operationalized as life satisfaction, in the case of Egypt. Regression results show that the smallest settlements have the highest positive effect on subjective well-being compared to the large urban centers. These results are persistent even after controlling for an extensive set of socio-demographic variables. Another important finding is that the two main urban and economic centers of the country, the Greater Cairo and Alexandria regions, generate lower subjective well-being when compared to the more rural regions (Lower and Upper Egypt). These results are both unexpected and compelling, especially in the context of a developing country such as Egypt—previous research argues that cities should generate higher subjective well-being when compared to rural and township settlements in such a context. Our empirical findings show otherwise and provide a novel and crucial contribution to the literature on the subjective wellbeing- urbanization nexus.

## 1. Introduction

Developing countries have experienced a dramatic increase in urban growth in the past few decades. For instance, the percentage of urban population increased from 37.5 % in 1995 to 49 % in 2015 with >75 % of the world's total urban population living in developing countries since 2015 (Smit, 2021). In addition, most of the urban growth in the future is going to happen in developing countries (Cohen, 2004, 2006; Smit, 2021). Out of the 793 million people that were expected to be added to the global urban population between 2015 and 2025, about 745 million, i.e., 93 % of the increase, will be registered in developing countries (Smit, 2021; see UNDESA, 2019). Currently, 27 out of the 33 megacities in the world are in developing countries (Smit, 2021). Egypt is one of those developing countries that have experienced significant changes in its demographic and urbanization trajectories. With its current population of 110 million people (see World Bank, 2022), Egypt is the most populous country in the Middle East and North Africa (MENA) region and the third most populous country in Africa. Egypt's population has almost tripled in half a century from 35 million in 1970 to 110 million in 2022 (World Bank, 2022). The urban population has almost doubled in three decades as it increased from 24.4 million in 1990 to 43.7 million in 2021 (Hatab et al., 2022). About 42 % of Egypt's total population lived in urban areas in 2013 compared to 37 % in 2006 (Khalil, 2021). But figures regarding urban population in Egypt as well as other developing countries should be treated with caution (Cohen, 2004). There is an apparent underestimation of the urban population in Egypt given how the Egyptian government classifies urban versus rural areas based on the administrative role of a settlement, and not the size of its population (Bayat & Denis, 2000; Sims, 2010; Zinkina & Korotayev, 2013). The population of many Egyptian villages has exceeded 10,000 people and they are still administratively and statistically considered villages and not small towns (Bayat & Denis, 2000; Zinkina & Korotayev, 2013). Interestingly, the UN data shows that the percentage of Egyptian population residing in urban areas has only increased from 41.5 % in 1970 to 42.8 % in 2015 (UNDESA, 2018). This seemingly flat trend of urbanization reflects the fact that international organizations count on national classifications of what is urban and what is rural (Cohen, 2004). In fact, in one of the publications of the United Nations, it is explicitly mentioned that the seemingly flat trend of urbanization in Egypt is "a reflection of the official definition of cities not accounting for the recent urbanization in rural settlements" (UNDESA, 2018, p.2). There are different practices of how to classify what is rural and what is urban in developing countries. For instance, in Benin, only settlements with >10,000 residents are considered urban (Cohen, 2006). On the other

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https://doi.org/10.1016/j.cities.2024.104804
Received 4 June 2023; Received in revised form 2 January 2024; Accepted 10 January 2024
Available online 27 January 2024

hand, in countries such as Angola, Ethiopia, and Argentina, all settlements with 2000 residents or more are classified as urban (Cohen, 2006). So, settlements that could be classified as urban in some countries could be considered rural in other contexts (Cohen, 2004, 2006). Hence, the classification of what is urban and what is rural could be a little bit blurred in different contexts based on the definition used and this makes the comparative study of urbanization in the context of developing countries a bit challenging (Cohen, 2004, 2006; Smit, 2021). Hence, the distinction between what is rural and what is urban in this context should be regarded as a continuum and not a dichotomy to reflect the urban-rural linkages (Cohen, 2006; Davis, 2004; Smit, 2021).

When it comes to the study of urbanization in developing countries, it is important to pay special attention to the capital cities where a large proportion of the national population usually reside and a large share of economic activities is located (Cohen, 2006). Hence, in the case of Egypt, the study of urbanization would be meaningless without zooming in on Cairo. The Greater Cairo Region (GCR), which delimits Egypt's administrative capital, "Cairo", houses about a quarter of the country's total population. The GCR is considered the biggest city in Africa (Fahmi & Sutton, 2008; Sims, 2010). In fact, in the year 2018, the GCR has been classified as the 6th largest urban agglomeration in the world ahead of other urban agglomerations such as Mumbai, Dhaka, and Beijing (Smit, 2021). About 56 % of Egypt's urban population is taking the GCR and Alexandria as their place of residence (Khalil, 2021). This over-dominance of the Greater Cairo Metropolis and Alexandria over the urban scene in Egypt is not new. In fact, it was one of the main features of the rapid urbanization process in Egypt in the 20th century (Abu-Lughod, 1965; Ibrahim, 1975). Such a phenomenon is still evident today (Khalil, 2021). Like many other capital cities in the global South[1] (see Cohen, 2006), Cairo has been described as the primate that dominated Egypt's urbanization and economic development (Abu-Lughod, 1965; Ibrahim, 1975; Sims, 2010). It is considered one of the largest and most crowded cities in the world, and still continues to grow steadily with a 2 % growth rate per year (Jaad & Abdelghany, 2021). Despite the important role of natural population growth, still internal migration continues to play an important role in driving this expansive urban growth in Cairo (David et al., 2019; Hatab et al., 2022). In fact, natural population growth and internal migration are the main drivers of urban growth in most cities of the global South in general (Cohen, 2006).

Like many other mega-cities in the global South, Cairo is not just the capital of the country, but it also acts as the main economic and financial center and a main hub of education, transportation, and commerce (El Araby, 2002). Still, many of Egypt's social and economic problems are concentrated within the extended limits of Cairo (Sutton & Fahmi, 2001). The uncontrolled and unsustainable growth of the city places significant pressure on the already dilapidated infrastructure and the suboptimal social services in a way that is stressing the overall wellbeing of its residents (Jaad & Abdelghany, 2021). Egyptian urban centers, including Cairo, have been neglected and the policies adopted by the Egyptian government have exacerbated their precarious situation. Despite the deterioration of existing public services and an increased demand for them, the government has directed scarce public funds to the construction of satellite cities and a new administrative capital in the hinterland of Cairo. Egyptian urban centers are not unique in terms of the resource constraints they are facing. Many cities in the global South have insufficient resources- a situation that limits their ability to provide a decent life to their residents (Cohen, 2006; Mabin, 2014). Urban growth in the global South has been "decoupled from industrialization, even development per se", i.e., urbanization without growth (Davis, 2004, p.9; see also Fox, 2012). Hence, the unsound public policies, especially urban planning policies, that divert scarce resources can exacerbate the problems found in cities of the global South including Egyptian cities.

Such enormous urban growth in the global South has important repercussions when it comes to the material and psychological well-being of people. Cities could be treated as a double-edged weapon where the cost of the material benefits provided by them might be the unhappiness or dissatisfaction of urbanites (Berry & Okulicz-Kozaryn, 2009, 2011; Fischer, 1973). Big cities provide a host of opportunities and amenities. But cities are also full of disamenities, including high inequality, longer commutes, pollution, noise, and higher rates of crime. Hence, the net effect of city life on Subjective well-being (SWB) is basically an empirical question. There is an increasing interest in the study of SWB (Ferrer-i-Carbonell & Frijters, 2004; Moro-Egido et al., 2022) to the extent that academicians, policy makers, and international organizations are proposing to use SWB measures as complements to the conventional objective measures in informing public policy deliberations and in assessing people's welfare (Diener et al., 2009; Diener et al., 2013; Stiglitz et al., 2009). A special attention in the SWB literature is paid to its determinants–one receiving increasing interest recently is the environmental milieu in which people live and whether place of residency matters (e.g., rural vs urban areas). However, most research has focused on the global North and there is a dearth of research on the SWB-urbanization nexus in the global South (Burger et al., 2020).

The lack of SWB research, especially on the SWB-Urbanization nexus, is evident in the case of Egypt. Most of the urbanization literature on Egypt focuses on issues such as urban informality (e.g., Bayat & Denis, 2000; Khalil, 2021; Sharp, 2022), the housing problem in big urban centers (e.g., Fahmi & Sutton, 2008; Ibrahim, 2017), failed urban planning schemes that aim at reducing the population density of large urban centers (e.g., Khalil, 2021; Sims, 2010; Sutton & Fahmi, 2001; Tarbush, 2012), or the impact of urban growth on the environment (e.g., El Araby, 2002; Robaa, 2011). Still, no attention has been given to the effect of urbanization and all its related problems in the Egyptian context on the perceived SWB. This is surprising considering the large population of Egypt and the fact that it is the most populous country in the MENA region and the third most populous country in Africa and has one of the top 10 largest urban agglomerations in the world. This paper aims to fill this gap by using the World Values Survey (WVS) data to test the direction and magnitude of the association between urbanization, operationalized as population size, and SWB, operationalized as life satisfaction. In the extant scholarship, it is argued that rural areas in developing countries have fewer resources and may fail to satisfy human needs; hence, cities are hypothesized to generate more SWB compared to rural areas (Veenhoven, 1994; Veenhoven & Ehrhardt, 1995). Our aim is to investigate whether the Egyptian case would fit such a hypothesis. Thus, this paper investigates the urban-rural SWB gradient in Egypt and contributes to the literature that explores urbanization-SWB relationship in developing countries. The importance and necessity of this study stem from its contribution to the scant and mixed evidence available on the relationship between SWB and urbanization in developing countries. Compared to the profusion of research on the urbanization-SWB relationship in the context of Western countries, there is a dearth of evidence on this relationship in developing countries (Burger et al., 2020). Hence, this study contributes to the literature by studying the case of a big developing country. What adds to the value of this study is that it is the first study of the urbanization-SWB relationship in Egypt according to the best of our knowledge. The study heeds to Burger et al. (2020)'s call for more case-study research on the topic to avoid the cultural and other contextual factors that complicate cross-country analyses of SWB. Also, the case study approach avoids the messiness of comparative analysis of urbanization given the different definitions of urban employed in different contexts (see Cohen, 2004, 2006).

The paper is organized as follows. First, we review the literature on SWB, urbanization, and the association between these two variables; then, we discuss the data and methods used in our analysis, followed by

[1] The term global South refers to nations of the world which are regarded as having a relatively low level of economic and industrial development and the term is used as synonymous to the developing countries term – a practice we adopt in this paper (see Smit, 2021).

a presentation of the results. Finally, we discuss the results, offer take-aways for practice, and draw together the main conclusions.

## 2. Subjective well-being

The interest in studying SWB has been on the rise during the past few decades (Ferrer-i-Carbonell & Frijters, 2004; Moro-Egido et al., 2022). Diener et al. (1985) show that there are two dimensions of the SWB construct: affective, either positive or negative, and cognitive. So, the term is usually used to refer to people's affective and cognitive evaluation of their lives (Diener, 1994). Two main measures of SWB are used in surveys: happiness and life satisfaction (Helliwell & Putnam, 2004). The happiness measure tends to reflect the affective dimension of the SWB construct while the evaluative cognitive dimension is captured by the life satisfaction measure (Campbell et al., 1976; Helliwell & Putnam, 2004). Usually, the terms SWB, life satisfaction, and happiness are used interchangeably in the literature (Easterlin et al., 2011; Knight & Gunatilaka, 2010; Piper, 2015).

As could be expected, the validity and reliability of SWB measures are occasionally questioned, especially that conventional objective measures (e.g., income), have been used in making inferences about well-being for a long time (Helliwell, 2006). But SWB self-reports are found to be valid, reliable, and sensitive to changes in quality of life (Diener, 1994; Diener et al., 2009; Diener et al., 2013; Veenhoven, 1996). Increasingly, subjective measures of well-being are gaining credence and are proposed to be good complements to the existing objective measures, used in assessing people's welfare, and in informing public policy deliberations (Di Tella & MacCulloch, 2006; Diener et al., 2009; Diener et al., 2013; Helliwell, 2006; Layard, 2010; Morrison, 2014; Stiglitz et al., 2009).

Much of the literature on the topic has been concerned with exploring the different determinants of SWB (for a comprehensive review, see Blanchflower & Oswald, 2011; Dolan et al., 2008; Helliwell & Putnam, 2004; Helliwell, 2006). Interest is specifically directed to those determinants defined by people's economic and social circumstances and not just the determinants that are more related to personality traits (Helliwell & Putnam, 2004). One of these factors receiving rising attention recently is urbanization. The question is usually whether there is an effect of the type and size of the human settlement in which people live on their SWB.

## 3. Urbanization

The rural setting remained the primary habitat for most of the world population for hundreds of years before the industrial revolution (Veenhoven, 1994). But this pattern was reversed by the fast-paced urbanization that was one of the characteristics of modern life. Urbanization as a concept could have different definitions but generally it means the massing of people in cities (Anderson, 1959). This definition is mainly concerned with the distribution of the population across human settlements of different sizes, where urbanization is defined as the percentage of the total population living in urban-size settlements (Abu-Lughod & Hay, 2007; Davis, [1965] 2016). But away from the limited focus on size or density variables to define what is urban, urbanization has implications that are usually related to changes in lifestyles, changes in social and economic relations, and changes in governance that stem from the need to deal with the repercussions of urbanization (Smit, 2021). One of the significant implications of urbanization has been the effect it had on the form of social organization within which humans interact with each other. As part of the urbanization process, people moved from a social setting where traditional social bonds were the center of social life and landed in a new social world where the money nexus controls most aspects of life, including social relations. Tönnies ([1887] 2001) shed light on such transformation in his discussion of the two ideal types of social organization, namely, Gemeinschaft (community) and Gesellschaft (society). In Gemeinschaft, human relations are predominantly personal face-to-face relationships based on natural will and regulated by traditional social bonds, including family, religion, and kinship. A setting that is believed to provide networks of social, and even economic, support that cannot be found in Gesellschaft, where human relations are more impersonal and indirect. Still, it is argued that communal life and warm face-to-face and direct communication and interaction do exist inside the city, but they take forms that could be different from those found in small settlements (Karp et al., 2015).

In England, which was the cradle of this social change, the introduction of machinery, and the rise of the factory system made the work of many village laborers and farmers redundant and forced them to leave their villages to search for livelihoods in newly erected industrial towns (Davis, [1965] 2016; Engels, [1845] 2001; Karp et al., 2015). So, it was in the first industrializing countries where economic development caused rural-urban migration to be the primary factor of the growth of cities (Davis, [1965] 2016). The situation is not very different in the late industrializing countries in the global South. Despite the important role of natural population growth in the urbanization process and the growth of cities (Abu-Lughod, 1961; Cohen, 2006; Davis, [1965] 2016), rural-urban migration was still the most crucial factor in driving the 20th-century urbanization in the global South (Abu-Lughod, 1961; Cohen, 2006).

Classical social theorists such as Marx and Weber believed that the trajectory of the cities of the future would be like that of the great early industrial cities such as Manchester, Chicago, or Berlin (Davis, 2004). But are the trajectory and implications of the urbanization process in the North and the South the same? The answer to this question is yes and no (see Davis, 2004; Mabin, 2014). Some cities in the global South resembled the trajectory projected by classical social theorists but others did not. For instance, although the Northern urban slums of the 19th century and early 20th century were the result of industrialization, the Southern urban slums in many big cities of the global South were actually the result of deindustrialization and the application of neoliberal policies especially in the agricultural sector that pushed the surplus rural labor force to urban centers despite the inability of cities to provide jobs for them (see Davis, 2004). Cities in different parts of the global South, especially in Africa, ceased to be engines of growth and prosperity and shifted into dumping sites for surplus population working in low-wage and unskilled informal sectors (Davis, 2004). Thus, despite the important role played by the rural-urban migration in forming the Southern and Northern cities alike, the main drivers and trajectory of this migration were a bit different in both cases.

With time, cities turned out to be the nucleus of modern life and the focus of scholarly debate on their advantages and disadvantages. On the macro level, cities facilitated, through density and proximity, the exchange of goods, services, and ideas, housed hundreds of thousands of people, and made labor specialization possible (Glaeser, 2011). Urbanization brought economic prosperity, and cities were considered the main engines of economic growth and innovation in both the global South and the global North alike (Cohen, 2006; Glaeser, 2011; Sampson, 2019). On the individual level, cities provide freedom from traditional social ties, found in small contexts (Milgram, 1970), and an abundance of choices and options (Milgram, 1970; Park, 2019). Cities also provide privacy that cannot be found in small settlements (Jacobs, 1961). Howard (1965) used one word, to sum up the causes that led people to aggregate in cities: attractions. But scholars, especially classical urban scholars (e.g., Simmel, [1903] 2013; Tönnies, [1887] 2001; Wirth, 1938), argue that city life has its own disadvantages. Big size, highly dense, and heterogeneous places were seen to be fertile environments for social problems, including "personal disorganization, mental breakdown, suicide, delinquency, crime, corruption, and disorder" (Wirth, 1938, p.24). Also, on the psychological level, city life overloads individuals' psychic capacity in an unbearable way (Milgram, 1970; Simmel, [1903] 2013). So, although cities were conceived as havens of freedom away from the repressions of the countryside, cities are also believed to be sites of anxiety and anomie (Harvey, 2000).

## 4. The association between urbanization and SWB: theoretical arguments and empirical findings

It could be argued that the city is sometimes conceived as a double-edged weapon. The cost of the opportunities provided by cities, especially in the developed countries context is the relative unhappiness or dissatisfaction of urbanites (Berry & Okulicz-Kozaryn, 2009; Berry & Okulicz-Kozaryn, 2011; Okulicz-Kozaryn & Mazelis, 2018) or what Fischer (1973:233) called “the emotional price for their economic well-being.” Big cities are full of opportunities (e.g., jobs and education) and amenities but cities are also full of disamenities (e.g., pollution, high cost of living, noise, longer commutes, and high inequality) (Burger et al., 2020; Cohen, 2006). In the global South context specifically, many “cities exhibit extremely high levels of inequality and are teetering on the edge of becoming uninhabitable” (Schindler, 2017, p.58–59).

It is generally hypothesized that urban-rural differential in SWB would be high at low levels of economic development and that this differential tends to decrease and even reverse at higher levels of economic development (Burger et al., 2020; Easterlin et al., 2011; Requena, 2016; Veenhoven, 1994; Veenhoven & Ehrhardt, 1995). This hypothesis was supported by a number of empirical cross-country investigations (e.g., Easterlin et al., 2011; Requena, 2016). For a group of 29 European countries, Requena (2016) reported that compared to other categories on the rural-urban continuum, big cities generated less SWB in developed countries but generated the highest level of SWB in less developed countries. Easterlin et al. (2011) reported a similar result for the group of 80 countries included in their investigation.

In the context of developed countries, the argument is that there is not much difference in living conditions between rural and urban areas. In addition, due to easy accessibility to cities, through transportation infrastructure, residents of the rural and peri-urban areas in developed countries are in a position that allows them to borrow the positive effects of cities but also avoid the disamenities of city life (Lenzi & Perucca, 2018). Because of that, rural and township areas are hypothesized to generate higher levels of SWB when compared to cities. This hypothesis has been intensively investigated in the context of developed countries. Compared to life in smaller settlements, city life was found to be associated with lower levels of life satisfaction in a number of cross-country studies (Berry & Okulicz-Kozaryn, 2009; Piper, 2015; Sørensen, 2014). In addition, this result has been confirmed in many country studies including the US (Berry & Okulicz-Kozaryn, 2009, 2011; Campbell et al., 1976; Okulicz-Kozaryn & Mazelis, 2018; Sander, 2011), New Zealand (Morrison, 2007, 2011), Italy (Lenzi & Perucca, 2019), Denmark (Sørensen, 2021), and United Kingdom (Dunlop et al., 2016; Hoogerbrugge & Burger, 2019).

In the context of developing countries, the argument goes that urban areas provide more public amenities and life opportunities compared to rural and township areas (Veenhoven, 1994; Veenhoven & Ehrhardt, 1995). Hence, cities are hypothesized to generate higher levels of SWB when compared to rural and township areas in developing countries. Compared to the profusion of literature on the urbanization-SWB relationship in the context of developed countries, there is a dearth of evidence on the effect of urbanization on SWB in developing countries (Burger et al., 2020). In addition, this scant research provides mixed results. For instance, place of residence was found not to be one of the determinants of life satisfaction in developing countries in a number of cross-country studies (Berry & Okulicz-Kozaryn, 2009; Valente & Berry, 2016). On the other hand, other cross-country studies reported positive urban-rural differential in SWB for the group of developing countries included in those studies (e.g., Burger et al., 2020; Easterlin et al., 2011). The same pattern of mixed results is also found in case studies. In the Indian context, rural dwellers reported lower SWB compared to urbanites (Deb & Okulicz-Kozaryn, 2023). On the other hand, in their investigation of the Chinese context, Knight and Gunatilaka (2010) reported rural-urban differences in SWB in favor of rural dwellers despite the gap in material conditions of life between the two groups.

This study contributes to this scant literature by focusing on a large developing country in the MENA region and the African continent. The question now is whether the association between urbanization and SWB in a developing country such as Egypt will support the hypothesis made in literature that urban areas generate higher SWB when compared to rural and town areas.

## 5. Data and methods

We use the World Values Survey (WVS) pooled cross-sectional dataset for Egypt to test the association between urbanization and SWB. There are seven waves of the survey, and it was conducted in Egypt in four of them (2001, 2008, 2013, and 2018). The sample used in all the survey waves is representative of the entire adult population in Egypt. The survey uses a multi-stage sampling procedure. Face-to-face interviews were conducted to obtain the data. The interviews were conducted using a questionnaire that was translated into the Arabic language.

The treatment of the SWB dependent variable differs across empirical studies. Some studies treat it as a cardinal variable, and hence the ordinary least squares (OLS) method is used in the analysis, while other studies treat it as an ordinal variable through the usage of ordered logit or ordered probit models (Dolan et al., 2008). We use the OLS method in our analysis as it is becoming the default method in SWB research, and its results do not differ from the case in which the SWB variable is treated as ordinal (Blanchflower & Oswald, 2011; Easterlin et al., 2011; Ferrer-i-Carbonell & Frijters, 2004; Graham, 2008; Knight et al., 2009; Sørensen, 2021). We also fit the data using an Ordered Probit technique, and the results are reported in Appendix B. In effect, the results of our Ordered Probit model show high similarity with the results of the OLS method. Our dependent variable is the life satisfaction measure of SWB. The life satisfaction question reads, “All things considered, how satisfied are you with your life as a whole these days? Using this card on which 1 means you are ‘completely dissatisfied’ and 10 means you are ‘completely satisfied’ where would you put your satisfaction with your life as a whole?”

The size of the place question (i.e., our main independent variable) is missing in one of the four waves: 2013. Hence, this wave was removed from our analysis. The place size variable is a categorical variable representing a continuum of urbanization with eight categories. Unlike in other surveys, the place size question in the WVS is answered by the interviewer herself and not the respondent. So, it is an objective measure of the degree of urbanization.[2] This is different from other surveys in which respondents provide self-reported answers to the type of settlement they live in, i.e., a perceived measure of urbanization (see Weckroth & Kemppainen, 2021).

We control for both personal characteristics and socio-economic variables that were found to be robustly correlated with SWB in the literature (see Blanchflower & Oswald, 2011; Dolan et al., 2008; Helliwell, 2006; Helliwell & Putnam, 2004). We also include two-year dummies as part of our equation to control for any time-related effects. We add a quadratic age term to our models as the relationship between SWB and age is found to be taking a U-shape in which people reach their minimum level of SWB in their middle age (Blanchflower & Oswald, 2008). A detailed description of all the variables used in our analysis is provided in Appendix A. The descriptive statistics of all the variables are presented in Table 1. The data of the three waves for which the dependent and independent variables are available are pooled to increase the sample size.

[2] According to the WVS survey section titled “Observations by the interviewer” which included the question on the size of the place, the technical codes of the questions in this section “are to be coded by the interviewer or fieldwork supervisor (not the respondent!) and must reflect actual, objective information.”

**Table 1**
Descriptive statistics.

| Variable | Obs. | Mean | SD | Min/max |
|---|---|---|---|---|
| Life satisfaction | 7245 | 5.61 | 2.93 | 1/10 |
| Income | 6839 | 4.68 | 2.17 | 1/10 |
| Age | 7247 | 39.5 | 14.58 | 16/90 |
| Health status | 7251 | 3.73 | 0.83 | 1/5 |
| Freedom | 7205 | 5.85 | 2.86 | 1/10 |
| Importance of religion | 7250 | 3.96 | 0.2 | 1/4 |
| Importance of God | 7245 | 9.79 | 0.81 | 1/10 |

| Variable | Categories | Frequency |
|---|---|---|
| Marital status | Married | 5181 |
| | Divorced | 94 |
| | Separated | 45 |
| | Widowed | 655 |
| | Single/never married | 1276 |
| Gender | Female | 3922 |
| | Male | 3329 |
| Subjective social class | Lower class | 1412 |
| | Working class | 1449 |
| | Lower middle class | 3065 |
| | Upper middle class | 1166 |
| | Upper class | 53 |
| Educational level | Lower level | 3385 |
| | Middle level | 2345 |
| | Upper level | 1519 |
| Employment status | Full time | 2410 |
| | Part time | 243 |
| | Self-employed | 372 |
| | Retired | 398 |
| | Housewife | 2905 |
| | Student | 309 |
| | Unemployed | 474 |
| | Other | 139 |
| Trust | Trusting | 1773 |
| | Not-trusting | 5434 |

## 6. Results

Table 2 shows summary statistics of the life satisfaction measure for the different categories of urbanization used in the analysis. The eighth category of the population size variable has the lowest average life satisfaction while the highest average life satisfaction is in the second category, followed by the sixth and first categories, respectively. The table shows that the first two categories of the population size variable (i.e., rural and village settlements) have higher average life satisfaction compared to the 7th and 8th categories (i.e., big cities). Also, except for the 6th category of urbanization, the first and second categories have the highest average life satisfaction compared to all other categories of urbanization. The question now is whether such differences in average life satisfaction across urbanization categories could be explained by the group of socio-economic variables that are usually controlled for in SWB multivariate analyses.

Table 3 shows OLS regression models of the life satisfaction measure on the different categories of the urbanization variable and control variables. In our bivariate model, model 1, we regress life satisfaction on our independent variable of interest. We keep adding control variables in models 2, 3, and 4 while still having our key independent variable in all three models. In model 2, we add the demographic group of variables. In model 3, the socio-economic group of variables is added. Finally, in model 4, we add the group of variables that relate to cultural and social capital aspects. In all models, the 8th urbanization category (>500,000) is our reference category for our urbanization variable.

**Table 2**
Summary statistics of life satisfaction by categories of urbanization.

| Population size | Summary of life satisfaction | | |
|---|---|---|---|
| | Mean | SD | Frequency |
| Category 1: <2000 | 5.81 | 2.86 | 119 |
| Category 2: 2000–5000 | 6.46 | 2.67 | 361 |
| Category 3: 5000–10,000 | 5.55 | 3.06 | 1463 |
| Category 4: 10,000–20,000 | 5.63 | 2.95 | 1142 |
| Category 5: 20,000–50,000 | 5.49 | 3.05 | 1147 |
| Category 6: 50,000–100,000 | 5.85 | 2.86 | 504 |
| Category 7: 100,000–500,000 | 5.52 | 2.45 | 313 |
| Category 8: >500,000 | 5.29 | 3.08 | 1071 |
| Total | 5.59 | 2.98 | 6120 |

The results of our bivariate model show that except for categories 1, 5, and 7, the residence in all other settlement categories would significantly positively affect life satisfaction compared to living in the 8th urbanization category. In our full specification, model 4, all urbanization categories, except for the 7th category (100,000–500,000), show a significant positive effect on life satisfaction compared to the reference category. The effect size is the greatest for category 2, followed by category 1 and category 6, respectively. The effect of living in category 2 settlements on life satisfaction is about three times the effect of the health status variable. The effect is equivalent to moving three steps up on the health scale from very poor to fair or from poor to very good. The sign of other significant control variables is congruent with previous empirical evidence in SWB literature.[3]

The rural-urban classifications are not standard in the literature. Some scholars would treat urbanization as a dichotomous variable (rural versus urban) while other scholars operationalize urbanization as a continuum or a gradient (see Sørensen, 2014, 2021). Furthermore, some studies would use a combination of rural-urban classifications to check the robustness of their results. For instance, in their investigation of rural-urban SWB differential in the Indian context, Deb and Okulicz-Kozaryn (2023) defined urbanization as both a dichotomy and a continuum (using the 8 categories of place size in the WVS). Sørensen (2014) recoded the 8 categories of the urbanization variable in the European Values Study to construct three categories (rural areas, town areas, and city areas) urbanization variable in a number of different ways through changing the cut-off values of the three categories. So, we test the robustness of our results by using other rural-urban classifications. First, we construct a dichotomous urbanization variable by treating places with <10,000 residents as rural and places with >10,000 residents as urban. Second, we construct a 3 categories urbanization variable. The three scales are as follows: (1) rural areas = places with fewer than 10,000 inhabitants; (2) town areas = places with 10,000–100,000 inhabitants; and (3) city areas = places with >100,000 inhabitants. The results of these robustness checks are shown in Tables 4 and 5 respectively. The two robustness checks show that our previous results are robust. As Table 4 shows, rural areas are more positively correlated with life satisfaction when compared to urban areas. Also, as Table 5 shows, rural and town areas show a positive effect on SWB when compared to city areas. The effect is higher in the case of rural areas followed by town areas.

But as previously mentioned, two regions have historically dominated over the urbanization and economic development trajectories of Egypt: Greater Cairo and Alexandria. Thus, we control for region dummies in our analysis. There are 5 regions for Egypt in the WVS: Greater Cairo, Alexandria, Upper Egypt, Lower Egypt, and Suez Canal. The results of this specification are shown in Table 6. As the findings show, the inclusion of the region dummies totally changes the results of the urbanization categories reported in Table 3. But when we look at the region dummies, we see that only the Lower and Upper Egypt regions show a positive effect on life satisfaction compared to the base category: Greater Cairo region. We obtained similar results when we made Alexandria the base category region. Interestingly, the Lower and Upper

[3] The sign of the trust variable is a bit unexpected as it is supposed to be positively correlated with SWB. Hence, future research could examine this unexpected relationship between trust and SWB in Egypt.

**Table 3**
OLS regressions: life satisfaction and urbanization.

| | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| Urbanization (Base case: category 8: >500,000) | | | | |
| Category 1: <2000 | 0.523 | 0.498 | 0.750** | 0.792** |
| Category 2: 2000–5000 | 1.170*** | 1.159*** | 1.373*** | 1.210*** |
| Category 3: 5000–10,000 | 0.263* | 0.252* | 0.506*** | 0.542*** |
| Category 4: 10,000–20,000 | 0.343** | 0.339** | 0.516*** | 0.522*** |
| Category 5: 20,000–50,000 | 0.203 | 0.195 | 0.372** | 0.398** |
| Category 6: 50,000–100,000 | 0.567*** | 0.571*** | 0.736*** | 0.684*** |
| Category 7: 100,000–500,000 | 0.235 | 0.229 | 0.399* | 0.236 |
| Female | | 0.194* | 0.297* | 0.307* |
| Age | | −0.00273 | 0.00246 | −0.0106 |
| Age squared | | −0.0000109 | 0.0000212 | 0.000161 |
| Marital Status (Base case: Married) | | | | |
| Divorced | | −0.0354 | 0.0601 | 0.136 |
| Separated | | −0.302 | −0.198 | −0.281 |
| Widowed | | 0.0381 | 0.189 | 0.126 |
| Single/Never married | | −0.131 | −0.234 | −0.212 |
| Income | | | 0.0969*** | 0.104*** |
| Social class (Base case: Upper class) | | | | |
| Lower class | | | 0.317 | 0.402 |
| Working class | | | 0.694 | 0.649 |
| Lower middle class | | | 0.721 | 0.654 |
| Upper middle class | | | 0.515 | 0.494 |
| Educational level (Base case: Lower level) | | | | |
| Middle level | | | 0.176 | 0.0222 |
| Upper level | | | 0.222 | 0.0983 |
| Employment status (Base case: Full time) | | | | |
| Part time | | | 0.0165 | −0.213 |
| Self-employed | | | −0.0969 | −0.0442 |
| Retired | | | −0.109 | −0.121 |
| Housewife | | | −0.0256 | −0.0214 |
| Student | | | 0.279 | 0.246 |
| Unemployed | | | −0.342 | −0.299 |
| Other | | | 0.126 | 0.2 |
| Health status | | | 0.342*** | 0.341*** |
| Freedom | | | | 0.185*** |
| Trust | | | | −0.227* |
| Importance of religion | | | | 0.469* |
| Importance of God | | | | −0.0104 |
| Year (Base case: 2001) | | | | |
| 2008 | | | | 0.243* |
| 2018 | | | | 0.438*** |
| Constant | 5.292*** | 5.349*** | 2.450*** | −0.166 |
| Observations | 6120 | 6118 | 5640 | 5585 |
| Adjusted R-sq | 0.007 | 0.007 | 0.03 | 0.071 |

All models use robust standard errors.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

**Table 4**
OLS regression: urbanization as a dichotomous variable.

| | Model |
|---|---|
| Urbanization (reference: urban) | |
| Rural | 0.314*** |
| Controls included | Yes |
| Constant | 0.187 |
| Observations | 5585 |
| Adjusted R-sq | 0.066 |

The model uses robust standard errors.
The controls included are the ones in our full specification "Model 4" in Table 3.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

**Table 5**
OLS regression: urbanization as a three categories variable.

| | Model |
|---|---|
| Urbanization (reference: urban areas) | |
| Rural areas | 0.629*** |
| Town areas | 0.449*** |
| Controls included | Yes |
| Constant | −0.183 |
| Observations | 5585 |
| Adjusted R-sq | 0.069 |

The model uses robust standard errors.
The controls included are the ones in our full specification in "Model 4" in Table 3.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

Egypt regions are the regions that delimit the rural areas of the country from which people migrate to Greater Cairo, Alexandria, and Canal regions which are predominantly urban. Therefore, it seems that the region dummies are picking up the effect of urbanization. This is not surprising since most of the urban population and economic activities of the country are concentrated in Alexandria and Greater Cairo Regions. So, the results reported in Table 6 support the findings in Tables 3, 4, and 5 and are aligned with what is reported in the literature given that

**Table 6**
OLS regression: life satisfaction and regionality.

| | Model |
|---|---|
| Urbanization (base case: category 8: >500,000) | |
| Category 1: <2000 | 0.236 |
| Category 2: 2000–5000 | 0.709*** |
| Category 3: 5000–10,000 | −0.00857 |
| Category 4: 10,000–20,000 | 0.0181 |
| Category 5: 20,000–50,000 | −0.0934 |
| Category 6: 50,000–100,000 | 0.241 |
| Category 7: 100,000–500,000 | −0.104 |
| Region (Base case: Greater Cairo) | |
| Alexandria | −0.193 |
| Lower Egypt | 0.521*** |
| Upper Egypt | 0.648*** |
| Canal | 0.168 |
| Control variables | Yes |
| Constant | −0.365 |
| Observations | 5585 |
| Adjusted R-sq | 0.075 |

The model uses robust standard errors.
The controls included are the ones in our full specification in "Model 4" in Table 3.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

urbanization and economic activities are concentrated in the Greater Cairo and Alexandria regions.

## 7. Discussion

The analysis of the data generated two main findings. First, the association between SWB and urbanization in Egypt is not congruent with the hypothesis that cities generate higher levels of SWB in developing countries. It seems that the long-assumed advantages enjoyed by urban masses in cities of many developing countries are coming into question especially with the uncontrollable growth of these cities without any concomitant expansion of public services provisioning (Cohen, 2006). In fact, it is not just that there is no expansion in the provisioning of social services and facilities, but there is a retrenchment of government subsidies and social spending because of the application of structural adjustment programs in many countries of the global South including Egypt itself (Cohen, 2006; Davis, 2004). Survival is becoming more and more difficult in many cities of the global South (Cohen, 2006), and it seems that Egypt is not an exception in this regard. Second, our results support the argument made by Morrison (2014:284) that, "the geography of well-being may not mirror the geography of growth." As shown previously, the two main urban and economic centers of the country, Greater Cairo and Alexandria regions, generate lower SWB when compared to the more rural regions, Lower and Upper Egypt regions. But what could be the main explanations for these findings?

One of the possible explanations for these intriguing results could be that there are more unhappy or unsatisfied people living in big Egyptian cities compared to smaller settlements, i.e., those who are more easily satisfied are sorted in rural and village contexts while those who are inherently less satisfied are sorted in big cities. Veenhoven (1994) argues that cities attract those already dissatisfied with their life in rural areas, and hence there could be an excess of unhappy/unsatisfied people in cities. For him, cities do not make unhappy people but attract them. Applying this logic to the Egyptian case, it could be argued that those who stayed behind and decided not to follow their peers who moved to urban centers in search of opportunities, are more content with the lives they lead. Thus, it would be important to control for personality traits to be sure that there is no sorting issue, and to determine whether it is cities per se that make people unhappy, or if those who are inherently unhappy are nested in cities. Unfortunately, the WVS dataset does not include any questions on personality traits of the respondents. Also, the cross-sectional nature of the data precludes a fixed effects analysis that would help in controlling for the fixed personality traits of the respondents.

The previous explanation raises another question about the role of expectations and aspirations in setting the levels of SWB especially among rural-urban migrants. In her study of life satisfaction in three urban towns in Sierra Leone, Peil (1984) finds that rural-urban migrants reported low satisfaction with their life overall compared to those who only experienced city life, and she attributes this to the unfulfilled high aspirations that led them to big towns in the first place. In the Chinese context, Knight and Gunatilaka (2010) reported rural-urban differences in SWB in favor of rural dwellers despite the gap in material conditions of life between the two groups. One of the explanations for the relative higher SWB of Chinese rural dwellers is their limited information sets and narrower orbits of comparison (Knight et al., 2009). On the other hand, the relative low happiness of Chinese urban dwellers is attributed to wider orbits of comparison and its affiliated high expectations characterizing urban life. So, another possible explanation of our findings could be that the narrow orbits of comparison in the village and township contexts, make people more content with their lives compared to individuals in Egyptian urban areas, where a broad spectrum of reference groups can be found, and signs of high inequality are conspicuous. Hence, the higher SWB of Egyptian rural dwellers could be a function of lower expectations and narrow orbits of comparison. On the other hand, the lower SWB of Egyptian urbanites could be a function of higher expectations and wider orbits of comparison. The problem could be exacerbated if the high expectations held by rural-urban migrants are not met or if the economic welfare of the migrants worsens. In fact, a study that utilizes the 2012 and 2018 waves of the nationally representative Egypt Labor Market Panel Survey (ELMPS) to investigate the effects of internal migration on the welfare of migrants, finds that rural-urban migration in Egypt was associated with a significant welfare loss for migrants (Hatab et al., 2022). Thus, it seems that the economic welfare gains expected by rural-urban migrants turned out to be net welfare loss. Hence, there is a need to control for the effects of moving to a city versus the effect of being in the city (Fischer, 1972). Unfortunately, the survey data used in our analysis does not include questions that would help us control for such a possible scenario. Also, the WVS data does not include information on expectations and social comparison variables, and this type of information is rarely collected in general.

Another possible explanation could be the problems or disamenities found in big cities specifically. Problems such as higher housing prices and higher cost of living in general, pollution, long commutes, and traffic congestion could account for the lower SWB reported in Egyptian urban centers. On the other hand, there might be a positive effect of the better access to nature and green spaces in rural areas on the SWB of rural dwellers (see Sørensen, 2021). Fischer (1972, 1973) has previously recommended that any investigation of the effect of urbanization/cities on satisfaction/happiness should control for the personal characteristics of individuals living in cities and problems that people usually attribute to cities. Our results showed robustness after the inclusion of personal characteristics in the regression models. Unfortunately, we cannot answer the question of whether it is the place of residence per se, or the problems associated with cities that generate dissatisfaction due to data limitations. Future research is needed in which multi-level analysis could be implemented, controlling for both personal characteristics and community-level variables, such as per capita income, crime, and housing. But there is evidence in the literature on the existence of such problems in Egyptian urban centers, especially in the Greater Cairo Region (GCR) and Alexandria. For example, urbanization and industrialization processes in the GCR were found to result in an increase in human discomfort because of the extreme local climatic changes, which leads to a hindrance to human activities (Robaa, 2011). In fact, urban growth in Egypt, especially in the GCR, happened at the expense of agricultural land (El Araby, 2002). Also, the effects of haphazard and unplanned urban growth on air quality and public health in the GCR are well documented in the literature (see El Araby, 2002). Like many mega cities around the world, Cairo is witnessing an urban flight by the wealthy who leave the city to gated enclaves located in the city's desert hinterlands (Denis, 1997; Elshestawy, 2016; Khalil, 2021; Kuppinger, 2004) and even the government is currently constructing a new, almost gated, administrative capital in the desert (Khalil, 2021). The goal of relocating government offices is a bit old and dates to the 1980s, with the main goal being the relief of traffic pressures (Sims, 2010). It seems that Cairo is no longer livable for the wealthy because of its many problems: density, eternal traffic congestion, pollution, and lack of privacy which hinder the practice of exclusive lifestyles (Kuppinger, 2004). There has been a continuous effort by the government to relocate inhabitants of Cairo to new satellite cities on its fringes, in hopes of fixing the maldistribution of the population in Cairo (Khalil, 2021; Sims, 2010; Tarbush, 2012). The main issue is that existing urban centers have entered an unfair competition over limited resources with new satellite towns that absorb many resources without achieving the goal of diverting the population of Cairo, or other urban centers. The only thing that these satellite towns and the new administrative capital have managed to do is to divert the scarce public money that could have been used in upgrading the environmental built-in of Cairo, and other existing urban centers, that suffer from increased demand on its already dilapidated public services (Khalil, 2021; Sutton & Fahmi, 2001). The question that both observers and Egyptian citizens ask is: what is the real value of a new administrative capital that would mostly house a few tens of thousands of the rich stratums but absorbs tens of billions of dollars that are badly needed to improve the provisioning of public services and facilities in urban and rural areas alike?

## 8. Takeaways for practice

There has always been a heated debate regarding the future of the great cities between modern urban planners and architects who considered old cities to be a hopeless case (e.g., Howard, 1965, Le Corbusier, [1929] 1986; see also Fishman, 1982) and those who considered modern urban planning to be very simplistic. Observers argue that modern urban planners who just care about how cities look did not try to understand the great cities and how they work because they demonized them at first sight (Jacobs, 1961; Scott, 1998). This debate and these visions about the future of cities were not confined to the borders of the Western world. It moved to other parts of the world including many countries in the global South in which modern urban planning ideas were imported. Egypt was not immune from such trend. As Sims (2010) succinctly argues, "decision makers, business elites, and even Cairo's intelligentsia seem to be engaged in a long love affair with modern technology and imported urban models, and this has been the case for decades" (p. 272). Cairo, as well as many other cities in the global South are being treated as pathological spaces "in need of salvation at the hands of Western experts" (Kanna, 2012, p.360). The Greater Cairo region is a very dense and congested metropolis where informal urban development has been, and it is still the norm. The metropolis is the home for almost a quarter of the total Egyptian population. The government's response to this reality was to launch a new towns program in the 1970s. The program entailed the construction of satellite towns in the desert hinterlands surrounding the Greater Cairo metropolis, with the primary aim is to divert population growth from the congested metropolis. Also, more recently, the Egyptian government has started the construction of other satellite towns along the desert hinterlands of many other urban centers in addition to the construction of a new administrative capital. The result of this new towns program was ghost cities and the accumulation of dead capital in the form of buildings and infrastructure in those new towns (Sims, 2010). The reasons are almost similar to what Jacobs (1961) detailed in her criticism of modernist urban planning. The Egyptian urban planners joined by the officials paid much attention to how the city should look but not how it should function. How the metropolis looks is the main priority for them. The main goal is to transform Greater Cairo into a modern city. They believe that the metropolis, in its current shape, is an embarrassment to Egypt's image. There are other political economic reasons behind the adoption of such urban planning policies including the fact that the construction sector is one of the main sectors in which the ruling elites are investing in heavily and extracting huge sums of rent from it. Arguably, part of the story is that the current ruling regime is embarking upon these mega construction projects to provide a source of rent to the main actors that guarantee the security and stability of the regime (Mandour, 2019; Sayigh, 2019). Hence, these projects could arguably be a tool used by the regime to guarantee loyalty and to consolidate support within the governing coalition[4] (Mandour, 2019; Sayigh, 2019). Regardless of the motive driving these urban planning policies, the result is that they divert scarce public finances that are supposed to be used in upgrading and improving public services in Cairo and other urban centers (Khalil, 2021; Sutton & Fahmi, 2001). The results of this study might reflect such long history of neglection of Egyptian urban centers and how such deteriorated situation is putting stress on the SWB of Egyptian urbanites. Given the resource constraints faced by many cities in the global South including Egyptian cities, priority should be given to the improvement of living conditions in existing urban centers where most of the urban growth is happening and will continue to happen.

## 9. Limitations and directions for future research

The main limitation of our analysis is related to our inability to control for the speculated explanations that we discussed above due to the data limitations of the WVS. Future research could utilize incoming survey data or could even design survey questions with specific hypotheses in mind. A good step in this direction could be the inclusion of a subjective well-being item in panel surveys such as the Egypt Labor Market Panel Survey (ELMPS). The inclusion of this item would help in conducting a panel analysis of the rural-urban SWB differential in the Egyptian context and would enable more robust causal inferences especially when it comes to the sorting issue. Future research should also look at the effect of migration on subjective well-being. Studies have found that rural-urban migration can harm SWB in developing countries such as China (Knight et al., 2009; Knight & Gunatilaka, 2010), Sierra Leone (Peil, 1984), and South Africa (Mulcahy & Kollamparambil, 2016). Also, internal rural-urban migration has been found to harm the welfare of migrants in the case of Egypt (Hatab et al., 2022). Again, a good step in this direction could be the inclusion of a SWB item in the ELMPS especially since this survey already includes a rich module on migration. The panel structure of the survey would also allow researchers to control for selective migration and sorting issues—things like personality traits or human values that might drive some people to move to cities (see Weckroth & Kemppainen, 2021).

## 10. Conclusion

Using the WVS data for Egypt, this study found a robust rural-urban difference in SWB. Rural and town dwellers were found to have higher life satisfaction compared to urbanites. These results are a bit unexpected, especially in a developing country context—the existing literature argues that SWB should be higher in cities compared to rural areas in such a context. Hence, our study provides a valuable contribution to the existing literature on the effect of urbanization on subjective well-being, especially in the context of developing countries. Our empirical results are robust and persistent even after the inclusion of personal characteristics in our models. Another interesting finding is that the two

[4] A full-fledged political economic analysis of urban planning policies in Egypt is beyond the scope of this paper and hence we leave it for future research.

main urban and economic centers of the country, the Greater Cairo and Alexandria regions, generate lower SWB when compared to the more rural regions, Lower and Upper Egypt regions. We discussed three speculated explanations for such rural-urban SWB differential. When warranted, future research using different datasets is required.

## CRediT authorship contribution statement

**Ebshoy Mikhaeil:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Adam Okulicz-Kozaryn:** Investigation, Methodology, Resources, Supervision, Validation, Writing – review & editing. **Rubia R. Valente:** Investigation, Methodology, Resources, Supervision, Validation, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## Acknowledgment

This research has benefited from the graduate summer stipend award provided by the research office at Rutgers University-Camden.

## Appendix A. Description of variables

| Variable | Description |
|---|---|
| Life satisfaction | Subjective report of life satisfaction by the respondent. Respondents are asked "All things considered, how satisfied are you with your life as a whole these days?". Responses are coded so that 1 means "completely dissatisfied", and 10 means "completely satisfied" |
| Place size (i.e., urbanization variable) | Observation of the interviewer of the size of place where the interview was conducted on a scale from 1 to 8. 1 means "under 2000 persons", and 8 means "500,000 and more persons". |
| Income | Subjective income scale from 1 to 10 where 1 means "the lowest income group", and 10 means "the highest income group. |
| Health | Subjective report of the state of health on a scale from 1 to 5. Responses are recoded so that 1 means "very poor", and 5 means "very good". |
| Female | Sex of the respondent where responses are recoded so that 1 means "female" and 0 means "male". |
| Age | Age of the respondent. |
| Age squared | Age squared. |
| Freedom | Respondents are asked "Some people feel they have completely free choice and control over their lives, while other people feel that what they do has no real effect on what happens to them. Please use this scale where 1 means "no choice at all" and 10 means "a great deal of choice" to indicate how much freedom of choice and control you feel you have over the way your life turns out". |
| Trust | Respondents are asked "Generally speaking, would you say that most people can be trusted or that you need to be very careful in dealing with people?" Reponses are recoded so that 1 means "most people can be trusted", while 0 means "need to be very careful". |
| Class | Social class (subjective). It is a 5-point scale. Responses are recoded so that 1 means "lower class", while 5 means "upper class". |
| Year | Year of the wave of the survey. |
| Education | Highest educational level attained. Three categories: Lower, Middle, and Upper. |
| Marital status | Respondents are asked about their marital status. 5 categories: married, divorced, separated, and single/never married. |
| Importance of religion | Respondents are asked to indicate how important religion is in their lives. It is a 4-point scale. Responses are recoded so that 1 means "Not at all important", while 4 means "Very important". |
| Employment status | Respondents are asked about their employment status. 8 categories: full time, part time, self-employed, retired, houswife, student, unemployed, and other. |
| Importance of God | Respondents are asked to indicate how important God is in their lives. It is a 10-point scale where 1 means "not at all important", and 10 means "very important". |

## Appendix B

| | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| Urbanization (Base case: category 8: >500,000) | | | | |
| Category 1: <2,000 | 0.209 | 0.197 | 0.303** | 0.339** |
| Category 2: 2,000 - 5,000 | 0.415*** | 0.411*** | 0.500*** | 0.475*** |
| Category 3: 5,000 - 10,000 | 0.0997* | 0.0937* | 0.196*** | 0.213*** |
| Category 4: 10,000 - 20,000 | 0.109* | 0.106* | 0.177*** | 0.187*** |
| Category 5: 20,000 - 50,000 | 0.0706 | 0.0658 | 0.139** | 0.149** |
| Category 6: 50,000 - 100,000 | 0.170** | 0.171** | 0.247*** | 0.239*** |
| Category 7: 100,000 - 500,000 | 0.0716 | 0.0692 | 0.141* | 0.104 |
| Female | | 0.0803** | 0.118** | 0.123** |
| Age | | -0.00283 | -0.00111 | -0.00495 |
| Age squared | | 0.0000106 | 0.0000213 | 0.0000624 |
| Marital Status (Base case: Married) | | | | |
| Divorced | | -0.0302 | 0.0191 | 0.0540 |
| Separated | | -0.125 | -0.0791 | -0.110 |
| Widowed | | 0.0160 | 0.0785 | 0.0565 |
| Single/Never married | | -0.0549 | -0.0814 | -0.0742 |
| Income | | | 0.0462*** | 0.0460*** |
| Social class (Base case: Upper class) | | | | |
| Lower class | | | 0.136 | 0.159 |
| Working class | | | 0.246 | 0.230 |
| Lower middle class | | | 0.292 | 0.270 |
| Upper middle class | | | 0.222 | 0.214 |
| Educational level (Base case: Lower level) | | | | |
| Middle level | | | 0.0464 | 0.000405 |
| Upper level | | | 0.0326 | -0.00136 |
| Employment status (Base case: Full time) | | | | |
| Part time | | | 0.00907 | -0.0726 |
| Self-employed | | | -0.000194 | 0.00769 |
| Retired | | | -0.00251 | -0.00239 |
| Housewife | | | -0.00723 | -0.000327 |
| Student | | | 0.0524 | 0.0407 |
| Unemployed | | | -0.133* | -0.124 |
| Other | | | 0.0418 | 0.0844 |
| Health status | | | 0.141*** | 0.142*** |
| Freedom | | | | 0.0658*** |
| Trust | | | | -0.0538 |
| Importance of religion | | | | 0.142 |
| Importance of God | | | | 0.00365 |
| Year (Base case: 2001) | | | | |
| 2008 | | | | 0.0442 |
| 2018 | | | | 0.130** |
| Observations | 6120 | 6118 | 5640 | 5585 |

* p<0.05, ** p<0.01, *** p<0.001

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### 4.5 Colombia: Unlivable but Happy? (Okulicz-Kozaryn and Martinez 2025)

Applied Research in Quality of Life
https://doi.org/10.1007/s11482-025-10518-x

# Colombia: Unlivable but Happy?

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Received: 3 January 2024 / Accepted: 17 October 2025


## Abstract
This theoretical and review-based article examines the Latin American phenomenon–the region's unexpectedly high levels of subjective well-being despite low income and persistent structural deficits. We focus on Colombia, a country often perceived as unlivable yet consistently reporting relatively high levels of subjective well-being, and contrast it with the United States, a nation widely regarded as highly livable but with comparatively lower happiness indicators. We argue that genuine happiness is possible even in contexts of limited material comfort, as conventional measures of livability often overemphasize material conditions that may not be essential for well-being once basic needs are met. Conversely, the excessive pursuit of material comfort–as exemplified in the US–may diminish happiness by fostering overwork, alienation, and depersonalized environments. While we cannot rule out the possibility that Colombia's reported happiness reflects adaptive responses to adversity, we critically assess alternative explanations and propose directions for future research.

**Keywords** Life satisfaction · Happiness · Subjective well-being · Quality of life · Livability · Best places to live · Colombia · Alienation · Degrowth

For a full earlier version see: https://theaok.github.io/docs/colQolSwb.pdf

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Published online: 24 November 2025

Traditional progress/development measures like income, production, and consumption cannot fully account for people's life experiences or emotions (Diener, 2009; Stiglitz et al., 2009; Lambert et al., 2020). While acknowledged in theory as inadequate, the predominant focus in world development is still on economy and material comfort.

Latin American countries consistently report higher levels of subjective well-being than most developed nations, despite lower average incomes and persistent structural challenges. This puzzle is recognized in the literature as the "Latin American Phenomenon" (Rojas, 2015, 2019).

Similar to other Latin American countries, Colombia reports high levels of subjective well-being (PNUD, 2023). This outcome often appears paradoxical when contrasted with the nation's enduring challenges. Poverty, inequality, violence, and informality remain widespread, and large segments of the population face precarious working conditions and limited access to social protection (Hurtado et al., 2017). Yet, despite these structural barriers, Colombians regularly report levels of subjective well-being (SWB) that are higher than would be predicted by income or security indicators (Krauss & Graham, 2013). This apparent contradiction between low livability and high reported SWB has made Colombia a case in point regarding the Latin American happiness puzzle (Graham & Lora, 2010; Velásquez, 2016). Scholars have proposed several explanations for this paradox. Research has highlighted the importance of social relationships and relational goods in Colombian society (Velásquez, 2016), the buffering role of social capital in contexts of insecurity and conflict (Wills-Herrera et al., 2011), and the influence of cultural patterns of optimism and expressiveness (Orozco & File-Muriel, 2012; Nasser, 2012; Woods & McColl, 2015). At the same time the average levels of happiness conceal marked inequalities across income groups, regions, and socio-demographic categories (Burger et al., 2021). While Colombians are happy, the distribution of happiness is uneven, and the conditions sustaining it may be fragile. In this article, we seek to contribute to the broader discussion of the Latin American happiness phenomenon by using Colombia as a study case. Colombia can be seen simultaneously as a "real paradise," where social ties and cultural resilience sustain surprisingly high levels of happiness, and as a "fool's paradise," where optimism masks deep structural deficits and inequalities. To contrast this paradox, we draw on comparisons with developed countries. While the US is often portrayed as materially prosperous but socially alienated (Frank, 2012; Manson, 2015), Colombia represents the opposite tension, a country facing significant deficits in livability yet consistently reporting high levels of subjective well-being (PNUD, 2023). This contrast allows us to situate Colombia within a larger debate on how well-being is perceived in societies with very different structural and cultural conditions.

Our study follows the classic happiness theorizing by Veenhoven (2000; 1995; 2014) and Michalos (2014); with Latin focus following Rojas (2015) and Yamamoto (2016). Further, we complement this traditional line of inquiry with rarely used perspectives in happiness research: folklore theory and Marx's alienation theory. We conclude that happiness and material underdevelopment can coexist, and even that underdevelopment may promote happiness in some ways.

The article is structured as follows. We start by documenting the Latin American phenomenon (Rojas, 2015, 2019) in Colombia: Colombians (like other Latinos) are satisfied with life ("Happy Colombia" section) despite low livability (high poverty, insecurity, inequality, etc) ("Unlivable Colombia" section). Having established an apparent paradox of high happiness despite low livability we move to the theoretical framework to understand it ("Unlivable but Happy–"Fool's Paradise"?" section): Veenhoven's 4 qualities of life (2000) are used to map happiness to livability, and Michalos 2 variable theory (Michalos, 2014) to describe the apparent contradiction as "fool's paradise," which is then explored with livability ("Livability Theory: Human Needs" section), folklore ("Folklore's Theory: Colombia's "Good Energy"" section), and Marx's alienation ("Theory of Alienation" section) theories. As a complement to fool's paradise in Colombia we turn to fool's hell (very livable but only moderately happy) in the US ("The US: Fool's Hell?" section). We finish with discussion and conclusion ("Discussion and Conclusion" section). Subjective Wellbeing (SWB)/happiness is defined in the first paragraph of "Happy Colombia" section. Livability is defined in the first paragraph of "Unlivable Colombia" section.

## Happy Colombia

Subjective Wellbeing (SWB), happiness, wellbeing, and life satisfaction are used interchangeably, and they all refer to the 4th quality of life, the satisfaction of the person (Veenhoven, 2000). Technically, we focus on evaluative or cognitive life satisfaction, as opposed to affective happiness (positive and negative emotions) or eudaimonia (flourishing and functioning). This section relies on evaluative/cognitive life satisfaction data, the only part of the article that uses survey-based happiness indicators. Further details and auxiliary points are provided in the online appendix.

The US Declaration of Independence guarantees not happiness itself, but the pursuit of happiness. In practice, some Americans succeed in this pursuit and others fail, with the result that the US tends to rank below the top quartile in international happiness comparisons (46th of 160 countries in the World Database of Happiness; see Table 1).

By contrast, the Colombian Constitution explicitly establishes happiness and quality of life as responsibilities of the state: "The general well-being and improvement of the population's quality of life are social purposes of the State."[1] Colombians also report substantially higher levels of happiness than Americans. However, existing evidence suggests that this is less an outcome of state action and more a reflection of cultural, social, and relational dynamics (Martínez, 2017; Martínez & Short, 2020). Like other Latin Americans, Colombians consistently report very high levels of life satisfaction, about 8.5 on 1-10 evaluative/cognitive life satisfaction scale, a score much higher than expected given economic, social, and institutional indicators (PNUD, 2023).

Colombia regularly appears among the happiest countries worldwide, both the World Values Survey and the World Database of Happiness rank it among the top

[1] Art 366, https://www.constituteproject.org/constitution/Colombia_2015

**Table 1** 10 happiest countries in the world

| WDH | | | WVS | | |
|---|---|---|---|---|---|
| Rank | Country | Happiness (1-10) | Rank | Country | Happiness (1-10) |
| 1 | Denmark | 8.2 | 1 | Puerto Rico | 8.4 |
| 2 | Mexico | 8.1 | 2 | Mexico | 8.3 |
| 3 | Colombia | 8.1 | 3 | Colombia | 8.3 |
| 4 | Switzerland | 8 | 4 | Qatar | 8.0 |
| 5 | Finland | 8 | 5 | Norway | 7.9 |
| 6 | Iceland | 8 | 6 | Nicaragua | 7.9 |
| 7 | Costa Rica | 7.9 | 7 | Tajikistan | 7.9 |
| 8 | Norway | 7.9 | 8 | Switzerland | 7.9 |
| 9 | Canada | 7.9 | 9 | Uzbekistan | 7.9 |
| 10 | Qatar | 7.8 | 10 | Ecuador | 7.8 |

Data from World Database of Happiness (WDH) 2010-2019 out of 160 countries at https://worlddatabaseofhappiness.eur.nl/rank-reports/satisfaction-with-life/; and World Values Surveys (WVS) 2005-2022 [waves 5-7] out of 88 countries at https://www.worldvaluessurvey.org. Technical information for WDH is at https://worlddatabaseofhappiness.eur.nl/reports/finding-reports-on-happiness-in-nations/technical-details-to-rank-reports-of-happiness-in-nations/technical-details-to-rank-report-average-happiness-in-nations-2010-2019 and WVS documentation is at https://www.worldvaluessurvey.org/WVSDocumentationWV7.jsp

three (Table 1). A World Values Survey (WVS) SWB item reads: "All things considered, how satisfied are you with your life as a whole these days?" on scale 1='dissatisfied' to 10='satisfied'. World Database of Happiness (WDH) is an aggregate of various surveys, and SWB survey items are very similar to that in WVS. Colombia is outstanding at achieving high happiness at low economic development. Colombia is happier than all other Latin countries in Table 1 and about as happy as Mexico, but Colombia is significantly poorer than Mexico, at least 25% poorer either in nominal or Purchasing Power Parity (PPP) terms.[2]

## Unlivable Colombia

Livability is the degree of fit between the environment and human needs (Veenhoven, 2014). Livability is measured in Table 2 in this section, with further discussion in "Livability Theory: Human Needs" section

In sharp contrast to high Colombian happiness, Colombia is not livable or has low objective quality of life (QOL), as measured with objective indicators in Table 2.[3]

Some of the objective indicators of quality of life from Table 2 are remarkably deficient: About a third of Colombians live on less than $5.50 a day (2019). Poverty (national benchmark) is at 42%–the whole nation has a higher poverty rate than

[2] See https://data.worldbank.org/indicator/NY.GDP.PCAP.CD and https://data.worldbank.org/indicator/NY.GDP.PCAP.PP.CD

[3] Colombia scores mediocre or low on all indicators in Table 2. Still, there are many other ways to measure QOL. UsNews, for instance, ranks Colombia 68/78 (https://www.usnews.com/news/best-countries/colombia). World Economic Forum provides indicators, too (World Economic Forum, 2017).

**Table 2** Livability/Quality Of Life (QOL): objective indicators

| Indicator | Value | Source |
|---|---|---|
| 2019 poverty (national benchmark) | 42% | https://data.worldbank.org/indicator/SI.POV.NAHC?locations=CO |
| 2011 median daily income/cap PPP USD | $7 | https://www.pewresearch.org/fact-tank/2015/09/23/seven-in-ten-people-globally-live-on-10-or-less-per-day/ |
| 2019 percent on <$5.5/day | 30% | https://data.worldbank.org/indicator/SI.POV.UMIC?locations=CO |
| 2017 R/P 10% | 40 | https://data.worldbank.org/indicator/SI.DST.10TH.10 |
| 2020 unemployment rate | 15% | https://data.worldbank.org/indicator/SL.UEM.TOTL.ZS?locations=CO |
| 2020 freedom rank | 96/210 | https://freedomhouse.org/countries/freedom-world/scores |
| 2021 corruption rank | 87/180 | https://www.transparency.org/en/cpi/2021/index/col |
| 2020 political stability, no violence/terrorism pctile | 20th | https://info.worldbank.org/governance/wgi/Home/Reports |
| 2020 rule of law pctile | 34th | https://info.worldbank.org/governance/wgi/Home/Reports |
| 2021 working conditions decile | bottom decile | https://www.globalrightsindex.org/en/2021/countries/col |
| 2018 quality of roads rank | 110/137 | https://reports.weforum.org/pdf/gci-2017-2018-scorecard/WEF_GCI_2017_2018_Scorecard_EOSQ057.pdf |
| 2021 victims of intentional homicide rank | top decile | https://dataunodc.un.org/dp-intentional-homicide-victims |
| 2023 criminality rank | 2/193 | https://ocindex.net/rankings?f=rankings&view=List |

For details on each indicator click the link under "Source" column

one of the poorest cities in the US, Camden NJ, at 36% (also national benchmark).[4] Median daily PPP per capita income in 2011 was at $7 (the US was at $56). Colombia is in the bottom decile of working conditions: there are murders and impunity, union-busting and dismissals. R/P 10% is the ratio of the average income of the richest 10% to the poorest 10%–Colombia ranks 3rd out of 70 at a remarkable 40–top decile of Colombians makes on average 40x the average of the poorest decile–even greater disparity than in the unequal US at 30. Unemployment rate is at 15%, with informal labor at about 50% of the workforce (Hurtado, 2016).

All these deficiencies–insecurity, precarious labor, poverty, and inequality are expected to result in unhappiness. Inequality in Latin America was found to have negative effects on happiness as it signals persistent unfairness (Graham & Felton, 2006)–unfairness seems to be more detrimental to happiness than inequality (Starmans et al., 2017). Inequality is a stark feature of Colombian life, and it is inequality that has sparked recent mass protests (International Crisis Group, 2021).

Colombia is still being haunted by violence and conflict, much of which is rural (Turkewitz, 2021). Colombia is less stable and more violent than 80 percent of the countries–in Table 2 metric "political stability and absence of violence and terrorism" is only at 20th percentile. Colombia is only partly free, ranking 96/210, and quite corrupt ranking 87/180. Rule of law is also problematic below about 2/3 of countries. Crime rates are high: in terms of homicides Colombia ranks in top decile, and by one

[4] https://www.census.gov/quickfacts/camdencitynewjersey

"criminality" index it ranks as the 2nd most "criminal" country in the world (after Myanmar; out of 193 countries).[5]

In terms of quality of roads Colombia ranks 110/137–part of the problem is mountains, yet, for example, equally mountainous Ecuador has relatively succeeded in road building. Transport is the blood of the society (e.g., De Vos et al. 2013)–roads are a basis for travel, commerce, and trade, especially that Colombia has no rail.

We agree with Krauss and Graham (2013); Wills-Herrera et al. (2011) that increasing happiness in Colombia should have to do with mitigating vulnerabilities and negative shocks that people face. As elaborated above there is a multitude of problems, some extreme. While, as argued later here, Colombia actually is already quite livable, there are some extreme problems like abject poverty that need to be addressed no matter how happy people may already be.

## Unlivable but Happy–"Fool's Paradise"?

In the previous two sections we have established that Colombia is happy, but unlivable. Hence, it could be labeled a "Fool's Paradise," a happy paradise, yet a fool's paradise, because it is unlivable and so there is no reason to be happy.

Now we will examine this apparent contradiction. The goal of this study is to try to explain this apparent mismatch or paradox, and spark the debate and future research.

It is instructive to start with Veenhoven's 4 qualities of life (2000) in Table 3. Life chances as an outer quality in first cell (livability of environment) should in theory correspond with life results as an inner quality in the last cell (satisfaction of the person). That is, a livable place should be happy.

Colombia's low livability should result in low satisfaction–it is unexpected for Colombia to be a happy country. Colombia scores mediocre or low on most livability indicators, but tops rankings of satisfaction. In other words, it appears to be unlivable but happy, a so called "Fool's Paradise," a place where people are subjectively happy, despite objective misery (Michalos, 2014). An intersection of livability and SWB can be visualized in a 2x2 matrix in Table 4–expected outcomes are low-low or high-high, but there can also be unexpected low-high "fool's paradise" or high-low "fool's hell."

There are a number of theories and explanations for high happiness despite low livability. The remainder of this paper is devoted to these: first two happiness theories (livability and folklore), second Marx theory of alienation, and finally a complemen-

**Table 3** Veenhoven's 4 qualities of life (2000)

| | Outer qualities | Inner qualities |
|---|---|---|
| Life-chances | Livability of environment | Life-ability of the person |
| Life-results | Utility of life | Satisfaction of the person |

[5] Although it is probably an overstatement in terms of how crime affects an average person–this "criminality" index weighs heavily organized crime networks: https://ocindex.net/report/2023/02-about-the-index.html.

**Table 4** Michalos 2 variable theory: fool's paradise and fool's hell (Michalos, 2014)

| | Low livability | High livability |
|---|---|---|
| Low SWB | Real hell [deprivation, unhappy poor] | Fool's hell [dissonance, unhappy rich] |
| High SWB | Fool's paradise [adaptation, happy poor] | Real paradise [well-being, happy rich] |

Cummins classification is shown in square brackets (Sirgy, 2002, p.61). For other examples of fool's paradise and fool's hell see Okulicz-Kozaryn and Valente (2019)

tary analysis from the opposite side (very livable but only moderately happy): fool's hell in the US.

## Livability Theory: Human Needs

Veenhoven's livability/needs theory is a major and ideally fitting happiness theory, specifically about the link between livability and SWB, as conceptualized in the last section in Tables 3 and 4 (Veenhoven & Ehrhardt, 1995; Veenhoven, 2014). Humans, like all animals, have needs, as those on the Maslow's Hierarchy of Needs (Maslow, [1954] 1987)–the more the needs are satisfied, the more happiness–places or societies that satisfy human needs well are livable:

> Societies are systems for meeting human needs, but not all societies do that job equally well. Consequently, people are not equally happy in all societies. Improvement of the fit between social institutions and human needs will result in greater happiness. (p. 3645 Veenhoven 2014)

The apparent Colombian chasm between livability and happiness may point to the limitations of livability theory. But we argue that, counter-intuitively, the livability theory may mostly hold true because: 1) Mediocre or even moderately poor development (physical and institutional infrastructure) is already good enough to satisfy most basic human needs and make a place livable; 2) Physical and institutional infrastructure mostly serves only first two steps of Maslow's pyramid (physiological and safety) (Maslow, [1954] 1987). Human physiological needs are simple and easily satisfied without much economic or institutional development. 3) Higher needs such as personal freedom and social connection that are critically important for livability are rarely properly captured by livability metrics. 4) Given always limited resources and attention, there is an opportunity cost. Excessive pursuit of money or consumption at person level, or economic growth at community or society level sacrifices non-commodities such as personal freedom and social connection notably through overwork and alienation as elaborated in later section "Theory of Alienation." This is an important mechanism that we cannot overemphasize, and it is often overlooked.

Next we provide examples of human needs that are overlooked by livability indices. These are human needs–they do count towards livability. In the examples we focus on Colombia, and also provide contrasts to the developed countries, and indicate trade-offs and opportunity costs.

Biophilia (Fromm, 1964; Wilson, 2021), a need for contact with nature is a fundamental human need, yet usually forgotten. There is a clear tradeoff between economic growth and nature abundance and preservation, for instance, the more urbanization, the less natural the human habitat. For instance, in Brazil–the more economic expansion and growth, the less Amazon rain forest. Climate change is a critical challenge for human needs as it endangers the very habitat of homo sapiens (Pachauri et al., 2014), and again, the more economic growth, the worse the environmental degradation (e.g., Klein 2014). A reasonable course of action is to de-grow the economies (Hickel, 2020; Kallis, 2011), especially the rich and carbon intensive ones such as the US. Per Colombian natural resources use and economic growth see discussion in Rubiano (2022).

Related to biophilia and climate change is biodiversity. Biodiversity improves happiness (Adjei & Agyei, 2015; Prescott & Logan, 2017). Nature is extraordinary in Colombia. Colombia has 2nd largest biodiversity after Brazil, despite being about 7x smaller in area. Colombia has just about any type of natural amenities. Exposure to nature (as opposed to urbanism) is the key ingredient for happiness (Pretty, 2012; Tesson, 2013; Thoreau, 1995 [1854]).

Social connection is a human need, and a key to happiness (Tönnies, [1887] 2002; Lane, 2000; McMahon, 2006; Putnam, 2001), and there is high degree of social connection in Colombia. Social gatherings, events and festivals are widespread, frequent and long-lasting. The high level of SWB in Latin America, including Colombia, is supported by a key SWB predictor: strong affective relationships. In Latin America, the strength of the affective relationships of its people and the ties they build in the communities are one of the primary sources of happiness–SWB in this region has social and affective foundations (Yamamoto, 2016; Rojas, 2015). High SWB is also predicted by satisfaction with family relationships and a higher frequency of positive emotions. The quantity and quality of interpersonal relationships are pivotal for the region's high levels of SWB. *Relational wealth*, which encompasses the strength and abundance of close and warm interpersonal relations with family, friends, neighbors, or colleagues, is a strong cultural characteristic in the region. On average, 85% of Latinos report having someone to count on in times of trouble. In some countries like Venezuela, Panama, Argentina, and Costa Rica, more than 90% of people report having a good social support network (Rojas, 2019). This is in contrast to developed countries where people report spending only six hours weekly with friends and family, almost half an hour less than in the previous decade (van Zanden et al., 2020).

Freedom is a human need. Colombia scores average on freedom listed earlier as a livability metric in Table 2, but that is one kind of freedom: "freedom from" (negative, objective): be no slave, live in a free country, have no coercion, be free from restrictions/impediments, lack obstacles. But there is another kind of freedom: "freedom to" (positive, subjective): be able to choose, control and direct one's own life, and be in charge. On scale 1-10, world's average is about 7; the US scores higher at 7.7, but Colombia scores higher yet at 8 (Okulicz-Kozaryn, 2014, 2015).

### Folklore's Theory: Colombia's "Good Energy"

Folklore theory is an attractive explanation of fool's paradise (Table 4). It is a theory put forth by Veenhoven as a competing explanation to his livability theory as discussed in the last section. The folklore theory states that happiness is a product of culture–tradition, national character and widely held notions about life determine one's happiness (Veenhoven & Ehrhardt, 1995). Happiness is the reflection of broadly held perceptions about life which are rooted in traditions and the culture of a society (Veenhoven & Ehrhardt, 1995). If a culture has an optimistic outlook on life regardless of circumstances, future generations will remain happy. Thus, a society may be happy, regardless of the socioeconomic situation, because of cultural influences (Veenhoven & Ehrhardt, 1995). In other words, one can wear so called "pink glasses" and be happy no matter the circumstances (or see through dark lenses, have a bleak outlook and be unhappy no matter the circumstances). This is in sharp contrast to livability theory, where happiness is a result of a person's experience, satisfaction of her needs.

A Peruvian social psychologist specializing in happiness argues that the origins of Latin happiness can be traced to:

> the minimalist well-being lessons of Andean and Amazonian small traditional communities which constitute the grounds of Latin American happiness, a life style that mimics the ancestral environment, the deep nature where the happiness brain wiring occurred; a physical and social environment that naturally activates the brain pleasure circuits. Culture resembles evolutionary needs; resources to achieve needs are available for everyone; positive, interdependent collectivistic interaction is ingrained in behavior, supporting, working, competing, and sharing. (Yamamoto 2016, p.45)

Then Colombian happiness is due to tradition and culture (folklore theory), but notably at the same time happiness stems from satisfying one's needs (livability theory) as tradition and culture fit human needs. This is a key point: while in principle folklore theory is in contrast to livability theory (Veenhoven & Ehrhardt, 1995), in practice, in specific cases, the two theories do not have to be contradictory. If tradition and culture help satisfy human needs, as is the case in Latin America, the two theories are not contradictory.

There is an indigenous concept of "Buen Vivir" (Good Living) (Hidalgo-Capitán & Cubillo-Guevara, 2017), similar to Aristotelian "Eudamonia"–both emphasize harmony and community. Buen Vivir is also about environment and food sovereignty, and it arguably contributes to SWB in Colombia, as it does in Ecuador (Guardiola & García-Quero, 2014). Notably, Buen Vivir helps to explain an apparent fool's paradise–economic poverty is relative–it depends on a specific way of life. For instance, households that grow their own food and are in an indigenous community depend less on money to be happy (García-Quero & Guardiola, 2018). Likewise, kibbutzniks (Morawetz et al., 1977) and Amish (Surowiecki, 2005) are able to be happy despite being poor. Francia Marquez, an indigenous Colombian vice president, proposed a similar concept to Buen Vivir, "Vivir Sabroso," to live in harmony with nature, traditions, and community: "vivir sabroso no es vivir con plata, vivir sabroso es vivir sin miedos" ("living joyfully is not living with money, living joyfully is living without fear.")[6]

[6] https://www.youtube.com/watch?v=AtpExO2e_0M

The folklore theory is about national disposition/trait/character. It does appear that Colombians have a slow-paced familial/social cheerful/happy disposition, which is conducive for happiness. Colombian happiness is arguably real, however, rather than just being due to cognitive cultural norms. And again, it is not just due to culture on its own, rather due to culture that is aligned with human needs and satisfies them, so that the environment is livable (livability theory).

Colombians celebrate over 3,000 festivals or carnivals in small towns or large cities and have about 20 holidays per year. Folklore and carnivals are part of the festive character of the country. The government uses them as tools to promote reconciliation and build social fabric and identity (Gutiérrez & Cunin, 2006). In Colombia, following a robust pattern in Latin America, collectivist values are ingrained in the culture (Mensing, 2002). Family is at the center of collective values, close friends follow, and in the exterior layer are neighbors and the community. This interdependent collectivism is at the core of Latin American happiness (Yamamoto, 2016).

Proper treatment of the folklore theory is left for future research as authors' anthropological and historical expertise is limited–but we discuss below popular explanations for Colombia's happiness–future research can test them properly and systematically.

According to accounts of foreigners living in Colombia, Colombian happiness stems from: family, friends, and optimism ; having less entitlement and appreciating what one has, having joy in small things; not worrying and not expecting much ("tranquilo," "no importa") (Bargent, 2016).

Bargent (2016) wonders further that in Colombia emotions change seamlessly and effortlessly between shame and pride, despair and hope, sorrow and happiness–but shouldn't they? Isn't being natural and simple a desirable quality? Surface acting (faking emotions that are deemed appropriate) is emotionally draining (Brooks, 2022).

Wallace (2017) offers many illustrative quotes by Colombians and about Colombians:

> Money is nice but it's not the most important thing. In general we are a culture that values what you have. [...] Colombians are innocent. They're curious. [...] Colombians have become indifferent to situations of war. In other words, if the problem does not touch me directly, I must feel grateful, satisfied, optimistic, lucky. [...] Colombians have always demonstrated incredible, Herculean and powerful resilience to war, death and to the harsh history of violence and diplomatic failures [...] Colombians feed this resilience through human connections and the communal experience. [...] We live for parties, holidays [...] The dance frees you. It is a way of expression and feeling. Here the music is carried in the blood, in the veins, in our heart. It's a great passion we carry throughout our lives.

### Theory of Alienation

From a Western perspective, Colombians appear remarkably connected. Indeed, there is a striking contrast between the high levels of social integration observed in

Latin America (Rojas, 2015; Woods & McColl, 2015) and the patterns of alienation and estrangement often described in Western societies (Scitovsky, 1976; Lane, 2000; Wilkinson & Pickett, 2010; Duany et al., 2001; Freud et al., 1930). This contrast resonates with Marx's theory of alienation. Marx defined alienation as the transformation of people's labor into a force that comes to dominate them. It implies separation from the conditions of meaningful agency, typically occurring when individuals lack ownership of the means of production (Horowitz, 2022). The overall alienation consists of alienation from: the product of labor, the activity of labor, one's own specific humanity, and others/society.[7]

In the Colombian context, there is often described a sense of "good energy" and "vitality" (Bargent, 2016; Wallace, 2017) that contrasts with the alienation documented in the West (Freud et al., 1930). It can be argued that the United States and other affluent societies have sacrificed elements of humanity in the pursuit of economic success. Latin America, by contrast, has retained higher levels of reported happiness but has not achieved the same levels of economic power. Fischer (1973, p.233) observed that urban residents pay "the emotional price for economic wellbeing," a dynamic that seems applicable to Americans, and other wealthy societies. Colombians' attitude and approach to life is spontaneous and optimistic –similar to what Marcuse and Fromm advocated (and what has been lost in the West) (Marcuse, 2015; Fromm, 2013, 2012, 1964, [1941] 1994), also reminiscent of Nietzsche's ideal of a child–curious, spontaneous, creative, and innocent (Nietzsche, 1896). It is present time orientation–not living in the past or worrying about future–in contrast to the West, where anxiety and materialism are widespread (Scitovsky, 1976; Lane, 2000; Wilkinson & Pickett, 2010; Duany et al., 2001; Freud et al., 1930).[8]

The US way of life has been characterized as fast paced and oriented toward material achievement (Easterlin, 1973). Busyness–being constantly occupied with work–is socially valued (Gershuny, 2005; Musk, 2018). Yet this lifestyle is also associated with stress, anxiety, and alienation, extending beyond the workplace. Cultural pressures to "keep up with the Joneses" and to strive for perfection amplify these stresses (Frank, 2012; Manson, 2015). This widespread pursuit of perfection and excellence creates competitive "arms race," generating cycles of stress, anxiety, shame, and guilt (Frank, 2012; Manson, 2015).

In highly capitalistic societies such as the US, social relationships also tend to be instrumental and oriented toward business rather than deep personal connection (Horowitz, 2022; Okulicz-Kozaryn, 2020). By contrast, Colombians are often described as spontaneous, closer to human nature, and more authentic in their interactions (Orozco & File-Muriel, 2012; Nasser, 2012; Woods & McColl, 2015). Evidence from Cali, Colombia's third largest city, reinforces this contrast. Annual surveys conducted since 2014 show that family and personal relationships are the most important sources of subjective well-being, while politics, corruption, and other adverse external circumstances are rarely mentioned in self-assessments of life satisfaction

[7] For elaboration see Horowitz (2022) and https://www.marxists.org/subject/alienation.

[8] On the other hand, there may be elements of culture of poverty (Banfield, 1967, 1974) in Latin America. Yet, some culture of poverty may counter-intuitively be desirable in wealthy societies such as the US–see a novel point of view in Manson (2015).

(Martínez, 2017; Martínez & Short, 2020). This helps explain why poor performance on objective livability indicators does not necessarily translate into lower subjective well-being.

#### The US: Fool's Hell?

The US, the very richest (per capita) country in the world (excluding small countries such as Norway), ranks 46/160 in the World Database of Happiness, not even ranking in the top happiness quartile. Recent evidence underscores the paradoxical nature of American prosperity. While the US economy has grown faster than nearly all other high-income countries since the 1990s, performance on most well-being metrics has deteriorated (Leonhardt & Wu, 2025). This resonates with the alienation framework, where the pursuit of economic dominance may have come at the expense of meaningful social integration and wellbeing.

One would imagine that the US must have top income mobility in the world or at least somewhere near the top. But many other countries have better income mobility, e.g., Norway, Denmark, and Finland (Corak, 2004, 2011, 2013; Economist, 2013, 2012b, a). In terms of mobility, the US is somewhat like a Fool's Paradise–people and especially immigrants think it is a paradise–you work hard, and you go to the top, but it is actually easier in other countries.

In the US, pursuit of money and pursuit of happiness are about the same thing (Easterlin, 1973). But we know that a lot of money does not buy much happiness, and if anything, excessive pursuit of it, such as that in the US, may actually decrease happiness (Kasser, 2016; Dittmar et al., 2014; Brown & Kasser, 2005; Kasser, 2003; Schmuck et al., 2000; Kasser & Ryan, 1993; Leonard, 2010).

### Discussion and Conclusion

The origin of the study has been the apparent paradox in the data of high happiness despite low livability. Conceptually and theoretically the article has followed and built on Veenhoven's 4 qualities of life (2000) and Michalos 2 variable theory (Michalos, 2014), livability and folklore theories (Veenhoven & Ehrhardt, 1995; Veenhoven, 2014), and Marx's theory of alienation. Our contribution lies in combining these conceptual approaches with Latin American evidence (Rojas, 2015; Hurtado, 2016; Velásquez, 2016; Yamamoto, 2016) to highlight the coexistence of resilience and vulnerability in Latin American context, using Colombia as a case of study. We also combine conceptual and theoretical approaches to offer new insights about high happiness despite apparently low livability in Colombia.

Is Colombia "unlivable but happy," a so called "fool's paradise"? The evidence suggests that this label only captures part of the story. Colombia is characterized by severe deficits–poverty, inequality, insecurity, weak institutions–yet Colombians consistently report high levels of life satisfaction (PNUD, 2023). Explanations include strong social and family networks, cultural orientations toward optimism and expressiveness, and adaptive responses to adversity (e.g., Roos 2019; Wallace 2017; Davies 2022; Woods and McColl 2015).

These factors buffer the negative effects of structural deficits, but they do not eliminate them. A key lesson is that traditional metrics of livability and progress–income, infrastructure, material comfort–capture only part of what matters for subjective well-being. Relational wealth, positive freedom, and cultural dispositions also play central roles. At the same time, average levels conceal inequalities across regions, income groups, and social categories (Burger et al., 2021).

In comparative perspective, the contrast with the United States illustrates the paradox. The US is materially prosperous yet often described as socially alienated, with happiness outcomes that are modest relative to its wealth. Colombia, by contrast, is materially constrained yet socially rich. This juxtaposition underscores that material progress is not linearly related to subjective well-being, and that excessive pursuit of growth and consumption may undermine rather than enhance quality of life. The broader conclusion is that Colombia is not simply an anomaly but part of a wider Latin American phenomenon of "high happiness at low income." Colombia shows that happiness and underdevelopment can coexist, and that social connection and cultural resilience can compensate for deficits in material conditions. Yet Colombia also illustrates the limits of this paradox–structural inequalities, persistent insecurity, and fragile institutions constrain the sustainability of well-being.

## Takeaway for Policy and Practice

The world has much to learn from the high subjective well-being in regions where economic growth has lagged, yet social relationships, cultural resilience, and adaptive capacities continue to sustain well-being. Our analysis suggests two sets of implications. First, for Colombia and Latin America, sustaining well-being requires addressing vulnerabilities that optimism and social ties alone cannot resolve. Reducing inequality, improving working conditions, and strengthening institutions remain central to the policy agenda (Burger et al., 2021). High subjective well-being in the country should not obscure the urgency of structural reforms.

Second, for developed countries such as the United States, the Colombian case illustrates the importance of relational goods, social connection, and positive freedom–dimensions often neglected in highly individualistic, consumption-oriented societies. Western societies are marked by materialism, consumerism, and the pursuit of status and wealth (Leonard, 2010; Frank, 2012). Although it is widely recognized that material comfort is not sufficient for happiness (Stiglitz et al., 2009), and that spending on experiences and social connection contributes more to well-being than spending on material possessions (Ware, 2012; Van Boven, 2005; Kumar et al., 2014; Bhattacharjee & Mogilner, 2014), the dominant pursuit of economic growth and consumption continues. Some scholars even argue that what may be needed in the West is economic degrowth rather than further expansion (Kallis et al., 2012; Kallis, 2011). The Western consumerist orientation contrasts with–and arguably comes at the expense of–the social connection, positive freedom, and joy often highlighted in Latin America (Orozco & File-Muriel, 2012; Nasser, 2012; Woods & McColl, 2015). However, these lessons must be applied cautiously. Cultural dispositions cannot be transplanted, and resilience in the face of adversity should not be romanticized as a substitute for material security and institutional quality. Colombia's paradox lies not in being a "fool's paradise," but in showing both the possibilities and limitations of happiness under adversity. It reminds us that subjective well-being is multidimensional, shaped as much by relationships and cultural orientations as by material comfort, and that both resilience and vulnerability must be recognized in order to design policies that foster sustainable human flourishing.

**Supplementary Information** The online version contains supplementary material available at https://doi.org/10.1007/s11482-025-10518-x.

**Author Contributions** AOK: Conceptualization, Methodology, Software, Writing-Original draft preparation, Visualization, Investigation, Supervision; LM: Writing- Reviewing and Editing

**Funding** Rutgers Camden Chancellor's Interdisciplinary Research Collaboration grant. Lina Martinez' research has received funding from the European Union's Horizon 2020 research and innovation program under grant agreement No 101094546. Views and opinions expressed are, however, those of the authors only and do not necessarily reflect those of the European Union. Neither the European Union nor the granting authority can be held responsible for them.

## Declarations

**Ethical approval** Complied

**Conflicts of interest** None declared

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